



वार्षिक प्रतिवेदन ANNUAL REPORT 2021



भाकृअनुप-राष्ट्रीय कृषि अनुसंधान प्रबंध अकादमी

राजेन्द्रनगर, हैदराबाद-500 030, तेलंगाना, भारत

ICAR-National Academy of Agricultural Research Management

Rajendranagar, Hyderabad-500 030, Telangana, India



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ABBREVIATIONS

AAU	Assam Agricultural University
ABF	Agri-Biotech Foundation
ABI	Agri-Business Incubator
ABS	Access and Benefit Sharing
ACP	Awareness Creation Program
ADG	Additional Director General
AERA	Agricultural Economics Research Association
AESSRA	Agricultural Economics and Social Science Research Association
AGTRI	Advances in Genomics Tools for Rice Improvement
AI	Artificial Intelligence
AICRP	All-India Coordinated Research Project
AICTE	All India Council for Technical Education
a-IDEA	Association for Innovation Development of Entrepreneurship in Agriculture
AIEEA	All India Entrance Examination for Admission
AIMA	All India Management Association
ALTENA	Asian Long Term Experimental Network for Agriculture
AMARA	Association for Management of Agricultural Research and Agripreneurship
ANGRAU	Acharya N. G. Ranga Agricultural University
ANRCM	Academy of Natural Resource Conservation and Management
APAR	Annual Performance Appraisal System
APMC	Agricultural Produce Market Committee
ARGM	Annual Rice Group Meeting
ARS	Agricultural Research Service
ASCI	Administrative Staff College of India
ASC	Agricultural Science Congress
ASEAN	Association of South East Asian Nations
ASRB	Agricultural Scientists Recruitment Board
ASTI	Agricultural Science and Technology Indicators
ATARI	Agricultural Technology Application Research Institute
AUs	Agricultural Universities
BARC	Bangladesh Agricultural Research Council
BASIX	Bhartiya Samruddhi Investments and Consulting Services Ltd.
BASU	Bihar Animal Sciences University
BAU	Bihar Agricultural University
BCIL	Biotechnology Consortium India Limited
BIOFIN	Biodiversity Finance Initiative

BIRAC	Biotechnology Industry Research Assistance Council
CAAST	Centre of Advanced Agricultural Science and Technology
CAI	Centre for Agri-Innovation
CAT	Common Admission Test
CAU	Central Agricultural University
CBI	Central Bureau of Investigation
CCS NIAM	Ch. Charan Singh National Institute of Agricultural Marketing
CCSHAU	Chaudhary Charan Singh Haryana Agricultural University
CDC	Career Development Centre
CDVL	Centre for Distance and Virtual Learning
CGIAR	Consultative Group for International Agricultural Research
CHRD	Center for Human Resource Development
CIFA	Central Institute of Freshwater Aquaculture
CIFE	Central Institute of Fisheries Education
CIFT	Central Institute of Fisheries Technology
CIIE	Centre for Innovation Incubation and Entrepreneurship
CIL	Coromandel International Limited
CIMMYT	International Maize and Wheat Improvement Center
CIRAD	The French Agricultural Research Centre for International Development
CIWA	Central Institute for Women in Agriculture
CJSC	Central Joint Staff Council
CMAT	Common Management Admission Test
CMFRI	Central Marine Fisheries Research Institute
COLLAge	Centre for Open and Lifelong Learning in Agriculture
CPI	Consumer Price Index
CRD	Completely Randomized Design
CRIDA	Central Research Institute for Dry land Agriculture
CRIJAF	Central Research Institute for Jute and Allied Fibers
CSAUA&T	Chandra Shekhar Azad University of Agriculture & Technology
CSIR	Council of Scientific & Industrial Research
CTCRI	Central Tuber Crops Research Institute
CTRI	Central Tobacco Research Institute
DAATTCs	District Agricultural Advisory and Transfer of Technology Centre
DAC&FW	Department of Agricultural Cooperation and Farmers Welfare
DFID	Department for International Development
DARS	Dryland Agriculture Research Station
DBT	Department of Biotechnology
DC	Data Centre
DCB	Development Credit Bank



DDG	Deputy Director General
DGR	Directorate of Groundnut Research
DIEO	Division for International and Educational Outreach
DKMA	Directorate of Knowledge Management in Agriculture
DRC	Disaster Recovery Centre
DST	Department of Science and Technology
DT	Design Thinking
DU	Delhi University
DWRP	Developing Winning Research Proposal
EDP	Executive Development Programme
EI	Extension Education Institute
e-NAM	National Agriculture Market
ENCORE	Enhancing Coastal And Ocean Resource Efficiency
EoDR	Ease of Doing Research
EP	Emeritus Professor
ERP	Enterprise Resource Planning
ES	Emeritus Scientist
ET	Education Technology
FADC	Food and Agriculture Division Council
FAI	Fertilizer Association of India
FAO	Food & Agriculture Organization
FCRI	Fisheries College and Research Institute
FDA	Food and Drug Administration
FDC	Faculty Development Center
FDP	Faculty Development Programme
FET	Field Experience Training
FICCI	Federation of Indian Chamber of Commerce and Industry
FLIs	Farmer Led Innovations
FOCARS	Foundation Course for Agricultural Research Service
FPC	Farmer Producer Company
FPOs	Farmer Producer Organizations
FSRC	Financial Services Regulatory Commission
FSSAI	Food Safety and Standards Authority of India
FTAPCCI	Federation of Telangana and Andhra Pradesh Chambers of Commerce and Industry
FTAs	Farm Tele Advisors
FTF-ITT	Feed the Future - India Triangular Training
FYM	Farm Yard Manure
GADVASU	Guru Angad Dev Veterinary and Animal Sciences University
GBPUA&T	Govind Ballabh Pant University of Agriculture and Technology

GDD	Growing Degree Days
GEF	Global Environment Facility
GHGs	Green House Gases
GHMC	Greater Hyderabad Municipal Corporation
GIS	Geographic Information System
GKVK	Gandhi Krishi Vigyana Kendra
Gol	Government of India
GPS	Global Positioning System
GST	Goods and Services Tax
HEIs	Higher Education Institutions
HICC	Hyderabad International Convention Centre
HRD	Human Resource Development
IARI	Indian Agricultural Research Institute
IAS	Indian Administrative Service
IASRI	Indian Agricultural Statistics Research Institute
ICAR	Indian Council of Agricultural Research
ICM	Institute of Cooperative Management
ICMR	Indian Council of Medical Research
ICRISAT	International Crops Research Institute for Semi-Arid Tropics
ICT	Information and Communication Technology
ICTA-A	Climate Action Transparency Project on Adaptation
ICZM	Integrated Coastal Zone Management
IDDC	International Dryland Development Commission
IDP	Institutional Development Plan
IFPRI	International Food Policy Research Institute
IFS	Indian Forest Service
IGFRI	Indian Grassland and Fodder Research Institute
IGH	International Guest House
IGKV	Indira Gandhi Krishi Vishwavidyalaya
IGNFA	Indira Gandhi National Forest Academy
IIIT	Indian Institute of Information Technology
IIM	Indian Institute of Management
IIMR	Indian Institute of Millets Research
IIOPR	Indian Institute of Oil Palm Research
IIOR	Indian Institute of Oilseeds Research
IIP	Indian Institute of Packaging
IIRR	Indian Institute of Rice Research
IISER	Indian Institute of Science Education and Research
IISS	International Institute for Strategic Studies



IIT	Indian Institute of Technology
IIITDM	Indian Institute of Information Technology Design and Manufacturing
IIVR	Indian Institute of Vegetable Research
IJSC	Institute Joint Staff Council
ILS	Institute of Live Sciences
IMC	Institute Management Committee
IoT	Internet of Things (IoT)
IPE	Institute of Public Enterprise
IPR	Intellectual Property Rights
IPTM	Intellectual Property & Technology Management
IRC	Institute Research Council
IRGS	Institute of Research and Graduate Studies
IRM	Insect Resistance Management
ISAP	Indian Society of Agribusiness Professionals
ISAPM	Indian Society of Animal Production and Management
ISB	Indian School of Business
ISO	International Organization for Standardization
IT	Information Technology
ITK	Indigenous Technical Knowledge
ITMC	Institute Technology Management Committee
ITMU	Institute Technology Management Unit
IWELA	Indian Women Excellence and Leadership Award
JAC	Joint Advisory Council
JAU	Junagadh Agricultural University
JNTU	Jawaharlal Nehru Technological University
JNU	Jawaharlal Nehru University
KAU	Kerala Agricultural University
KCCs	Kisan Call Centers
KJWA	Koronivia Joint Work on Agriculture
KPIs	Key Performance Indicators
KVKs	Krishi Vigyan Kendras
LMS	Learning Management System
LSD	Latin Square Design
MABIF	Madurai Agribusiness Incubation Forum
MAFSU	Maharashtra Animal and Fishery Sciences University
MANAGE	National Institute of Agricultural Extension Management
MDP	Management Development Programme
MIDH	Mission for Integrated Development of Horticulture
MLA	Member of Legislative Assembly

MoEFCC	Ministry of Environment, Forest and Climate Change
MOOC	Massive Open Online Courses
MORE	Marketing Officers' Resurgence & Excellence
MoU	Memorandum of Understanding
MPUA&T	Maharana Pratap University of Agriculture and Technology
MVIRDC	M. Visvesvaraya Industrial Research and Development Centre
NAARM	National Academy of Agricultural Research Management
NAAS	National Academy of Agricultural Sciences
NABARD	National Bank for Agricultural and Rural Development
NADMP	National Agriculture Disaster Management Plan
NAEP	National Assessment of Educational Progress
NAHEP	National Agricultural Higher Education Project
NAIF	National Agricultural Innovation Found
NAPCC	National Action Plan on Climate Change
NARC	Nepal Agricultural Research Council
NARES	National Agricultural Research and Education System
NARS	National Agricultural Research System
NASF	National Agricultural Science Fund
NASI	National Academy of Sciences, India
NBA	National Biodiversity Authority
NBA	National Board of Accreditation
NBFGR	National Bureau of Fish Genetic Resources
NBPGR	National Bureau of Plant Genetic Resources
NBSSLUP	National Bureau of Soil Survey and Land Use Planning
NCCT	National Council for Cooperative Training
NCSCM	National Centre for Sustainable Coastal Management
NDDI	Normalized Difference Drought Index
NDRI	National Dairy Research Institute
NDVI	Normalized Difference Vegetation Index
NDWI	Normalized Difference Water Index
NERC	North Eastern Regional Centre
NET	National Eligibility Test
NF	National Fellow
NFDB	National Fisheries Development Board
NGOS	Non-Government Organizations
NIAM	National Institute of Agricultural Marketing
NIAP	National Institute of Agricultural Economics and Policy Research
NIDHI PRAYAS	National Initiative for Developing and Harnessing Innovations - Promoting and Accelerating Young and Aspiring technology entrepreneurs
NIF	National Innovation Foundation



NIOT	National Institute of Ocean Technology
NIPHM	National Institute of Plant Health Management
NIRD &PR	National Institute of Rural Development and Panchayat Raj
NITI	National Institution for Transforming India
NMSA	National Mission on Sustainable Agriculture
NP	National Professor
NPTEL	National Programme on Technology Enhanced Learning
NRC	National Register of Citizens
NRCB	National Research Centre for Banana
NRCC	National Research Centre on Camel
NRCM	Natural Resource Conservation and Management
NRM	Natural Resource Management
NRRI	National Rice Research Institute
NSTEDB	National Science & Technology Entrepreneurship Development Board
NSTMIS	National Science and Technology Management Information System
NTC	National Training Course
OPAC	Online Public Access Catalog
OPD	Outpatient Department
OUAT	Odisha University of Agriculture and Technology
PARC	Pakistan Agricultural Research Council
PAU	Punjab Agricultural University
PCK	Pedagogical Content Knowledge
PCT	Patent Cooperation Treaty
PDKV	Panjabrao Deshmukh Krishi Vidyapeeth
PDP	Project Directorate on Poultry
PGDETM	Post Graduate Diploma in Education Technology Management
PGDMA	Post Graduate Diploma in Management-Agriculture
PGDM-ABM	Post Graduate Diploma in Agribusiness Management
PGDTMA	Post Graduate Diploma in Technology Management in Agriculture
PJTSAU	Professor Jayashankar Telangana State Agricultural University
PME	Priority Setting, Monitoring and Evaluation
PPV&FRA	Protection of Plant Varieties & Farmer Rights Authority
PRA	Participatory Rural Appraisal
PVNRTVU	PV Narasimha Rao Telangana Veterinary University
QRT	Quinquennial Review Team
RAC	Research Advisory Committee
RAG	Regional Advisory Group
RAWE	Rural Agricultural Work Experience
ReMS	Rashtriya e-Market System

RKVY	Rashtriya Krishi Vikas Yojana
RMP	Risk Management Plan
RSMs	Retail Sales Managers
RTE	Ready to Eat
RVSKVV	Rajmata Vijayaraja Scindia Krishi Vishwa Vidyalaya
SAARC	South Asian Association for Regional Cooperation
SAMETI	State Agriculture Management & Extension Training Institute
SAS	Statistical Analysis System
SAUs	State Agricultural Universities
SBSTA	Subsidiary Body for Scientific and Technological Advice
SDGs	Sustainable Development Goals
SHUATS	Sam Higginbottom University of Agriculture, Technology and Sciences
SKLTSHU	Sri Konda Laxman Telangana State Horticultural University
SKNAU	Sri Karan Narendra Agriculture University
SKUAST	Sher-e-Kashmir University of Agriculture Sciences and Technology
SLAs	Students' Learning Approaches
SLTP	State Level Technical Programme
SMD	Subject Matter Division
SNRM	Strengthening Natural Resource Management
SOC	Senior Officers Committee
SSS	Skilled Support Staff
STEM	Society for Technology Management
SVVU	Sri Venkateswara Veterinary University
TAAS	Trust for Advancement of Agricultural Sciences
TAFE	Tractors and Farm Equipment
TANUVAS	Tamil Nadu Veterinary and Animal Sciences University
TARCMES	Technological Approaches for Resource Conservation and Management for Environmental Sustainability
TBI	Technology Business Incubator
TEFR	Techno-Economic Feasibility Report
TEL	Technology Enhanced Learning
TELAge	Technology Enhanced Learning in Agricultural Education
TERI	The Energy and Resources Institute



TISS	Tata Institute of Social Sciences
TLSOE	Teaching Learning Strategies of Online Education
TNAU	Tamil Nadu Agricultural University
TNAU UIC	Tamil Nadu Agricultural University University Innovation Cluster
TNFU	Tamil Nadu Fisheries University
TNJFU	Tamil Nadu Dr. J. Jayalalithaa Fisheries University
ToT	Training of Trainers
TPACK	Technology Pedagogy and Content Knowledge
TQIMC	Training Quality Improvement Measures Committee
TRIPS	Trade-Related aspects of Intellectual Property Rights
UAS	University of Agricultural Sciences
UASR	University of Agricultural Sciences, Raichur
UBKV	Uttar Banga Krishi Viswavidyalaya
UK	United Kingdom
UNDP	United Nations Development Programme
UNFCCC	United Nations Framework Convention on Climate Change
UoH	University of Hyderabad
USA	United States of America
USAID	U.S. Agency for International Development
VCD	Video Compact Disc
VNMKV	Vasantrao Naik Marathwada Agricultural University
WDP	Wheat Development Programme
WSSV	White Spot Syndrome Virus
WTO	World Trade Organization
ZBNF	Zero Budget Natural Farming
ZTMC	Zonal Agro-Technology Management Center
ZTMU	Zonal Technology Management Unit



Dr. Ch. Srinivasa Rao

Director

I am delighted to be a part of NAARM, and pen this foreword for the annual report-2021. NAARM is the premier institute of excellence in agricultural research management in the country. Despite the challenges posed due to COVID pandemic, NAARM has been successful in converting these challenges into opportunities by conducting online programmes, and made tremendous contribution to the National Agricultural Research and Education System (NARES). With the mantra of 'Reaching the Unreached', NAARM successfully conducted 95 programmes reaching 10,131 participants through various online digital platforms. These include two leadership development programmes, 54 Need-Based Programmes, 13 workshops/conferences/webinars, 24 off-campus programmes and two MOOCS programmes. Additional 15 Sensitization programmes on Entrepreneurship Development were organized for 2411 students of 13 agricultural universities including Central Agricultural University, Imphal and Sher-e-Kashmir University of Agricultural University, Jammu towards building start-up ecosystem. The Academy organized an Induction Course on Project Management and Research Methodology in virtual mode for 30 newly recruited Scientists of Indian Council of Forestry Research and Education (ICFRE), Dehradun.

Agribusiness Management is another important academic activity of NAARM. It gives me immense pleasure to share that all the 48 students of PGDM-

ABM 2019-21 (11th batch) were placed successfully in 20 reputed agribusiness companies. New batches of Diploma in Educational Technology and Management (DETM) with 32 candidates, and Diploma in Technology Management in Agriculture (DTMA) with 29 candidates were started during 2021.

a-IDEA (Association for Innovation Development of Entrepreneurship in Agriculture), the Technology Business Incubator of ICAR-NAARM organized the fourth edition of AGRI UDAAN, India's most sought after food and agriculture accelerator programme. Dr. G.R. Chintala, Chairman, NABARD launched the programme wherein 700 participated in the virtual mode.

It gives me immense pleasure to share that the year 2021 began with the valedictory programme of 111th batch of FOCARS who completed their training on Jan 02, 2021. During the year, the NAARM also witnessed several new initiatives including collaborations with VAMNICOM, Pune; NCDEX, Mumbai and empanelment of eight IT companies.

The high-quality scientific research of our colleagues has been recognized in terms of peer-reviewed policy papers besides 70 research papers in reputed national and international journals with high impact factors. We have also published eight books (authored/edited), 32 book chapters including seven chapters in Hindi, 16 technical reports/bulletins, 15 popular articles and nine conference papers. Kudos to our faculty members who bagged seven copyrights this year.

Many of our faculty have been recognized nationally and internationally for their outstanding

research work. Dr. P. Krishnan is elected as Fellow of NAAS; Dr. MB Dastagiri was awarded “ISDS Fellow” by the International Society for Development and Sustainability (ISDS), Japan; and Dr. Surya Rathore bagged the team award for “Most Impactful Training Delivery” during the Teach-back of GIZ, to name few. I would like to extend my heartiest congratulations to all the scientists and staff members who have contributed immensely to the growth of the Academy and Farming fraternity.

During the year, the Academy added few more infrastructure facilities in its green campus.

These are Agro-biodiversity Park; Herbal Garden, Common Reception Centre and Maintenance Office (Civil) office. The Academy conducted virtual study tours for 760 students from different colleges of state agricultural universities during 2021. I am sure that Team NAARM will work continuously towards improving the capacity building and competency enhancement of human resources of NARES, in turn enhancing livelihood of farmers of the country, and all other stakeholders with the same commitment and zeal.

Date: 28.02.2022

Place: Hyderabad

A handwritten signature in black ink, appearing to read 'Ch. Srinivasa Rao', is positioned above the printed name.

(Ch. Srinivasa Rao)

Director

EXECUTIVE SUMMARY

National Academy of Agricultural Research Management is ICAR's nodal Institution for capacity building, to improve the capability of professionals at all levels of India's National Agricultural Research and Education Systems (NARES) and to serve as a 'Think Tank.' Despite the challenges of COVID-19 pandemic, the Academy made substantial contributions to NARES and its stakeholders such as Agricultural Universities, Krishi Vigyan Kendras (KVKs), Farmers' Producers Organizations (FPOs), the Agri-startup ecosystem, and others. In 2021, the Academy hosted 95 capacity-building programmes reaching 10,131 beneficiaries including 54 need-based programmes, 13 workshops and 24 Off-campus programmes, 02 leadership development programmes and 02 Massive Open Online Courses (MOOC).

The FOCARS has been the flagship programme of the Academy for Agricultural Research Service (ARS) probationers. The 111th FOCARS with 37 ARS probationers completed their training program and received their completion certificates during the valedictory program held on Jan 02, 2021. Dr. R.S. Paroda, Former Secretary, DARE and DG, ICAR was the Chief Guest of the event, which was attended by many eminent researchers and research managers from across NARES. The Academy also organized a tailor made training programme for the 165 Indian Forest Service (IFS) probationers (2020-22 batch) of Indira Gandhi National Forest Academy (IGNFA), Dehradun and a collaborative capability programme for young agribusiness professionals of Coromandel International Limited. The Academy has now evolved into a Think Tank of ICAR by providing advisory role and research policy advocacy in critical areas of importance in

agriculture with changing needs. The Academy has produced three policy documents as a result of this process, including Circular Carbon Economy, Strategies for Enhancing Soil Organic Carbon for Food Security and Climate Action and Mainstreaming Biodiversity in Public Finance. These policy documents resulted from multiple rounds of brainstorming sessions and stakeholder engagements in the various areas, and major policy prescriptions for further improvement of NARES.

The Post-Graduate Diploma in Management-Agribusiness Management (PGDM-ABM) is another flagship activity of the Academy. The AICTE - approved 2-year fully residential programme admitted 57 students for its 12th batch during the academic year 2021-22 whereas 11th batch of PGDM-ABM got success in employment through campus placement with top agribusiness industries in India. Other academic programmes of the Academy, viz., Diploma in Technology Management in Agriculture (DTMA) and Diploma in Education Technology Management (DETM) are one-year distance education programmes offered in online mode. Both the programmes are being offered in collaboration with University of Hyderabad (UoH). In these programmes, a total of 62 students took admission in the year 2021. On the research front, the Academy had 30 research projects, which comprised of 22 Inhouse projects, and 8 externally funded contract research and consultancy projects. The sponsored projects are from various organization including ICAR Schemes, GIZ, NSTMIS, DST, DBT World Bank, NITI Aayog, IFPRI, NIPHM, ITC Pvt. Ltd., NASF, NDRI, UNDP-BCIL & NCSCM.

Agricultural Business Management for Inclusive Growth, Education Systems Management for Enriched Learning Quality and Environment, Enhancing Capacities for Leadership and Governance, Extension Systems Management in Market-driven Environment, Information and Communication Management for Promoting Innovation and Governance, and Mobilizing Science & Technology for Innovation and Sustainable Development are the six thematic areas covered by the research projects. The faculty members of the Academy published 70 more than peer-reviewed research papers in high impact journals. The other publications include books, book chapters, popular articles, training manuals, seminar proceedings, research reports etc. ICAR-NAARM is one of the consortia partners of Farmers FIRST Initiative launched by ICAR.

The Academy was essentially involved in the capacity building activities and organized several workshops under the Farmers FIRST programme. The Centre for Agri-Innovation (CAI) is located in the campus provides hand-holding support to the agri-startups through a-IDEA, the technology business incubator of the Academy. During the year, 31 incubatees have been inducted. Besides, 20 incubatees who were inducted during earlier years have been graduated in 2021. a-IDEA organized several Start-Up – FPO Immersion programmes and training programme for ABI units. It has launched Nidhi Prayas inviting innovators working on Idea/POC/Prototype in the field of agriculture and allied sectors to apply for “Nidhi PRAYAS” grant up to INR 10 Lakhs. It has organized series of webinars on various topics from sector specific to AMA (Ask Me Anything) sessions for startups, specifically focusing on finance and legal/compliances related issues. Among others, the Technology Enhanced Learning in Agricultural Education (TELAge) is

state-of-the-art Agricultural Education Lab, which was established in the Academy to develop digital courses, offer distance and online programmes, conduct virtual classes and support in-house programmes. With its technologically advanced infrastructure, the facility is now being used to offer MOOCs consultancy support to other organizations in developing and hosting digital content. Other neighboring institutions like MANAGE, PJTSAU and NIPHM have been using this facility for generating their digital output. The facility is now renamed as Centre for open and Lifelong Learning in Agricultural Education (COLLAge). The Academy celebrated National and International Days such as National Science Day, World Water Day, International Women’s Day, International Yoga Day, ICAR Foundation Day, Constitution Day, Agricultural Education Day and National Farmers’ Day.

During the year 2021, the Academy signed 10 MoUs in areas of mutual interest with VAMNICOM, Pune; ANGRAU, Lam, A.P, NCDEX e-Markets, Mumbai; and with eight IT companies. Several dignitaries graced different events of the Academy through online and offline platform namely Dr. T. Mohapatra, Secretary, DARE & DG, ICAR; Dr. R.C. Agrawal, DDG (Ag. Edn); Dr. R.S. Paroda, Ex-DG, ICAR; Prof. G. Padmanaban, Chairman, RAC; Dr. Krishna Ella, CMD, M/s Bharat Biotech, Dr. GR Chintala, Chairman, NABARD. During the year, many of our faculty have been recognized nationally and internationally for their outstanding work i.e. Dr. P. Krishnan is elected as Fellow of NAAS, Dr. MB Dastagiri was awarded “ISDS Fellow” by the International Society for Development and Sustainability (ISDS), Japan and many were nominated as Member/Chairman for various committees of ICAR and other organizations.

The Indian Council of Agricultural Research (ICAR) founded the ICAR-National Academy of Agricultural Research Management (NAARM) in Hyderabad in 1976 with a principal mandate to enhance capacity in agricultural research, education, and extension education systems, as well as to offer policy advocacy. The Academy has been working with the mission of enhancing leadership, governance and innovation capacities of the National Agricultural Research and Education System (NARES) through capacity strengthening, education, research, consultancy, and policy support, to emerge as an institution par excellence to facilitate and support a culture of dynamic management in agricultural research, education and frontline extension. The Academy aims to improve individual and institutional capacity for innovation for the transformation of NARES into a more pluralistic innovation system, given the strategic importance of agricultural research in the country's food security and economic prosperity. The Academy aims to foster a sense of fraternity among agricultural scientists and faculty across the country by instilling a scientific work culture.

To develop and nurture students, the Academy has leveraged its capacity for excellent human resource management in agricultural research into agribusiness management, technology management and education technology management. In collaboration with the University of Hyderabad (UoH), the Academy offers higher education programmes such as the Post Graduate Diploma in Management-Agribusiness Management (PGDM-ABM) with AICTE approval and accreditation from the National Board of Accreditation (NBA), the Diploma in Technology Management in Agriculture (DTMA) and the Diploma in Educational Technology and Management (DETM) in distance mode. The Academy's Technology Business Incubator, a-IDEA (Association for Innovation Development of Entrepreneurship in Agriculture), has won national recognition for its innovative approach to incubating Agri-Startups.

The Academy is well-positioned to serve as an agricultural think tank, proposing policies to strengthen the National Agricultural Research and Education System and ensure the long-term viability of Indian agriculture.

VISION

A global knowledge institution enabling the National Agricultural Research and Education System (NARES) to adapt to change through continuous innovation.

MISSION

To enhance leadership, governance and innovation capacities of National Agricultural Research and Education System (NARES) through capacity strengthening, education, research, consultancy and policy support.

1.1 Strategic framework

The Academy built a six-tier strategic framework by its Vision, Mission, and Mandate to guide various activities toward the defined goals and targets. These are:

1. Agribusiness Management for Inclusive Growth

- Enhance capacities for food and Agribusiness management education and research in NARES.
- Enhance capacities for entrepreneurship development in NARES and strategic management of agribusinesses.

2. Education System Management for Enriched Learning Quality and Environment

- Enhance capacities for agricultural education policy, planning and evaluation in institutions of NARES.
- Enhance capacities for faculty excellence and technology-enhanced learning for increasing learning effectiveness and continuous learning by deploying emerging media to create new and vibrant education environments.

3. Enhancing Capacities for Leadership and Governance

- Institutionalize a framework for leadership development at all levels of NARES - early, mid-career, senior professional and manager across all functions (research, education, extension).
- Institutionalize good governance in NARES by building capacities for effective management of institutions, performance assessment, accountability, and developing

strong organizational value systems and work culture that promote innovation.

4. Extension Systems Management in a Market-driven Environment

- Enhance operational, adaptive and generational capacity of front-line agricultural extension systems to address emerging challenges.
- Enhance capacities for use of ICTs to provide customized knowledge, skills and solutions to farmers, farmer groups, rural communities and for social networking.

5. Information and Communication Management for Promoting Innovation and Governance

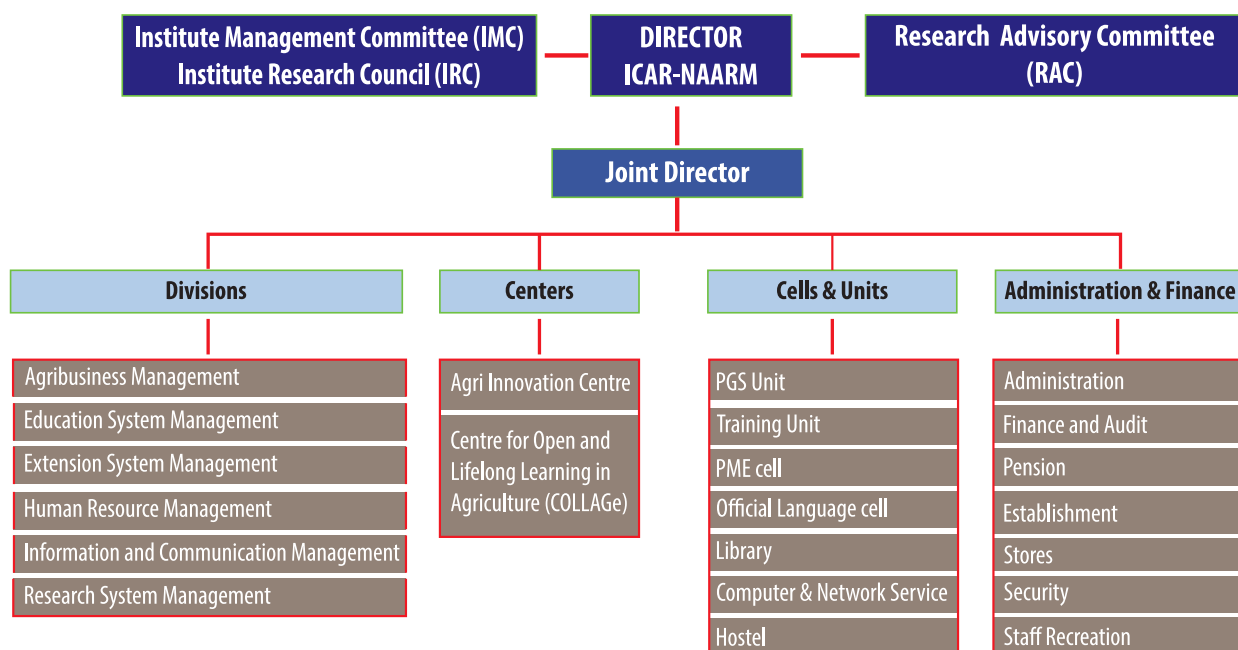
- Enhance capacities for integrating ICTs and Internet of Things (IoT) in institutional management and governance.
- Enhance capacities for managing data, information and knowledge in agriculture, scientific communication and institutional governance.

6. Mobilizing Science and Technology for Innovation and Sustainable Development

- Enhance capacities for research policy, strategy, priority setting, planning, management, monitoring and evaluation for a more pluralistic NARES, in the emerging contexts of food security, climate change and globalization.
- Enhance capacities for technology foresight and strategic management of intellectual property and commercialization of technologies



1.2 Organization and Management



1.2.1 Research Advisory Committee (RAC)

The Indian Council of Agricultural Research (ICAR) constituted the Research Advisory Committee (RAC) for three years from 14-05-2020 to 13-05-2023 as given below:

Prof. G. Padmanaban President NASI and Former Director IIS, Bangalore	Chairman
Dr. B. Venkateswarlu , Former Vice-Chancellor, Vasantrao Naik Marathwada Krishi Vidyapeeth, Parbhani, Maharashtra	Member
Dr. Pratibha Jolly , Science and Society Fellow, NASI Former Principal Miranda House, Delhi University	Member
Prof. V.K. Unni Public Policy Management Indian Institute of Management, Calcutta	Member
Dr. R.P. Singh Regional Manager-Operation & Resource, Breeding & Trait Development-Asia Pacific, Bayer BioScience Pvt. Ltd. Madhapur	Member
Dr. S.L. Goswami Former Director, National Academy of Agricultural Research Management, Rajendranagar, Hyderabad	Member
Dr. Ch. Srinivasa Rao Director, ICAR-NAARM, Hyderabad	Member
DDG (Agril. Education) , ICAR KAB-II, Pusa, New Delhi	Member
Shri Vallabhaneni Asha Kiran , Vijayawada	Member
Shri Gone Shyamsunder Rao Seven Hills High School Mancherial, Telangana	Member
Dr. P. Krishnan , Principal Scientist ICAR-NAARM, Rajendranagar, Hyderabad	Member Secretary

1.2.2 Institute Management Committee (IMC)

The following officials were nominated as members of the Institute Management Committee of NAARM for 3 years. (24-07-2020 to 23-07-2023).

Dr. Ch. Srinivasa Rao , Director, ICAR-NAARM, Hyderabad	Chairman
Director of Agriculture Department of Agriculture, Government of Telangana	Member
Spl Commissioner and Director , Department of Agriculture, Government of Telangana	Member
Director (Research) Prof. Jayashankar Telangana State Agricultural University, Rajendranagar, Hyderabad	Member
Shri Gone Shyamsunder Rao Seven Hills High School Mancherial, Telangana	Member
Shri Vallabhaneni Asha Kiran , Vijayawada	Member
Dr. R.K. Mathur , Director, ICAR-IIPR, Andhra Pradesh	Member
Dr. V. Tonapi , Director, ICAR-IIMR, Hyderabad	Member
Dr. Raghavendra Batta , Director, National Institute of Animal Nutrition and Physiology, Bangalore	Member
Dr. G. Venkateshwarlu , Joint Director, NAARM, Hyderabad	Member
Dr. Seema Jaggi , ADG (HRD), ICAR, KAB-II, New Delhi	Member
Addl. Secy. DARE & FA, ICAR/his nominee	Member
Joint Director (Admn.) & Registrar NAARM, Hyderabad	Member Secretary

1.3 Linkages/Networking/Collaborations

NAARM has a strong strategic and functional network with 108 ICAR Research Institutes and 75 Agricultural Universities (AUs) and plays a crucial role through the capacity building of these institutions, agribusiness industry, scientists, and academia. Its network spans various other National as well as international institutions.

1.3.1 National Institutes/ Organizations

1. All ICAR institutes and Agricultural Universities.
2. Various departments of the Government of India like NITI (National Institution for Transforming India) Aayog, Department of Science and Technology (DST), Department of Biotechnology (DBT), National Science & Technology Entrepreneurship Development Board (NSTEB).
3. University of Hyderabad, Osmania University, Jawaharlal Nehru Technological University (JNTU), Indian Institute of Information Technology (IIIT), Council of Scientific & Industrial Research (CSIR), Indian Council of Medical Research (ICMR) and various institutions located at Hyderabad.
4. Ministry of Environment, Forest and Climate Change (MoEF & CC), National Board of Accreditation (NBA), Protection of Plant Varieties & Farmer Rights Authority (PPV&FRA), National Fisheries Development Board (NFDB) and State Biodiversity Boards.
5. Management Institutions located at Hyderabad viz. the Administrative Staff College of India (ASCI), Indian School of Business (ISB), National Institute of Agricultural Extension Management (MANAGE), National Institute of Rural Development and Panchayat Raj (NIRD & PR), and Institute of Public Enterprise (IPE); and across the country such as Indian Institutes of Management (IIMs).

6. State Departments of Agriculture of various States (Telangana, Andhra Pradesh).
7. Private Sector Organizations (Agri Input Companies, Service Companies, etc.).
8. Non-Government Organizations (NGOs).
9. Other organizations viz. National Innovation Foundation (NIF), Federation of Indian Chamber of Commerce and Industry (FICCI), National Bank for Agricultural and Rural Development (NABARD) Centre for Innovation Incubation and Entrepreneurship (CIIE) etc.
10. Krishi Vigyan Kendras (KVKs).

1.3.2 International Organizations

1. CGIAR Institutions (CIMMYT, ICRISAT, IFPRI, etc.)
2. World Bank, FAO, UNDP, GIZ and other United Nations Organizations
3. Department of International Development (DFID, UK).
4. SAARC Agricultural Centre (SAC).
5. NARS of South Asian Countries such as Bangladesh Agricultural Research Council (BARC), Nepal Agricultural Research Council (NARC), Pakistan Agricultural Research Council (PARC), and other NARS Institutions in Afghanistan, Sri Lanka and other ASEAN countries.
6. NARS in other African and South-East Asian countries such as Kenya, Mali, Tanzania, Nigeria, Malawi, Liberia and the Philippines.
7. Leading Land Grant and State Universities of the USA and other Universities in Europe, Australia and other developing and developed countries.

Innovation, governance and knowledge form the fulcrum of the Academy's mission, which is driven by its vision documents (2030 and 2050) as well as those of ICAR. Recognizing the enabling nature of emerging science and technologies across diverse sectors, it voices for their integration into agricultural value chain. Further, the evolving changes in Agri-education sector shifting focus from 'teaching to learning outcome' signifies the requirement to develop newer institutional mechanisms and capacities of the system. The Academy also identifies the crucial need for a continuous stakeholder engagement to bring out effective and efficient performance in the NARES with more transparency and accountability in place. Therefore, the activities of research, capacity building and education have been aligned to the six-tier strategic approach. Through a structured deployment of this strategy, the Academy envisages to play a catalytic role in the planned transformation process of the NARES. The subsequent sections present the annual progress and achievements in research, capacity building and the other activities of the Academy under the six strategic areas during the year 2021.

2.1 Agribusiness Management for Inclusive Growth

With the shift in agricultural development strategies from a pure production-oriented approach to a broader systems perspective, the importance of the agribusiness sector has grown significantly. Seen as an engine for growth, agribusiness and its related industries are receiving increased attention in policies and strategies that aim to promote investments in agro-enterprises

and agro-based value chain development. This has prompted a need for deeper understanding of agribusinesses, agro-industries and Agri-food value chains. The theme on 'Agribusiness Management for Inclusive Growth' encompasses research, education, capacity building and consulting for developing future leaders in the field of agribusiness. It broadly aims to enhance capacities for food and agribusiness management education and research in NARES and to enhance capacities for entrepreneurship development in NARES and strategic management of agribusinesses. Rapid urbanization, rising disposable income of households, expanding and diversifying food demand and globalization of agricultural economy have given impetus to research and capacity building in the area of Agribusiness Management. To meet the broader objectives of the theme, the Academy focuses on the following thrust areas for research, capacity building and education:

- **Agribusiness environment:** With a major focus on value chain analysis for agri-commodities and services; consumers demands; price behavior and forecast analysis.
- **Public policy and impact:** With focus on market reforms like e-NAM, GST, etc.; financial inclusion and its spillover; spread and convergence of crop insurance.
- **Markets and institutional innovation:** Covering market integration of commodities; producer companies and FPOs; inputs and service delivery systems; diversification and intensification of agriculture for high value commodities; business analytics.

- **Nurturing future leaders in agribusiness:**

Through imparting quality education in agribusiness management and capacity building in agri-preneurship.

2.1.1 Techno-Economic Feasibility Studies for ICAR Agro-Processing Technologies / Value Added Products

Techno-Economic Feasibility Report (TEFR) is critical for understanding the implications of setting up any industry using agro-processing technology. It provides complete picture of the targeted products starting from the product technical parameters and the process outline involved, capital requirement, nature and source of the machinery to install, plant economics, project technical feasibility and the economic viability. It will act as a base material for any entrepreneur to venture in any new project/industry. Techno- Economic Feasibility report or DPR is also a bankable study which the financing institutions look for making decision about funding such project. ICAR-NAARM has taken steps to develop such reports for selected technologies developed by different ICAR institutes. These reports are important tools in the commercialization process of technologies through Agrinnovate India Limited (AgIn) of ICAR. Identified ten products for the preparation of TEFRs in consultation with Agrinnovate and the ICAR institutes involved in developing the technology viz., super absorbent polymer from cassava starch; process for the production of freeze-dried sugarcane juice; dehydration of tubers; live fish carrier system; lac-based fruit coating formulation for kinnow; microbial method for production of protein isolate/concentrate from oilseed cakes/meals; preparation of milk-based spray dried nano encapsulated curcumin formulation; technology

of sour dahi using prolific acidifying lactic cultures; VL paddy thresher; and FRP demand fish feeder.

Data on technology, machinery required for pilot model were collected for the preparation of TEFR for the super absorbent polymer (SAP) from cassava starch in consultation with the scientists from ICAR-Central Tuber Crops Research Institute who developed this technology. This polymer can be used as a soil additive especially under the prevailing situation of reduced moisture availability. Under controlled conditions such as in green houses for plant nurseries and ornamental plants, it can be effectively used. It absorbs over hundred times their weight in liquids and the absorbed liquid is released slowly when put in the dry air to maintain the moisture of the environment. Based on the machinery and materials required for pilot plant of 175 l capacity final product, estimated cost of production of Super Absorbent Polymer from cassava starch (for pilot plant) was ₹ 94 per kg.

2.1.2 Agri Startups in India: Decoding the Determinants and Success Factors

Startups are widely recognized as important engines for growth and jobs generation. Government has launched many initiatives for promoting startups across sectors in India. The present study was undertaken with the objectives of investigating the motivation of entrepreneurs for endeavoring into agri startup; examining their business processes; and determining the factors influencing their success. Thirty-two startups incubated at aIDEA of ICAR-NAARM were interviewed during the period. It was found that majority of the founders were engineers (50%) followed by management graduates (31%). The findings also revealed that entering into startup was not accidental for majority of the founders

(Fig. 2.1). In fact, foreseeing future business opportunities have motivated them to foray into the identified business activity. The challenges faced by the sample startups include lack of growth capital, lack of guidance and planning, problem in hiring right people and problems in

maintaining customer relationship. All the sample startups are in the early traction or scaling-up stage. Organizational structure becomes more stabilized and well defined as the startup moves up the stage. In this process, innovation, either in product or process, is quite obvious.

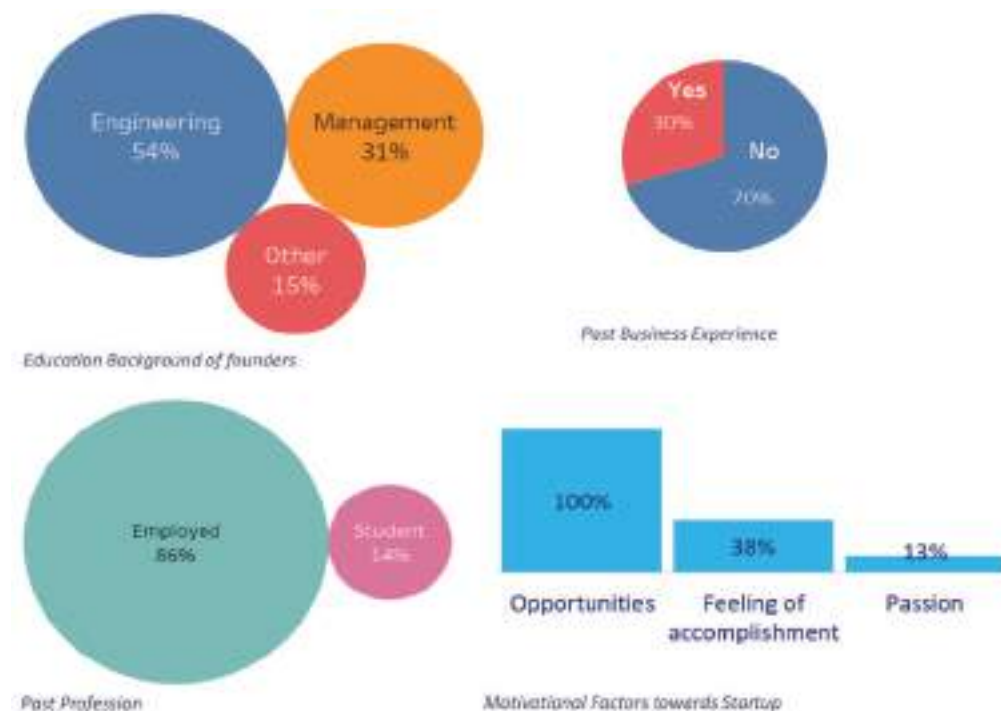


Fig. 2.1: Agri- Startups Background

2.1.3 Farmer Producer Companies in India: A Study on their Management Practices and Business Potential

Farmer Producer Companies (FPC) are promoted by NABARD and various other agencies in India to help small land holders to effectively connect to the market, ensuring better bargaining power through collective actions. This study was taken up to estimate the benefits of FPC to the farmer-members and to identify the critical success factors for the FPOs. During the reported year (2021), four FPCs based in Telangana were studied. They are Mathota FPC Ltd, Shanti FPC Ltd, Swakrushi FPC Ltd and Eruvaka FPC Ltd. These FPCs were established during 2016 at Warangal

with members ranging from 297-1000 with an average of 526 per FPC. The annual turnover of these companies ranged from ₹ 31- 70 lakhs. An indicator of success, Minimum Business Potential (MBP) was calculated for the select FPCs (Table 2.1). This indicator reflects the potential available with the FPC if all member-farmers are transacting their farm related businesses through FPC. The MBP of the companies in the study area ranged between 2.95% and 9.19% depending on the crop focussed by FPC which indicate huge potential for scaling up the businesses to the extent of by 11 to 34 times, provided the critical inputs of FPCs functioning are not limited. The pain points of FPC identified in the study are liquidity crunch,

lack of participation of members in FPC, no proper planning and low level of member involvement while decision making. Almost all the FPCs were facing liquidity crunch, which limit their scope of operations and threaten their survival. The study also found that FPC specializing in specific commodity and moving along the value chain through innovations are performing well.

Table 2.1: Commodity-wise minimum business potential in Telangana

Crops	Value of Input + Output / Ha (Rs. '000)	Minimum Business Potential of FPC (Rs. Crores)
Paddy	67.2	3.96
Cotton	156.0	9.19
Maize	67.5	3.98
Peanut	113.6	6.69
Soybean	97.1	5.72
Chickpea	58.5	3.45
Pigeonpea	50.0	2.95

The critical success factors were identified based on the responses from stakeholders and experts involved in the FPCs. In the first stage, the factors were populated based on the responses from various stakeholders including members of FPCs, and office bearers of FPCs. Then in the second stage, experts were asked to rate the statements based on their influence on performance of FPCs using Likert scale ranging from 1 to 5. These experts were from associated organizations like Resource Institutions, Agricultural Department, and Advisors who are connected with FPCs. The results show that providing right advice, conducting skill based training programme and ensuring members' participation through purchase of input and sale of their produce through FPC are critical for the success of FPC (Table 2.2).

Table 2.2: Rating of statements on Critical Success Factors of FPC

Statements	Avg. score	Rank
Membership strength		9
Less than 250	2.79	
Range of 251-500	3.63	
Range of 501-1000	3.28	
More than 1000	3.19	
Being multi-commodity FPC	3.70	6
Providing credit facilities to members	3.70	6
FPC having sufficient cash reserves for scaling up	3.72	5
Ability to source credit facilities through financial institutions	3.67	8
More number of members buying their inputs through FPC	3.84	3
More number of members selling their produce through FPC	3.79	4
Providing advisory services to members	4.00	1
Conducting more training/ skill development programmes to members	3.98	2

Source: Primary Survey, 2021

2.1.4 Mainstreaming the Small Ruminant Farmers into Efficient Value Chains in India

The main objectives of the study are to project the socio-economic profile of the goat farmers, to study production and marketing practices of goat farmers, to identify the constraints in goat value chains in the study area, and to suggest suitable strategies and policies to increase the income of goat farmers through efficient value chains. The survey was carried out at Barmer district of Rajasthan and 80 farmers were contacted telephonically with the help of AU, Jodhpur to elicit required information.

Profile of goat farmers

A majority goat farmers were of medium type having a flock size of 5-15 animals (78%). The

average flock size of small, medium and large farmers was 4.2, 11.7 and 19 respectively. Similarly, a majority of the farmers were earning in the range of Rs. 50,000 to Rs. 1 lakh annually from goat farming (75%) in the study area.

Cost of goat rearing

The goat was reared in the selected area under semi-intensive type. The study reveals that the average feed cost per animal in the study area was Rs.45, health/ treatment cost was Rs.161, labour cost was Rs.1101, cost related to sheds, equipment and land lease was Rs.238, and cost related to miscellaneous items was Rs.41. Thus, the average cost of production of a goat in the study area worked out to be Rs. 1,586 per annum (Table 2.3).

Table 2.3: Cost of production per goat in Rajasthan

S. No.	Items	Average Cost (Rs.)
1.	Feed	1090
2.	Health/Treatment	150
3.	Labour, including family labour	100
4.	Costs related to sheds equipment and land lease	577
	Overall	1917

Source: Primary Survey, 2021

Marketing of goat

A majority of the farmers marketed their goat at 9-12 months of age (69%) (Fig. 2.2).



Source: Primary Survey, 2021

Fig. 2.2: Marketing age of goat in Rajasthan

It was found that about 66% of the farmers sold their goat three or more times a year, followed by 30 per cent of them annually and only 4 per cent sold the goats annually once. Further, it was observed that 71% of the farmers sold their goats on the basis of age, followed by 21% of them on the basis of sex. Only 8% of the farmers could sell their goats on the basis of weight, which is the profitable and recommended method in fixing the price for goats at the primary market. Marketing pattern revealed that about 61% of the farmers sold their goats to the local traders, followed by 20%, 14% and 5% of them found selling their goats to outstation traders, butchers and consumers, respectively. A perusal of alternative marketing channels adopted in goat farming revealed that farmer gets greater share in consumer rupee in traditional channel involving butcher (Table 2.4). Contrary to the expectations, producer gets least benefit in e-commerce market.

Table 2.4: Farmers' share in the consumer rupee regarding goat rearing in Rajasthan

Parameters	Value	Farmers' Share (%)
Cost of production	Rs. 1917 / 20 kg (adult goat in live weight) Rs. 96 / kg	--
Price received by them	Rs. 5000 / Goat (20 kg live weight) Rs. 250 / kg	--
Price at butcher	Rs. 400 / kg	62.5
Price at online market	Rs. 460 / kg (Mutton curry cut)	54.0
Price at e-commerce market	Goat curry cut (450 gm): Rs. 382	29.0
	Goat biriyani cut (500 gm): 525	24.0
	Goat ribs & chops (500 gm): 599	21.0
	Goat boneless (500 gm): 699	18.0

2.1.5 Financial Accessibility and Occupational Risks of Vegetable Street Vendors in India

In all Indian cities, street vendors provide a low cost, decentralized and highly efficient system of distribution of a wide variety of goods of daily needs ranging from fruit and vegetables to clothes, utensils and many others. They reach the consumers at convenient locations, without whom people would have to travel long distances to procure these goods. The middle and lower income groups buy a substantial proportion of daily goods from these vendors at reasonable rates often lower than the formal stores. This study aims to assess the socio-economic status of vegetable street vendors and the specific objectives of the project are understanding socio-economic characteristics of street vegetable vendors and examine their issues and challenges, assess the contribution of vegetable vendors in vegetable value chain, and examining their extent and level of financial accessibility. The primary survey was carried out in four weekly vegetable markets namely Rajendranagar, Budwel, Chintalmet and Attapur at Hyderabad in which a total of 120 vendors were interviewed. A perusal of profile of the vendors revealed that about three-fourth of them were young with age less than 50 years and a majority of them were relatively new to the business with less than 10 years of experience (91.67 percent). A majority of them had education up to higher secondary level. Lack of employment opportunities was the key reason for the vendors to venture into this business.

2.1.6 New Market Reforms and E-NAM in India: Effect on Marketing Choices and Price Realization for Smallholders

In April 2016, eNAM was introduced to integrate all the physically dispersed APMC mandis, with an ambition of bringing both farmers and different trade functionaries on a single platform, resulting in

more transparent and competitive price discovery. Since then, a total of 1000 APMC mandis have been integrated with e-NAM platform across 18 states and 3 UTs. However, the benefits expected from the e-NAM for the smallholder farmers could not be realized. Therefore, in the year 2020, New Farm Acts were brought by the Government of India, viz., 'The Essential Commodities (Amendment) Act 2020', 'The Farmers' Produce Trade and Commerce (Promotion and Facilitation) Act 2020' (FPTC Act), and 'The Farmers (Empowerment and Protection) Agreement on Price Assurance and Farm Services Act 2020'. The FPTC Act assumed that mandis controlled by Agricultural Produce Marketing Committee (APMC) are more in favour of traders and middlemen than the farmers. Therefore, the Act would facilitate in establishment of many private markets and farmers would be free to sell their produce to any buyer, anywhere, thus farm gate price would rise. However, after the Acts came into force, there were protests against the law in many places of the country and finally, the Government of India repealed all three Acts through The Farm Laws Repeal Act, 2021 on November 30, 2021.

The current project attempted to examine the effect of New Farm Laws on functioning of e-NAM. The study analysed the commodities transacted on e-NAM platform during May 2020-April 2021 in Rajasthan state, which has the largest number of e-NAM mandis (144) in India. Daily transaction data for whole year in the mandi on both the platforms were collected from eNAM and Agmarknet portal. In Rajasthan, Kota APMC adopted e-NAM in the first phase itself during 2016 and has the highest daily arrival of commodities (12% of state total) in the state. In Kota mandi, major commodities traded on eNAM platform were mustard (19%), wheat (15%), paddy (8%), cotton (6%) and pearl millet (5%).

The analysis of transaction data collected from Agmarknet observed that the difference between

eNAM and non-eNAM platform for weekly average of arrival as well as modal price for both the commodities (mustard and wheat) were statistically not significant. The weekly average modal price of mustard ranged from Rs. 4,231 to Rs. 5,808 per quintal under eNAM, whereas Rs. 4,018 to Rs. 6,600 per quintal in case of non-eNAM. In case of wheat, the weekly average modal price ranged from Rs. 1,559 to Rs. 1,851 per quintal under eNAM

and Rs. 1,623 to Rs. 1,864 per quintal in case of non-eNAM. From the analysis, it also appeared that when the same mandi is offering both the options- eNAM and non-eNAM platform for selling the commodities, the buyers are indifferent in giving differential prices for the commodities. Possible reasons may be that in both the cases, buyer-traders remain the same, trading is happening only after physical verification of the commodities by the buyers (Fig. 2.3).

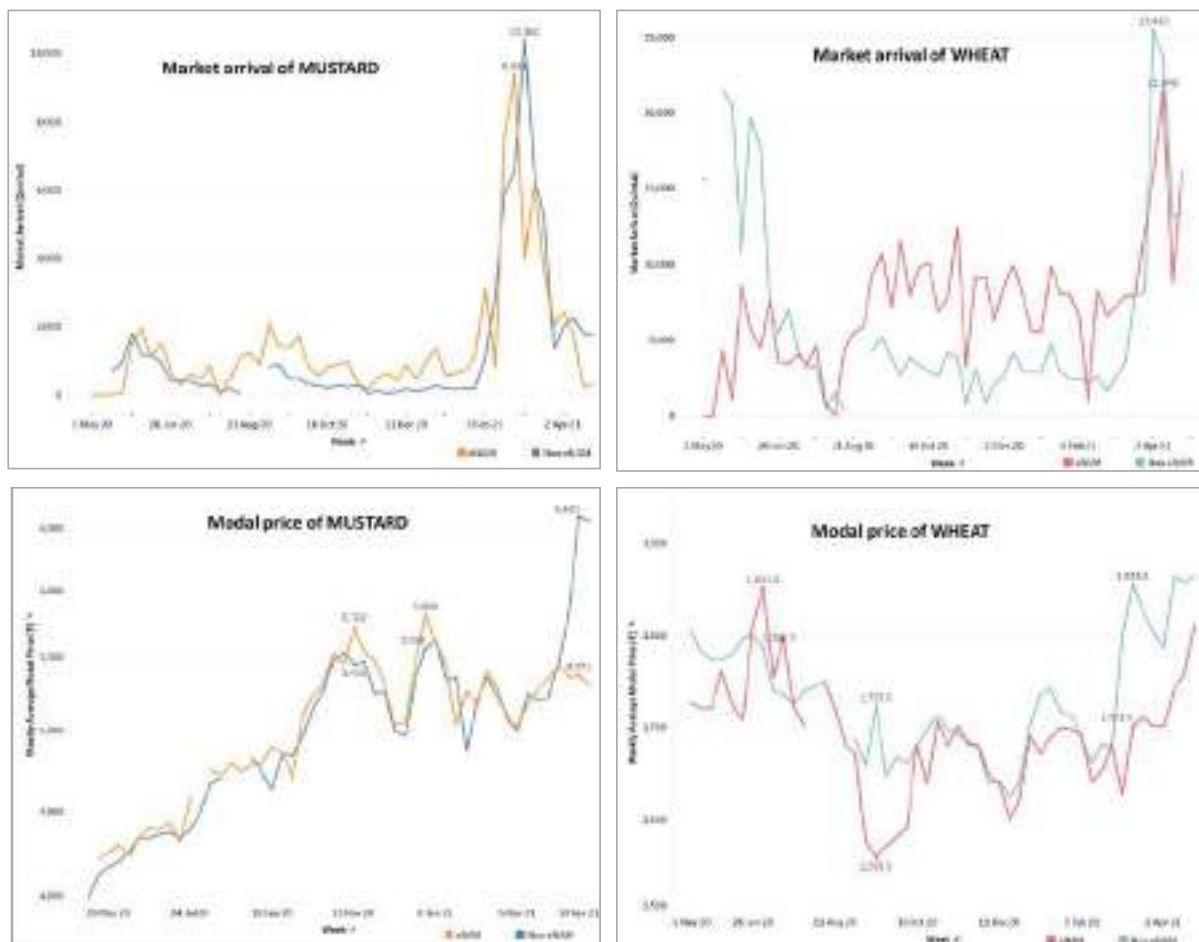


Fig. 2.3: Weekly average market arrival and modal price of mustard and wheat in Kota APMC mandi, Rajasthan traded on eNAM and non-eNAM platform

2.1.7 Entrepreneurship Development through Farmer Led Innovations- A Case Study in Plantation Sector

Under the research project on “Entrepreneurship Development through Farmer Led Innovations - A Case Study in Plantation Sector” field verification and validation at farmer’s field level were

conducted for nine innovations by ICAR-NAARM, ICAR-CPCRI and IIPM, Bengaluru. The proposed hypothesis between identified variables and Entrepreneurship/Innovation was tested using Chi-Square analysis. The result indicated that there is a strong association between innovation and entrepreneurship and there is no relationship

between age, size of the plantation and education with entrepreneurship/innovations. Among various existing innovation models, identified Cyclic Model of Innovation with Interconnected cycles is a suitable model for plantation sector through factor analysis. This model exhibited six characteristics namely, customer value, market factors, commercial aspects, societal value, technological aspects and economic aspects. Developed Entrepreneurship Assessment Index (EAI) based on OECD2015 framework using 17 indicators. The EAI value was 120 and it indicated that 75% of innovators are ready to take up the entrepreneurial activities. As part of validation process, field experiments were organized for the selected FLIs in 20 farmers' field with five replications. The validation was compared with the traditional method. The analysis indicated that out of 22 FLIs prioritized, 14 innovations are under commercialized category, four are potential to commercialize and remaining four need support for commercialization. Strategies are being prepared to upscale the FLIs based on the existing schemes in promotion of entrepreneurship development and competitor analysis of the FLIs.



Fig. 2.4: VINPEPT Pepper Thresher Innovation

2.1.8 ICAR-NAARM and Coromandel International Limited (CIL) Collaborative Capability Building Programmes for Sales and Marketing Managers

Marketing Officers' Resurgence & Excellence (MORE) programme organized during 8-12 February 2021 (32 Participants). The participants of the programme are exposed with advances in agriculture technologies, soil health and nutrition management and to provide knowledge in Coromandel product portfolio and to bring attitudinal & behavioral changes to enhance business management skills to make them ready to take higher responsibilities. The topics included advances in agronomy practices of major crops, climate change and balanced nutrition landscape, Farmer Producers Organizations. (FPOs), Data Analytics & Forecasting. Agribusiness Management Skill development sessions on Business Model Canvas, Marketing Plan and Brand Management & Promotion strategies. The soft skills in Relationship Management, Negotiation and Conflict Management also covered during the training programme.



Fig. 2.5: Collaborative Capability Building Programme for Sales and Marketing Managers of Coromandel International Limited

2.1.9 ICAR-NAARM and Coromandel International Limited (CIL) Collaborative Capability Building Programmes for Sales and Marketing Managers



ICAR-NAARM and Coromandel International Limited (CIL) Collaborative Capability Building Programmes for Sales and Marketing Managers organized during 15-19 February 2021 (25 Participants). The objectives of the program are to expose the participants with current trends in agriculture, soil health and nutrition management, to provide knowledge in Coromandel product portfolio, and to bring about attitudinal and behavioral changes to enhance efficiency and effectiveness. Providing a basic understanding of crop production and crop protection practices, and upgrading the knowledge and skill of the participants in the sales and marketing domain were the other motives behind the training programme. The topics covered were agriculture and Indian economy, agronomy aspects of major field crops, soil health-based nutrient marketing, basics in plant health management and plant protection chemicals, irrigation and fertigation management, organic fertilizers, ICT and social media in sales, and value-added selling skills. The behavioral aspects such as personality development and emotional intelligence also covered in the training programme.

2.1.10 Data Visualization in Agribusiness and Agricultural Research

An online training programme on “Data Visualization in Agribusiness and Agricultural Research” was organized by the Academy during 22-27 February, 2021. There were 30 participants including 9 female participants from various ICAR institutes and State Agricultural Universities. The programme covered various visualization tools and techniques which can be used for presenting data in various publications as well as presentations. It was a hands-on training in which participants were exposed to two popular softwares, namely, R and Tableau Public for generating graphic visuals. Creating advance

visuals including dashboard preparation was covered during the programme. Speaking on the occasion, Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM emphasised the role of visualization in today’s era. He also urged the participants to use the large data sets which are already available in agriculture for analysis.



Fig. 2.6: Training programme on “Data Visualization in Agribusiness and Agricultural Research”

2.1.11 Entrepreneurial and Communication Skill Development at PVNRTVU, Telangana

A training programme on “Entrepreneurial and Communication Skill Development” was organized during March 08-12, 2021, jointly with PV Narsimha Rao Telangana Veterinary University (PVNRTVU) at College of Veterinary Science, Hyderabad as a part of sub-plan activity of ICAR. There were 23 MVSc students participated in the programme including 8 girl students. The programme was designed to promote and nurture entrepreneurship abilities in students along with equipping them with management tools needed for making successful entrepreneurs. Other objective was to impart communication skill in the students. Accordingly, there were modules on entrepreneurship, livestock sector opportunities and personality development. Majority of the resource persons were faculty members of NAARM. Few guest faculty were also invited to deliver sessions. Dr. Ch. Srinivasa Rao was the chief Guest during the valedictory session. Other dignitaries include

Dr. S.T. Viroji Rao, Dean of Faculties, PVNRTVU and Dr. D. Pramod Kumar, Associate Dean, College of Veterinary Science, Hyderabad. Speaking on the occasion, Dr. Ch. Srinivasa Rao highlighted the importance of entrepreneurship in the agricultural area including veterinary sector. He also briefed about the government initiatives which can be utilized by the students for entrepreneurship.



Fig. 2.7: Participants of training programme with the faculty at PVNRTVU, Telangana

2.1.12 Training Programme on “Entrepreneurship Development for Horticulture Students” of CoH, Rajendranagar, SKLTSU

ICAR-NAARM organized five-day training programme on “Entrepreneurship Development for Horticultural students” College of Horticulture, Rajendranagar of Sri Konda Laxman Telangana State Horticultural University during June 14-18, 2021. The programme covered topics such as entrepreneurship concepts, emerging opportunities in agribusiness sector, Business Idea Generation, Business Plan Preparation, Business Model Canvas, Business Pitch Presentation, Market Research, IP Rights and regulations and success stories in Agripreneurship. The programme, sponsored by Indian Council of Agricultural Research, was attended by 105 students comprising undergraduates, postgraduates and doctorates in Horticulture. The sessions were handled by faculty of ICAR-NAARM, business managers of a-IDEA, and guest faculty from ICAR HQ, Corporates and Startups. The valedictory programme was held on 18th

June 2021 virtually in which Dr. G Venkateswaralu, Joint Director gave a detailed note on different programmes such as Skill India, Student READY, Stand-up India and ACABC and IDEA which aims to bring changes in the mindset of student from job seekers to job providers. The Chief Guest, Dr. B Neeraja Prabhakar, Vice Chancellor of SKLTSU has reiterated student to imbibe entrepreneurial qualities to succeed in their careers. The Dean of SKLTSU Dr. M Padma has appreciated the programme and expressed that it will help the students to step into entrepreneurship in future.



Fig. 2.8: Training programme on “Entrepreneurship Development for Horticultural Students”

2.1.13 Entrepreneurship Development in Veterinary Sector for Students of TANUVAS

The Academy organized a sensitization programme on Entrepreneurship Development in Veterinary Sector for students of Tamil Nadu Veterinary and Animal Sciences University (TANUVAS), Chennai on June 19, 2021. The programme was organized to orient the students of TANUVAS and other colleges about the entrepreneurship, understand the scope of Startups in Veterinary sector and create awareness on Startup eco-system. About 160 students and faculty members virtually participated in the programme. During the program, Dr. G Venkateswarlu, Joint Director, ICAR-NAARM

highlighted the need for transformation of students as job providers instead of job seekers. Dr. B Mohan, Dean, Veterinary College and Research Institute, TANUVAS, Namakkal gave inspirational address to the students on various opportunities available in dairy, sheep, goat and poultry enterprises. Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM outlined the potential areas of Agripreneurship which are yet to be tapped and indicated the students need to venture into agri-business and plug the gap through viable business propositions. Dr. Vivek Unnikrishnan, Co-Founder, Amalgo Healthcare Private Ltd., shared his experiences in starting of Veterinary Startup and given the way forward.



Fig. 2.9: Sensitization programme on Entrepreneurship Development in TANUVAS

2.1.14 MDP on Business Plan Development and Accelerating FPOs/FPCs

The Academy organized an online MDP on “Business Plan Development and Accelerating FPOs/FPCs” during June 21-26, 2021. Total 58 participants from 15 different states attended the training, out of which 16 were from different ICAR institutes/ SAUs, 17 represented FPOs, and 25 were from promoting/resource institutions who were working with FPOs. Among them, 15 participants were women. The programme covered all the aspects of FPO, right from mobilization of farmers, FPO formation & registration as

producer company, and all aspects of business plan development for FPOs, like Business Model Canvass (BMC), market research, marketing plan, etc. The participants were also made aware about different regulatory compliances needed for producer companies. Resource persons from different organizations viz., NABARD, Samunnati Financial Intermediation Ltd., and a supply chain company, Waycool Foods Ltd were invited for taking sessions. The founders of two successfully federated FPCs- Sahaydri Farms, the largest grapes exporter; and Sahaja Aharam, organic products company also shared their experiences with the participants. The participants were grouped in eight different teams to practice and prepare their own business plan for one FPO. In his remarks, Dr. G. Venkateshwarlu, Joint Director, ICAR-NAARM expressed that FPOs are the gateway towards doubling farmers’ income of the country. In his valedictory address, Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM complemented all the participants and emphasized that in coming years, FPOs are going to be the vehicle of prosperity for the rural India by transferring the benefits to the farmers through aggregation and collectivization.



Fig. 2.10: MDP on Business Plan Development and Accelerating FPOs/FPCs

2.1.15 MDP on Business Planning and Strategy Formulation for FPOs/FPCs

The Academy organized an online Management Development Program (MDP) on “Business Planning and Strategy Formulation for FPOs/FPCs” during July 05-09, 2021. Total 46 participants, all working as Chief Executive Officer (CEO) of recently registered Farmers Producer Companies (FPCs) from 8 different states attended the training. Majority of the participants were from Tamil Nadu state.



Fig. 2.11: MDP on Business Planning and Strategy Formulation for FPOs/FPCs

Total 11 participants were women. The Program covered different issues related to FPO, right from mobilization of farmers, FPO formation & registration as producer company, and all aspects of business plan development for FPOs, like business model canvass (BMC), market research, marketing plan, etc. Resource persons from organizations engaged in FPO financing viz. NABARD and Samunnati Financial Intermediation Ltd., market linkages and supply chain viz.

Ninjacart and NCDEX e Market Ltd. shared their model of FPO engagement with the participants. The founder of Sahaja Aharam, an FPO federation involved in organic products also shared the experiences with the participants. Finally, the participants grouped in different teams prepared and presented their own business plan.

2.1.16 Sensitization Programme on Entrepreneurship Development & Start-Up Ecosystem

ICAR-NAARM organized a 5-days Sensitization Programme on Entrepreneurship Development & Start-Up Ecosystem during 5-9 July 2021 for scientists and faculty in agriculture and allied disciplines. The objectives of the MDP programme are to orient scientists on the process of technology transfer and co-development of technology with Start-Ups and to sensitize scientists about Start-ups and Incubators working on Agriculture & prospects of collaboration. The major topics covered in the programme include Agri Business Opportunities & Challenges, Introduction to Entrepreneurship and Incubation Ecosystem, Process of incubation & Acceleration, Funding opportunities for Start-ups, Setting up & Managing Incubation Centres and IP & Technology Management. In total, there were 18 lectures and the guest lectures were delivered by directors of agri-incubators, industry professionals and successful agri-based entrepreneurs. In total, 37 participants took part in the programme, of them, 80 per cent were doctorates and rest were Post-Graduates. About one-sixth of them were in associated with incubators and half of them were associated with technology transfer activities in their respective institutes.

In the valedictory programme, the participants expressed that they gained more insights into Start-up ecosystem and the process of technology

commercialization which entails research yield better outcomes. They also voiced that this programme would help them to improve their incubation centres and foster collaboration with actors in start-up ecosystem. Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM reiterated that every scientist and faculty in National Agricultural Research and Education System should motivate graduates to step into entrepreneurship, identify budding entrepreneurs and support them to solve the problems of resource poor farmers which propels Indian agriculture towards another green revolution. Dr. Ranjit Kumar expressed that supporting technology transfer and start-ups should be part of the research plan, which scientists commit to speed up the technology adoption process and promoting agripreneurship.



Fig. 2.12: Sensitization Programme on Entrepreneurship Development & Start-Up Ecosystem

2.1.17 MDP on Formation of FPO/FPC and Preparing its Business Plan for KVK

ICAR-NAARM organized online MDP on “Formation of FPO/FPC and Preparing its Business Plan” during Aug 17-24, 2021, based on the request received from ATARI, Zone-II, Jodhpur. It was a customized need-based training of officials of the KVKs from Rajasthan, Haryana and Delhi states. Total 66 participants including 2 female participants representing ICAR, SAU and NGO

supported KVKs; and few FPO members also attended the programme.

Sessions on formation and registration of FPO, business plan development including marketing and financial plan, regulatory compliances for FPO, experience sharing were covered during the programme. Guest speakers from NABARD, NEML and Sahaja Aharam, an FPO were invited to enlighten the participants about various facets of FPOs. Dr. AK Singh, DDG (Agricultural Extension), ICAR was the Chief Guest during valedictory session. He highlighted various initiatives of the Government regarding FPO and alignment of priorities for the KVKs along these initiatives in the interest of the smallholder farmers.



Fig. 2.13: MDP on Formation of FPO/FPC and Preparing its Business Plan for KVKs

2.1.18 MDP on Market Research & Value Chain Management of Agricultural Commodities

A MDP on ‘Market Research & Value Chain Management of Agricultural Commodities’ was held at ICAR- National Academy for Agricultural

Research Management (NAARM), Hyderabad during September 21-25, 2021 on online mode. The programme was conducted to orient the participants on different aspects of market research such as market integration, market segmentation, consumer survey and so on; to expose them with various concepts of value chain management in different sectors of agriculture; and to equip them with the knowledge about various methodological approaches employed in the value chain management and research. The programme covered the following broad areas, viz. Concepts in market research; Business planning process; Market segmentation & product differentiation; Consumer survey; Concepts in value chains; Value creation in the existing supply chains; Facilitating and strengthening of value chains; Descriptive analysis of market parameters; Operations and logistics management; Demand planning and forecasting; Enterprise Resource Planning (ERP); ICT in value chains; Commodity trading and Spot exchanges; Analytical tools applied in value chain management; Engaging Public Private Partnership along with people in VC; Developing business plans; Financing value chains and improving Business Environment of value chains. There were total 29 registered participants including 8 female participants which mainly included participants from KVKs and State Agricultural Universities. There were 2 non-ICAR participants. The programme consisted of a blend of lectures, case analysis, discussions and hand-on exercises.

2.1.19 Training Programme on “Data Analysis in Social Sciences Research”

A five-day online training programme on “Data Analysis in Social Sciences Research” was conducted by ICAR-National Academy of Agricultural Research Management, Hyderabad

during Oct. 04-08, 2021. A total of 58 participants including 29 female participants took part. Out of these, 40 were from ICAR institutions, and rest were from SAU and other institutions. The training programme comprised of theory as well as practical classes on data analysis. The topics covered in the training programme include Researchable Issues for Social Scientists in Agriculture; Learning R & R Studio; Graphics in R; Statistical Tests of significance; Correlation & Regression Analysis; Logistic Regression; Impact Assessment Tools & Techniques etc. This training programme also introduced advanced topics like Artificial intelligence (AI) and Machine Learning (ML) which has tremendous applications in Agricultural Research and Agribusiness. The programme had several hands-on sessions. Guest faculty from top institutions and corporate companies having expertise in artificial intelligence and machine learning have delivered their lectures and conducted practical. The valedictory ceremony of the training programme was held on 08th Oct, 2021. In his valedictory address, Dr. Ranjit Kumar, Head, ABM Division, ICAR-NAARM insisted that the trainees should continuously work on the software to maximize the benefits from this programme. He also expressed that a training programme for students pursuing Ph.D. and Post-doctoral should be conducted every year to strength the data analysis skill at the entry level.

2.1.20 Training Programme on Sustainable Management of Small Scale Seed Enterprises

To address the training needs of the small scale seed enterprises which have an important role in making available quality seed to farmers, a training programme on Sustainable Management of Small Scale Seed Enterprises was conducted by



ICAR-National Academy of Agricultural Research Management during October 25-29, 2021. A total of eight training participants representing Private Industries, Krishi Vigyan Kendras (KVKs), State Agriculture Universities (SAUs) and Institutes of Indian Council of Agricultural Research from seven states participated in the training programme. The training programme included sessions such as Overview of Seed Business Scenario: Global and India; Types of Seed Enterprises; Understanding Economic and Institutional Aspects of Seed Production and Marketing; Risk Analysis in Seed Production: Social and Environmental Aspects of Commercial Seed Business; Developing Bankable projects for seed business; Market Supply and Estimation of Market Margins for a Seed Enterprise; Seed Processing Cost Estimation; Market Segmentation, Targeting and Positioning; Breeders Seed Production & Supply to Companies/ Departments; Pricing Strategies and Mechanisms; Market Promotion; Financial Viability Analysis; Seed Quality Regulation in India; Seed Export Procedures and Market Distribution. The training included interactive sessions, hands-on exercises, case studies and experience sharing. Dr. Ch. Srinivasa Rao, Director, NAARM highlighted the importance of the small scale seed enterprises in the seed sector of India in its roadmap to attain the Sustainable Development Goals.

2.1.21 Training Programme on Entrepreneurship Development for Horticultural Students

ICAR-NAARM organized training programme to orient young minds towards entrepreneurship through a six days training programme for students of Dr. YSR Horticultural University on 'Entrepreneurship Development for Horticultural Students' during November 22-27, 2021. Total 60 final year graduate students attended the

programme comprising 39 girls and 21 boys from four constituent colleges of the University, viz College of Horticulture, Venkataramannagudem, College of Horticulture, Anantharajupet, College of Horticulture, Parvathipuram and College of Horticulture, Chinalataripi. The programme was sponsored by Dr. YSR Horticultural University, Venkataramannagudem. The training programme included sessions such as Agribusiness in India: Opportunities & Challenges; Introduction to Entrepreneurship & Startups; Fostering Entrepreneurship in Students; ICT based Agribusiness and Entrepreneurship; Strategies for Promotion of Agri-Clinics and Agribusiness Centre; Business Idea Generation Process; Introduction to Business Plan; Business Plan Preparation; Business Model Canvas; IP Issues and Technology Management; Incubators and Incubation Ecosystem in India; Business Pitch Presentation; Market Research; Success Stories in Agripreneurship; Branding and Advertisement; Skillset for Entrepreneurship and Pitch Your Startup etc. Presentations of all the sessions including hands on exercises were shared with the participants.



Fig. 2.14: Training Programme on Entrepreneurship Development for Horticultural Students

Dr. G. Venkateshwarlu, Joint Director, ICAR-NAARM who was the chief guest during the programme stressed that students should focus

on secondary agriculture and is possible only through entrepreneurship which is integral part of National Education Policy and Atmanirbhar Bharat. Dr. T. Janakiram, Hon'ble Vice-Chancellor, Dr. YSR Horticultural University, emphasized that, the students to focus more on hortibusiness. Further he highlighted on self-employment avenues in horticulture sector for hortibusiness management.

2.1.22 Training Programme on “Management Development Programme on Leadership Development” for the Faculty of Dr. YSR Horticultural University, Andhra Pradesh (a pre-RMP Programme)

Management Development Programme on Leadership Development for the Faculty of Dr. YSR Horticultural University was conducted by ICAR-National Academy of Agricultural Research Management during November 29- December 03, 2021. A total of 40 faculty of Dr. YSRHU nominated for the training programme including 32 Associate Professors & Senior Scientists and 08 Principal Scientists from various research stations, KVKs and Horticulture Colleges. The training was given on different aspects such as Personality Profiling for Leadership Development; Research Data on Leadership; Quality Assurance in Higher Agricultural Education; Emotional Intelligence; Priority Setting, Monitoring and Evaluation (PME); Goal Setting for High Performance; Consultancy & Collaborative Projects-Issues and Concerns; Stress Management; Role Perception; Communication for Leadership; Conflict Management Skills for Leadership and Positive Thinking and How to face Selection Process for Leadership. Presentations of all the sessions including hands on exercises were shared with the participants. Dr. G. Venkateshwarlu, Joint Director, ICAR-NAARM who was the chief guest during the programme stressed on different training programmes are conducted by ICAR and NAARM. In his address,

Dr. T. Janakiram, Hon'ble Vice-Chancellor, Dr. YSR Horticultural University, highlighted the need to develop second line of leadership and qualities that required for organizational growth.



Fig. 2.15: Training Programme on Management Development Programme on Leadership Development for the Faculty of Dr. YSR Horticultural University

2.1.23 Sensitization Programmes (SPs) on Entrepreneurship Development

During this year 2021, the academy organized eight online Sensitization Programmes (SPs) with a participation of 1769 students across the country on Entrepreneurship Development in Agriculture, Animal Husbandry and Fisheries sectors (Table 2.5). These sensitization programmes were supported by National Bank for Agriculture and Rural Development (NABARD to rejuvenate the student's minds for taking entrepreneurship as a profession with the following objectives. (i) to orient the students about entrepreneurship (ii) to make the students understand the scope for startups in agriculture. (iii) to spread awareness about the startup ecosystem in agriculture and the process of incubation.

The sensitization programmes received an overwhelming response from universities across the country under the skill development initiative for agri-graduates to take entrepreneurship in general and agri-preneurship in particular.

Table 2.5: Online sensitization programmes organized by a-IDEA

Sl. No.	Name of the University	Dates
1	Rani Lakshmi Bai Central Agricultural University (RLCAU), Jhansi	1-Jun-2021
2	Tamil Nadu Veterinary and Animal Sciences University (TANUVAS), Chennai	19-Jun-2021
3	Central Institute of Fisheries Education (CIFE), Mumbai	17-Jul-2021
4	Sardar Vallabhbhai Patel University of Agriculture and Technology (SVPUA&T), Meerut	22-Jul-2021
5	Pandit Jawaharlal Nehru College of Agriculture and Research Institute (PAJANCOA), Karaikal	23-Jul-2021
6	Sher-e-Kashmir University of Agricultural Sciences and Technology (SKUAST), Jammu	9-Sep-2021
7	Rajmata Vijayaraje Scindia Krishi Vishwa Vidyalaya (RVSKVV), Gwalior	20-Sep-2021
8	Central Agricultural University (CAU), Imphal	22-Sep-2021



Fig. 2.16: Director participated in Online Sensitization Programmes

2.1.24 Enhancing Resilience through Entrepreneurship

The course was designed to provide a comprehensive understanding of the process of entrepreneurship development (stimulation, support and sustenance). It also provided an opportunity to experience and

explore various entrepreneurial skills and competencies required for an entrepreneur to start and manage entrepreneurship. It had the following specific objectives: to train the participants with adequate knowledge and skill for starting and managing entrepreneurship and to expose them to the various concepts of entrepreneurship in the startup ecosystem. The programme was implemented through a total of 24 sessions including lectures and practical by eminent experts on entrepreneurship. A total of 20 participants from various states and union territories viz., Uttar Pradesh (4), Karnataka (2), Tamil Nadu (1), Lakshadweep UT (1), Haryana (1), Puducherry (1), Andhra Pradesh (9), Telangana (1) representing from ICAR, State Agricultural Universities and NGOs supported KVKs virtually participated in the training programme.



Fig. 2.17: Enhancing Resilience through Entrepreneurship

2.1.25 ICAR-NAARM and Coromandel International Limited (CIL) Collaborative Capability Building Programme-New Entrants Assimilation Training (NEAT) for Young Agribusiness Professionals

NAARM-Coromandel Collaborative Capability Building program-New Entrants Assimilation Training (NEAT) of 5 days (Phase-I) was organized on-line by the academy for 29 sales and marketing professionals of CIL during May 18-22, 2021. The objective of the program is to provide the basic understanding of crop production and crop protection practices and to upgrade the knowledge and skill in sales and marketing domain. The topics covered include Agriculture and Indian economy, Agronomy aspects of major field crops, Soil health based nutrient marketing, Basics in plant health management and plant protection chemicals, Irrigation and fertigation management, Organic fertilizers, ICT and social media in sales, and Value added selling skills. The behavioral aspects such as personality development and emotional intelligence also covered during training programme.



Fig. 2.18: Participants of NEAT programme for the officials of Coromandel International Ltd.

2.1.26 Capability Building Program on New Entrants Assimilation Training (NEAT Phase-II)

The Academy, in collaboration with Coromandel International Limited (CIL), conducted a capability

building program 'New Entrants Assimilation Training (NEAT)' phase-II 31 sales and marketing officers of CIL during November 16-20, 2021. The training was conducted in-person at the Academy. The objective of the program was to provide the basic understanding of agri-ecosystem and to upgrade the knowledge and skill in sales and marketing domain. The topics covered include Buying behaviour of farmers, Market research, ICT application, Target setting, Team building, Opportunities for fertilizer industry, Customization of nutrient inputs, team building, FPO, Soil health based nutrient application, etc. The behavioral aspects such as personality quotients, team building and group dynamics, Health & lifestyle management were also covered during training programme. During valedictory session, Dr. G. Venkateshwarlu, JD highlighted the importance of such partnership between academia and corporate sector for dissemination of latest knowledge and tools in the interest of farming community.



Fig. 2.19: Capability Building Program on New Entrants Assimilation Training (NEAT Phase-II)

2.2 Education System Management for Enriched Learning Quality and Environment

During the year, the Academy continued to focus on the broad theme of enhancing the capacities of the National Agricultural Research and Education System (NARES) for agricultural education policy, planning, and evaluation in

agricultural higher education institutions and to enhance capacities for faculty excellence and technology-enhanced learning for increasing learning effectiveness and continuous learning by deploying emerging media to create new and vibrant education environments. The sub-themes are Lifelong Learning for Professional Development; Education beyond Higher Educational Institutions; Technology Enhanced Learning; Policy Advocacy on Education Systems Management. Research has been carried out on student learning approaches, digital teaching evaluation, upscaling up e-learning (Ag-Academy), apart from the enhancement of quality and relevance of agricultural education through a World Bank Funded Project. On the capacity building front, many training programmes, workshops, seminars, brainstorming workshops and international webinars were organized through online mode. The major themes for capacity building revolved around educational technology and digital teaching techniques, facilitative teaching, digital content development and management, Artificial Intelligence etc. The Academy amply demonstrated the broadening the horizon of online capacity building on a larger scale encompassing thousands of learners through MOOCs (Massive Open Online Courses), which assumed high significance during COVID 19 pandemic.

2.2.1 Digital Teaching Evaluation in Agricultural Education Using TPACK Model

The study attempted at evaluating the various teaching aptitude traits of teaching faculty using a standard TPACK model which was modified to suit the context. Respondents included the teachers who were exposed to TEL, teachers from AUs where ecourses were developed and participants of MOOC. A total of 585 participants responded for these 50 items. Participants rated

their knowledge on TPACK using the following five-level Likert scale. Data analysis was done to assess the internal reliability and relationships between each factor of TPACK subdomain and demographic factors like gender, age, subject, education and teaching experience. Various parameters of modified TPACK model touching Technological (TK), Pedagogical (PK) and Content Knowledge (CK) aspects were analyzed. The studies revealed that the teachers' Technological Knowledge was less as compared to Pedagogical and Content Knowledge. The study also showed that variables such as gender, age, education and teaching experience affected the teachers' digital competencies in terms of certain dimensions. Faculty with higher education had more knowledge of TK and CK. Experienced and Senior faculty were found to have higher levels of competency with respect to PK, PCK and TPK. The study suggested that more technological and practical oriented professional development opportunities for agricultural higher education teachers should be provided to exploit the digital competencies.

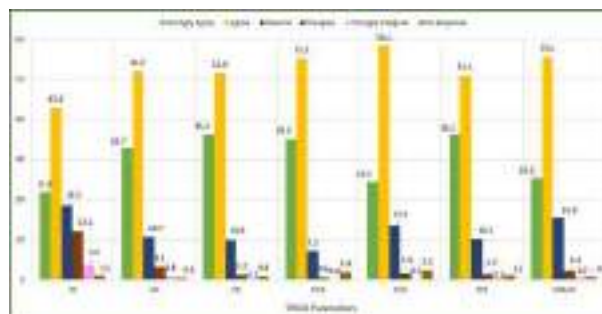


Fig. 2.20: Overall TPACK Assessment of Teachers

The results indicated that further effort must be taken to encourage more technology-related learning and confidence among female teachers. Furthermore, it is suggested to provide customized and need-based capacity development programs to enhance the faculty TPACK competencies level.

2.2.2 Students' Learning Approaches (SLAs) for enhanced learning outcomes in Agriculture Education: A Critical Analysis

An applied research project was taken up with the objectives: to analyse Students' Learning Approaches (SLAs) among agriculture and allied sciences for enhanced learning outcomes; to analyze the changes in Students' Learning Approaches over time; to investigate the factors affecting Students' Learning Approaches for effective learning outcomes and to develop strategies for enhancing the quality of learning outcomes in agriculture education. By using the instrument ASSIST (Approaches and Study Skills Inventory for Students (Tait, Entwistle & McCune (1998), the data were collected from 2161 students, representing 30 State Agricultural Universities of 18 states to analyse the student learning approaches. The students represented all streams of agricultural and allied sciences i.e. Agriculture (43.7%), Vety. Sciences (13.6%), Horticulture (6.73%), Community Science (8.19%), Ag Engineering (6.73%), Forestry (4.35%), Food Science and Technology (8.65%), Agri-Business Management (1.32%), Fisheries (0.73%) and others (1.25%) and among the pooled sample, 56.3% were female students. A very few Postgraduate were considered for the study (5.5%). Critical Analysis on Students' Learning Approaches (SLAs) for enhanced learning outcomes in Agriculture Education indicated

that the strategic learning approach (41.1%) was predominant, followed the deep approach (40.3%) and surface approach (15.5%) (Table 2.6). A significant relationship between students learning approaches and gender ($p < 0.001$), especially in 'strategic' and 'surface' approaches was also found. However, the study establishes no significant association (Chi-square statistic = 24.106, $p = 0.156$) between student learning approaches and disciplines. But there was a significant difference among postgraduate and undergraduate students in the case of 'deep approach' (t-statistic = 2.248, $p = 0.028$). The data were collected twice for temporal dimensions from PGDMA students of NAARM at an interval of one year. Marginal temporal variation among PGDMA students over a one-year time observed from the Deep approach (19 to 4.7%) to the Strategic approach (71.42 to 85.71%). Among the components of Student Learning Approaches, three items viz. Seeking Meaning, Relating ideas and Time management of PGDMA students significantly differed over a year. Wilcoxon test statistic (142) indicated changes in the deep approach of PGDMA students. The study also analyzed the quality of teaching for better learning outcomes by adopting a Course Experience Questionnaire (Ramsden, 1991; McInnis *et al.*, 2001) which comprises of five components - Clear Goals, Good Teaching, Appropriate Assessment, Generic Skills, Appropriate Workload and the data were collected from 180 final year students.

Table 2.6: Student Learning Approaches across Agricultural Universities

Sl. No.	Variable	Category	Student's Learning Approaches			
			Deep	Strategic	Surface	Combination
1	Gender	Female	340 (39.9)	382 (44.8)	110 (12.9)	20 (2.3)
		Male	270 (40.8)	240 (36.3)	124 (18.7)	28 (4.2)
2	Degree	PG	41 (48.8)	31 (36.9)	8 (9.5)	4 (4.8)
		UG	569 (39.8)	591 (41.3)	226 (15.8)	44 (3.1)
	Overall		610 (40.3)	622 (41.1)	234 (15.5)	48 (3.2)



The generic skills were given top priority followed by good teaching, clear goals, appropriate workload and appropriate assessment, in that order, by students while evaluating the quality of teaching. Good teaching, clear goals, generic skills were correlated with both deep and surface approaches of students. While appropriate assessment is negatively correlated with deep and surface approaches and appropriate workload was negatively correlated in the case of surface approach. The component 'Good teaching' was significantly differing among PG and UG students. Clear goals and Appropriate

workload are significantly differing among the three age groups (Table 2.7). The Student Learning Approaches should be considered in the development and implementation of curriculum and also in designing assessment methodologies. The factors affecting the quality of teaching are to be given due weightage. One of the objectives has been integrated as an activity into the National Agricultural Higher Education Project on 'Investments in Indian Council of Agricultural Research Leadership in Agricultural Higher Education'.

Table 2.7: Factors Affecting the Quality of Teaching

S. No.		Gender		Degree		Age	
		Mean Square	F	Mean Square	F	Mean Square	F
1	Clear Goals	21.006	1.780 NS	6.434	0.541 NS	26.260	2.493**
2	Good Teaching	0.118	0.003 NS	172.626	4.159*	56.574	1.382 NS
3	Appropriate Assessment	15.288	1.104 NS	1.938 NS	0.139 NS	21.919	1.670 NS
4	Generic Skills	26.251	0.729 NS	0.615 NS	0.017 NS	54.681	1.597 NS
5	Appropriate Workload	17.551	2.303 NS	13.600	1.779NS	15.819	2.282**

2.2.3 Investments in Indian Council of Agricultural Research Leadership in Agricultural Higher Education

The World Bank funded National Agricultural Higher Education Project (NAHEP) component 2 titled "Investments in ICAR Leadership in Agricultural Higher Education" is implemented at ICAR-National Academy of Agricultural Research Management, Hyderabad. Under NAHEP Component 2 of ICAR-NAARM, 09 trainings/ workshops were organized and a total of 414 (Female: 96) were benefitted during January – December 2021. Three student's awareness workshops titled "Development of Soft skills for Entrepreneurship among Agri Graduates" were organized at 03 State Agricultural Universities covering more than 300 final year agricultural

graduates. As part of coordinating the External Advisory Panel of ICAR-NAHEP, two External Advisory Panel Meetings were organized during Feb, 2021 and November, 2021. ICAR-NAARM established Career Development Centers at 05 State Agricultural Universities i.e. CAU, Imphal; IGKV, Raipur; SKANU, Jobner; SVVU, Tirupati; UBKV, Cooch Behar; and Faculty Development Center at GBPUAT, Pantnagar. During this period 44 Career Development workshop were organized at 05 SAUs benefitting 4327 students, 03 Faculty Development workshops were organized at GBPUAT, Pantnagar benefitting 345 faculties of SAUs. In the process of improving the quality and relevance of agricultural higher education in India, a training workshop on Academic Leadership and Education Management was organized

during September 2021 and 47 faculties from SAUs were trained. NAARM also improving the capacities of Social Safeguards by organizing two days training workshop on Effective Implementation of Social Safeguard plans during July, 2021 and 39 nodal officers of NAHEP were involved and enriched their knowledge on Social Safeguard of NAHEP project. To strengthen the collaborations between SAUs and global educational institutions, an International Webinar on Fostering Global Collaborations among Agricultural Higher Educational Institutions was organized during November 2021 and around 200 delegates participated online and benefitted from the webinar. Major ongoing activities under NAHEP Component 2 are- facilitating global collaboration among agricultural HEIs, collecting alumni information from SAUs, updating the information of network of inspired teachers of NARES, creation of database of abroad trained scientists of NARES, Web portal and databases for CDC/FDC monitoring systems, network meetings in India and abroad with identified institutions for MoUs, consultations with Indian Agriculture University Association and National Academy of Agricultural Sciences, maintenance of Disaster Recovery Centre, coordination of External Advisory Panel of globally renowned educational experts and Pilot testing of a model for nurturing academia- industry linkages, etc.

2.2.4 Ag-Academy: A Framework for scaling up e-learning platform

Ag-Academy is a cognitive edtech platform with a vision to create a National Agricultural Education Platform that enables educators to create content and engage with learners. Basically it enables educators to develop courses with a Web Application and can be shared with different stake holders. Ag- Academy's vision is to get the best minds from the country to share

their knowledge in an easy-to-understand form the following objectives 1) to develop an Artificial Intelligence driven framework for scaling up of electronic learning platform for Indian Agriculture 2) to develop the robust cloud based platform for hosting the courses developed by end users 3) Developing proper strategic, marketing and communication methodologies for popularizing the developed platform and 4) to develop Learning path construction for individual and customized learning of each and every end user. The significant achievements include - development of learning path construction algorithms using Artificial Intelligence technologies, commissioning of Learning Management System at www.agacademy.net; documenting communication methodologies for better digital diffusion and plugging the Learning Path construction module with developed platform at agacademy.net, by ensuring compatibility of PHP, MySQL etc and initiation of scientific communication on learning path methodologies.

2.2.5 Teaching Competency Enhancement of Middle & Senior Faculty of AAU

An off-campus training programme on "Teaching Competency Enhancement of Middle & Senior Faculty of AAU" was conducted by the Academy during January 19-23, 2021. The programme was organized at College of Veterinary Science, Khanapara, Guwahati under Sub-plan. Thirty-two participants attended the programme which included 5 women and belonging to 22 disciplines. The program encompassed all aspects that a middle and senior level faculty in an educational environment is expected to be equipped with. During the programme, a vast range of topics were covered starting from the modern methods of teaching/learning to the oral communications, importance of AI and IOT applications in Agriculture, Bioinformatics and



its applications e-learning practices, Emotional Intelligence, Overview of Teaching Management and Innovative Teaching Methods for Quality Veterinary Education, that can be adopted in the system having immediate relevance to their job. The participants were engaged mostly through discussions, demonstrations, personality development analyses. Design Thinking in Research Project Formulation and Implementation, Building Right Perspectives in Science Communication and Counselling and Mentoring of Students. Resource Generation through Research, Software in open source and freeware environment like whiteboard, and Cloud-Based Digital Teaching were demonstrated with appropriate examples. In his valedictory address, Dr. K.M. Bujarbaruah, Ex Vice-Chancellor, AAU, outlined the critical role of advanced teaching and technology, Teaching Management and Innovative Teaching Methods for Quality of education in agricultural research and education that can be adopted in the system having immediate relevance to the education and research. Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM urged the participants for undertaking the collaborative research to improve agricultural productivity and Education.



Fig. 2.21: Training programme on “Teaching Competency Enhancement of Middle & Senior Faculty of AAU, Assam

2.2.6 Competency Enhancement and Student Mentorship at Acharya NG Ranga Agricultural University Campuses, AP

The Academy conducted two 2-day training programme on “Competency Enhancement and Student Mentorship in Agricultural Sector Opportunities” under Sub-plan in collaboration with Acharya N.G Ranga Agricultural University for the students of the Agricultural College, Mahanandi and Agricultural College, Rajamahendravaram, Rajahmundry.

The training program at Agricultural College, Mahanandi was held during 5-6 March, 2021. Dr. G Prabhakar Reddy, Associate Dean of the college inaugurated the program. The training program sensitized the students regarding the career opportunities, prospects in agricultural education, agri-preneurship and other competitive examinations. Dr. Ch. Srinivasa Rao, Director, NAARM sensitized the students on agri-preneurship and higher education prospects. The training programme at Rajamahendravaram was conducted in collaboration with NAAS, Hyderabad Chapter during 9-10 March, 2021. On behalf of the NAAS, Hyderabad chapter Dr. D. Damodar Reddy, Director, CTRI graced the inaugural function and delivered the address and motivated the students through his real life struggles and experiences. Dr. Ch. Srinivasa Rao, Director, NAARM highlighted the role of students in the nation building and sensitized them for higher education besides motivating them to pursue higher education from reputed national and international universities. The Associate Dean Dr. G. Subba Rao stressed the students to make use of the training to help them pursue their future career avenues.

2.2.7 Design Thinking in Research & Education with Focus on Veterinary Sciences

A virtual training programme on “Design Thinking in Research and Education with focus

on Veterinary Sciences” was organized by the Academy jointly with Rajasthan University of Veterinary and Animal Sciences, Rajasthan under NAHEP during March 22-27, 2021. The programme was aimed at improving the design thinking in veterinary research & education and competence enhancement in education management and policy issues in the veterinary sciences. Around 49 faculty members of RAJUVAS, Rajasthan participated in the six-day virtual training programme. During the closing ceremony, the Chief Guest, Prof. Dr. Vishnu Sharma, Vice-Chancellor, RAJUVAS, Rajasthan emphasized on the integration of intellectual skills and scientific competencies in design thinking in research and education. Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM, Hyderabad stressed on the need for winning external grants in research with multidisciplinary perspectives in emerging areas.



Fig. 2.22: Virtual training programme on “Design Thinking in Research and Education with Focus on Veterinary Sciences, Rajasthan

2.2.8 Brainstorming Workshop on Framework for Strengthening Academia Industry Partnership

As a part of National Agricultural Higher Education Project (NAHEP) on “Investments in Indian Council of Agricultural Research Leadership in Agricultural Higher Education” a brainstorming workshop was organized on May 31, 2021 with an objective to

develop a framework/ model for strengthening the relationship between Agri Universities and Agri Industry to have greater impact on various stakeholders of Indian Agriculture. 1. How to nurture the Agri-Academia-Industry relations in a sustainable way through institutionalization 2. Identifying various pathways of partnership 3. the Process/Framework required for implementation. Dr. V.V. Sumanth Kumar presented the draft Academia-Industry framework as a combination of pathways and governance structure. Later the deliberations were moderated by Dr. D. Thammi Raju, Co-PI, NAHEP Comp 2, ICAR NAARM and were held with inputs given by the experts & Team from Industry and Academia.

Dr. Vijay Raghavan informed that, in India, around 70 per cent seed industries are small or not having any good and big research laboratory so there is huge demand for the industry and academia to develop linkage between them for the better result. Dr. Karuppan Chetty underscored the need for the commercialization of Research at university level. Dr. Tarak Dhurjati emphasized that practical approach is the most important for any model.

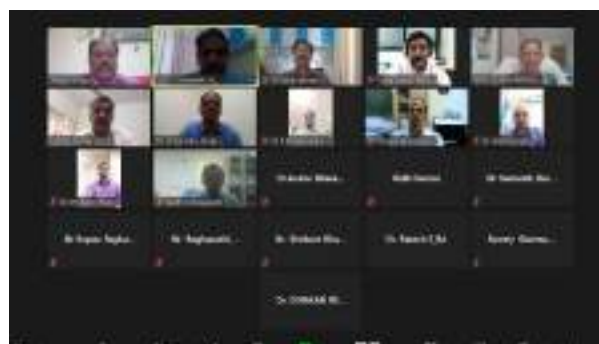


Fig. 2.23: Brainstorming Workshop on Framework for Strengthening Academia Industry Partnership under NAHEP

Dr. S.T. Viroji Rao opined that there should be university approach industry for the CSR Funding. In his concluding remarks Dr. S.K. Soam



underscored the need of studying existing models implemented at TNAU, IIT, IIM and other organizations of national importance.

2.2.9 Training Workshop on Effective Implementation of Social Safeguard Plans under NAHEP

An online training workshop on “Effective Implementation of Social Safeguard Plans” was organized during July 23-24, 2021 with an objective to develop awareness regarding the meaning, need for and importance of social safeguard plans in international research projects. The workshop was organized as part of the World Bank-sponsored National Agricultural Higher Education Project Component-2A. Thirty-nine enthusiastic Deans, Directors, and faculty members from agricultural universities including 06 lady participants from 13 states participated in the online training workshop. Dr. Ch. Srinivasa Rao, Director ICAR NAARM, the Chief Guest of the Inaugural programme stressed the importance of awareness and the need for social justice plans in the institutions for weaker sections like physically challenged people. He also stressed upon the implementation part of these social safeguard activities. Dr. Hema Tripathi, National Coordinator (NAHEP) M&E and ESS, the Guest of the honour of the inaugural programme highlighted the importance and status of social safeguard activities and equity action plans performed in 58 NAHEP institutions throughout the country. She also talked about the integrated social safeguard activities.

2.2.10 Hands on Training on CDC and FDC Monitoring Systems

A Virtual Hands on Training on Career Development Centre (CDC) and Faculty Development Centre (FDC) Monitoring Systems was organized by the Academy on August 24, 2021. The training was

organized as a part of the World Bank-sponsored National Agricultural Higher Education Project Component-2A with an aim to maintain the uniformly and focus in organizing various activities at 5 CDCs established *i.e.*, at CAU, Imphal; IGKV, Raipur; SKNAU, Jobner; SVVU, Tirupati; UBKV, Cooch Behar and FDC at GBPUAT, Pantnagar.

Around 25 participants from CDCs and FDC of SAUs participated in this virtual hands on training. Dr. N. Srinivasa Rao demonstrated the features of CDC/ FDC Monitoring Systems. Dr. Alok Kumar discussed all the verticals of CDC and FDC along with monitoring indicators and explained the need of standardization in organizing various activities at each CDC and FDC. This programme is coordinated by Dr. N. Srinivasa Rao and Dr. Alok Kumar, Co-PIs, NAHEP Component 2, ICAR-NAARM.



Fig. 2.24: Hands on Training on Career Development Centre (CDC) and Faculty Development Centre (FDC) Monitoring Systems

2.2.11 Competency Enhancement in Agricultural Research and Education

During the year, three off-campus faculty development programmes on ‘Competency Enhancement in Agricultural Research and Education’ were organized for the faculty of Assam Agricultural University, Tamil Nadu Dr. J. Jayalalithaa, Fisheries University and Central Agricultural University during September 27- October 01, 2021, October 04-08, 2021 and November 08-12, 2021, respectively, under Sub-plan of NAARM. In

these three programmes 108 faculty members belonging to fisheries, veterinary science, community science, sericulture was benefitted. The training program covered all aspects into an educational environment is expected to be equipped with. There is a vast range of topics starting from the Modern methods of teaching/learning to the oral communications, Importance of AI and IoT applications in fisheries, Bioinformatics and its Applications in fisheries, New Vistas in Technology Enhanced Learning, Psychology principles of teaching and learning, Overview of Teaching Management and Innovative Teaching Methods for Quality Education, Farm based programmes through

broadcast media that can be adopted in the system having immediate relevance to their job. The participants were engaged mostly through discussions, demonstrations, personality development analyses. Design Thinking in Research Project Formulation and Implementation, Building Right Perspectives in Science Communication and Counselling and Mentoring of Students. Resource Generation through Research, Software in open source and freeware environment like whiteboard and Cloud-Based Digital Teaching, Recent trends in ET-ICTs, Social Media, OER etc, Developing Entrepreneurial skills among faculty were demonstrated with proper examples.



Fig. 2.25: Training programme on “FDP on Competency Enhancement in Agricultural Research and Education for the Faculty of TNJFU, Tamil Nadu

2.2.12 Facilitative Teaching Technologies for TANUVAS Faculty

Two programmes on Facilitative Teaching Technologies for the faculty of TANUVAS, Chennai were organized during May 17-21, 2021 and October 26-30, 2021 in which 60 faculty participated. The programmes included 29 female participants. The objectives of the programme are (i) to sensitize and update on the latest teaching methods and its management. (ii) to provide theoretical/practical experience on



Fig. 2.26: FDP on Competency Enhancement in Agricultural Research and Education for the Faculty of CAU, Manipur

Content Capturing and Content Development. (iii) to impart various pedagogy tools and innovative technology-based methodology in agricultural education. The programme was Inaugurated by Dr. K.N. Selvakumar, Vice-Chancellor, TANUVAS, Chennai. The program had touched various contemporary aspects of teaching management, teaching methods, technologies and digital resources that every participant needs to be updated about.



Fig. 2.27: Dr. Ch. Srinivasa Rao, Director, NAARM during the valedictory programme on Facilitative Teaching Technologies (2nd programme)

There were 14 sessions of 75 minute duration which were spread over five days. Everyday an assignment was given to seek the reflections of the participants about the session/topic through handwritten form.

2.2.13 Innovations in Educational Technology

The Academy organized an online training program on “Innovations in Educational Technology” during October 04-08, 2021. A total of 26 participants participated in the program in which about 13 sessions covering both theory and practical were held. More emphasis was given on hands on training on Digital Content Editing and LMS. Additional knowledge was given on application of Education Technology in Clinical Science to enable them to develop similar module in agriculture/veterinary science.



Fig. 2.28: Training Program on Innovations in Educational Technology

2.2.14 Education Management and Academic Leadership

The programme was organized under National Agricultural Higher Education Project to enhance the efficiency of the Indian Agricultural Education System through Education Management and Academic Leadership. The training workshop focused on key areas of education management which includes Teaching Management, New vistas in Technology Enhanced Learning, Impact of Curriculum on the Quality of Agricultural Education, National Education Policy Education Management for Building Excellence, Agricultural Research Management in India-Contemporary Issues and Challenges, Academia-Industry Linkages for Academic Excellence etc. Academic Leadership focused on, Leadership Challenges in Agri Universities, Collaboration for Academic Leadership and Leadership Attributes apart from communication, Stress management, Emotional Intelligence etc. coupled with perspectives of Extension Management agricultural research management. The program encompassed all aspects that leaders or prospective leaders are expected to be equipped with. The programme witnessed the participation of 47 faculty and administrators of 22 Universities.



Fig. 2.29: Dr. Ch. Srinivasa Rao Director, NAARM delivering the Valedictory Address

2.3 Enhancing Capacities for Leadership and Governance

Human Resource Management Division has taken the strategic approach to the effective and efficient management of people in the organization such that they help to gain a competitive advantage to the organization. This will help in maximizing employee performance in the service of an employer's strategic objectives. The Division is involved in the development and implementation of human resource policies and processes for the ICAR. Human Resource Management deals with issues related to Performance Management, Organizational Development, Safety, Wellness, Employee Motivation, Teamwork and training individuals and groups on different soft skills required for individual and organizational development in ICAR. The Main objective of the HRM Division is maintenance of better human relations in the organization by the application and evaluation of policies, development and procedures and programs relating to human resources. One of the important domains of HRM is Management Development and Leadership. The Strategic framework of the thematic area is to enhance capacities for leadership and governance by

- Institutionalize a framework for leadership development at all levels of NARS - early, mid-career, senior professional and manager across all functions (research, education, extension)
- Institutionalize good governance in NARS by building capacities for effective management of institutions, performance assessment, accountability and developing strong organizational value systems and work culture that promote innovation

Human Resource Management Division focuses on Management Development in a highly-interactive way for the development of personal and group leadership skills. To expand managerial ability in an organization, to achieve its mission and goals by improving individual and organizational performance is usually done by conducting a training programme called Management Development Program (MDP). The training focuses on developing an employee's skill levels and has a more long-term focus designed to help an employee to develop for future challenges. Major objectives of these include effectiveness, learning outcome, and performance of participants in their professional careers. Both in Society and public administrations there is a genuine demand for leadership development programme which helps for finding innovative solutions for the existing and future problems. Leadership Development program was initiated to track the responses of the senior managers to the successes, setbacks, and outcomes in the system. Management Development Program is one of the most desired methods of developing leadership skills and managerial within an organization. Leaders and managers are considered as a highly powerful group in terms of creating high-performance organizations. The Management Development is a systematic awareness, ability, attitude and behavior process that helps in the development future leaders.

2.3.1 Identification of Best Practices developed by the Leadership in Scientific and General Administration in NARES

Based upon input given by NARES, 15 Best Practices identified (Practiced by Leadership in Scientific and General Administration of NARES during their respective period and it was recognized by the different stakeholders) and



simultaneously 5 practices identified which needs to avoided for better work environment.

2.3.2 Happiness and Psychological Well-Being of Scientific Personnel in NARES

This study was conducted to investigate the personality traits of agricultural scientists and to find out their relationship with well-being attributes and to predict their happiness and life satisfaction. Data was collected from 622 respondents (424 males, 198 females; mean age 41.8 ± 6.3 years) by administering Big-Five Inventory (BFI-10), Oxford Happiness Questionnaire (OHQ), Satisfaction with Life Scale (SWLS), and Psychological Well-Being (PWB-18) questionnaires. Findings revealed significant gender differences in neuroticism and openness, favouring female scientists. Such gender differences were not observed in their scores on happiness, life satisfaction and Psychological Well-Being (PWB). Most of the scientists reported higher levels of happiness (86.2% moderately to very happy), life satisfaction (68.9% satisfied to extremely satisfied) and PWB (57.4%). With age, extraversion, agreeableness and conscientiousness were increased; neuroticism and openness were decreased. Similarly, environmental mastery, positive relations, and total PWB were improved with age. Except for neuroticism, personality traits were positively correlated with happiness, life satisfaction, and PWB. The regression model on happiness captured 63.7% of variation by considering personality traits (neuroticism, being the strongest contributor) and traits of PWB (self-acceptance and environmental mastery, being the largest contributors). In the model for life satisfaction, neuroticism among the personality traits; self-acceptance, environmental mastery, positive relations and purpose in life among the PWB captured 35.3% of the variation in life satisfaction.

The implications of the study include selection, training, and create possible organizational interventions to build a happy workforce environment, which is a pre-requisite for any scientific endeavour. Since personality and PWB are relatively stable over time and predict employee's happiness and productivity, these traits can also be used in the selection and recruitment criteria. Another option is to alter workers by serving to them learn to be happier. There's sensible proof that varied sorts of stress management coaching will have positive effects on employee happiness. A variety of ways exist wherever individual workers will proactively self-monitor or manage their perceptions to reinforce positive, and discourage negative, displays of fleeting mood and feeling. The third approach to potential interventions involves ever-changing the surroundings so it promotes or doesn't impair worker PWB.

2.3.3 Emotional Intelligence and Organizational Effectiveness

Emotional intelligence (EI) is known to contribute for the success of individual or group, and plays an important role in organizational effectiveness through positive behavior. The objective of the study was to find out the relationship between EI and different components of organization effectiveness viz., teamwork effectiveness, conflict-handling modes and leadership competencies of the scientific personnel in National Agricultural Research and Education System (NARES) in India. The data was collected from 234 respondents by administering EI Test (EIT), Teamwork Skill Questionnaire, Thomas-Kilmann Conflict Mode Instrument (TKI Test) and Leadership Effectiveness Profile. The results indicated that 93.2% of the participants recorded an average level of EI; whereas, only 6.8% have a high level of EI. Majority of participants (78.6%)

recorded lower levels of teamwork effectiveness as against 19.2 and 2.2% having average and high levels of team work effectiveness, respectively. Among the conflict-handling modes, avoiding (37.2%), compromising (20.5%) and accommodating (20.1%) are the dominant modes employed by the participants. Among the leadership competencies, people enablement, directional clarity, driving persistence and change orchestration are the dominant reported competencies. EI is significantly and positively correlated with teamwork effectiveness and all the attributes of leadership effectiveness. Among the domains of EI, self-awareness, motivating oneself and handling relationships are significantly related to the teamwork effectiveness and leadership competencies. The findings of the study revealed the importance of EI in contributing the overall success of the organization through building up the teamwork effectiveness and contributing to the leadership competencies. The implication of the study includes the role of EI as important criteria in selection and recruitment and a competency to be trained in employee's in-service capacity building programs in order to improve the overall organizational effectiveness.

2.3.4 Assessment & Development of Competency Framework for Agricultural Research and Extension Scientists

Identified the functional (6 Nos) & Behavioural competencies (2 Nos) for research & extension scientists and competency domains are ranked for different scientific cadres. This would help HRD Nodal officers to nominate/guide the scientists for attending training programs. Eight competencies are identified for the self-development of scientists for superior performance. Seven competencies are identified for assessing the

scientists of different cadres. All the competencies are described, and indicators are developed for ready reference.

2.3.5 HR Interventions and their influence on Institutional Innovations

Major HR Interventions that influence Institutional Innovation in ICAR are Participative Decision Making, Reward Systems, Team Spirit, Delegation, Training & Development, Career Planning and Development, Motivation, Remuneration and Talent Management. A Model is fit for HR interventions and institutional innovation with Knowledge Management Practices (KM Creation, Storage, Transfer and Application) as mediation and Innovative climate as moderation effects. The major strategies to nurture HR interventions for institutional innovation are Recognition programmes that help to encourage and inspire staff to discuss and develop ideas; Reinforcing the value of innovation activity and outcomes; Create career development frameworks that enable resource sharing to break down internal silos and foster idea-sharing; Develop leaders to "horizon scan" and be dissatisfied with the status quo regularly; Determine which positions contribute disproportionately to innovation value; Train the employees who occupy these responsibilities; Ensure complete competency in the innovation process; Design the organization (structures, processes, roles, capabilities, and so on) to support the innovation strategy; Reducing barriers and encouraging cooperation to speed up the idea lifecycle; Assisting in the adoption of new working habits that promote innovation; Institutional venturing where employees receive budget in the form of money and time and support from management and experts



enabling them to pursue their business idea; Selective hiring; Extensive and well-designed training; Self-managed teams; Decentralized decision making; Sharing information and remuneration, and Talent Management.

2.3.6 Training Programme on Leadership Development for Women Scientists in NARES

A 3-days online training programme on “Leadership Development for Women Scientists in NARES” was organized in online mode from during March 08-10, 2021 on the occasion of International Women’s Day 2021. In his inaugural address, Chief Guest Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM emphasized the United Nations theme of International Women’s Day-2021; Women in Leadership: Achieving an equal future in a COVID-19” In this context; Dr. Pratibha Jolly, Former Principal, Miranda House, made a very good deliberation on her journey as a scientist and later as Principal. She stressed on fostering the next generation women science leaders. On this occasion, a booklet entitled “Women led Agri-start-ups: Courage to Innovate for Change” was also released by director NAARM. A panel discussion was organized on women and governance in which Dr. Neelam Grewal (VC, Guru Kashi University, Bhatinda), Dr. Meena Kumari Former DDG (Fisheries) ICAR and Dr. Kalpana Sastry, MD, Ag-hub, PJTSAU, Hyderabad narrated their stories being in governance to set examples for the participants to follow. This programme was organized with the objectives to increase the participation of women in agricultural research and educational leadership for better gender balance; for qualitative development of research and higher education by utilizing the untapped potential of women as leaders and to develop confidence and facilitate networking among participants

for better collaborations and governance. The programme was attended by 75 (70 women and 05 men) participants representing around 25 SAUs, 18 ICAR, four Krishi Vigyan Kendras and five other universities.



Fig. 2.30: Training Programme on Leadership Development for Women Scientists in NARES

2.3.7 Management Development Program on Leadership Development (A pre-RMP programme)

Two weeks virtual Management Development programme on Leadership Development (Pre-RMP) organized during June 14-25, 2021 & December 13-24, 2021. A total of 91 participants including 09 women represented the ICAR & State Agricultural Universities virtually participated in this research management programme. The programme provided an opportunity for the pre-RMPs, who are in the “middle zone”, to develop smooth and gradual transformation to step into future leadership position. The major objectives of the programme was to help participants understand their own personality dimensions; to explore the leadership challenges and core competencies needed in present day context; to develop understanding and skills in research management, Information, knowledge management, administration and finance management; and to adapt ideas of diverse skills for sharpening capacities

to effectively execute various roles in the organization. The programme covered the areas of Leadership and personality development; Research management; Information and knowledge management; and Administration, finance, vigilance management & lessons from epics. Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM, Hyderabad, in his Valedictory address explore the leadership challenges and core competencies needed in present-day context; and given tips to participants, how to face selection process for leadership position in NARES.



Fig. 2.31: Online Management Development Program on Leadership Development (A pre-RMP programme)

2.3.8 Training Programme on Stress Management

An Online Training Program on Stress Management was conducted by the Academy during July 6-9, 2021. A total of 19 participants including 17 Principal Scientists from ICAR and 2 Professors from SAUs participated in the program. This program provided in-depth understanding of personal and psychological causes of stress/anxiety and its effects-physical, social and organizational behaviour and to develop constructive means for coping up with distress and achieve productive and enriched life-style. The participants were exposed to various methods and techniques

of stress management viz., health and life style management, emotional intelligence, time management, inter-personal relationship management and the role of diet in stress mitigation through interactive lectures, practical sessions and experiential sharing.



Fig. 2.32: Training Programme on Stress Management

2.3.9 Workshop for Vigilance Officers of ICAR Institutes

The Academy organized 3-days Training Workshop for Vigilance Officers of ICAR Institutes during August 16-18, 2021. The training workshop benefited 42 Vigilance Officers of various institutes of ICAR who were nominated by ICAR.

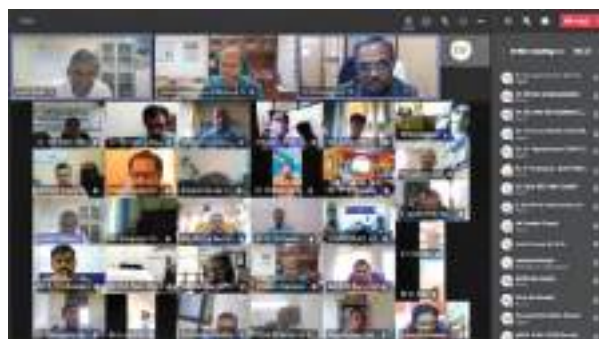


Fig. 2.33: Online Training Workshop for Vigilance Officers of ICAR Institutes

The participants were sensitized on different vigilance management activities such as preventive vigilance, duties and responsibilities of vigilance officers, disciplinary procedures, recruitment process, RTI and purchase procedures etc. The Program was inaugurated by Shri. Shaleen Agrawal, Director (DARE) & CVO (DARE & ICAR).



Dr. G. Venkateshwarlu, Joint Director, ICAR–NAARM addressed in the valedictory session.

2.3.10 Management Development Program on Leadership Development

The Academy organized a Management Development Program on Leadership Development for the Faculty of Dr. YSR Horticulture University during November 29–December 03, 2021. A total of 40 faculty of Dr. YSRHU nominated for the training Program including 32 Associate Professors & Senior Scientists and 08 Principal Scientists from various research stations, KVKs and Horticultural Colleges participated in the program. The training was given on different aspects such as Personality Profiling for Leadership Development; Research Data on Leadership; Quality Assurance in Higher Agricultural Education; Emotional Intelligence; Priority Setting, Monitoring and Evaluation (PME); Goal Setting for High Performance; Consultancy & Collaborative Projects-Issues and Concerns; Stress Management; Role Perception; Communication for Leadership; Conflict Management Skills for Leadership and Positive Thinking and How to face Selection Process for Leadership. Presentations of all the sessions including hands on exercises were shared with the participants.



Fig. 2.34: Management Development Program on Leadership Development for the Faculty of Dr. YSR Horticulture University

Dr. G. Venkateshwarlu, JD, the Chief Guest during the valedictory session stressed on different training programmes conducted by ICAR and NAARM. In his address, Dr. T. Janakiram, Hon'ble Vice-Chancellor, Dr. YSR Horticulture University, highlighted the need to develop second line of leadership and qualities that required for organizational growth.

2.3.11 Training Programme on Executive Development Programme (EDP) on Leadership Development for the Senior Officials of YSRHU, AP

An Online training programme on “Executive Development Programme (EDP) on Leadership Development” for Senior Officials of Dr. YSR Horticulture University, AP was organized during Nov 30– Dec 04, 2021 at the Academy to upgrade the knowledge and skills of agricultural professionals for sharing of knowledge management process in research, education pertaining to agriculture education and research. The aim of the programme is to enhance leadership competencies of the newly recruited Research Management Positions in DRYSHU, AP. The programme covered the topics such as Leadership Development; Leadership experiences and Positive thinking, Teaching Management, New Vistas in Technology Enhanced Learning, Emotional Intelligence for organizational effectiveness, Enhancing Capabilities in IPRs, Information and Knowledge Management; Science Communication, AI and IoT Applications in Agriculture and allied sectors, Managing Holistically by Design Thinking, Perspectives of Extension Management for Academic Leadership, Contemporary Issues and Challenges and resource generation etc., There are 32 participants from Dr. YSRH University.



Fig. 2.35 Online Training Programme on Executive Development Programme (EDP) on Leadership Development for the Senior Officials of YSRHU, AP

2.4 Extension Systems Management in Market-driven Environment

Timely dissemination of agricultural innovations through outreach and educational services is vital for farmers' empowerment. It is a core research management competency that requires concerted inputs in terms of policy, research and capacity building besides other support systems. It is here that effective management of extension systems assumes critical importance within the larger domain of agricultural research management. Agricultural extension systems at national, regional, state and district levels have played significant role in the growth of agricultural and allied sectors and farmers' welfare. However, the changing context of agriculture needs innovative approaches and methods to manage extension systems to align their efforts towards sustainable livelihood security of millions of peasant families. Sustainable livelihood security and farmer's empowerment through management of frontline extension systems is one of the six thematic areas of the Academy's strategic plan for achieving its vision and mandate. This thematic area is addressed through research and policy, capacity development and outreach services. The core functional areas under this theme are extension policy and management, institutional

innovations, knowledge management including digital extension and gender mainstreaming. Brief account of the work done and achievements pertaining to research and policy, capacity development and outreach activities during the reported period is presented below.

2.4.1 Development of a Framework for Gender Mainstreaming in the Development Schemes of Agriculture Sector in India

This research project is being carried out to develop methodological framework for assessment of gender attribution in developmental schemes of agriculture across ministries and to recommend reforms in identified government schemes for gender mainstreaming. Methodological framework was finalized including selection of programmes and status of each programme is documented through a series of online consultation of project team with domain experts. A working draft of farm women's handbook is ready with details on programmes and schemes pertaining to farm women of seven ministries and is being reviewed internally.

2.4.2 Management and Impact Assessment of Farmer FIRST Projects

This research project is being implemented from the year 2017 with funding support from ICAR. NAARM is a partner institution in the project management consortium mandated to support management of FFP through capacity building, documentation, information showcasing and impact assessment. During the reported period, 'outcome-based assessment' taken up to evaluate the programme in terms of identified outcome indicators. Data were collected from 40 farmers from the adopted village and 10 farmers from non-adopted village (as control group), randomly selected as respondents. Thus, a total of 2500 farmers from 50 FFP centres were interviewed

with the pre-tested interview schedule to assess the outcome of FFP in the villages. Two prominent technological interventions from each FFP centre were also assessed in terms of change in net income and B:C ratio. The data so collected was analyzed using descriptive statistics and other statistical tests. Salient findings of this evaluation study are

- There was significant increase in adoption of improved cultivation practices and productivity due to the technological interventions implemented across the centres under FFP in over 80% of farms.
- Further, this programme has resulted in significant gains of over 25 per cent in asset creation, especially farm assets and per capita consumption of various food items.
- Significant improvement in nutrient intake was recorded as outcomes of the interventions. Adoption of poultry and increase in the farm animals have led to increase in the income as well as nutritional security.
- The perceived impact of the FFP programme was significantly above the average and over 95 per cent of farms have impact score of over 3 out of 5. Farmers from adopted villages reported that the programme is beneficial as

observed from their mean score of perceived effectiveness which was significantly higher than that of their counterparts in the non-adopted villages.

- FFP centres have contributed immensely towards climate resilient agriculture. Various interventions across all components of agriculture including animal husbandry, dairy, etc. coupled with concerted efforts to create sense of ownership by mobilizing farmers into viable groups like Farmer Producer Organizations have resulted in significant increase in farm income

Onsite field visits were made to FFP villages of ICAR-IIVR, Varanasi; CAU, Imphal and ICAR-NIANP, Bengaluru. Our team conducted focused group discussions with farmers and FFP teams and documented good practices for scaling up. Project team facilitated scouting eleven farmers (six from FFP villages) and awarded them with “Innovative Farmers Award” on the occasion of 46th Foundation Day ICAR-NAARM on 1 September 2021.

The output was released as book in English and Hindi languages. Hindi version of the book was released by Sri Narendra Singh Tomar, Hon’ble Union Minister of Agriculture & Farmers Welfare, GoI during 93rd Foundation Day of ICAR on 16 July 2021.



Fig. 2.36: Book Released by Sri Narendra Singh Tomar, Hon’ble Union Minister of Agriculture & Farmers Welfare, GoI during 93rd Foundation Day of ICAR on 16 July 2021



2.4.3 Impact Assessment of SC Sub Plan (SCSP) implemented by Agricultural R & D Institutes on Socio-economic Status of Beneficiaries

The project proposed for study the impact of the SC peoples is socially and economically very backward in our country as compared to other category of population. About 71% of SC farmers are agricultural labourers. They work for wages on others' land. In rural areas, they live as tenants and marginal farmers. Their literacy level is only 66.1% as compared to the all-India level of 73 per cent. Female literacy among them is as low as 56.5 per cent against all- India female literacy level of 64.6 per cent (Census Report, 2011). They are mostly poor in many states and in spite of affirmative action by the Govt., they continue to be marginalized from the rest of the society. It might be due to their very backwardness itself that they find it difficult in accessing the schemes and subsidies for their benefits and ultimate upliftment. Towards the objective of improving their socio-economic conditions, Govt. implements Schedules Caste Sub-Plan (SCSP) since 2007-08 as an umbrella strategy to ensure flow of targeted financial and physical benefits from all the general sectors of development for the benefit of Scheduled Castes. As a constituent of ICAR, NAARM is implementing various activities and interventions under SCSP for the target beneficiaries from SC farmers, including youth and women in selected districts such as Mahbubnagar, Adilabad, Nagar Kurnool and Ranga Reddy in Telangana state by adopting the villages having at least '50% and above' SC population since 2019. Interventions in the form of providing agri-inputs and imparting skill development programmes to enhance their income and livelihood. Any such intervention is

expected to bring about the desired impact after a considerable period. In this project, the objective wise survey schedule and questionnaire has been prepared and collection of primary data on socio economic status of beneficiaries and the base line survey data collection and tabulation of the data is under process.

2.4.4 Training Programme on Soft Skills for Extension Professionals

This 10-days online training programme was designed to enable participants to benchmark their select soft skills, provide opportunities to practice and develop soft skills through structured exercises, and inculcate strategies to leverage soft skills for personal and professional excellence. The programme covered important soft skills like communication (listening, reading, presentation, assertiveness, non-verbal) skills, personality development, team-work and group dynamics, time management, leadership, conflict resolution, motivation, interpersonal relations, life style management, professional values and ethics, and networking and negotiation skills, each with structured exercises and experiences to enable benchmarking and strategizing for improvement.



Fig. 2.37: Group Photo of the Participants

These contents were imparted using a combination of lectures, structured self-assessment experiences, and individual and group presentations. Analysis of participants' feedback revealed that the

programme was very well organized to meet their learning expectations. It was rated very good to excellent on most of the indicators pertaining to training design and delivery.

2.4.5 Training Programme on Advances in Extension Research and Evaluation

This 10-days online training programme was designed to enable participants to equip participants with the latest knowledge, skills and innovative practices needed for conducting impact-oriented and quality extension research and evaluation; to orient the participants with the advanced methodology used in social science research and to facilitate networking among the participants for inter-institutional research in extension. Thirty-seven participants representing NARES institutions from fifteen states attended the programme. Progressive and experiential learning strategies were followed to allow participants to progress in learning from known to unknown and immediately applying the learnt knowledge, attitude and skills (KAS) through structured exercises (individual and team). Participants were organized in to 6 teams to work on writing concept note on an assigned theme integrating 2-3 good practices covered in the workshop and the guidance of mentors. Overall, it was an insightful learning experience for the participants, resource persons and coordinators alike.



Fig. 2.38: Online Training on Advances in Extension Research and Evaluation

2.4.6 Skill Development Training on Organic Cultivation of Nutrition Garden

As a part of SC Sub-Plan implementation, ICAR-National Academy of Agricultural Research Management, Hyderabad organized Skill Training on Organic Cultivation of Nutrition Garden in collaboration with SAKVK, Gaddipally for creating awareness on organic nutritional garden and its health benefits especially for SC women during 18, 19 & 25th September 2021. The program was organized at SAKVK, Gaddipally and 300 farm women and farmers represented from 12 villages from three mandals in Suryapet district of Telangana were benefited with the training. The training cum demonstration was given on different aspects such as Preparation of vermicompost, Organic practices and preparation of inputs, Production techniques in Vegetable nursery seedlings, Tips in cultivation of back yard nutritional garden and its establishment and also explained the importance of cultivating Nutrition garden and distributed vegetable garden kits to farm women. Dr. G. Venkateshwarlu, Joint Director, ICAR-NAARM urged rural women to start cultivating nutrition garden so as to provide a balanced diet to family members and to overcome the deficiency of micro nutrients especially of iron, Vitamin-A, Iodine etc.

In his address, Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM highlighted the need for organic cultivation of nutrition garden through which rural women can produce pesticide free vegetables for domestic purpose and can also reduce the expenditure on vegetable purchases.



Fig. 2.39: Training on organic cultivation of nutrition garden

2.4.7 Stakeholders Training and Demonstration of Farm Implements

As a part of SC Sub-Plan implementation 2020-21, the Academy carried out Training and Demonstration on use and distribution of farm implements, farm hand tool kits, power sprayers, grain storage bins and tarpaulin covers in five adopted under sub-plan. The Academy also created custom hiring centre in all adopted villages with bush cutters, computers, multi-function printers, UPS, computer tables, computer chairs, almirah, book racks, plastic chairs and tables in village panchayat in the presence of village secretary, Agriculture officer and other officials. In every adopted village, the activity was taken-up under the guidance of Dr. Ch. Srinivasa Rao, Director, and Dr. G. Venkateshwarlu, Joint Director, ICAR-NAARM. Dr. M. Balakrishnan Chairman SC Sub-Plan committee and Dr. M. Ramesh Naik, Member secretary SC Sub-Plan committee explained the importance of Sub plan scheme and implementation of scheme by ICAR-NAARM. Along with NAARM faculty & staff, the officers of local institutes like KVK Heads, Project Director (ATMA), Assistant Director of Agriculture, Agriculture Officer, Agriculture Extension Officer were also present during the events.

Table 2.8 Stakeholders Training and Demonstration of Farm Implements

Village	Date	Beneficiaries
Vonguronipally & Kalmulonipally, Nagarkurnool district	26-27 July, 2021	259
Lachampur, Adilabad district	30 July, 2021	65
Galiguda, Mahabubnagar district	6 August, 2021	45
Agraharam, Mahabubnagar district	11 August 2021	52

2.4.8 Livelihood Enhancement Training for Farmers

The Academy organized a 3-day training on Vermicompost Production Technology for Entrepreneurs and farmers of North Bank Plain Zone (NBPZ) of Assam at Biswanath College of Agriculture (BNCA), AAU, Assam during December 8-10, 2021. The Program was organized under the SCSP in which 111 farmers from six different districts of NBPZ viz., Udalguri, Darrang, Sonitpur, Biswanath, Lakhimpur and Dhemajj of Assam participated. During his talk, Dr. RN Barman, Associate Dean, BNCA stressed upon sustainable agriculture and described about the importance and proper management of organic resources in agriculture. Mr. Kulen Sarma, Additional Deputy Commissioner (ADC), Biswanath district informed about the state government initiatives on waste management undertaken at district level. Er. KK Nagle, Director, NER-FMTI expressed his gratification over the enthusiastic attendance of farmers particularly female farmers and mentioned that vermicomposting will bring revolution in the country which will give enough employment to the youths.



In his inaugural address, Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM highlighted different models of Integrated Farming Systems; which are launched in the NER region for the benefit of farming community and described about the importance of organic resources in agriculture. He encouraged the farmers to take-up vermicompost production in entrepreneurial approach and to generate the income. He highlighted the mitigation strategies for climate change, need of farm mechanization practices in small holdings, custom hiring models etc.



Fig. 2.40: Training on Vermicompost Production Technology for Entrepreneurs and farmers of North Bank Plain Zone (NBPZ) of Assam

2.5 Mobilizing Science and Technology for Innovation and Sustainable Development

Innovation in agricultural science and technology is critical for good governance in the national agricultural research system and overall sustainable development in agriculture. Evidence based research therefore assumes greater significance to leverage science based interventional tools for effective research and technology management in agriculture. Infusing innovative ideas and bringing in a new milieu in the agricultural research system with the integration of advancement in information technologies plays a critical role for better governance and online management by developing digital products and services, which will greatly improve decision making and reduce the redundancy

in research. Special focus is also accorded to agricultural technology management with the application of management fundamentals for efficient management of research outputs and management of Intellectual Property for gaining competitive advantage. In order to achieve the objectives under this theme, the Division of Research Systems Management carves out the following three niche areas in which the mandated activities of research, academic and capacity building activities revolve around: (1) Research Management and Policy Advocacy; (2) Technology and Innovation Management; (3) Research in the context of Food security, Sustainability and Globalization.

2.5.1 Impact Assessment of Research and Development Programs on Food and Nutritional Security: Pilot Study

The pilot project started with an aim to profile key research and developmental programs on food and nutritional security. Initially the profiling of the programs on food grains were targeted and during which inventory of major schemes on food with special emphasis on cereals and nutritional security was prepared. The schemes under National Food Security Mission, viz., NFSM-Wheat, NFSM-Pulses and NFSM-Coarse Cereals were selected for the preparation of inventory. The food supply status in terms of availability at national level and at household level was also worked out.

2.5.2 IP Management and Transfer/Commercialization of Agricultural Technologies Scheme

NAARM-ZTMU is involved in capacity building in IP management and tech transfer/commercialization domain by providing trainings and also serve as “help desk” for Agri Business Incubators/Startups/SAUs. NAARM-ZTMU helps other ICAR institutes in conducting patent search as prior art for patent filings, with

proper institutional mechanism to protect/ manage innovations/intellectual properties (IPs) generated within institute, unit has facilitated the filing of 11 copyright applications has been facilitated. As a part of capacity building in the areas of IP and Technology Management, two training programs were conducted for the faculty of State Agriculture Universities.

2.5.3 IPR, Copyright, Industrial Design and Plagiarism

To raise the level of awareness and knowledge among academicians and researchers about the Intellectual Property Rights issues and Plagiarism, the Academy conducted two five-day online training programmes on “IPR, Copyrights, Industrial Design and Plagiarism” for the staff of TANUVAS, Chennai during March 22-26, 2021 and August 24-28, 2021. The participants, in total, comprised of 60 faculty members from TANUVAS, Chennai.

The training programmes were designed to cover the various aspects of intellectual property rights viz., Patent, Copyrights, Trademarks, Industrial designs and Geographical Indications and Plagiarism in scientific writing. The training also dealt with the aspects of management of innovations, valuation of intellectual properties, business planning, biodiversity issues, etc. The importance of innovations and IPRs in Indian agriculture was highlighted and impressed upon the participants to disseminate the learnings to the students of the University it was emphasized on the importance of IPRs and asked the faculty undergoing the training to make use of learnings of the training for the benefit of university. The training programmes were sponsored by National Higher Education Project (NAHEP).



Fig. 2.41: Gallery of participants during virtual training on “IPR, Copyright, Industrial Design and Plagiarism”

2.5.4 Science Communication Workshop for Indian Forest Service (IFS) Probationers

The Academy organized a virtual workshop on Science Communication for IFS Probationers belonging to 2019-21 and 2020-22 batches from Indira Gandhi National Forest Academy (IGNFA), Dehradun during April 23, 2021. The participants comprised of 165 probationers including the officers belonging to Royal Bhutan Forest Service and the faculty members of the Academy. The workshop was organized as a part of the Memorandum of Understanding (MoU) signed between ICAR-NAARM and IGNFA. Shri Bharat Jyothi, IFS, Director, IGNFA highlighted the need to build adequate skills in technical and scientific writing in the light of changing trends in the professional services of the officers, some of which requiring full-time research. Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM during the programme emphasized on how science communication for IFS Officers is essential for societal advancements. Illustrations were given with various real-life situations and the issues they may encounter while communicating on different platforms. The need for building right perspectives for scientific communication and

sensitization various online tools. As well as practical tips for writing research papers and popular articles were provided.



Fig. 2.42: Workshop on Science Communication for IFS Probationers, IGNFA, Dehradun

2.5.5 Induction Training on Project Management and Research Methodology for Newly Recruited Scientists of ICFRE, Dehradun

The Academy organized a virtual induction course on Project Management and Research Methodology for the Scientists of Forest Research Institute (FRI), Dehradun during May 03-14, 2021. The participants comprised of 30 newly recruited scientists attached with FRI as part of their Foundation Course conducted by the Indian Council of Forestry Research and Education (ICFRE), Dehradun. The program was organized as a part of the Memorandum of Understanding (MoU) signed between ICAR and ICFRE. During the induction course, the participants were introduced to the concept of Design Thinking (DT) and its use in various aspects related to research management, with relevant illustrations and case studies. Innovative exercises were designed to instill DT perspectives among the participants in research problem identification, prioritization of research solutions, research concept development and project logframe construction. They were

introduced to experimental designs, statistical methods and data visualization tools and provided hands-on exercises using contemporary statistical software. During his inaugural address, Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM highlighted the significance of winning research grants and science communication to be a successful researcher. Dr. G. Venkateshwarlu, Joint Director, ICAR-NAARM highlighted the need for strengthening the partnership between both agricultural and forestry scientists by leveraging existing institutional mechanisms. Earlier, Dr. HS Ginwal, Dean, (Academic), FRI contextualized the training module in their overall foundation course.



Fig. 2.43: Induction Training on Project Management and Research Methodology

2.5.6 Integrated Watershed Management (IWM) for the IFS Probationers from IGNFA, Dehradun

The Academy organized a virtual training on Integrated Watershed Management (IWM) for the IFS Probationers from Indira Gandhi National Forest Academy (IGNFA), Dehradun during May 24-29 2021. The participants comprised of 90 probationers including the officers belonging to Royal Bhutan Forest Service belonging to 2018-20 and 2019-21 batches. The program was organized as a part of the Memorandum

of Understanding (MoU) signed between ICAR-NAARM and IGNFA.

Shri Bharat Jyothi, IFS, Director, IGNFA in his opening remarks highlighted the relevance of watershed management in the light of evolving responsibilities of the IFS officers. Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM took a lead session on the current status of watershed in India and provided practical tips on participatory watershed management. He highlighted the interface between the forestry and agriculture sector and the need for stakeholder engagement for successful watershed development. During the program, eminent speakers like Dr. Ashok Dalwai, CEO, National Rainfed Area Authority (NRAA), New Delhi; Dr. Suhas P. Wani, Former Director, ICRISAT Development Centre and Dr. JS Samra, Former DDG (NRM), ICAR & Former CEO, NRAA interacted with the participants. Dr. KV Rao and Dr. Manoranjan Kumar from ICAR-CRIDA, Hyderabad handled the practical sessions on developing detailed project reports (DPR) on IWM. Dr. PD Sreekanth, faculty member from ICAR-NAARM coordinated the hands-on sessions on use of GIS in IWM.



Fig. 2.44: Training on Integrated Watershed Management (IWM) for the IFS Probationers from Indira Gandhi National Forest Academy (IGNFA)

2.5.7 MDP on Biodiversity and Environmental Laws for Agricultural Researchers (BELAR '21)

India is a front runner in terms of implementation of biodiversity and environmental laws including forest and wildlife conservation. The biodiversity laws and laws pertaining to environment, wildlife and forest conservation have significant implications for the agricultural scientists in ICAR and agricultural universities (AUs), working on various bio-resources, including microbes. Recognizing this, the Academy organized an MDP on Biodiversity and Environmental Laws for Agricultural Researchers (BELAR '21) during June 07-09, 2021 in partnership with Indira Gandhi National Forest Academy (IGNFA), Dehradun; National Biodiversity Authority (NBA), Chennai; Biotechnology Consortium India Limited (BCIL), New Delhi; UNDP and GIZ. The participants (40 including 07 women) comprised of representatives from ICAR (30), agricultural universities (08), state biodiversity board (01) and NGO (01). The MDP provided an overview of various biodiversity and environmental laws, highlighting the process, status and challenges with special reference to agricultural research in India. The sessions were held in interactive and relevant case studies from NARES were discussed.

Dr. VB Mathur, Chairman, NBA during his address emphasized the need for agricultural researchers being partners in strengthening the biodiversity laws concerning them. Dr. Bharat Jyothi, Director, IGNFA complemented ICAR-NAARM for the initiative and suggested that scientists from ICAR and ICFRE should get an opportunity to work across the Councils to appreciate the mutually complementing areas of research. Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM highlighted that



Fig. 2.45: MDP on Biodiversity and Environmental Laws for Agricultural Researchers

the program was the first in-house training program being organized in collaboration with other national and international organizations and hoped that the partnership would aid in harnessing better synergy.

2.5.8 Virtual Training Programme on “IPR and Technology Commercialization for the Faculties of Kamdhenu University”

The three-day Virtual Training Programme on “Intellectual Property Rights and Technology Commercialization for the Faculties of Kamdhenu University” was organized by the ICAR-National Academy of Agricultural Research Management, Hyderabad from 15th to 17th July, 2021. The Faculty Members from the ICAR-NAARM and ICAR Headquarters, Agri-Incubation Centres of ICAR Research Institutes, IIM - Calcutta, Officials from the Intellectual Property Office of the Government of India and Patent & Trademark Attorney registered their participation as the speakers for the Capacity Building initiative. The main objective of the programme was to create a complete ecosystem with the upcoming knowledge networks for infrastructural development. About 50 Young Faculty Members and Researchers from 11 Colleges and 3 Post Graduate Research Institutes in Gujarat under

the Kamdhenu University were virtually trained during the programme.



Fig. 2.46: Programme on “IPR and Technology Commercialization for the Faculties of Kamdhenu University, Gujarat

2.5.9 Design Thinking in Research Project Formulation and Implementation

An online training program on “Design Thinking in Research Project Formulation and Implementation” was organized by the Academy during August 24-28, 2021. Thirty-three scientists/faculty from NARS attended the programme. The participants were from Kerala to Kashmir; Assam to Maharashtra and represented diverse background (14 females, 25 agricultural research disciplines, 09 ICAR and 11 Agricultural Universities). The participants were shortlisted from around 100 applicants from across India.

The participants were introduced to the concepts of 'Design Thinking (DT)' and its use in various aspects related to project formulation & implementation, and research management. Practical sessions on design thinking practice and breakout exercise were taken using Research Concept Writer, Project Logframe Writer and AHP Analyser, YouTube exercise on 'Juice Extraction Machine: understanding the hierarchy of objectives', and also other online services

such as Mendeley and Web of Science. Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM and Chief Guest during valedictory session appreciated the program as 'innovative and creative'. He explained the utility of such training during early years as researchers and stressed that for young scientists, there can be a provision of tiny grants as 'seed money', which help them to grow to apply proposals for funding by national and international agencies.



Fig. 2.47: Training program on "Design Thinking in Research Project Formulation and Implementation"

2.5.10 MDP on Priority Setting, Monitoring & Evaluation

The Academy organized the 9th Management Development Program on 'Priority setting, Monitoring and Evaluation (PME) of agricultural research projects' for the in-charges of PME Cell of ICAR institutions during October 25-30, 2021. The in-charges/ members of PME Cells from 37 different ICAR institutions, representing 7 subject matter divisions, nominated by the Director General of ICAR, participated in the program. The online MDP comprised of modules on principles of priority setting, monitoring, evaluation and impact assessment of agricultural research projects. The participants were sensitized on various metrics and tools used for measuring individual and institutional research performance, global and national

frameworks for measuring research efficiency and ICAR guidelines pertaining to the various functions of PME Cell like managing research publications, consultancy services, contract research, IP and technology management. The training included interactive sessions, hands-on exercises case studies and experience sharing. The participants were sensitized on various online tools and applications for research review, project proposal preparation, checking plagiarism and choosing appropriate journals for research communication. Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM during his valedictory address urged all the participants to institutionalize the learnings in their day-to-day activities in their respective institutions and assured all support from NAARM in their endeavours.



Fig. 2.48: MDP on Priority Setting, Monitoring & Evaluation

2.5.11 Impact Assessment of Agricultural Research and Technologies

A training programme on 'Impact Assessment of Agricultural Research and Technologies' was organized by ICAR-NAARM for five days to the Scientists and Faculty of ICAR and State Agricultural Universities during December 18-22, 2021. The main objectives of the programme were to orient the participants to the concepts and description of impact assessment of agricultural research, to acquaint the participants with techniques/tools and methodological approaches of impact assessment of agricultural research, and also to discuss empirical studies and to synthesize experiences in application of Impact Assessment methods in the domain of agricultural research and technologies. A total of 24 Scientists and Faculty from ICAR and State agricultural universities participated in the programme. Among those eight are female participants who participated from different institutes. The methodology and tools for assessing the impact of technologies in agriculture and allied sector were discussed and the results of empirical studies on impact assessment were also shared with the participants during the programme.



Fig. 2.49: Training programme on Impact Assessment of Agricultural Research and Technologies

2.5.12 Workshop on NAARM's DRIVE Dashboard & National S&T Survey 2020-21

The Academy conducted a series of Sensitization Workshops during November 27-29, 2021 on the Dashboard, DRIVE-Data-driven Research and Innovation Virtual Ecosystem for the identified nodal officers from various institutions under NARES for the survey. The Dashboard is developed by ICAR-NAARM and it facilitates visualizing temporal changes in the research output, expenditure incurred, manpower available, etc; aggregate the datasets for institute, SMD, ICAR, NARES; compare the institutes based on over 20 metrics-research output, age of institute, size of institute, budget used, productivity per person, etc and generate reports of respective institutes.

Department of Science and Technology (DST), Government of India, conducts national surveys to collect data on resources devoted to S&T activities in the country, which are used for S&T assessments and policy formulations. DST has collaborated with ICAR-NAARM, Hyderabad for the Government of India's National S&T Survey for 2018-21 (3 years), wherein information on various aspects such as expenditure on S&T (R&D) activities, R&D Manpower and R&D Output are solicited from the members of

NARES. The DRIVE Dashboard access is provided from NAARM with the funds received from DST for this year, which will aid in data entry and also visualization.



Fig. 2.50: Workshop on NAARM's DRIVE Dashboard & National S&T Survey

2.6 Information and Communication Management for Promoting Innovation and Governance

Enhancing the research, capacity building and consultancy in the areas of IT based decision support systems; GIS in agricultural management; ICT applications for village knowledge centers, Development of digital multimedia and e-learning resources for agriculture; Geospatial knowledge discovery and management; ICTs in agri-supply chains; ICT policy for NARS; Emerging information technology and systems; ICTs in participatory technology development and transfer; Designing e-learning systems and Distance training are the objectives to carryout activities related to research and capacity building programmes. In order to advance the objectives of the broad theme, the Academy focused in the following broad areas for its research, academic and capacity building activities. Application of GIS, GPS and remote sensing technologies; Development of information systems and knowledge management; Statistical computing and data analytics.

2.6.1 Training Needs Assessment in Data Analytics for Scientists and Faculty

ICAR has a scientific strength of 6304 scientists working in various institutions located all over the country under various subject matter divisions. Of the 6304 scientists, only a total of 258 scientists are from the disciplines of statistics and computer applications. i.e. for every scientist from agricultural discipline, only 0.041 scientist having analytical background viz. statistics and computer applications is available in the system. Out of the total 258 scientists having analytical background, 56% are from statistics background and the remaining 44% are from the computer applications. Of the 258 scientists in Analytics, 47% are only from one Institute of ICAR viz. IASRI which is basically involved in the development of applications in Statistics and Computer based systems. These 47% of scientist are also involved in imparting training in data analytics besides few other organizations such as NAARM. NAARM has been conducting training programme since 2009 regularly and few trainings for the trainers are also being conducted regularly. It was found out that from 24 training programmes conducted and 1434 scientists trained in data analytics programme since 2013, it was found out that the knowledge level of scientists in data analytics is not very high although the candidates are highly qualified. 1% of the candidates have Post-Doctoral qualification, 60% have Ph.D. and 34% in Post-Graduation. Barring scientists from statistics and engineering background, 95% of the scientists are from science and agriculture backgrounds and the knowledge level in analytics is very minimal. These 75% of the scientists are having very limited knowledge of data analytics. Of the participants, 62% are working in various institutions in ICAR and 34% work as faculty members from State Agricultural Universities. 4% of the participants



are from Private Institutions, and NGOs. It was also found out that majority of the participants viz. 44% and 24% from Education Division (including Universities) and Crop Sciences divisions. Only 1% of the faculty from Extension SMDs are trained in data analytics and hence more participation from this division will help to enhance the analytical skills of the participants. The presentation from other divisions can be seen from Fig.

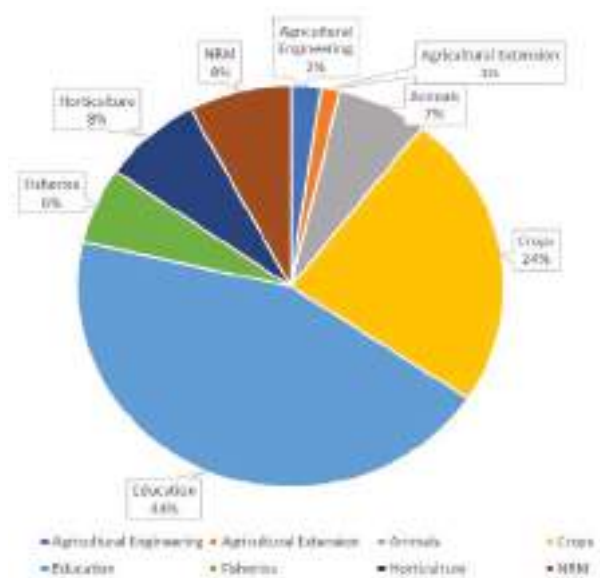


Fig. 2.51: SMD-wise Trainees

It was found out that the trainings have improved their basic understanding of statistics and data analytical skills. Through these trainings, majority of the faculty members felt they will be able to make of use of their data analytical in solving their problems. It was found out that 95% of the trainees are from non-analytical background and hence trainings of this nature will help them to gain knowledge in Statistics.

2.6.2 Geo-Spatial Knowledge System for Monitoring Climate Change in Agro Climatic Zone-X: Southern Plateau and Hills region

The extent of surface water spread influences the hydrogeology and ecology of waterbodies. Remote sensing technology provides spatial and temporal data sets, which aid in mapping the

dynamics of surface waterbodies at the regional and global scale. In the present study, the temporal changes in the surface area of waterbodies in the Telangana state were monitored through index methods, using indices like Normalized Difference Vegetation Index (NDVI), Normalized Difference Water Index (NDWI) and modified NDWI (mNDWI), and machine learning algorithm like a random forest using Landsat -8 data. Google Earth Engine (GEE) cloud computing platform was used for processing earth observation data, based on the time series images of Landsat satellite and compared with real time groundwater levels.

The results showed a significant increase ($p < 0.01$) in both surface water area and groundwater levels in the Telangana state for analyzing surface water of the Telangana state, GEE - a web-based cloud computing platform, which provides currently over 11 PB of earth observation data was used. GEE platform offers various advantages; large volume remote sensing image data processing and dedicated cloud storage; simple and rapid programming and graphic interface and possibility for assessing surface water dynamics at planetary scale for a prolonged period. A total of 874 top of atmosphere (TOA) scenes of Landsat 8 with 30-meter resolution data were acquired during monsoon and post-monsoon seasons for the years 2013 to 2019 for the above study area (Table 2.9) following United States Geological Survey (USGS).

The district-wise estimated area of annual surface water extent of the Telangana state using Random Forest (RF) algorithm is given in Table 2.10. The highest surface water extent was observed in districts like Nalgonda (533.63 km²), Mahabubnagar (336.92 km²), and Adilabad (303.06 km²) during the year 2013 and proportionately the ground water table also rose in the above districts. The higher surface water area of these

districts may be attributed to the presence of a large number of water storage structures like reservoir, lakes and ponds. For instance, major reservoirs such as Nagarjunasagar and Srisailem in Krishna river basin are situated in Nalgonda and Mahabubnagar district, respectively. Likewise, Sriram Sagar dam in Godavari river basin is situated in Adilabad district. The lowest value of the extent of surface water was observed in Hyderabad (25.89 km²), Ranga Reddy (140.62 km²) and Warangal (178.83 km²) districts. Hyderabad is one of the fastest growing mega city in India in terms of its expansion on urban agglomeration along with the Ranga Reddy district. The estimated surface water area declined during 2014 in all the districts due to less rainfall recorded in the Telangana state i.e., 746.4 mm compared to the previous year

(1348.7 mm). During the year 2016, there was a good rainfall received by the state (around 1043.4 mm). Accordingly, the estimation of surface water area also increased and the ground water recharge rose proportionately. A declining trend in both surface water area and ground water table was noticed during the year 2017 due to the spell of drought. Again, an increasing trend of the surface water spread can be seen in both variables in the year 2018 as compared to the previous year. The extent of surface water area and ground water level are depicted in Figure.

(1. Adilabad; 2. Karimnagar; 3. Khammam; 4. Mahabubnagar; 5. Medak; 6. Nalgonda; 7. Nizamabad; 8. Ranga Reddy; 9. Warangal; 10. Hyderabad)

Table 2.9: Specification of satellite data used in the study

Year	Satellite	Sensor	Interval	Resolution (m)	Image count	Wavelength (µm)
2013-19	Landsat 8	OLI	16 days	30	874	Band 1: 0.43 - 0.45 Band 2: 0.45 - 0.51 Band 3: 0.53 - 0.59 Band 4: 0.64 - 0.67 Band 5: 0.85 - 0.88 Band 6: 1.57 - 1.65 Band 7: 2.11 - 2.29 Band 8: 0.50 - 0.68 Band 9: 1.36 - 1.38

Table 2.10: Surface water area estimated using RF algorithm

Districts*	2013		2014		2015		2016		2017		2018	
	GWL	RF	GWL	RF	GWL	RF	GWL	RF	GWL	RF	GWL	RF
Adilabad	3.12	303.06	4.15	192.08	4.68	151.60	4.83	333.10	4.75	197.68	4.21	259.91
Karimnagar	3.59	237.04	7.1	125.20	8.7	98.42	4.77	220.91	7.25	144.44	5.84	227.27
Khammam	3.26	298.36	4.84	138.67	4.67	140.06	4.64	80.19	3.59	159.36	5.26	229.74
Mahabubnagar	9.55	336.92	13.25	299.77	16.81	206.17	14.42	425.83	11.97	241.75	5.71	206.66
Medak	5.43	286.76	10.55	106.21	15.47	30.73	8.87	243.74	10.23	141.70	8.15	148.42
Nalgonda	2.61	533.63	8.00	252.97	8.97	191.32	6.23	218.06	5.69	262.39	5.00	283.10
Nizamabad	4.22	282.77	8.40	126.05	12.19	40.52	5.66	326.47	6.17	156.89	5.20	367.86
Ranga Reddy	6.25	140.62	9.93	38.62	11.44	20.50	6.92	55.53	4.54	27.18	9.17	40.63
Warangal	3.85	178.83	7.12	145.15	7.95	141.86	5.80	130.90	5.26	155.00	4.47	138.05
Hyderabad	5.85	25.89	11.58	9.47	10.88	11.43	6.73	2.57	6.33	4.81	8.19	5.30
Total	47.73	2623.88	84.92	1434.19	101.76	1032.61	68.87	2037.30	65.78	1491.20	61.20	1906.94

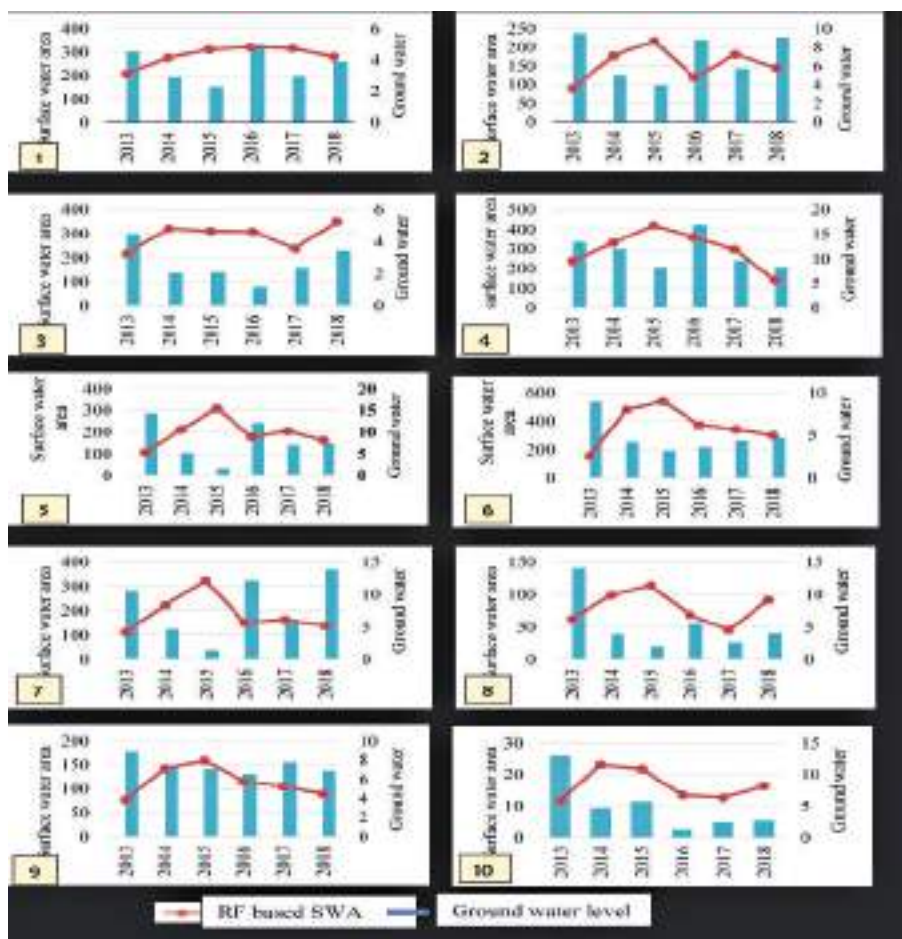


Fig. 2.52: District-wise estimation of surface water area (km²) using RF algorithm vs ground water level

*Estimation as per the pre-partition district boundaries

GWL - Ground water level (m bgl); RF - Random forest algorithm surface water area estimation (km²)

2.6.3 Establishment of Skill Development in Bioinformatics infrastructure facility (SKILL-BIF) on Agricultural Biotechnology at NAARM

Under this project, online training programme on “Application of Bioinformatics in Agricultural Research and Education was conducted “from 20-24 September 2021 at SKILL-BIF, ICAR-NAARM. This programme was designed to train the participants to recent advances in Bioinformatics applications in agriculture. 47 participants participated the training programme from all over India. Under this project, a study conducted on the Comparative Molecular docking analysis of Target fruit ripening enzyme Tomato Beta galactosidase (TBG-4).

Tomatoes comprise a high level of TBG4 (Tomato Beta galactosidase-4) enzyme activity that plays a key role in fruit softening by significant changes in the galactosyl content in the pericarp cell wall. In the present work, in silico docking studies of beta galactosidase with specific elucidated ligands were carried out. For the better understanding of protein ligand interactions, a set of 16 ligands were used for docking studies. In the present study, two different comparative docking softwares, Autodock4.0 and iGEMDOCK were used to study the protein–ligand interactions and performed to get the best docking scores. PLIP software was used for visualization of protein ligand complex and their interactions. Binding energies

of 16 ligands were predicted among which 5 ligands 151, 2FL; B2G, EPE and LAT were analyzed and confirmed as best ligands. Among them 151(2S)-3-Methyl-2-((2R, 3S)-3-[(Mehtylsulfonyl) amino]-1-[2-(Pyrolidin-1-ylmethyl)-1, 3-Oxazol-4yl} Butanoic acid is the best inhibitor of TBG4 enzyme activity leading to significant enhancement in fruit shelf life. These analyses were useful for further studies on controlling the fruit softening and generate genetically modified fruit crops with increased shelf life. Two insilco approaches were used for interpreting the accurate docking results.

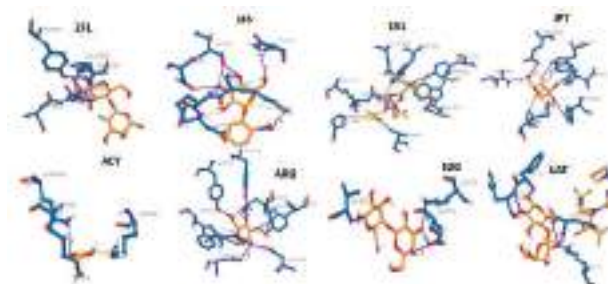


Fig. 2.53: Enzyme ligand complexes and their amino acid interactions



Fig. 2.54: The interaction profiles of the compounds

2.6.4 Mega Environment Analysis Tools for AICRP Trials

The project aims at utilizing the data collected through All-India Coordinated Research Projects for crop improvement to understand the Environmental effects in the performance of lines and develop models to predict genotype performance based on environmental and

genotype covariate information available. The prediction model would help in predicting the genotype performance of future time points based on predicted weather situation or at other locations for the same year. The weather data for the year 2019 and 2020 were collected from 19 locations from June to November month. About 176 trials data were analyzed initially and 50 trials were selected for model building purpose. After data cleaning, preliminary locations and experiments were identified for further processing. The models attempted are to use individual environmental covariate such as Rainfall, Maximum Temperature, Minimum Temperature and Relative Humidity either as singly or together. The initial analysis suggest that it will be possible to estimate the genotype responses to environmental effects which can be useful for prediction purpose. Further analysis are under progress to investigate the usefulness of these responses for prediction purposes.

2.6.5 KRISHI: ICAR Research Data Management for Knowledge Management

The project is aimed at operationalizing the ICAR Data Management Policy by creating tools, workflows, dashboards, sensitization of scientists, research managers. The Academy is one of the institute with ICAR-IASRI as lead center and ICAR-IARI, ICAR-NBSSLUP as other institutes involved in this project. During the year, four more All-India Coordinated Research Projects (AICRP) viz. Castor, Sunflower, Safflower and Linseed have started using the AICRP Information Systems to plan and randomize their trials for the year 2021-22. The number of trials created using the system are in Castor-26, Sunflower 34, Safflower 17 and Linseed 32. With this new additions, the number of AICRPs using the automated system is increased to seven. The feasibility of using estimates of heritability of individual locations



to assess the quality was examined for different traits in Sorghum experiments and it was found based on 17 trials that 60 per cent of locations have estimated heritability more than 0.5. For physiological traits, more than 70 per cent of locations crossed the threshold value. Based on the similar results obtained for Maize, it is recommended to have the values of heritability along with the usual criteria such as coefficient of variation (CV) and general mean for selecting the locations for State/Zone/All-India analysis for the performance of the new lines. The measure will be made available along with the other statistical summary measures in the analysis module of AICRP information system.

Use of bi-plot for assessing the individual replication data to identify potential human errors in the data, a macro was created for off-line analysis. It is expected that it will be available in data securitizing state to principal investigator before accepting the data in AICRP information system.



Fig. 2.55: AICRP on Castor

2.6.6 Workshop on Analysis of Multi-Location Experiments

The online training workshop was conducted to train the participants on analyzing multi-location experiments and how to infer the interaction effects using various techniques particularly in the context of All-India Coordinated Crop Improvement Programmes. Being a training programme of the first kind organized by the Academy focusing solely on a single topic, participants have attended the online

training programme. The workshop focused on differentiating the different levels of multi-location data and the analysis options available for each level. For each levels, using actual datasets, the analysis options were demonstrated in R. Biplots and other visualization tools for exploring Genotype x Interaction Analysis were also explained in detail. Use of stability measures and selection indices were the other important topics covered in the programme.

2.6.7 Workshop on Response Surface Methodology

The response surface methodology is one of the least explored area in Agricultural Sciences. The (On-line) workshop was conducted to sensitize and train the participants in use of response surface methodology for optimization of quantitative levels of treatments in different types of experiments. The workshop focused initially on introduction and idea behind response surface methodology, the choice of designs available and how to choose based on the experimental situation. The subsequent steps of how to collect data and analyze them were explained with multiple real-life situations. The interpretation of results and how to proceed were explained in detail. All the codes and examples were in open source packages of R software.

2.6.8 Applications of Artificial Intelligence and Cloud Computing in Agriculture

The course was designed to sensitize and apprise the NARES researchers/faculty on use of Information and Communication Technology (ICTs) Specifically Artificial Intelligence technologies for in their research domain. All the resources required for the programme were shared in Online mode, which includes

ecosystem of digital resources required for hands-on practical session as well. The programme was designed in to 3 sections of 2 days each which covers 1). Concepts of Artificial Intelligence, Machine Learning, Deep Learning & IoT; 2). Python programming with hands-on; 3). AI models used various start-ups in Indian Agriculture. The programme was attended by 32 participants from 9 states and belongs to 17 agricultural disciplines.

2.6.9 Training programme on “Time Series Data Analysis”

The objective of the training programme was to orient the researchers with the basics of time series data analytical tools and techniques mostly utilized in agricultural research. Researchers would also be exposed to practical aspects of time series data analysis. Main aim of this programme was to train the participants in performing statistical analysis of data generated from agricultural research experiments over time. The topics covered include: Introduction to Time Series Analysis and Forecasting & Sources of time series data in India, Introduction to R programming, Forecasting using Exponential Smoothing Methods, Autoregressive Integrated Moving Average Models & Seasonal ARIMA Models, Artificial Neural Networks (ANN models), Structural Time Series Modelling and Forecasting, Non-linear models (Logistic, Gompertz, Richards, MMF), Nonlinear Time Series Models (ARCH group of models), Multivariate Time Series Analysis and Forecasting Using Vector ARMA Models, Cointegration & Causality Analysis, Machine learning and other topics in Time Series data analysis. The training consisted of both lecture cum practical demonstration followed by hands-on by the trainees themselves. A softcopy of training manual on various topics was prepared consisting of lecture materials,

hand-outs, datasets and codes was provided to all the participants through Google Drive and the softcopy of the manual was also given to the participants. All presentations, research papers, datasets and codes were also provided. There was a session on communication of research results wherein research papers published using various statistical techniques was discussed and were distributed to the trainees and results from few were discussed during the session. During the feedback, participants have expressed about the heterogeneous group. In future, this point may be kept in mind while formulating training programmes and also selection of the participants. Trainees expressed that the conduct of the programme through online mode is really good as many participants could attend the programme by sitting in their respective office chambers and without travelling to the Academy as this has saved lot of time and money.

2.6.10 Geospatial Analysis using QGIS & R

A training programme on “Geospatial Analysis using QGIS & R” was conducted by the Academy during 1-6 February, 2021. Total 32 participants attended this training programme from different ICAR and Non-ICAR institutes including 8 female participants. The training class schedule and topics covered mainly on geospatial technologies in agriculture, GIS, GPS and remote sensing, exploring data from open source platforms, spatial database design and implementation using quantum GIS open source software, spatial analysis like clipping, joining, buffering, interpolation, krigging etc., image processing and land use land cover using QGIS and R studio. During the valedictory function, Dr. SK Soam, JD (I/C), ICAR-NAARM suggested the participants to continue to work hard for improving the productivity of NARES system. He also opined that this programme should be

conducted both in online and in-person mode so that it is beneficial to a larger audience. It was also emphasized that the Academy would be continuing to conduct such capacity building programmes in the future with concerted efforts towards open-source softwares. The softcopies of resource materials, datasets & R codes and session videos were shared with the participants through shared one-drive folder.



Fig. 2.56: Training programme on “Geospatial Analysis using QGIS & R”

2.6.11 E-Resources, E-Learning and IoT for Students of UAS, Raichur

Three programmes were organized during February 15- 19, 2021, March 23-27, 2021 and August 23-27, 2021 benefitting 90 participants comprising students and faculty of UAS, Raichur through online mode. These programmes included 28 female participants. The objectives of these programmes are (i) to apprise the trainees about agricultural scenario, and

the agricultural research, education in the country, (ii) to enhance the knowledge on IOT in agriculture, Precision agriculture and AI, IoT Policies & Case Studies in Agriculture, (iii) to sensitize on digital learning and digital resources. During this online programme, pre and post knowledge test were included and also day-wise reflections about every session. The programs had touched various contemporary aspects of education, technologies and digital resources that every students needs to be updated about. Different topics related to the broad theme viz. Digital Initiatives in Agricultural Education, documentation & data techniques through MS Office, Library Information Systems & Academic Publishing, Competency Enhancement for Digital Teaching, Current Challenges and Future Perspectives of Digitalization of Agricultural Education, E-Learning, Teaching Management, IoT applications in precision agriculture, AI & Cloud in Education, Breaking Myths in Science Communication, and Open Source tools for building personal websites etc. were covered. There were 14 sessions of 75-minute duration which were spread over five days in each programme. In-house faculty and one Guest faculty covered all the sessions. Everyday an assignment to seek the reflections of the participants about the session/topic through



Fig. 2.57: Valedictory session on EResources, E-Learning and IoT

handwritten form. Pre and Post knowledge test was done before and after the training respectively.

2.6.12 Advanced Bioinformatics Tools and its Applications in Agriculture

An online training programme on “Advanced Bioinformatics tools and its Applications in Agriculture” for the Faculty of College of Biotechnology (CoB), SVPAUT, Meerut-UP was organized by the Academy during 22-24 February, 2021. Bioinformatics is an innovative developing field that helps in collecting, linking, and manipulating different types of biological information to discover new biological insights. This specific genetic knowledge could then be used to produce stronger, more drought resistant, disease resistant, insect resistant and more productive crops. The participants were provided the insight about the enormous potential and application of advanced Bioinformatics Tools in the domain of agriculture during the programme. They were also provided with the theory and hands-on-practical sessions on various Bioinformatics softwares and tools on the different topics related to the emerging requirements of Bioinformatics Tools in the agriculture and allied sectors. In his Inaugural address, Dr. R. K. Mittal, Vice Chancellor, SVPUAT, Meerut outlined the critical role of Bioinformatics and related the applications in Agriculture, NGS, biological data analysis, AI and big data applications. Highlighting the importance of Bioinformatics to undertake the frontier research like genome editing, CRISPR-CAS, etc., Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM urged the participants for undertaking the collaborative research to improve agricultural productivity.

About 14 Professors, including 4 women, participated in the programme.



Fig. 2.58: Online Training Programme on “Advanced Bioinformatics tools and its Applications in Agriculture”

2.6.13 Digital Content Development and Management

Two online programmes on Digital Content Development and Management for the faculty of TANUVAS, Chennai were organized during April 16-21, 2021 and September 14-18, 2021 benefitting 60 participants. The programs included 29 female participants. The objective of the programme is to sensitize on latest concepts in Digital Learning, Content development and Learning Management. The programme is conducted through online mode and tweaked to suit teaching faculty of the university for enhancing teaching effectiveness. Innovations like using Google Classroom for continuous interactions were introduced for these programmes. In his valedictory address, Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM called participants to adopt to digital mode in education especially in the wake of Covid scenario. He emphasized on online methodologies to reach the target audience that include students, farmers and other stakeholders. Dr G Sarat Chandra, Dean, Madras Veterinary College thanked the Academy for timely offering of the program

and called for harmonization of technologies into learning.



Fig. 2.59: Dr. Ch. Srinivasa Rao, Director, delivering the valedictory address

2.6.14 Training Programme on Development of Web & Mobile Applications using Low-code Platforms

The ICAR-National Academy of Agricultural Research and Management (NAARM), Hyderabad organized the Virtual Training Programme on “Development of Web & Mobile Applications using Low-code Platforms” during 7-12 July 2021. About 45 scientists and officers including 12 women representing 18 ICAR Institutions and 8 Agricultural Universities participated in the programme organized in the virtual mode. The aim of the programme to sensitize the Researchers and Faculty on the future of instill Coding abilities and to provide a hands-on opportunity to build their confidence in this new arena. Organize the program jointly with M/s. Threshold Software Solutions (P) Ltd, an IT firm empaneled with NAARM. The participants were provided an overview about the changing trends in mobile and web applications development and the increasing use of low-code platforms for developing IT solutions. They were introduced to the ZEROCODE platform and each of them were provided hands-on support through the learning management system to develop a comprehensive mobile and web application by taking an illustrative case of COWIN application.

Mr. Bharath Kumar, CEO of M/s Threshold Software thanked NAARM for all the support in organizing this joint program and assured the participants post-training support for one month to develop an application of their choice on the ZEROCODE platform.



Fig. 2.60: Online Training Programme on Development of Web & Mobile Applications using Low-code Platforms

2.6.15 Application of Bioinformatics in Agricultural Research and Education

The training programme was conducted under the (SKILL-BIF), DBT Project, The Virtual Training Programme on “Application of Bioinformatics in Agricultural Research and Education” organized by the ICAR-National Academy of Agricultural Research and Management, Hyderabad from 20-24, September, 2021. The main aim of the programme to create an awareness with respect to the basic tools and techniques used in bioinformatics in agriculture. This training programme was conceived with an idea to provide an insight to the participants about the enormous potential and application of Bioinformatics and its tools management in the domain of agriculture. The participants were provided the insight about the enormous potential and application of advanced Bioinformatics Tools in the domain of agriculture during the programme. They were also provided with the theory and hands-on-practical sessions on various Bioinformatics softwares and tools on the different topics related to the emerging requirements of Bioinformatics

Tools in the agriculture and allied sectors. About 45 Agricultural Researchers including 20 women representing 18 research disciplines from 17 ICAR Institutions and 9 Agricultural Universities virtually participated in the programme.



Fig.2.61: Training programme on Bioinformatics in Agricultural Research and Education

2.6.16 Advances in Web and Mobile Application Development

The five-day Capacity Building Training program on “Advances in web and mobile application development” organized by the ICAR-National Academy of Agricultural Research and Management, Hyderabad, from 6-10th December, 2021 in virtual mode. The objective of the programme was to raise awareness and build the essential skills in mobile app development. The programme included 42 agricultural researchers, technical personnel from 22 ICAR institutes, and 8 KVKs. There were 11 females and 31 male participants, with 25 having a computer science or information technology background and the other 17 having a background in agriculture or allied areas. During the virtual training, the participants were provided with theory and hands-on practical sessions on an overview of the Android platform, Java programming, Android Studio installation, development of Android applications, database connectivity with android apps, and insight into no-coding technologies for android app development during the virtual training.



Fig. 2.62: Advances in Web and Mobile Application Development

2.6.17 Short certificate course “E-Resources, E-Learning and IoT for Faculty of University of Agricultural Sciences, Raichur

The off-campus short certificate course on “E-Resources, E-Learning and IoT for Faculty of University of Agricultural Sciences, Raichur” organized by the ICAR-National Academy of Agricultural Research Management, Hyderabad at UAS Raichur, Karnataka from 13 to 17th December, 2021. The aim of the programme to sensitize on digital learning, digital resources and IoT. The programme has been offered under the NAHEP- IG Project initiative of UAS Raichur with a purpose to expose the clientele to the latest trends in digital information resources, learning and cutting-edge technologies like Internet of Things. The program had touched various contemporary aspects of Education, technologies and digital resources that every students needs to be updated about. There were 15, sessions of 75-minute duration, which were spread over five days. In-house faculty and one Guest faculty covered all the sessions. Everyday an assignment to seek the reflections of the participants about the session/topic through handwritten form. Pre and Post knowledge test was done before and after the training respectively.



Fig. 2.63: Certificate course “E-Resources, E-Learning and IoT for Faculty of University of Agricultural Sciences, Raichur

2.7 Research Projects (Completed & On-going)

2.7.1 Completed Research Projects during 2021

S. No.	Project Code	Project Title	Team	Duration	
				From	To
In-House					
1.	AGED/NAARM/SIL/2018/003/00142	Digital teaching evaluation in Agricultural Education using TPACK Model	GRK Murthy, Senthil S Vinayagam, M Balakrishnan	October 18	September 21
2.	AGED/NAARM/SIL/2018/009/00148	Mainstreaming the Small Ruminant Farmers into Efficient Value Chains in India	B Ganesh Kumar, Sanjiv Kumar, D Thammi Raju, PC Meena, Tavva Srinivas, A Dhandapani	October 18	September 21
3.	AGED/NAARM/SIL/2018/010/00149	HR Interventions and their influence on Institutional Innovation	Alok Kumar, RVS Rao, P Ramesh, Surya Rathore, A Dhandapani	October 18	December 21
4.	AGED/NAARM/SIL/2018/011/00150	Farmer Producer Companies in India: A Study on their Management Practices and Business Potential	N Sivaramane, Ranjit Kumar, PC Meena, Bharat S Sontakki, K Srinivas, P Venkatesan, Tavva Srinivas	October 18	September 21
5.	AGED/NAARM/SIL/2020/006/00172	Assessment of training (online and offline/ physical) needs in data analytics	S Ravichandran, Surya Rathore	June 20	May 21
6.	AGED/NAARM/SIL/2020/008/00174	Impact Assessment of Research and Development Programs on Food and Nutritional Security	H Umesh, M Ramesh Naik, P Krishnan, Tavva Srinivas, BS Yashvanth, Vijaykumar	June 20	May 21

7.	AGED/NAARM/SIL/ 2020/009/00175	Dynamics of Agriculture and Food Systems in different agro-ecoregions of India	M Ramesh Naik, H Umesh, P Krishnan, BS Yashavanth, N Subash Monoranjana, Mohanty, Nishant K Sinha, Ch Srinivasa Rao	June 20	May 21
Extra-Mural					
1.	AGED/ NAARM/CCL/ 2019/007/00161	Strengthening Natural Resource Management (SNRM): A framework for mainstreaming biodiversity conservation in the development schemes of agricultural sector in India	Ch Srinivasa Rao, P Krishnan, SK Soam, I Sekar	July 19	September 21
2.	AGED/ NAARM/CCL/ 2019/010/00164	Rodents and Household Pest Management under MOOC Platform	Senthil S Vinayagam, GRK Murthy	April 19	March 21
3.	AGED/ NAARM/CCL/ 2019/011/00165	Development of Digital Content for MOOC on "Commercial Dairy Farming and Milk Processing and Value Addition"	Senthil S Vinayagam, GRK Murthy	April 19	March 21

2.7.2 Ongoing Research Projects

2.7.2.1. In-House Projects

S. No.	Project Code	Project Title	Team	Duration	
				From	To
1.	AGED/ NAARM/SIL/ 2017/009/00129	Developing a framework for enhancing agricultural research management efficiency	Ch Srinivasa Rao, P Krishnan, I Sekar, SK Soam, B Ganesh Kumar, K Kareemulla, VV Sumanth Kumar	April 17	March 22
2.	AGED/ NAARM/SIL/ 2017/013/00133	Assessment & Development of Competency Framework for Agricultural Research and Extension Scientists	K H Rao, Alok Kumar, RVS Rao, P Ramesh, BS Sontakki, SK Soam, Ch Srinivasa Rao	April 17	March 22
3.	AGED/ NAARM/SIL/ 2018/002/00141	Ag-Academy: A Framework for Scaling Up E-Learning in Agriculture	VV Sumanth Kumar, P Krishnan, SK Soam, GRK Murthy, NS Rao	April 18	March 22
4.	AGED/ NAARM/SIL/ 2018/012/00151	Students' Learning Approaches (SLAs) for enhanced learning outcomes in Agriculture Education: A Critical Analysis	D Thammi Raju, P Ramesh, Bharat S Sontakki, P Krishnan	October 18	March 22
5.	AGED/ NAARM/SIL/ 2019/008/00162	Economic Analysis of Food prices and Policies for the development of Indian Agriculture	MB Dastagiri, B Ganesh Kumar, A Dhandapani	June 19	July 22



S. No.	Project Code	Project Title	Team	Duration	
				From	To
6.	AGED/NAARM/SIL/2019/012/00166	Agri Startups in India: Decoding the Determinants and Success Factors	Sanjiv Kumar, K Srinivas, Ranjit Kumar, Umesh Hudedamani, KH Rao	June 19	May 22
7.	AGED/NAARM/SIL/2020/001/00167	New Market Reforms and E-NAM in India: Effect on Marketing Choices and Participation and Price Realization for Smallholders	Ranjit Kumar, Sanjiv Kumar, A Dhandapani, K Srinivas, PC Meena, KH Rao	June 20	May 22
8.	AGED/NAARM/SIL/2020/002/00168	Techno-Economic Feasibility Studies for ICAR Agro-Processing Technologies/ Value Added Products	Srinivas Tavva, B Ganesh Kumar, GRK Murthy, K Kareemulla, Sudha Mysore	June 20	May 23
9.	AGED/NAARM/SIL/2020/003/00169	Financial accessibility and occupational risks of vegetable street vendors in India	PC Meena, Sanjiv Kumar	June 20	May 22
10.	AGED/NAARM/SIL/2020/004/00170	Identification of best practices developed by the leadership in scientific and general administration in NARES	RVS Rao, P. Ramesh, Alok Kumar, K Kareemulla, Senthil Kumar, S Ravichandran, D Thammiraju, Senthil S Vinayagam, Ch Srinivasa Rao	June 20	February 23
11.	AGED/NAARM/SIL/2020/005/00171	Happiness and Psychological Well-Being of Scientific Personnel in NARES	P Ramesh, RVS Rao, BS Yashavanth	June 20	May 22
12.	AGED/NAARM/SIL/2020/007/00173	Mega Environment Analysis Tools for AICRP Trials	A Dhandapani	June 20	May 22
13.	AGED/NAARM/SIL/2020/010/00176	Development of a framework for gender mainstreaming in the development schemes of Agriculture sector in India	Surya Rathore, P Krishnan, Lipi Das, Veenita Kumari, BS Yashavanth	December 20	March 23
14.	AGED/NAARM/SIL/2021/001/00177	Social Media in Agriculture Marketing : A study on its role & potential	NS Rao	January 21	December 22
15.	AGED/NAARM/SIL/2021/002/00178	Strategy for Strengthening National Water Policy	K Kareemulla, S Ravichandran	January 21	March 23
16.	AGED/NAARM/SIL/2021/02/179	Impact Assessment of SC Sub Plan (SCSP) implemented by Agricultural R & D Institutes on Socio-economic Status of Beneficiaries	M Balakrishnan, B Ganesh Kumar, P Venkatesan, Ramesh Naik, P Supriya, Yashavanth, BS Sontakki, D Thammi Raju, N Sivaramane, PC Meena	July 21	June 24

2.7.2.2 Externally Funded Contract Research Project & (Sponsored/Collaborative/Grant-in-Aid)

S. No.	Project Code	Project Title	Team	Budget (in Rs. Lakhs)	Sponsored by	Duration	
						From	To
1.	AGED/NAARM/SOL/2008/003/00067	IP Management and Transfer/ Commercialization of Agricultural Technologies	H Umesh, Manju Gerard, K Srinivas	19.15	ICAR Scheme	April 14	September 21
2.	AGED/NAARM/SOL/2015/007/00110	KRISHI- ICAR Research Data Repository for Knowledge Management	A. Dhandapani, S.K. Soam	76.0	ICAR	November 15	March 21
3.	AGED/NAARM/SOL/2016/002/00112	Establishment of Agri-Business Incubation (ABI) Centers under XII Plan Scheme for NAIF	K Srinivas, Manju Gerard	79.15	ICAR Scheme	January 16	September 21
4.	AGED/NAARM/COP/2017/001/00121	Management & Impact Assessment of Farmer FIRST Project	P Venkatesan, BS Sontakki, N Sivaramane	144.05	ICAR	March 17	March 22
5.	AGED/NAARM/ COP/2019/001/00155	Investments in Indian Council of Agricultural Research Leadership in Agricultural Higher Education	SK Soam, D. Thammi Raju, N Srinivasa Rao, Alok Kumar, VV Sumanth Kumar, Surya Rathore, Sanjiv Kumar, M Balakrishnan, S Senthil, Vinayagam P Krishnan, BS Yashavanth	1704.17	ICAR - NAHEP	February 19	December 22
6.	AGED/NAARM/SOL/2019/002/00156	Entrepreneurship Development through Farmer Led Innovations - A Case Study in Plantation Sector	S Vinayagam, GRK Murthy, B Ganesh Kumar, P Venkatesan	84.66	NASF	April 19	March 22
7.	AGED/NAARM/SOL/2019/005/00159	Establishment of Skill Development Bioinformatics Infrastructure Facility (Skill-BIF) for Agricultural Biotechnology (DBT)	M. Balakrishnan, SK. Soam, P. Supriya	44.46	DBT, New Delhi	December 18	March 22
8.	AGED/NAARM/CCL/2022/001/180	Developing and Pilot Testing Knowledge Products for Agriculture and Veterinary Sector on Human Wild Life Conflict	Ch. Srinivasa Rao, Ranjit Kumar, G. Venkateshwarlu, Tavva Srinivas, Bharat S. Sontakki, Sanjiv Kumar	96.11	GIZ	December 21	September 22

2.8 Educational Programmes

2.8.1 Project Reports Submitted by 11th Batch (2019-21) PGDM-ABM Students

Sl. No.	Roll No.	Student Name	Title of the Project	Guide Name
1	PGDMA-1901	Aayush Awasthi	Export strategy for Indian rice trade	Ch. Srinivasa Rao
2	PGDMA-1902	Abhishek Singh	Problem identification of FPOs in the state of Uttar Pradesh	M Balakrishnan



Sl. No.	Roll No.	Student Name	Title of the Project	Guide Name
3	PGDMA-1904	Aiswarya Santhosh	Perception of different stakeholders on farm laws 2020 and its effects on agri-ecosystem in Tamil Nadu	S K Soam
4	PGDMA-1905	Ajith Kumar	Agriculture market linkage: market linkage for small farmers and producers	Alok Kumar
5	PGDMA-1906	Akansha Gupta	Study on value chain analysis	P Venkatesan
6	PGDMA-1907	Aman Kumar Sharma	International trade scenario of cotton	M B Dastagiri
7	PGDMA-1908	Annett Skinner	Value chain analysis of vegetables in southern India	S Senthil Vinayagam
8	PGDMA-1909	Anshu Singh	Value chain analysis of red gram in Telangana	K H Rao
9	PGDMA-1910	Arvind Mehara	Impact of covid-19 on households' consumption	Surya Rathore
10	PGDMA-1911	Ashish Pandey	Economic analysis of goat value chain in Rajasthan	B Ganesh Kumar
11	PGDMA-1912	Aurodeep Kamal	Scope of private agriculture extension in India	S K Soam
12	PGDMA-1913	Badush Salim	Cardamom trading and marketing scenario in Idukki district of Kerala	N Sivaramane
13	PGDMA-1914	Byri Siva	Value chain analysis of cashew nut in Srikakulam	Ch Srinivasa Rao
14	PGDMA-1915	Celine Chacko	Understanding and assessing different agritech startup involved in input market linkage segment	Sanjiv Kumar
15	PGDMA-1916	Chaithra P M	Value chain analysis of ginger in major growing areas of Karnataka	Mude Ramesh Naik
16	PGDMA-1917	Chatakonda Kusumanjali	Digitalisation in Indian agriculture	P C Meena
17	PGDMA-1918	Chelumalla Ranjith	Turmeric arrivals and price trend analysis in major markets of Telangana	Tavva Srinivas
18	PGDMA-1919	Devadi Hari Prasad	Value chain analysis of pineapple in Andhra Pradesh	K Kareemulla
19	PGDMA-1920	Divyanshu Yadav	Consumer purchase behaviour through online portal in case of fruit and vegetables in Hyderabad	B S Yashavanth
20	PGDMA-1921	G Sai Vijita	Analysing the current status and growth of frozen food segment in India food	Ranjit Kumar
21	PGDMA-1922	Ganesha T B	Start-ups in post-harvest cold storage technologies	Alok Kumar
22	PGDMA-1923	Gourishetty Nikhil	Analysis of start-ups in the urban farming domain and formulating a business plan by identifying the gaps in supply chain.	P D Sreekanth
23	PGDMA-1924	Gowtham A N	Value chain analysis of maize in Karnataka	P Krishnan

Sl. No.	Roll No.	Student Name	Title of the Project	Guide Name
24	PGDMA-1925	Gunjan Sharma	Analysis of consumer behaviour on app based meat purchase	B Ganesh Kumar
25	PGDMA-1926	Hardika Mehta	Assessing the FPOs in Chhattisgarh & Madhya Pradesh	N Sivaramane
26	PGDMA-1927	Harshana M T	Textures vegetable protein- local and global opportunities	I Sekar
27	PGDMA-1928	K Shravya	International trade scenario of ginger	M B Dastagiri
28	PGDMA-1929	Karlapudi Venkatesh	Happiness index of employees working at ICAR-NAARM	R V S Rao
29	PGDMA-1930	Kokate Utkarsh Kiran	Study on role of agritech players in output marketing	Sanjiv Kumar
30	PGDMA-1931	Kola Haritha	Study on value chain efficiency of sweet orange	V.V. Sumanth Kumar
31	PGDMA-1932	Koushik S	Sustainability of farmer producer organisations in India: conspectus of government and institutional finance support	B S Sontakki
32	PGDMA-1933	Kshitiz Saxena	Supply chain management of Indian textile industry	P Venkatesan
33	PGDMA-1935	Nora Mohanty	Estimation of post harvest losses and marketing of tomato in Odisha	S Ravichandran
34	PGDMA-1936	Prakshyati Satpathy	Emerging trends of agritech startups in India	Manju Gerard
35	PGDMA-1937	Pranjali Siwal	Risk management in agri-input supply chain in Maharashtra region	K Kareemulla
36	PGDMA-1938	Priyanka	Consumer behaviour towards urban farming technology and growbags in Hyderabad region.	P D Sreekanth
37	PGDMA-1939	Renuka S	Mapping of pepper production	S Ravichandran
38	PGDMA-1940	Rishav	Indian trading scenario of sugar and impact analysis of covid-19	A Dhandapani
39	PGDMA-1941	Sachin Kattimani	Value chain analysis of green gram	K H Rao
40	PGDMA-1942	Shivanand Masali	Understanding and evaluating fertiliser deep placement technology in India.	V.V. Sumanth Kumar
41	PGDMA-1943	Shubham Rawat	Farm mechanization scenario and farmer perception towards farm machinery in Haryana	N Srinivasa Rao
42	PGDMA-1944	Siddanagouda Patil	Value chain analysis of groundnut in Karnataka	P C Meena
43	PGDMA-1945	Sree Bala V P	Plant by-products used as raw materials for animal feed industry in India	B S Yashavanth



Sl. No.	Roll No.	Student Name	Title of the Project	Guide Name
44	PGDMA-1946	Supriya T D	Study on Indian retail coffee market and understanding urban youth consumer behaviour towards coffee consumption in Karnataka	Umesh Hudedamani
45	PGDMA-1947	Sushant Chandra	Exploring potential of contract farming in West Bengal state	Ranjit Kumar
46	PGDMA-1948	Vedavyas V Pandurangi	Study on efficient agricultural market information system for farmers	Manju Gerard
47	PGDMA-1949	Vishakha Wankhade	Relationship of product quality and innovation with the components of value chain in fruits and vegetables of Maharashtra's FPOs	Surya Rathore
48	PGDMA-1950	Yenugula Dorayya Naidu	Exploratory study of fisheries potential and aqua feed industry in Andhra Pradesh and Telangana	M Balakrishnan

2.8.2 Summer Internship of 12th batch (2020-22) PGDM-ABM Students

Sl. No.	Roll No.	Student Name	Title of the Project	Guide Name
1	PGDM-ABM-2001	A J Anupriya	Inland fisheries in Thanjavur, Tamil Nadu	Tavva Srinivas
2	PGDM-ABM-2002	Ajasr	Identifying opportunities and model for new farm services platform in Haryana	Ch Srinivasa Rao
3	PGDM-ABM-2003	Akhil C	Emanate sourcing and demand generation for pomegranate in central Kerala	Sanjiv Kumar
4	PGDM-ABM-2004	Akshaykumar D	Micro level planning for procurement of pulses	P C Meena
5	PGDM-ABM-2005	Ambala Anila	Understanding sustainable farm mechanisation needs for climate smart agriculture	S K Soam
6	PGDM-ABM-2006	Amrit Kaur	Evaluation of horticultural opportunities and migration strategy for jivagro in the states of Punjab, Haryana and Himachal Pradesh	Alok Kumar
7	PGDM-ABM-2007	Ankit Kumar	Pan 2007 performance and hybrid potential for pan seeds	S Senthil Vinayagam
8	PGDM-ABM-2008	Apoorva Marandi	Use of digital platform for customer recall(Jharkhand) & study of OP, improved & hybrid paddy market in Bihar and its opportunity for pan seeds	M BalaKrishnan
9	PGDM-ABM-2009	Ashish Ranjan	Product performance of PAN 2423 and hybrid potential for PAN portfolio, use of digital platform for customer recall	P Venkatesan

Sl. No.	Roll No.	Student Name	Title of the Project	Guide Name
10	PGDM-ABM-2010	Barsha Agrawal	Understanding the go to market approach for the raagi value chain in India	Sivaramane N
11	PGDM-ABM-2011	Bhanu Shree C	Processing the trade of agricultural commodities & document verification for e-fresh FDC	P Krishnan
12	PGDM-ABM-2012	Bharatha Vinaykumar	Study on existing IT tools used by PRIDEs (FPO's/FPC's) and design the procurement and warehousing module	A Dhandapani
13	PGDM-ABM-2013	Chichchula Mythreyi	Comprehensive market analysis of lepidopteran pest segment and strategy for 'ENGAGE' launch	DThammi Raju
14	PGDM-ABM-2014	Divya	Understanding scope and business opportunities for stayfresh packaging product at different levels of supply chains by associating with KVK's across India	Umesh Hudedamani
15	PGDM-ABM-2015	Gaikwad Kirti Balkrishna	Understanding scope and business opportunities for stayfresh packaging product among the FPOs dealing in fruits and vegetables across the supply chain	B Ganesh Kumar
16	PGDM-ABM-2016	Gangula Akshitha Reddy	Understanding MAP and working for certification (nutrilock and shelflife) of stayfresh packaging solutions from recognized institutes like CFTRI, NIN	N Srinivasa Rao
17	PGDM-ABM-2017	Gangula Soundarya	Value chain analysis of oilpalm in Badradri Kothagudem	Surya Rathore
18	PGDM-ABM-2018	Hasika Yeruva	Deploying mass farmer connect through technology	Bharat S Sontakki
19	PGDM-ABM-2019	Ippili Bhaswini	Increase penetration of nouriture feeds in underpenetrated and penetrated markets	V.V. Sumanth Kumar
20	PGDM-ABM-2020	Jagannadham Kireeti	Development and implementation of crop calendar of maize for Mulkanoor farmers	G R Ramakrishna Murthy
21	PGDM-ABM-2021	Janakiramachandran R	Development of risk model based on key agricultural factors that can be integrated to the rating tool	RV S Rao
22	PGDM-ABM-2022	Jayasurya R	Value chain mapping of red gram and green gram	Ranjit Kumar
23	PGDM-ABM-2023	Kaithoju Vivek	Identify opportunities for agronomic interventions in seed production for optimal production yield and cost reduction	Surya Rathore



Sl. No.	Roll No.	Student Name	Title of the Project	Guide Name
24	PGDM-ABM-2024	Kaleru Nikhil	Market research and generate a repository on government schemes & international agencies which the foundation may leverage for a better reach of its initiatives for North East India	Mude Ramesh Naik
25	PGDM-ABM-2025	Kalla Pardha Teja	Impact analysis of the training programs conducted by ICAR-NAARM	P Ramesh
26	PGDM-ABM-2026	Keerthana Menon	Scope of jackfruit value chain in Thrissur, Kerala	K Kareemulla
27	PGDM-ABM-2027	Koka Lalitha	Identify products from potato waste and accessing their market potential	K H Rao
28	PGDM-ABM-2028	Korlagunta Yagana Sri	Design a traceability system for agriculture ecosystem	S Ravichandran
29	PGDM-ABM-2029	Kotha Prudhvi	Secondary research and analysis of life cycle stages of high value horticultural crops at clusters identified by national horticultural board	M B Dastagiri
30	PGDM-ABM-2030	Masku Nishanth Reddy	Evaluation of perceived value of the existing products and acceptance of new products in districts of Madhya Pradesh with special focus on Hoshangabad	Sreekanth P.D
31	PGDM-ABM-2031	Medi Saiprakash	Development and execution of paddy crop calendar in kalgudi app (for Mulkanoor cooperative society)	P C Meena
32	PGDM-ABM-2032	Megha Raju	Establishment of supply chain for pomegranate	I Sekar
33	PGDM-ABM-2033	Nalli Sandeep	Implementation of quality management systems for model farmer franchise across states	Yashavanth BS
34	PGDM-ABM-2034	Nidhin. J. K	Awareness generation on climate smart agriculture and preparing a business plan for the state of Sikkim	Tavva Srinivas
35	PGDM-ABM-2035	Nivas Bharathi P	Establishment of quality and operational aspects in the pomegranate supply chain	Sivaramane N
36	PGDM-ABM-2036	P Prasanth Kumar	Understanding various clusters and scouting the opportunities mainly around delegate and fungicide portfolio	Sanjiv Kumar
37	PGDM-ABM-2037	Pallavi S	Rice seed treatment in India and develop a business model for seed treatment service in India	S Ravichandran
38	PGDM-ABM-2038	Panimalar T	Study on the feasibility of modern farming methods and the role of IT in these methods	S K Soam

Sl. No.	Roll No.	Student Name	Title of the Project	Guide Name
39	PGDM-ABM-2039	Pasala Satya Praveen	Understanding sustainable farm mechanisation needs for climate smart agriculture	Alok Kumar
40	PGDM-ABM-2040	Pramod G B	Comprehensive study on onion seed production in Karnataka	S Senthil Vinayagam
41	PGDM-ABM-2041	Rajender Kumar	Business development and cluster identification of northern fruits	M BalaKrishnan
42	PGDM-ABM-2042	Renukareddy	Survey on dealer satisfaction and competitive analysis for the regions in South Karnataka	P Venkatesan
43	PGDM-ABM-2043	Rohan S H	Protection to production-a survey regarding nematode management in pomegranate by focusing crop program	V.V. Sumanth Kumar
44	PGDM-ABM-2044	Sagar Satpathy	Impact assessment of combine harvester in Nabarangpur, Odisha	P Krishnan
45	PGDM-ABM-2045	Shaik Ayathullah Siddique	Value chain analysis of groundnut and redgram in YSR Kadapa, Andhra Pradesh	A Dhandapani
46	PGDM-ABM-2046	Shivani Chidambar Huddar	"Analysing the market potential for establishing general trade model vs modern trade for FnV in Hyderabad.	DThammi Raju
47	PGDM-ABM-2047	Shivangi Gupta	Market research on customer requirements to establish a dynamic model and formation of SKU's."	Ranjit Kumar
48	PGDM-ABM-2048	Shrija	Future of hybridisation in mustard in next 2-5 years	B Ganesh Kumar
49	PGDM-ABM-2049	Sourabh Katake	Onboarding customers in pulses for technology platform and commodity market assessment	N Srinivasa Rao
50	PGDM-ABM-2050	Sunil J Malagimani	Mapping of stayfresh packaging product among the top exporters dealing in fruits and vegetables	Yashavanth B S
51	PGDM-ABM-2051	Sushma S	Survey on dealer satisfaction and competitive analysis of kisankraft in North Karnataka region	Bharat S Sontakki
52	PGDM-ABM-2052	Thimmareddy Bharath Kumar Reddy	Development and execution of cotton crop calendar in kalgudi app.	Ch Srinivasa Rao
53	PGDM-ABM-2053	Vasanth Kumar K M	Seed processing processes and supervision checkpoints	G R Ramakrishna Murthy
54	PGDM-ABM-2054	Yerramsetti Sai Mohana Kavya	Emanate sourcing and demand generation for pomegranate, sweet lime, mango and apple in Tamil Nadu	R V S Rao



2.8.3 Students (M.Sc. & Ph.D) Guided by NAARM Faculty as Major Guide/Supervisor

Sl. No.	Name of the faculty	Student Name	Title of the thesis	Programme Name (M.Sc./Ph.D)	University /Institute Name
1	M. Balakrishnan Supervisor	D. Karunakaran	Development of algorithm for prediction of fish yield in reservoirs using artificial neural network	Ph.D.	Bharathiar University, Coimbatore
2	B.S. Yashavanth	Jammugani Rakesh	Spatio-temporal study of agricultural labour wages in Telangana and neighbouring States	M.Sc. (Ag Stat)	PJTSAU Hyderabad
3	V.V. Sumanth Kumar	Mr. Pettem Kamalnath	The study of farmer's perceptions on use of pesticides on cotton in Telangana	MBA (ABM)	PJTSAU, Hyderabad
4	Bharat S Sontakki (as Co-guide)	N Bhuvana	Ecosystem analysis of krishi vigyan kendras of Southern India	PhD	PJTSAU, Hyderabad
5	K H Rao	Madne Mahanama	A Study on supply chain of vegetables and fruits in B2B in Hyderabad and Ranga Reddy districts of Telangana	MBA (ABM)	PJSTAU Hyderabad

3.1 Policy Research and Advocacy

The Academy undertakes policy research on various contemporary and futuristic issues and networks with various experts across the NARES. To publish policy papers/briefs. A gist on the documents developed during 2021 is summarized below:

3.1.1 National Framework for Mainstreaming Biodiversity Conservation in Agricultural Development Schemes in India

The policy document was prepared based on a detailed assessment of the implementation status of biodiversity relevant public schemes of the agriculture sector. Over 40 success stories have been documented from those states with significant contribution to biodiversity conservation.

A Policy brief on, “Mainstreaming Biodiversity in Public Finance: A Case for Agriculture Sector in India” was prepared jointly with MoEFCC-Gol, National Biodiversity Authority (NBA), Biotechnology Consortium India Limited (BCIL) under the aegis of UNDP-BIOFIN Project. The report recommends reforms in identified public schemes to ensure enhanced biodiversity outcomes for agrobiodiversity conservation and identifies key entry points. It also calls for the creation of a separate Mission on Agrobiodiversity focused on Plant Genetic Resources as a Sub-mission of National Mission on Sustainable Agriculture (NMSA) under National Action Plan on Climate change (NAPCC). A trainers’ guide on “Capacity Building of Senior Officials from Central and State Governments on Mainstreaming Biodiversity in

Agriculture Sector” was also prepared.



Fig. 3.1: Policy brief on Mainstreaming Biodiversity in Public Finance: A Case for Agriculture Sector in India

3.1.2 Circular Carbon Economy in India: Efficient Crop Residue Management for harnessing Carbon, Energy and Manure with Co-benefits of Greenhouse Gases (GHGs) Emissions Mitigation

Producing carbon-neutral renewable bioenergy from surplus agriculture and other organic wastes is the main emerging strategy of a typical circular economy. Larger areas in different states have witnessed the field burning of crop residues such as rice and wheat in irrigated systems; maize, cotton, pigeon pea, castor, rainfed rice residues in rainfed drylands; vegetable, fruit, cut flower wastes under both the conditions underlines to implement this strategy in the country. Both the benefits of green energy production and improving soil health management are met from a win-win strategy if implemented in regions with high potentials. Unless farmers get some economic benefits, it is difficult to promote a hitherto stressed strategy of crop residue recycling for soil health improvement as a matchstick costing

a few paise clears the field within few hours of burning including nutrients. It vacates the field for immediate preparation and cultivation of next crop of the system immediately. MoA & FW, MNRE and others should reconsider all tiny scale family type Gobar Gas plants being subsidized since 1962 and fortification of manures under different names of the schemes. The economics of these labour intensive and long-duration traditional practices do not fit into the present competitive market scenario. The Government should also promote alternative options on high priority under its Waste to Wealth policy.

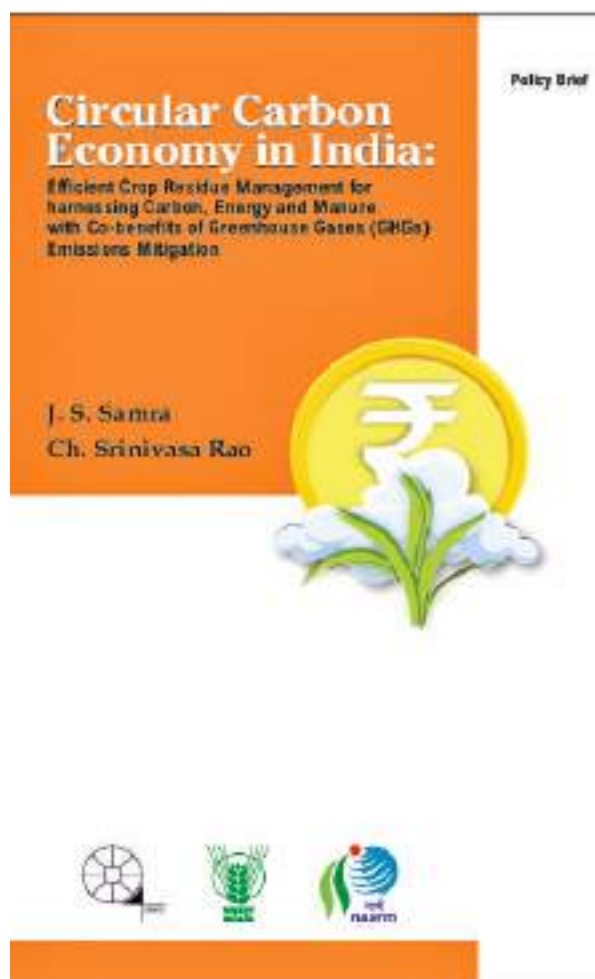


Fig. 3.2: Policy brief on Circular Carbon Economy in India: Efficient Crop Residue Management for harnessing Carbon, Energy and Manure with Co-benefits of Greenhouse Gases (GHGs) Emissions Mitigation

3.1.3 Strategies for Enhancing Soil Organic Carbon for Food Security and Climate Action

Soil organic carbon (SOC) governs the native productive capacity of soils and regulates climate change and ecosystem services that are critical for sustainable growth of agricultural productivity and food supplies. Management practices influence soil carbon build-up and its subsequent sequestration. Intensive tillage accelerates decomposition and diminishes carbon content while integrated nutrient supply promotes its restoration and retention. Since India has a predominantly tropical agro-ecosystem, the scientific efforts hover around understanding SOC dynamics in terms of spatial and temporal distribution. Indian soils are low in carbon, hence the strategies on its restitution through the addition of organic manure, legume intercrops and conservation tillage must be central to the smart soil management practices. This becomes more important in view of the declining factor productivity and rising cost of inputs.

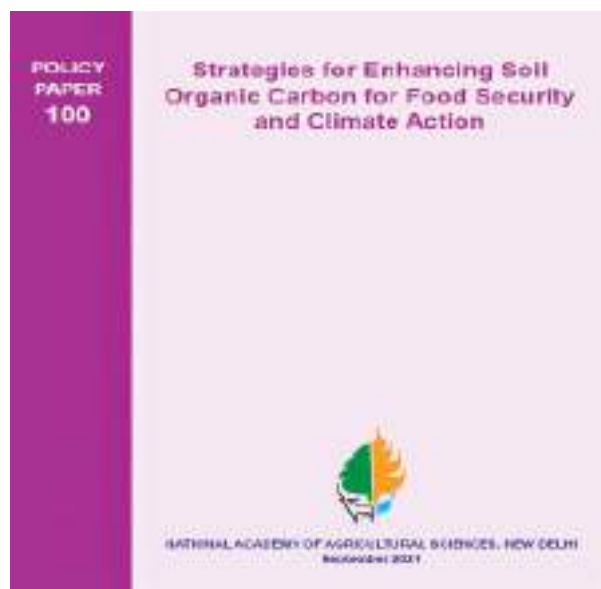


Fig. 3.3: Policy brief on Strategies for Enhancing Soil Organic Carbon for Food Security and Climate Action

India has pledged in the 2015 Paris Agreement to reduce greenhouse gas emission intensity by 33-

35% of the 2005 level by 2030. For this, a carbon sink of 2.5-3 billion tonnes of CO₂ equivalent has to be created by 2030. Another action agreed upon in the Paris Agreement is to implement the '4 for 1000' initiative, which means raising the SOC level at a rate of 0.4% per annum in the top 1 m soil layer.

3.1.4 Enhancing Efficiency of Agricultural Research Management in ICAR

3.1.4.1 National Methodological Framework for Assessing Ease of Doing Research (EoDR)

Human resource and finance which are the major tangible inputs (drivers) for research. But research productivity is also influenced by intangible inputs like research facilitation, leadership, governance, culture, etc. Most global and national ranking frameworks capture only the quantitative and easily measurable tangible inputs.



Fig. 3.4: Methodological Framework for Assessing 'Ease of Doing Research' (EoDR)

To address this crucial gap, a methodological framework for assessing 'Ease of Doing Research'

(EoDR) in research organizations was developed based on an extensive review and expert consultations. EoDR refers to the perceived ease with which the researchers can carry out their research work in respective institutions. The framework takes a major departure from existing frameworks and focuses on the intangible inputs and process dimensions of the research ecosystem, which could be significant enablers of the tangible inputs.

3.1.4.2 DRIVE DashBoard: A tool for Evidence-based Science & Technology Governance

NAARM Developed the architecture of the Dashboard, which provides for the collection of R&D input (human resource, budget) and output data (publications, technology, varieties, etc.), visualization of institutional research efficiency and also benchmarking of institutional performance.



Fig. 3.5: Snapshot of DRIVE DashBoard: A tool for Evidence-based Science & Technology Governance

3.1.5 Economic Analysis of Food prices and Policies for the Development of Indian Agriculture

The study suggests that New Global price policy for developed countries is that they must equate international prices to domestic prices and for low and middle-income countries is that international prices including 20% in order to protect farmer's welfare coupled with WTO should discipline global trade with stable prices. The study found that even during WTO modern period also developing

countries and continents; Africa, Latin America and South America trade economies depend more on agriculture. It is necessary to address the challenges of societal, demographic, and political drivers of poverty. Developing countries welfare is possible only through Farmers' Welfare through modern agriculture. Multilateral trade relationship with high CAGR countries would help in smooth trade of agricultural crops. There is need to drive the economic growth with environmental sustainability so that nonrenewable resources do not trigger irreversible damage to the environmental resources. Integration of the environment policies, trade practices and individual actions must be enhanced for environment reliable technology and innovative solutions, which are necessary to achieve economic viability and sustainability in future.

3.1.6 India's National Water Policy – Issues, Suggestions & Strategies

A workshop on India's National Water Policy – Issues & Suggestions was organized as part of the project during 25-26 November 2021. Eminent water experts participated and deliberated on the issues and the needful reforms in the existing water policy. Dr. Mihir Shah, former Planning Commission member and the Chairman of the Committee on water policy reforms inaugurated and spoke at length on the key suggestions made for reforming the existing water policy of 2012. Other prominent speakers included Ms. Debashree Mukherjee, Dr. Nandakumaran, Prof. K. Palanisami, Prof. M.G. Chandrakanth, Prof N. Nagaraj, Prof. A. Narayanamoorthy, Dr. Dinesh Kumar, Dr. R.C. Srivatsava and Dr. G.R. Rao besides the project team. The topics covered by the speakers included issues of water management, groundwater status, costing of water, approaches for sustainable use of ground water, micro irrigation, water resources and utilization, water

use efficiency etc. Over 60 participants from the NARS system participated in this workshop. As part of the project online survey of stakeholders such as department officials, farmers and other civil society representatives from across the country is being done.

3.1.7 National Workshop and Consultations on "Sustainability of Buffalo Meat Sector in India"

A National workshop and consultations on "Scientific Interventions and Policies for Strengthening of Buffalo Meat Sector: Catalyst for Boosting India's Agro-Economy" was held at the Academy on 19 January, 2021. This event was jointly organized by ICAR-NAARM, ICAR-NRC on Meat, NAAS-Hyderabad Chapter and Indian Meat Science Association. Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM chaired the inaugural session. Dr. Inderjeet Singh, Vice-Chancellor, GADVASU, Ludhiana acted as Chief-guest while Dr. Laxma Reddy, Director, Animal Husbandry & Fisheries, Govt. Telangana acted as Guest of honor. Dr. S.B. Barbuddhe, Director, NRC on Meat convened the workshop. During the inaugural session, all the experts highlighted the potential of Indian buffalo meat sector, contribution of buffaloes through milk and meat towards nutritional security, value addition and diversification of food, livelihood opportunities for farmers, scope for salvaging of male buffalo calves, issues related productivity, sustainability and requirements for augmenting farmers' income while doubling buffalo meat exports and contribution to *Atmanirbhar* Bharat. He has also released a policy document on "Policy issues and stakeholder requirements in Indian meat sector" jointly developed by ICAR-NRCM and ICAR-NAARM. Different presentations on various aspects of unlocking the potential of buffalo meat sector in India, buffalo sector value chain



analysis, sustainability of buffalo production and export meat sector requirements and doubling of farmer's income were delivered by experts. As a part of the workshop, a separate panel discussion involving more than 30 experts from Research Institutes, Universities, APEDA, Animal Husbandry Departments, and Export meat plants was also organized. Various Technological, Institutional, Regulatory and Policy requirements viz, uniform slaughter policies, removal of restrictions on male buffalo calf slaughter, value addition to buffalo meat, issues related to animal marketing, transportation, slaughter infrastructure, food safety requirements to ensure nutritional security and public health and contribution towards meeting UN-Sustainable Development Goals were discussed. Dr. BM. Naveena, Principal Scientist, NRC on Meat and Dr. P. Krishnan, Principal Scientist, NAARM acted as organizing secretaries for the program.



Fig. 3.6: National workshop and consultations on "Scientific Interventions and Policies for Strengthening of Buffalo Meat Sector"

3.1.8 Policy Perspectives on Farm Subsidies in India: Issues and Way Forward

The Academy organized one-day brainstorming session on "Policy Perspectives on Farm Subsidies in India Vis a Vis Other Countries: Issues and Way Forward" in virtual mode on 11 March, 2021 considering the frequent objections from other developed countries in the Dispute Settlement Committee in the World Trade Organization (WTO) on various support measures extended

by the Government of India. The Brainstorming session was envisaged to draw the attention on various issues pertaining to farm subsidies in India in comparison to other countries by sharing the thoughts of eminent speakers in this field viz., Dr. C Ramasamy, Former Vice Chancellor, TNAU; Dr. D Narasimha Reddy, Former Professor in Economics and Dean, School of Social Sciences, University of Hyderabad; Dr. A Narayanamoorthy, Sr. Professor, Alagappa University & Former Member, Commission on Agriculture Costs and Prices; Dr. Seema Bathla, Professor, Centre for the Study of Regional Development, School of Social Sciences, Jawaharlal Nehru University (JNU); and Dr. SK Sharma, Associate Professor, Centre for WTO Studies, Indian Institute of Foreign Trade (IIFT). Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM during his introductory remarks mentioned that this brainstorming session is the first in series of such sessions planned to be organized while playing the role of ICAR Think Tank. A total of 76 stakeholders including 29 Agricultural Economists from ICAR Institutes, State Agricultural Universities; CGIAR institutions and NGOs participated in the event. After the talks, there was group discussion on three topics of importance and a Q&A session. The deliberations have given the options for way forward and the policy perspectives on the issue.



Fig. 3.7: Brainstorming session on "Policy Perspectives on Farm Subsidies in India Vis a Vis Other Countries: Issues and Way Forward"

4.1 Centre for Agri Innovation (CAI)

The Centre for Agri Innovation of the Academy functions through the technology business incubators from ICAR-NAIF (National Agriculture Innovation Fund), Department of Science & Technology (DST), National Bank for Agriculture and Rural Development (NABARD), Government of India (GoI) supported Business incubator a-IDEA (Association for Innovation Development

of Entrepreneurship in Agriculture) which also supports activities of BioNest an Agri- Bio-incubator for Agri-Biotechnology Startups. BioNest has received financial support from BIRAC, DBT, GoI, which runs in partnership with Agri-Biotech Foundation (ABF), Rajendranagar, Hyderabad.

a-IDEA fosters innovation and entrepreneurship in agriculture in India.



Fig. 4.1: Focus areas of a-IDEA

a-IDEA, since 2014 has been supporting startups across 14 domains of agriculture and allied sectors. It has helped entrepreneurs to sensitize, ideate, incubate and accelerate their innovative early-stage startups that are scalable to become competitive food and agri-business ventures through capacity building, mentoring, networking and advisory support. These startups supported by a-IDEA are in turn disrupting the agriculture value chains along the farm gate to food plate, these innovations which are influencing upon

the lives of farmers, rural and urban stakeholders. The a-IDEA is presently focusing on different sub-sectors of agriculture innovations across the value chain in agriculture:

4.1.1 Agri Udaan 4.0

Agri Udaan, Food and Agribusiness Accelerator 4.0 is the 4th edition of the flagship accelerator program of a-IDEA, NAARM. It brings on board a range of diverse partners to create an inclusive and collaborative ecosystem focusing on catalyzing

scale-up stage Food and Agribusiness startups through rigorous mentoring, industry networking and investor pitching. It is a unique platform for scale-up stage innovators, entrepreneurs and startups in Food and Agribusiness sectors to showcase their products/services and to receive valuable inputs from Mentors, Incubators, R&D Institutions, Agribusiness Industry, and Investors.

AGRI UDAAN 4.0, was launched on May 17, 2021. About 500+ startups and faculty members took part in this programme in virtual mode, including honourable VCs, former DDGs, ADGs, Directors of ICAR Institutes, agri start-ups, agri based private companies, ZTMU and ITMU members of ICAR, Progressive farmers, Ministry representatives, agri business management students, faculty and scientists of ICAR & State Agriculture

Universities across India and press and media. Dr. GR Chintala, Chairman, NABARD launched the programme virtually and congratulated a-IDEA for the launch of fourth edition of this national event. The top 10 startups cohort of Agri Udaan 4.0 (Table 4.1) selected from a pool of 308 applications. A capacity building programme for the selected startups was organized during Dec 20-23, 2021 by inviting the Mentors from successful Agri Startups, Industry Experts to refine the Business Model and help them setting up milestones for their companies. This will facilitate the startups for presentation during the demo day, as the demo days will serve as the first point of the contact for the startups to liaison with investors across Food and Agribusiness Industry that can result in investment over time.



Fig. 4.2 : Capacity Building Workshop under Agri Udaan 4.0



Table 4.1: Agri Udaan Cohort Startups of Agri Udaan 4.0

S. No.	Name of the Start-up	Name of the Founder	Company Profile
1.	Sadabahar Greens Pvt Ltd	Shishir Agrawal	Urban Plants (Sadabahar Greens Pvt Ltd) is building India's First Gardening Hyperlocal Marketplace and Hobbyist Forum with AI-based shopping experience
2.	Chimertech Innovations LLP	Chimertech Innovations	Chimertech Innovations LLP is a Farmer focused product-oriented veterinary and Healthcare startup focusing on DairyTech and the poultry industry to increase milk, and egg production with better livestock disease management solutions. Our product Quadmastest (QMT) offers a completely reagent-free solution for very early detection of subclinical mastitis in cattle, thereby reducing cattle death and increase milk production by 20 to 50%
3.	TrayamBhu Tech Solutions Private Limited	Prabir Mishra	TRST01 (TrayamBhu Tech Solutions Private Ltd) a multi-chain block chain platform providing solutions on Agri and Food traceability, connected supply chain and document authentication
4.	Qlikchain International B.V.	Ayon Hazra	Borlaug Web Services™(BWS), a purpose-built block chain SaaS platform providing Transparency and Traceability solutions for Agri value chains. BWS provides sub-tier visibility ensuring provenance and compliance information to be shareable, accountable and verifiable
5.	GenFlow AI Pvt. Ltd.	NITISH GAURAV	GenFlow AI is digitizing livestock care through tele-medicine where our software (AHB Suraksha) connect dairy farmers to qualified vets just through one tap with facilities like online advisory, video consultation, book vet visit, AI visit, endoscopy and medicine at door step or from AHB Suraksha Centre
6.	ATFEM Kheti Gaadi Private Limited	Pravin Shinde	KhetiGaadi - One stop e-commerce for Farm Mechanization and Crop Nutrition with Knowledge and Advisory
7.	Dharaksha Eco solutions	Arpit Dhupar	Dharaksha Eco solutions has created a process to naturally grow packaging material out of crop stubble waste that is superior in strength, finish, and cost-competitive to conventional packaging materials. This would go a long way to reduce pollution across India from crop wastes to turn them into cost-effective bio-degradable packaging materials. Here we are ultimately incentivising the farmers on the Waste they produce. More to it is Fire Proof, Water Resistant & Anti-Static too
8.	Fermentech Labs Pvt Ltd	Sidharth Arora	Fermentech Labs Pvt Ltd uses novel SSF Bioreactor technology to transform agro-industrial wastes into industrial enzymes and nutraceuticals
9.	Farm Sathi	Susanth Masana	Farm Sathi develops Fully autonomous affordable robotic solutions for farmers to eliminate/reduce chemicals, fuel and labour by 90%. They identify the pest and weeds among the crop, till/ cut weeds and spray targeted pesticide, leading to a potential savings of about 17,500 INR per acre in a year. Our Robotics-as-a-Service and Ownership business models cater to different farmer segments with a Total Addressable Market of 6 Billion Dollars in Southern India, and initial target market as Andhra Pradesh
10.	BMH Transmotion	Ravindra Dekate	BMH Foldable and Portable Bulk Agriculture and Horticulture Products Storage Technology to reduce post-harvest losses

4.1.2: Webinars Organized Under Agri-BioNEST

Topic	Date
Understanding BIRAC BIG Scheme Guidelines & Eligibility in Agri and Agri-Biotech Sector	11 Jan, 2021
How to write a winning proposal for BIRAC BIG grant	16 Jan, 2021
Role of ICAR-NAARM in promoting agri startups in BIG grant	18 Jan, 2021
BIG awareness session with Bangalore Bio innovation Centre	19 Jan, 2021
BIRAC BIG for Agritech & Allied Sector Startups: Sky is the limit	28 Jan, 2021
Opportunities for Agri-Biotech Startups in BIG scheme: Efforts of ABF-a-IDEA	03 Feb, 2021
TEP Mentoring Session	03 Feb, 2021
Tips for filling application for BIRAC BIG 18 th call	10 Feb, 2021
Application Form Submission tips for startups under BIG	25 Feb, 2021

4.1.3 Web Shows

a-IDEA conducted a series of webshows under Agri Udaan 4.0. The webshows had speeches from industry, investors, successful startups and

government functionaries. These webshows helped in connecting experienced and professional leaders from Corporates and Startup Incubators.

Date	Theme
May 21, 2021	Reflections on earlier accelerators
May 28, 2021	Investors Corner for Agri-Startups
June 4, 2021	Collaboration Corridor between Corporate & Agri-Startups
June 11, 2021	Assessing and Monitoring Social Impact for Inclusive Growth: Agri-Startup Journey
June 18, 2021	Enabling Ecosystem for Nurturing Agri-Start

4.1.4 Aggnite

A two-days national entrepreneurship ideation competition “Aggnite” was organized during September 24-25, 2021, to stimulate entrepreneurship among students. It was attended 211 teams from 150 top Institutes like agricultural universities, IITs, IIMs and other renowned institutions.



Fig. 4.3: Aggnite programme during September 2021

The programme was organized jointly by a-Idea, Technology Business Incubator of ICAR-NAARM funded by NABARD and National Academy of Agricultural Sciences (NAAS)-Hyderabad Chapter. Shri Y. Krishna Rao, CGM, NABARD, Telangana Regional, Chief Guest of the Program expressed that it is imminent to promote entrepreneurship to solve major challenges faced by Indian agriculture. He also stressed that investment in Agri-Startups has huge multiplier effect and can bring disruptive changes in the livelihood of farmers and rural community. Dr. Ch. Srinivasa Rao in his presidential address, He highlighted the critical role of sensitization of the agri students in fostering their growth and guiding them to become successful entrepreneurs. Shri Vasudevan Chinnathambi, Co-Founder, Ninjacart, who was the Guest of Honour, gave valuable tips for the students to succeed in their entrepreneurial



endeavour. He explained the practical issues in promoting agri start-ups in India. After three rounds of selection process, the top 3 teams were awarded cash prizes and other benefits like mentoring support.

4.1.5 Training Programme on Entrepreneurship Development for Agricultural Graduates

a-IDEA organized a five day training programme on “Entrepreneurship Development for Agricultural Graduates” for students of College of Horticultural, Rajendranagar of Sri Konda Laxman Telangana State Horticulture University, Mulugu during June 14-18, 2021. Dr. G. Venkateshwarlu, Joint Director, ICAR-NAARM gave a comprehensive overview on different programmes such as Skill India, Student READY, Stand-up India, ACABC and IDEA (Internship for Development of Entrepreneurship in Agriculture) which are expected to bring changes in the mindset of student from job seekers to job providers.



Fig. 4.4 : EDP Programme for SKLTSU Students

Dr. B. Neeraja Prabhakar, VC, SKLTSU reiterated students to imbibe entrepreneurial qualities to succeed in their careers. Dr. M. Padma, Dean, SKLTSU appreciated the programme and expressed that it will help the students to step into entrepreneurship in future. The participants rated the programme as excellent. About eighty three students participated and 18 sessions were conducted, of which two were handled by guest

faculty and three by entrepreneurs.

4.1.6 NIDHI PRAYAS Scheme

a-IDEA, invited innovators working on innovative Idea/POC/Prototype stage to apply for “NIDHI PRAYAS” grant up to Rs. 10 Lakhs in the field of agriculture and allied sectors. During the period, 10 startups were supported under NIDHI PRAYAS.



Fig. 4.5 : Webinar organized for NIDHI PRAYAS

4.1.7 BIRAC BIG Scheme

BIG (Biotechnology Ignition Grant) is the flagship program of BIRAC, which provides grant funding support to young start-ups & entrepreneurs to the tune of up to INR 50 lacs to build best in class innovative ideas to proof-of-concept/product. As a BIRAC BIG Partner, a-IDEA, NAARM is supporting nine startups on different domains.

a-IDEA, NAARM, a BIRAC’s Biotechnology Ignition Grant (BIG) scheme partner specialized in the Agri and Allied sector opened the 19th Call for applications and received 78 proposals, of which 72 were selected for the Online Evaluation Round after the Technical and Legal Eligibility Check. Out of 72 proposals, 19 were selected for TEP round.

4.1.8 Global Bio India Roadshow 2021

Agri-BioNEST, an Agri-Biotech-focused Bio-Incubator of a-IDEA, NAARM & Agri-Biotech Foundation organized the Global Bio-India

2021 Roadshow Virtually on 25th February, 2021. Aspiring entrepreneurs, students, researchers, startups, Innovators, and ecosystem stakeholders attended the roadshow. The event had speeches from industry experts, innovators, startups and government functionaries.

4.1.9 KRISHIBOOT

It is an incubation program aimed to attract startups or individuals who are in the ideation/early startup stage to get incubated at a-IDEA. Out of 119 Pan India, applications received, 22 were selected in the final round for incubation. (Table 4.2) Out of 22, 14 on-boarded by signing MoU.

Table 4.2: List of Incubates

1	Isaayu Farms Pvt Ltd
2	Datair Technology Private Limited
3	Inventohack Innovations Pvt Ltd
4	FarmOR Agri Solutions Pvt. Ltd.
5	Social Well Technologies (OPC) Pvt. Ltd.
6	Water N Spices Foods Pvt Ltd., Epanipuriikartz
7	Quick Solution Project Consultancy Pvt. Ltd.
8	Microbial Research Lab Pvt. Ltd.
9	Navshali Innovations Private Limited, Pyotam
10	Visron Pvt. Ltd.
11	ELAI Agritech Pvt. Ltd.
12	Farm Pool Pvt. Ltd. (Farm 2 Families)
13	Vattam Agro and Dairy Industries Pvt. Ltd.
14	Starskill Consultancy Pvt. Ltd. (Animal ICU)
15	Pasidi Panta Pvt. Ltd.
16	Contrivation Labs Pvt. Ltd.
17	MLIT Sol Pvt. Ltd.
18	RR Animal Health Care
19	Celebrating Farmers Edge International Pvt. Ltd.
20	Amvicube Private Limited



Fig. 4.6: Announcement of Incubation Support under KRISHI BOOT

4.1.10 Seed Investments

a-IDEA has been supported under the two umbrella programmes namely NIDHI SSS under NSTEDB, DST and BIRAC SEED FUND under DBT. The support of NSTEDB, DST have yielded significant results and a visible change in the investment scenario of the Agri Startup ecosystem. Three out of six portfolio startups funded by a-IDEA have raised follow on funding from investors within 1-2 year of seed funding. One among the startups (AgNext) had an exit which was 5 times the investment made by a-IDEA.

During the year, a-IDEA received an extension of duration till 24 months towards the implementation of BIRAC SEED FUND. The investments made in 3 startups to the tune of



INR 25 Lakh each, second tranche of INR 10 Lakh was disbursed to Fishy farmers and first tranche of INR 15 Lakh was disbursed to Fermentech Labs in April 2021.

4.1.11 NABARD District Level Officers Visit a-IDEA

The a-IDEA, organized one-day visit for the senior officials of NABARD, Andhra Pradesh to acquaint with the various incubation activities performed a-IDEA namely, sensitization, ideation, innovation, incubation and acceleration on October 1, 2021. During the event, Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM and President, a-IDEA greeted the officials of NABARD and appreciated the efforts taken by NABARD to visit the institution with the purpose of gaining knowledge in start-ups and the process of incubation. He insisted the importance of developing success models in secondary agriculture for better agri-preneurship. He also briefed various ways in which NABARD can collaborate with a-IDEA, NAARM for capacity building of emerging start-ups. Shri. SK Jannawar, CGM, NABARD highlighted that want to replicate similar models of start-up ecosystem practiced and adapted by ICAR-NAARM. He also extended his best wishes to a-IDEA and expressed his desire to send more NABARD teams in future to learn the challenges and complexities in incubation sector.

Shri B. Uday Bhaskar, General Manager, Shri N S Murthy, General Manager, Shri RIA Selvan, Deputy General Manager, Shri B Ramesh Babu, Deputy General Manager, 8 officials from AP Regional Office and 10 District Level Officials of NABARD stationed in various districts of Andhra Pradesh, staff of ICAR-NAARM, officials from PJTSAU and the entire a-IDEA team were present during the occasion.



Fig. 4.7: Visit of NABARD Officials to a-IDEA

4.2 Centre for Lifelong Learning in Agricultural Education (COLLAge)

The centre, which has been carved out of the efforts under Niche Area project on Technology Enhanced Learning, has been spearheading the activities of Educational Technology that support various online programs like MOOCs and Distance Education programs like PGDETM.

The Centre actively supported conducting the online assessments for PGDETM programs. The centre developed 17 educational modules with interactive format (H5P) which were used effectively in MOOCs programs. The details of MOOC taken up through COLLAge during the year is given below:

4.2.1 Massive Open Online Courses (MOOCs)

Keeping in view the rising demand for online programmes, Academy continued its efforts to reach wider trainee-base through its innovatively designed Massive Open Online Courses (MOOCs). A total of 2 MOOCs were conducted during the year, - first one on Theoretical Foundations of Educational Technology in June 1-30, 2021 and second one on Digital Teaching Techniques in December 1-31, 2021. Both programs were aimed at laying foundations and honing the skills on use of technology in education. The programs

were well received by the participants as it was customized for teaching fraternity and those who are interested to improve their teaching skills using technology.

The MOOCs were combination of weekly released lesson modules with inbuilt exercises and periodic evaluation. Participants were expected to submit an assignment after completing all modules, which test their overall understanding of the subject. There was an online exam at the end of MOOCs. Every activity was assigned weightages for participants to qualify for certification.

First MOOC had an impressive turnout of 1008 eligible for certification out of a total of 2355 registered participants with a very high course completion rate of 42.8 per cent. The program had 1844 active learners. Second MOOC had 1469 active learners out of a total of 1870 registered participants. The evaluation of participants for eligibility to certification will be carried out during January 2022.

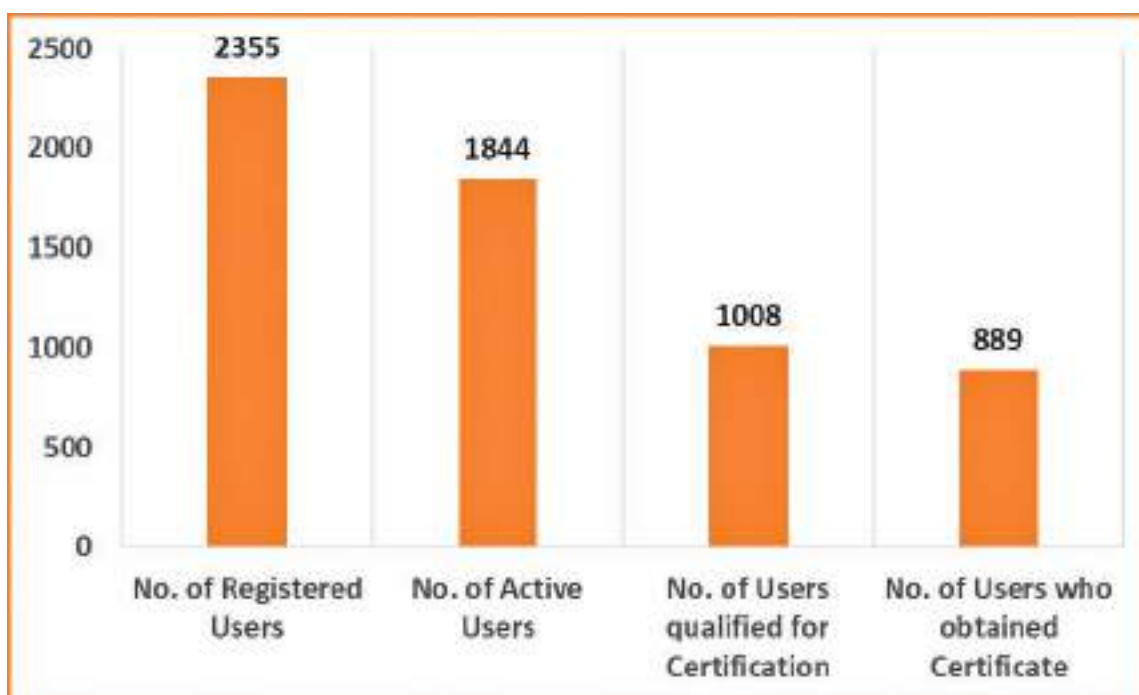


Fig. 4.8: MOOC Overview

5.1 Training Programmes attended by ICAR-NAARM Employees

Sl. No.	Name	Programme	Duration	Organizer
1.	M Balakrishnan	Training of Trainers on Effective Delivery of Trainings using participatory Training Methods	May 11-14, 2021	IGNFA and GIZ, under the Indo-German HWC Project, Dehradun
2.	S Senthil Vinayagam	GIZ -Dale Carnegie Training of Trainers for Effective delivery of courses on Holistic Approach to Human-Wildlife Conflict Mitigation	May 11-14, 2021	IGNFA, Dehradun
3.	Surya Rathore	Training of Trainers for effective delivery of courses on holistic approach to Human – Wildlife Conflict (HWC)	May 11-14, 2021	GIZ–Dale Carnegie & IGNFA, Dehradun
4.	N A Vijay Avinashilingam	Training of Trainers for effective delivery of courses on holistic approach to Human – Wildlife Conflict (HWC)	May 11-14, 2021	GIZ – Dale Carnegie & IGNFA, Dehradun
5.	D Thammi Raju	Training on Agri-DIKSHA Web Education Channel	May 21, 2021	IASRI, New Delhi
6.	Surya Rathore	Agridaksh training organized under NAHEP, Component 2 A	May 28, 2021	IASRI, New Delhi
7.	Alok Kumar	Agridaksh training organized under NAHEP, Component 2 A	May 28, 2021	IASRI, New Delhi
8.	Surya Rathore	Training Programme on effective health management for increasing efficiency of ICAR employees	June 10, 2021	ICAR – NAARM, Hyderabad
9.	Alok Kumar	Training Programme on effective health management for increasing efficiency of ICAR employees	June 10, 2021	ICAR – NAARM, Hyderabad
10.	D Thammi Raju	Mainstreaming of Agriculture as a subject in the school curriculum	June 16, 2021	IASRI, New Delhi
11.	M Balakrishnan	Online training Programme on Reservation in Service for SC/ST/OBC/EWS/PWD	Jun 28-30, 2021	ICAR-NAARM
12.	S Ravichandran	Designing better courses: Blending the best of pre- and post-pandemic pedagogy	July 21, 2021	Harvard Business Publishing Education
13.	S K Soam	Designing Better Courses: Blending the Best of Pre- and Post-Pandemic Pedagogy	July 21, 2021	Harvard Business School, USA
14.	N Srinivasa Rao	Training on effective implementation of social safeguard plans	Jul 23-24, 2021	ICAR – NAARM and NAHEP
15.	P Supriya	Developing Work Life Well Being for Farm Women Extension Functionaries	Aug 2-5, 2021	National Institute of Agricultural Extension Management (MANAGE), Hyderabad

Sl. No.	Name	Programme	Duration	Organizer
16	S Ravichandran	Open Science-challenges, possible solutions and the way forward	Aug. 13, 2021	ICAR-IIHR, Bangalore
17.	M Ramesh Naik	Different Modeling approaches in Agriculture	Sep 2021 5-10,	ICAR-IISS, Bhopal
18.	M Ramesh Naik	Dynamics of Agriculture and Food Systems	Dec 2021 06-17,	ICAR-IIFSR, Modipuram

5.2 Capacity Building Programs Organized by ICAR-NAARM

Sl. No.	Title of the Programme	Period	Programme Directors	No. of Participants	Programme Type
1.	Training Programme on Time Series Data Analysis (Online Mode)	Jan 04-09, 2021	S Ravichandran N Sivaramane BS Yashavanth	72	Need Based Programme
2.	National Workshop on Scientific Interventions and Policies for Strengthening of Buffalo Meat Sector: Catalyst for Boosting India's Agro-Economy	Jan 19, 2021	P Krishnan	73	Workshop / Seminar/ Conference
3.	Off-Campus Programme on Teaching Competency Enhancement of Middle and Senior Faculty of AAU	Jan 19-23, 2021	M Balakrishnan Atul Borgohain	32	Off-Campus Programme
4.	Online Training Programme on Geospatial Analysis using QGIS & R	Feb 1-6, 2021	PD Sreekanth BS Yashavanth	32	Need Based Programme
5.	Coromandal International Ltd.-NAARM Collaborative Programme for New Entrants Assimilation Training (NEAT)-Phases-I	Feb 8-12, 2021	KH Rao Ranjit Kumar N Sivaramane	25	Need Based Programme
6.	Online Training Programme on E-Resources, E-Learning and IoT for Students of UAS, Raichur	Feb 15-19, 2021	S Senthil Vinayagam GRK Murthy VV Sumanth Kumar	30	Need Based Programme
7.	Coromandal International Ltd.-NAARM Collaborative Programme for New Entrants Assimilation Training (NEAT)-Phase-II	Feb 15-19, 2021	KH Rao Ranjit Kumar N Sivaramane PC Meena	25	Need Based Programme
8.	Advanced Bioinformatics Tools and its Applications in Agriculture for the faculty of College of biotechnology, SVPUAT, Meerut	Feb 22-24, 2021	M Balakrishnan	14	Need Based Programme
9.	Training Programme on Data Visualization in Agribusiness and Agricultural Research	Feb 22-27, 2021	Sanjiv Kumar BS Yashavanth	26	Need Based Programme



Sl. No.	Title of the Programme	Period	Programme Directors	No. of Participants	Programme Type
10	Annual Conference on Visionary Innovations in Statistical Theory and Applications (VISTA – 2021)	Feb 24-28, 2021	A Dhandapani S Ravichandran	170	Workshop / Seminar/ Conference
11.	Training Programme on Competency Enhancement & Student Mentorship in Agricultural Competitive Exams under SCSP at Agricultural College, Mahanandi	Mar 5-6, 2021	M Ramesh Naik Umesh Hudedamani	50	Off-Campus Programme
12.	Training Programme on Leadership Development for Women Scientists	Mar 8-10, 2021	Surya Rathore	49	Need Based Programme
13.	Training Programme on Entrepreneurial and Communication Skill Development in collaboration with College of Veterinary Sciences, PVNR Veterinary University Hyderabad (SC Sub-Plan)	Mar 8-12, 2021	PC Meena Sanjiv Kumar	21	Need Based Programme
14.	Training Programme on Competency Enhancement & Student Mentorship in Agricultural Competitive Exams under SCSP at Agricultural College, Rajahmundry	Mar 9-10, 2021	M Ramesh Naik Umesh Hudedamani	50	Off-Campus Programme
15.	National Workshop on Policy Perspective on Farm Subsidies in India vis a vis Other Countries: Issues and Way Forward	Mar 11, 2021	Tavva Srinivas K Kareemulla Ch Srinivasa Rao	25	Workshop / Seminar/ Conference
16.	Training Programme on Applications of Artificial Intelligence and Cloud Computing in Agriculture	Mar 15- 20, 2021	VV Sumanth Kumar D Thammi Raju	32	Need Based Programme
17.	Training on Design Thinking in Research and Education with Focus on Veterinary Sciences (An online sponsored Programme for RAJUVAS-Bikaner, Rajasthan)	Mar 22-27, 2021	SK Soam D Thammi Raju B Ganesh Kumar	49	Need Based Programme
18.	Training Programme on IPR, Copy Right, Industrial Design and Plagiarism for Staff of TANUVAS, Chennai	Mar 22-26, 2021	K Srinivas	30	Need Based Programme
19.	Online Training Programme on E-Resources, E-Learning and Internet of Things (IoT) for Faculty of UAS, Raichur	Mar 23-27, 2021	GRK Murthy	30	Need Based Programme
20.	Skill Development Programme on Tailoring for rural women/youth under SC Sub-Plan (batch-1)	Mar 23-27, 2021	Laxman M Ahire	30	Need Based Programme
21.	Skill Development Programme on Tailoring for rural women/youth under SC Sub-Plan (batch-2)	Mar 30-Apr 3, 2021	M Ramesh Naik	32	Need Based Programme

Sl. No.	Title of the Programme	Period	Programme Directors	No. of Participants	Programme Type
22.	Training Programme on Content Development and Management TANUVAS Faculty (Sponsored by TANUVAS)	Apr 16-21, 2021	GRK Murthy	30	Need Based Programme
23.	Virtual Workshop on Technical & Scientific Writing (IFS Probationers of IGNFA, Dehradun)	Apr 23, 2021	P Krishnan	165	Workshop / Seminar / Conference
24.	Virtual Induction Training On Project Management & Research Methodology (ICFRE), Dehradun	May 3-14, 2021	P Krishnan A Dhandapani	30	Need Based Programme
25.	Training Programme on Facilitative Teaching Technologies for TANUVAS Faculty (Online mode)	May 16-21, 2021	S Senthil Vinayagam	30	Need Based Programme
26.	Coromandel Training Programme for New Entrance Assimilation Training (NEAT) Phase-I (Online mode)	May 18-22, 2021	Ranjit Kumart KH Rao	29	Need Based Programme
27.	Sensitization Programme on Entrepreneurship Development in Agriculture for students of Rani Lakshmi Bai Central Agricultural University, Jhansi (Online mode)	Jun 1, 2021	N Sivaramane	195	Off-Campus Programme
28.	MOOC on Principles of Educational Technology (Online Mode)	Jun 1-30, 2021	GRK Murthy	2355	MOOC
29.	MDP on Biodiversity and Environmental Laws for Agricultural Researchers (Online Mode)	Jun 7- 9, 2021	P Krishnan	38	Need Based Programme
30.	Entrepreneurship Development for Agri Students of SKLTSHU (Online Mode)	Jun 14-18, 2021	N Sivaramane	83	Need Based Programme
31.	MDP on Leadership Development (a Pre-RMP Programme) (Online Mode)	Jun 14-25, 2021	P Ramesh	37	MDP Pre-RMP
32.	Sensitization Programme on Entrepreneurship Development in Veterinary Sector for students of TANUVAS, Chennai	Jun 19, 2021	N Sivaramane	150	Off-Campus Programme
33.	MDP on Business Plan Development and Accelerating FPOs/FPCs (Online Mode)	Jun 21-26, 2021	Ranjit Kumar	58	Need Based Programme
34.	Training Programme on Administrative and Finance Management for Section Officers/ AAO/ AFAOs/Assistants of ICAR Headquarters/Institutes (Online Mode)	Jun 24-26, 2021	BD Phansal	46	Need Based Programme



Sl. No.	Title of the Programme	Period	Programme Directors	No. of Participants	Programme Type
35.	Training programme on Reservation in Service including Roster and Reservation Register (Online Mode)	Jun 28-30, 2021	BD Phansal	42	Need Based Programme
36.	Training Programme on Business Planning and Strategy Formulation for FPOs/FPCs (Online Mode)	Jul 5-9, 2021	Ranjit Kumar	48	Need Based Programme
37.	Sensitization Programme on Entrepreneurship Development and Start-up Ecosystem (Online Mode)	Jul 5-9, 2021	N Sivaramane	37	Need Based Programme
38.	Training Programme on Stress Management (Online Mode)	Jul 6-9, 2021	P Ramesh	19	Need Based Programme
39.	Programming of Web & Mobile Applications using Low-code Platforms (Online Mode)	Jul 7-12, 2021	M Balakrishnan	48	Need Based Programme
40.	Training Programme on RTI Act 2005 (Online Mode)	Jul 15-16, 2021	BD Phansal	34	Need Based Programme
41.	Online Training Programme on Intellectual Property Rights and Technology Commercialization in Agriculture' for Faculty members of Kamdhenu University, College of Dairy Science, Amreli, Gujarat	Jul 15-17, 2021	SK Soam	50	Need Based Programme
42.	Online Sensitization Programme on Entrepreneurship Development in Fisheries Sector for students of Central Institute of Fisheries Education, Mumbai	Jul 17, 2021	Vijay Avinashlingam	125	Need Based Programme
43.	Online Sensitization Programme on Entrepreneurship Development in Agriculture for students of Sardar Vallabhbhai Patel University of Agriculture and Technology, Meerut	Jul 22, 2021	Vijay Avinashlingam	319	Need Based Programme
44.	Online Sensitization Programme Entrepreneurship Development in Agriculture for students of Pandit Jawaharlal Nehru College of Agriculture & Research Institute, Karaikal	Jul 23, 2021	Vijay Avinashlingam	490	Need Based Programme
45.	Training Workshop on Effective Implementation of Social Safeguard Plans (Sponsored under NAHEP) (Online Mode)	Jul 23-24, 2021	Surya Rathore	39	Workshop / Seminar/ Conference
46.	Online Training Programme on Budget Utilisation Procedure	Aug 9-11, 2021	ZH Khilji	39	Need Based Programme

Sl. No.	Title of the Programme	Period	Programme Directors	No. of Participants	Programme Type
47.	Training Workshop for Vigilance Officers of ICAR	Aug 16-18, 2021	S Senthil Vinayagam	42	Workshop / Seminar/ Conference
48.	Training Programme on MDP on Formation of FPO/FPC and Preparing its Business Plan for KVKs of Rajasthan, Haryana and Delhi (Online Mode)	Aug 17-24, 2021	Sanjiv Kumar	66	Need Based Programme
49.	Training Programme on E-resources, E-learning, and IoT in Agriculture, UAS Raichur (Online Mode)	Aug 23-27, 2021	D Thammi Raju	30	Need Based Programme
50.	Training Programme on Advances in Extension Research and Evaluation Methodology (Online Mode)	Aug 23-Sep 1, 2021	P Venkatesan	37	Need Based Programme
51.	Training Workshop on Response Surface Methodology (Online Mode)	Aug 24-26, 2021	A Dhandapani	40	Workshop / Seminar/ Conference
52.	Virtual Hands on Training on Career Development Centre and Faculty Development Centre Monitoring Systems	Aug 24, 2021	N Srinivasa Rao Alok Kumar	26	Workshop / Seminar/ Conference
53.	Training Programme on Intellectual Property Rights, Copyright, Industrial Design and Plagiarism, TANUVAS Chennai (Online Mode)	Aug 24-28, 2021	Umesh Hudedamani	30	Need Based Programme
54.	Training Programme on Design Thinking for Research Project Formulation and Implementation (Online Mode)	Aug 24-28, 2021	SK Soam	33	Need Based Programme
55.	National Summit on Building organized sheep and goat meat sector in India towards ensuring Atmanirbhar Bharat	Sep 3, 2021	P Krishnan	125	Workshop / Seminar/ Conference
56.	Training Programme on Implications Of New Labour Codes, 2020 for ICAR for Administrative and Finance Officers of ICAR (Online Mode)	Sep 6-8, 2021	ZH Khilji	31	Need Based Programme
57.	Sensitization Programme Entrepreneurship Development in Agriculture for students of Sher-e-Kashmir University of Agricultural Sciences and Technology, Jammu (Online Mode)	Sep 9, 2021	Vijay Avinashlingam	137	Need Based Programme
58.	Training Programme on Content Development and Management, TANUVAS Chennai (Online Mode)	Sep 14-18, 2021	GRK Murthy	30	Need Based Programme



Sl. No.	Title of the Programme	Period	Programme Directors	No. of Participants	Programme Type
59.	Training Programme on "Organic Cultivation of Nutrition Garden for SC Womens"	Sep 18, 2021	M Balakrishnan	100	Under Sub-Plan
60.	Training Programme on "Organic Cultivation of Nutrition Garden for SC Womens"	Sep 19, 2021	M Balakrishnan	100	Under Sub-Plan
61.	Sensitization Programme Entrepreneurship Development in Agriculture for students of Rajmata Vijayaraje Scindia Krishi Vishwa Vidyalaya, Gwalior (Online Mode)	Sep 20, 2021	Vijay Avinashlingam	111	Need Based Programme
62.	Training Programme on Entrepreneurial Skill Development for Agricultural Graduates for Students of Agricultural College, Bapatla, ANGRAU (Under SCSP)	Sep 20-22, 2021	M Ramesh Naik	170	Off-Campus Programme
63.	Training on Application of Bioinformatics in Agricultural Research and Education (Online Mode)	Sep 20-24, 2021	M Balakrishnan	45	Need Based Programme
64.	Competence Enhancement Programme on Soft Skills and Personality Development for T1/T4 Staff of ICAR (Online Mode)	Sep 20-25, 2021	Danam Murahari	52	Need Based Programme
65.	MDP on Market Research & Value Chain Management of Agricultural Commodities (Online Mode)	Sep 21-25 2021	B Ganesh Kumar	28	Need Based Programme
66.	Sensitization Programme Entrepreneurship Development in Agriculture for students of Central Agricultural University, Imphal (Online Mode)	Sep 22, 2021	Vijay Avinashlingam	290	Need Based Programme
67.	Training Programme on Education Management and Academic Leadership (under NAHEP) (Online Mode)	Sep 23- 28, 2021	D Thammi Raju	47	Need Based Programme
68.	Training Programme on "Organic Cultivation of Nutrition Garden for SC Women"	Sep 25, 2021	M Balakrishnan	100	Under Sub-Plan
69.	FDP on Competency Enhancement in Agricultural Research and Education for the Faculty of AAU College of Veterinary Science, Khanapara, Assam (under SCSP)	Sep 27, Oct 1, 2021	M Balakrishnan	38	Off-Campus Programme
70.	Training Programme on Entrepreneurial Skill Development for Agricultural Graduates at Agricultural College, Naira.(under SCSP)	Sep 29- Oct 1, 2021	M Ramesh Naik	160	Off-Campus Programme

Sl. No.	Title of the Programme	Period	Programme Directors	No. of Participants	Programme Type
71.	FDP on Competency Enhancement in Agricultural Research and Education for the Faculty of TNJFU, Nagapatnam	Oct 4-8, 2021	M Balakrishnan	31	Off-Campus Programme
72.	Training Programme on Innovations in Educational Technology (Online Mode)	Oct 4-8, 2021	S Senthil Vinayagam	26	Need Based Programme
73.	Training Programme on Data Analysis in Social Sciences Research (Online Mode)	Oct 4-8, 2021	N Sivaramane	56	Need Based Programme
74.	Training Programme on Soft Skills for Extension Professionals (Online Mode)	Oct 21-30, 2021	Bharat S Sontakki	23	Need Based Programme
75.	Training Programme on Sustainable Management of Small-Scale Seed Enterprises (Online Mode)	Oct 25-29, 2021	T Srinivas	7	Need Based Programme
76.	MDP on Priority Setting Monitoring and Evaluation (PME) of Agricultural Research Projects (Online Mode)	Oct 25-30, 2021	P Krishnan	37	Need Based Programme
77.	Training Programme on Facilitative Teaching Technologies, TANUVAS Chennai (Online Mode)	Oct 26-30, 2021	VV Sumanth Kumar	30	Need Based Programme
78.	Training Workshop on Analysis of Multi/ Location Experiments (Online Mode)	Oct 28-Nov 1, 2021	A Dhandapani	85	Workshop / Seminar/ Conference
79.	International Webinar on Fostering Global Collaboration Among Agricultural Higher Educational Institutions (Under NAHEP)	Nov 8-9 2021	SK Soam	250	Workshop / Seminar/ Conference
80.	Third Meeting of External Advisory Panel of NAHEP	Nov 10, 2021	SK Soam	30	Workshop / Seminar/ Conference
81.	FDP on Competency Enhancement in Agricultural Research and Education for the Faculty of CAU (under SCSP)	Nov 8-12, 2021	M Balakrishnan	40	Off-Campus Programme
82.	Training NEAT Phase- II of Coromandel International Ltd.	Nov 16-20, 2021	Ranjit Kumar KH Rao	26	Need Based Programme
83.	Entrepreneurship Development for Horticulture Students (Online) of Dr. YSRHU	Nov 22-27, 2021	Tavva Srinivas	60	Need Based Programme
84.	Workshop on National Water Policies Implementation and Impact (Online Mode)	Nov 25-26, 2021	K Kareemulla S Ravichandran	56	Workshop / Seminar/ Conference
85.	Training Programme on Motivation, Positive Thinking and Communication Skills for Technical Officers of ICAR (T5 and above) (Online Mode)	Nov 25-30, 2021	MA Basith	16	Need Based Programme



Sl. No.	Title of the Programme	Period	Programme Directors	No. of Participants	Programme Type
86.	MDP on Leadership Development (a Pre-RMP Programme) for Dr. YSRHU (Online)	Nov 29 - Dec 3, 2021	Tavva Srinivas	40	Need Based Programme
87.	Executive Development Programme on Leadership Development for Dr. YSRHU (Online)	Nov 30 - Dec 4, 2021	M Balakrishnan	30	Need Based Programme
88.	MOOC on Digital Teaching Techniques	Dec 1-31, 2021	GRK Murthy S Senthil Vinayagam	1870	MOOC
89.	Training Programme on Enhancing Resilience through Entrepreneurship (Online Mode)	Dec 6-10, 2021	Vijay Avinashlingam	20	Need Based Programme
90.	Training Programme on Advances in Web and Mobile Application Development (Online Mode)	Dec 6-10, 2021	N Srinivasa Rao	42	Need Based Programmes
91.	Training Programme on Administrative and Finance Management for Dr. YSRHU	Dec 6-10, 2021	BD Phansal	20	Need Based Programme
92.	Training Programme on Vermicompost Production Technology for Entrepreneurs and Farmers for NBPZ of Assam under SCSP	Dec 8-10, 2021	M Balakrishnan	111	Off-Campus Programme
93.	Training Programme on E-resource, e-learning & internet of things (IoT) for the Faculty of UAS, Raichur Karnataka	Dec 13-17, 2021	M Balakrishnan	30	Off-Campus Programme
94.	MDP on Leadership Development (a Pre/RMP Programme) (Online Mode)	Dec 13-24, 2021	Alok Kumar	54	MDP Pre-RMP
95.	Training Programme on Impact Assessment of Agricultural Research and Technologies (Online Mode)	Dec 18-22, 2021	I Sekar	23	Need Based Programme

5.3. Participation in Seminar/Symposium/ Conferences/Workshops/Meetings

5.3.1 National

Month	Name of staff	Name of the Programme	Venue / Organized by	Period
January	Ch. Srinivasa Rao	Review Meeting of Officers & Staff of DARE & ICAR under the Chairmanship of Secretary (DARE) & Director General, ICAR	Virtual/ICAR	Jan 1, 2021
	Ch Srinivasa Rao	Preliminary Meeting to finalize work plan with food industry and other related activities of IPTM, ICAR in association with ICAR-NAARM	Virtual	Jan 2, 2021
	S Senthil Vinayagam	Seventh Annual Review Meeting on Entrepreneurship Development through Farmer Led Innovations (FLIs)-A Case Study in Plantation Sector	ICAR - NASF, New Delhi	Jan 6, 2021
	Ch Srinivasa Rao	Hyderabad Cluster Virtual Launch Event	Virtual/	Jan 8, 2021
	D Thammi Raju	Review of NAHEP by National Director, NAHEP	ICAR-NAARM	Jan 9, 2021
	Ch Srinivasa Rao	ICAR SOC Meeting	Virtual/ICAR	Jan 12, 2021
	Ch Srinivasa Rao	NAAS Hyderabad Chapter Meeting	Virtual/ NAAS Hyderabad Regional Chapter	Jan 16, 2021
	Ch Srinivasa Rao	National Workshop on Scientific Interventions and Policies for Strengthening of Buffalo Meat Sector: Catalyst for Boosting India's Agro-Economy	ICAR-NAARM, Hyderabad (Hybrid mode) jointly organized by ICAR-NRC-Meat, ICAR-NAARM and NAAS Hyderabad Chapter	Jan 19, 2021
	S K Soam	Webinar on Supply Chain and Logistics Management	IIM Ahmedabad	Jan 18, 2021
	S K Soam	Online Review Meeting to review the progress of ZTMC activities in ICAR Crop Institutes in Southern Indian Zone (As Chief Guest)	ICAR-IIMR	Jan 18, 2021
	Ch Srinivasa Rao	Inaugural Session and Closing Ceremony of National Dialogue - Indian Agriculture Towards 2030 - Pathways for Enhancing Farmers' Income, Nutritional Security and Sustainable Food Systems (Inaugurated by Hon'ble Vice President of India)	Virtual	Jan 19&22, 2021
	Ch Srinivasa Rao	Meeting on Development of Tech Dome for Central Vista Project, held under the chairmanship of Director General, ICAR	Virtual/ ICAR	Jan 22, 2021
	S Senthil Vinayagam	IMC Meeting	ICAR-National Research Centre On Meat, Hyderabad	Jan 22, 2021
	Bharat S Sontakki	92 nd Annual General Meeting of ICAR Society	ICAR, New Delhi (Virtual Meeting)	Jan 27, 2021
	Ch Srinivasa Rao	92 nd Annual General Meeting of ICAR Society	Virtual/ ICAR	Jan 27, 2021
	S K Soam	Meeting of NAARM-Zonal Technology Management Committee (ZTMC)	ICAR-NAARM	Jan 28, 2021
February	D Thammi Raju	SP Award Committee meeting	ICAR-NAARM	Jan 30, 2021
	Ranjit Kumar	Reinventing the Agriculture Business Management (ABM) Programme of its School of Agriculture Business Management (SABM) of PJTSAU	Indian School of Business (ISB), Hyderabad	Feb 1-2, 2021
	D Thammi Raju	Domestic External Advisory Panel Meeting under NAHEP	ICAR-NAARM	Feb 2, 2021
	D Thammi Raju	SP Award Committee meeting	ICAR-NAARM	Feb 2, 2021
	Ch Srinivasa Rao	ICAR SOC Meeting	Virtual/ICAR	Feb 2, 2021
	Ch Srinivasa Rao	Meeting of ICAR Committee to review Career Advancement Scheme (CAS) and Score Card for Scientists of ICAR	Virtual/ ICAR	Feb 8, 2021
	D Thammi Raju	Internal Audit Party Meeting under NAHEP	Virtual/ ICAR	Feb 9, 2021
	D Thammi Raju	Board of Studies Meeting of PGDMA	Virtual/ ICAR	Feb 10, 2021
	Ranjit Kumar	Executive Committee meeting of the Indian Society of Agricultural Economics	CARDS, Coimbatore	Feb 11, 2021
	Ch Srinivasa Rao	Consultation meeting by Addl Secretary, EF&CC, MoEFCC, Govt. of India, regarding partnership with academic/research institutes for achieving India's voluntary target of land restoration.	Virtual/ MoEFCC, Govt. of India	Feb 12, 2021



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February	Ch Srinivasa Rao	Consultation meeting by Addl Secretary, EF&CC, MoEFCC, Govt. of India, regarding partnership with academic/research institutes for achieving India's voluntary target of land restoration.	Virtual/ MoEFCC, Govt. of India	Feb 12, 2021
	Ch Srinivasa Rao	25 th Board of Management Meeting of PJTSAU	Virtual/PJTSAU	Feb 12, 2021
	Ch Srinivasa Rao	Meeting of all the Conveners of NAAS Regional Chapters	Virtual/NAAS	Feb 15, 2021
	Ch Srinivasa Rao	First Meeting of NAAS Organizational Committee to frame guidelines relating to organizational, personnel, finances and investments as well as delegation of powers	Virtual/NAAS	Feb 15, 2021
	V V Sumanth Kumar	Second PMRC meeting of NABARD a-IDEA ABIC	ICAR-NAARM	Feb 15, 2021
	Vijay Avinashilingam	India CII Global Startup Summit – Enabling Robust Entrepreneurial and Innovation Ecosystem	Online platform	Feb 15-17, 2021
	Ch Srinivasa Rao	2 nd Meeting of ICAR Committee to review Career Advancement Scheme (CAS) and Score Card for Scientists of ICAR	KAB-II, Pusa, New Delhi/	Feb 17-18, 2021
	Ch Srinivasa Rao	116 th Executive Council Meeting of NAAS	NAAS, New Delhi/	Feb 19, 2021
	Ch Srinivasa Rao	Chief Guest for the International Webinar #4 on Breeding for efficient root system for water and Nutrient uptake in crops	Virtual/ ICAR	Feb 21, 2021
	Ch Srinivasa Rao	Chief Guest for the DST sponsored online National Training Programme on Soil & Water Conservation Technologies for Climate Smart Agriculture in the Context of Extreme Weather Events	Virtual/IISWC RC, Udhagamandalam, Tamil Nadu	Feb 22, 2021
	Bharat S Sontakki	54 th Meeting of Institute Management Committee	ICAR-CTRI, Rajahmundry	Feb 24, 2021
	Ch Srinivasa Rao	Fellows Meeting and Annual General Body Meeting of the NASI	Virtual/	Feb 27, 2021
	Ch Srinivasa Rao	Chaired the Valedictory Session of the of 23 rd Annual Conference(online) of the Society of Statistics, Computer and Applications – Web Conference on Visionary Innovations in Statistical Theory and Applications (VISTA-2021)	ICAR-NAARM, Hyderabad (Hybrid mode)/ jointly organized by Society of Statistics, Computer and Applications (SSCA) & ICAR-NAARM	Feb 28, 2021
March	Ranjit Kumar	Study Management Committee Meeting of Co-Benefits of Large scale Organic Farming on Human Health (BLOOM): An evaluation of Andhra Pradesh Community-managed Natural Farming	University of Edinburgh	Mar 01, 2021
	V V Sumanth Kumar	General Body meeting of a-IDEA	a-IDEA, ICAR-NAARM	Mar 01, 2021
	Ch Srinivasa Rao	ICAR SOC Meeting	Virtual/ICAR	Mar 05, 2021
	Ch Srinivasa Rao	ICAR Meeting on Budget Utilization held under the chairmanship of Secretary, DARE & DG, ICAR	Virtual/ICAR	Mar 05, 2021
	Ch Srinivasa Rao	International Women's Day Programme of ICAR	Virtual/ICAR	Mar 08, 2021
	Bharat S Sontakki	International Women's Day	ICAR-NAARM, Hyderabad	Mar 08, 2021
	Surya Rathore	Online Joint Advisory Committee (JAC) meeting of NAHEP	ICAR –NAARM, Hyderabad	Mar 8, 2021
	S K Soam	Meeting of Institute Management Committee of ICAR-NAARM, Hyderabad	ICAR-NAARM	Mar 10, 2021
	Ranjit Kumar	Brainstorming Session on Contract Farming for Transforming Agriculture in India: Challenges and Way Forward	NAAS	Mar 10, 2021
	Ch Srinivasa Rao	NAAS Executive Council Meeting	Virtual/NAAS	Mar 11, 2021
	P Venkatesan	Review cum Action Plan Workshop on Entrepreneurship Development through Farmer Led Innovations – A Case Study in Plantation Sector under National Agricultural Science Fund	ICAR-NAARM, Hyderabad	Mar 15-16, 2021
	Ch Srinivasa Rao	ICT Steering Committee Meeting of ICAR	Virtual/ICAR	Mar 16, 2021
	Ch Srinivasa Rao	Regional Meeting of Regional Advisory Committee (RAC) on Off Farm/Non-Farm Sector Development in Telangana	Virtual/ Telangana	Mar 16, 2021
	Ch Srinivasa Rao	Meeting of senior officers from NAARM and CRIDA to discuss issues related to Pension	ICAR-NAARM	Mar 18, 2021
	Ch Srinivasa Rao	Preparatory Meeting for CoP-26 of UNFCCC organized by MoEFCC, Govt of India	Virtual/ UN Climate Change Secretariat	Mar 18, 2021
	V V Sumanth Kumar	Empanelment Committee Meeting to Review Proposals for Product Development/ Strategies to Strengthen.	ICAR-NAARM	Mar 19, 2021

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March	Bharat S Sontakki	Interactive Meeting and Discussion with Farmers on World Water Day	ICAR-NAARM, Hyderabad	Mar 22, 2021
	P Venkatesan	Interactive Meeting and Discussion with Farmers on World Water Day	ICAR-NAARM, Hyderabad	Mar 22, 2021
	Ch Srinivasa Rao	World Water Day Programme Lecture by Mark Smith, Director General, International Water Management Institute	Virtual/ICAR	Mar 22, 2021
	Ch Srinivasa Rao	25 th Meeting of Food and Agriculture Division Council (FADC), Bureau of Indian Standards, Govt. of India, held under the chairmanship of Secretary, DARE & DG, ICAR	Virtual/FADC, New Delhi	Mar 23, 2021
	Ranjit Kumar	Planning meeting of NAHEP-CAAST sponsored subproject on Mapping and Performance Analysis of FPOs in Western Region of India	Anand Agricultural University, Anand	Mar 23, 2021
	Ch Srinivasa Rao	Final Review Meeting of the NAAS Task Force on Compendium on Impact of Agricultural Research	Virtual/NAAS	Mar 24, 2021
	Ch Srinivasa Rao	First Meeting of Apex Committee for Implementation of Paris Agreement (AIPA) under the Chairmanship of Secretary, MoEFCC, Govt. of India	Virtual/MoEF&CC	Mar 24, 2021
	Ch Srinivasa Rao	Meeting on Bharat ka Amrut Mahotsav – Commemoration of 75 years of India's Independence - Lecture on Precision Mechanization by K Alagasundaram, DDG (Ag Engg), ICAR.	Virtual/Education Division, ICAR	Mar 25, 2021
	S Senthil Vinayagam	Advisory Committee meeting of a-IDEA	ICAR-NAARM, Hyderabad	Mar 25, 2021
	V V Sumanth Kumar	Advisory Group meeting of a-IDEA	a-IDEA, ICAR-NAARM	Mar 25, 2021
	Ch Srinivasa Rao	Meeting of the Programme Committee of NAAS	Virtual/NAAS	Mar 25, 2021
	Ch Srinivasa Rao	Zoom Webinar - Policy Dialogue on Biofortification to Address Hidden Hunger and Nutritional Security: Present Status & Way Forward under the Convenorship of U.S. Singh, South Asia Regional Coordinator and Country Manager, CIP-South Asia, organized by NAAS	Virtual/NAAS	Mar 25, 2021
	S Senthil Vinayagam	Standing Advisory Committee of RTP, NIRD	NIRD & PR, Hyderabad	Mar 30, 2021
April	Ranjit Kumar	Study Management Committee Meeting of the study on Co-Benefits of Largescale Organic Farming on Human Health (BLOOM): A Cluster randomized controlled evaluation of Andhra Pradesh Community managed Natural Farming	University of Edinburg & PHFI	Apr 12, 2021
	Ch Srinivasa Rao	Bi-annual Review Meeting of the XXV Meeting of ICAR Regional Committee No. II for West Bengal, Odisha, Telangana, Andhra Pradesh and the Union Territory of Andaman & Nicobar Islands under the Chairmanship of Deputy Director General (Crop Science), ICAR, New Delhi	Virtual/ICAR	Apr 12, 2021
	Ch Srinivasa Rao	UNFCC Meetings - SBI Consultations on matters related to the Fourth Review of the Adaptation Fund and First Expert Group Meeting on ACE	Virtual/ UN Climate Change Secretariat	Apr 14, 2021
	Ch Srinivasa Rao	Chaired the Meeting of Organizing Committee for ICAR National Convention for Industry-Institute Interface for Food Processing Technologies	Virtual/ICAR	Apr 15, 2021
	Ch Srinivasa Rao	UNFCC Informal Consultations on Santiago Network on Loss and Damages	Virtual/ UN Climate Change Secretariat	Apr 15, 2021
	Ch Srinivasa Rao	118 th Meeting of the Executive Council of National Academy of Agricultural Sciences, New Delhi	Virtual/NAAS	Apr 16, 2021
	Ch Srinivasa Rao	Fifth Convocation of PJTSAU	Virtual/PJTSAU	Apr 17, 2021
	Ch Srinivasa Rao	Meeting on Agriculture Loss and Damages under the Chairmanship of Secretary, MoEFCC, Govt. of India for briefing on Climate Change Negotiation Issues	Virtual/MoEF&CC	Apr 17, 2021
	Ch Srinivasa Rao	Chaired the Meeting of Organizing Committee for ICAR National Convention for Industry-Institute Interface for Food Processing Technologies	Virtual/ICAR	Apr 23, 2021



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April	S K Soam	Meeting of Organizing Committee for ICAR National Convention for Industry-Institute Interface for Food Processing Technologies	ICAR	Apr 24, 2021
	S K Soam	5 th meeting of PRAYAS Monitoring Committee (PMC) of Association for Innovation Development of Entrepreneurship in Agriculture (a-IDEA)	ICAR-NAARM	Apr 26, 2021
	V V Sumanth Kumar	Empanelment Committee Meeting to Review Proposals for Product Development/ Strategies	ICAR-NAARM	Apr 28, 2021
	S K Soam	Review Meeting of Career Development Centres and Faculty Development Centres	ICAR-NAARM	Apr 30, 2021
	D Thammi Raju	Review Meeting of CDCs and FDCs	Online	Apr 30, 2021
	Surya Rathore	Review Meeting of Career Development Centres (CDCs) and Faculty Development Centre (FDC) under NAHEP	ICAR –NAARM, Hyderabad	Apr 30, 2021
May	D Thammi Raju	NAHEP Review Meeting of Component 2A	Online	May 19, 2021
	Surya Rathore	Review meeting of NAHEP with PI (NAHEP)	ICAR – IASRI, New Delhi	May 19, 2021
	S K Soam	Review Meeting of NAHEP- Component 2 by PI, NAHEP Component 2	ICAR-NAARM	May 19, 2021
	S K Soam	Scoping Meeting on Building Synergies & Exploring Areas of Partnerships between ICAR-NAARM & GIZ	ICAR-NAARM	May 21, 2021
	Ch Srinivasa Rao	Preparatory Briefing Meeting with the Negotiators for the UNFCCC Jun Subsidiary Body Meetings under the Chairmanship of Ms Richa Sharma, Additional Secretary, MoEFCC, Govt. of India.	Virtual/MoEF&CC	May 28, 2021
	D Thammi Raju	Brainstorming workshop on Framework for strengthening Agri Academia Industry partnership under NAHEP	ICAR-NAARM	May 31, 2021
	S K Soam	Brainstorming Workshop on Framework for strengthening Agri-Academia-Industry partnership under NAHEP	ICAR-NAARM	May 31, 2021
	Yashavanth BS	Brainstorming Workshop on Framework for strengthening Agri Academia Industry partnership under NAHEP	ICAR-NAARM	May 31, 2021
	Surya Rathore	Brainstorming workshop on framework for strengthening Academia-Industry Partnership under NAHEP	ICAR – NAARM	May 31, 2021
	Alok Kumar	Framework for strengthening Agri Academia Industry partnership	ICAR-NAARM	May 31, 2021
	Ch Srinivasa Rao	Programme on Observance of World No-Tobacco Day and to take pledge under the Chairmanship of Secretary, DARE & DG, ICAR	Virtual/ICAR	May 31, 2021
	N Srinivasa Rao	Brainstorming workshop on Framework for strengthening Academia-Industry partnership	ICAR – NAARM	May 31, 2021
	Ch Srinivasa Rao	Inauguration of UNFCCC Meetings - Opening Plenary Meetings of the SBSTA and SBI (UNFCCC) & Joint SBSTA and SBI Plenary for statements (UNFCCC)	Virtual/UN Climate Change Secretariat	May 31, 2021
	Ch Srinivasa Rao	Part 1 of the Koronivia Intersessional Workshop as part of UNFCCC 2021 SB Sessions (Koronivia Joint Work on Agriculture on the provisional agendas of SBSTA and SBI during the 2021 sessions of the subsidiary bodies)	Virtual/UN Climate Change Secretariat	Jun 1-2; 9-11 & 15-16, 2021
June	S Senthil Vinayagam	IRC Meeting	IISR, Indore	Jun 2, 2021
	Ch Srinivasa Rao	Meeting to discuss on SFC/EFC of Agril. Education Division under the Chairmanship of DDG (Education), ICAR	Virtual/ICAR	Jun 3, 2021
	Ch Srinivasa Rao	Workshop on EFC Presentation and Preparation under the Chairmanship of Secretary, DARE and DG, ICAR	Virtual/ICAR	Jun 4, 2021
	Ch Srinivasa Rao	Meeting regarding EFC of Education Division	Virtual/Education Division, ICAR	Jun 4, 2021
	Ch Srinivasa Rao	NAAS Foundation Day Lecture by Shakuntala Haraksingh Thilsted, 2021 World Food Prize Laureate and Global Lead for Nutrition and Public Health, World Fish Centre, Malaysia	Virtual/NAAS	Jun 4, 2021
	Ch Srinivasa Rao	Koronivia Informal Consultations in UNFCCC 2021 SB Sessions	Virtual/UN Climate Change Secretariat	Jun 4, 2021
	Ch Srinivasa Rao	26 th Meeting of Board of Management of PJTSAU	Virtual/PJTSAU	Jun 5, 2021

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June	Ch Srinivasa Rao	SB-Koronivia Joint Work on Agriculture	Virtual/UN Climate Change Secretariat	Jun 7-8; 10-11 & 15, 2021
	Ch Srinivasa Rao	Post-2020 Global Biodiversity Framework - Thematic Stakeholder Consultations Theme: Agriculture, Fisheries and Allied Sectors, organized by MoEFCC	Virtual/MoEF&CC	Jun 7, 2021
	Ch Srinivasa Rao	ICAR SOC Meeting	Virtual/ICAR	Jun 8, 2021
	N Srinivasa Rao	Workshop on Effective health management for enhancing work efficiency of ICAR employees	ICAR – NAARM	Jun 10, 2021
	Bharat S Sontakki	Second Meeting of VIII Research Advisory Committee	ICAR-NAARM (Online)	Jun 11, 2021
	Ch Srinivasa Rao	Interface Meeting on Enhancing the Preparedness for Agricultural Contingencies during Kharif 2021 for Telangana organised by CRIDA	Virtual/CRIDA	Jun 11, 2021
	N Srinivasa Rao	Brainstorming workshop on mainstreaming of Agriculture as a core subject in the school curriculum	ICAR - IASRI	Jun 16, 2021
	N Srinivasa Rao	Brainstorming workshop on Mainstreaming of Agriculture as a subject in the school curriculum	ICAR – IASRI (Virtual)	Jun 16, 2021
	V V Sumanth Kumar	Brainstorming workshop on Mainstreaming of Agriculture as a core subject in the school curriculum	ICAR-IASRI	Jun 16, 2021
	Ch Srinivasa Rao	Meeting under the Chairmanship of Secretary, DARE & DG, ICAR to discuss the policy framework on Continuation of Association in Teaching, Student Guidance and Research Projects for those freshly joined at ICAR Hqrs. in various positions	Virtual/ICAR	Jun 16, 2021
	Ch Srinivasa Rao	Informal meeting by the COP 25 Presidency and the COP 26 incoming Presidency on the Santiago Network on Loss and Damage, organized by UNFCCC	Virtual/UN Climate Change Secretariat	Jun 16, 2021
	Ch Srinivasa Rao	UNFCCC Meetings - Closing plenary meetings of the SBSTA and the SBI Joint SBSTA and SBI Plenary for statements	Virtual/UN Climate Change Secretariat	Jun 17, 2021
	N Srinivasa Rao	Brainstorming meeting on Advanced technologies in agriculture with focus on AI, ML and IOT	TIFAC, New Delhi (Virtual)	Jun 21, 2021
	Ranjit Kumar	Study Management Committee Meeting of the study on Co-Benefits of Largescale Organic Farming on Human Health (BLOOM): A Cluster randomized controlled evaluation of Andhra Pradesh Community managed Natural Farming	University of Edinburg & PHFI	Jun 21, 2021
	Umesh Hudedamani	National dialogue on innovative food for hospitality industry.	IPTM Unit, ICAR	Jun 22, 2021
	Ch Srinivasa Rao	National Dialogue on Innovative Foods for Hospitality Industry, organized by ICAR	Virtual/ICAR	Jun 22, 2021
	Ch Srinivasa Rao	Meeting of all the Conveners of NAAS Regional Chapters	Virtual/NAAS	Jun 23, 2021
	B Ganesh Kumar	XXV th Meeting of ICAR Regional Committee No. IV comprising the States of Bihar, Jharkhand and UP through Video Conferencing	ICAR-Indian Institute of Vegetable Research, Varanasi	Jun 25, 2021
	Ch Srinivasa Rao	Mid-term Meeting to review the action taken on different points of 25 th Regional Committee-IV under the Chairmanship of DDG (Extension), ICAR, New Delhi	Virtual/ICAR	Jun 25, 2021
	Ch Srinivasa Rao	Review Meeting on Azadi ka Amrut Mahotsav Celebrations under the Chairmanship of Secretary, DARE and DG, ICAR	Virtual/ICAR	Jun 25, 2021
	Ch Srinivasa Rao	Brainstorming Session on Regenerative Agriculture for Soil Health, Food and Environmental	Virtual/TAAS	Jun 26, 2021
	D Thammi Raju	Board of Studies Meeting of PGDMA	Online	Jun 28, 2021
	Ch Srinivasa Rao	Chaired the Meeting of Committee to develop a Roadmap for creation of Technology Incubation Centres in all Institutes including KVKs	Virtual/ICAR	Jun 28, 2021
	Ch Srinivasa Rao	Informal meeting on the Santiago Network on Loss and Damage	Virtual/UN Climate Change Secretariat	Jun 28, 2021
	Ch Srinivasa Rao	Webinar on Adaptation Communications: Draft supplementary guidance for voluntary use by Parties in communicating adaptation information through Adaptation Communications.	Virtual/ UN Climate Change Secretariat.	Jun 30, 2021



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July	Ch Srinivasa Rao	Meeting of India EU Task Force on Agritech under the Chairmanship of Secretary (DARE) & DG, ICAR	Virtual /ICAR	Jul 5 2021
	Ch Srinivasa Rao	ICAR SOC Meeting	Virtual/ICAR	Jul 6 2021
	Ch Srinivasa Rao	Asia Pacific Climate Week (APCW) Side Event on KJWA, Transformation and Ambition in Food Systems in Asia and the Pacific.	Virtual/ FAO Regional Office for Asia and Pacific, Bangkok, Thailand	Jul 9 2021
	Ch Srinivasa Rao	Meeting with all the lead authors to finalize the outline and content of the proposed book 'Indian Agriculture after Independence' under the Chairmanship of the Secretary, DARE and DG, ICAR	Virtual/ICAR	Jul 9 2021
	S Ravichandran	Online workshop on High-Performance Computing (HPC) in the Agriculture Domain	IIT, Indore	Jul 14 2021
	Ch Srinivasa Rao	Meeting on Consultation on Adaptation and Loss & Damage under the chairpersonship of Ms. Richa Sharma, Additional Secretary, Ministry of Environment, Forest and Climate Change.	Virtual/MoEFF&CC	Jul 15, 2021
	Ch Srinivasa Rao	ICAR 93 rd Foundation Day Celebration	Virtual/ICAR	Jul 16, 2021
	Ch Srinivasa Rao	Aatma Nirbhar Krishi: Interaction with Padma Awardee Farmers 2021 organised by Secretary (DARE) & DG (ICAR)	Virtual/ICAR	Jul 16, 2021
	Ch Srinivasa Rao	Preparatory virtual meeting under Quad Climate Working Group - Adaptation, Resilience and Preparedness Pillar under the Chairmanship of Ms. Rajasree Ray, Economic Advisory, MoEFCC to discuss and finalise the potential areas of cooperation and collaboration under this pillar.	Virtual/MoEF&CC	Jul 20, 2021
	I Sekar	Webinar on Building Resilience against climate change: Role of technologies, Policies, and Institutions	ICAR, New Delhi	Jul 20, 2021
	Bharat S Sontakki	28 th Meeting of ICAR-IIMR, Hyderabad	ICAR-IIMR, Hyderabad	Jul 23, 2021
	Ch Srinivasa Rao	Meeting of all Task Force Members regarding Compendium on Impact of Agricultural Research on the National Economy	NASC Complex, New Delhi/NAAS	Jul 26, 2021 & Jul 27, 2021
	Ch Srinivasa Rao	Opening Ceremony of 54 th Session of Intergovernmental Panel on Climate Change – Invitation from Global Strategic Communications Council (GSCC) and nomination approved by Secretary (A&FW)	Virtual/IPCC	Jul 26, 2021
	P Venkatesan	Interaction Meeting with Farmers and Demonstration of Farm Implements (organized under the SCSP)	Lachampur village of Adilabad district of Telangana state	Jul 30, 2021
August	Ch Srinivasa Rao	Multilateral Heads of Delegations Consultations on Loss and Damage with Chairs and Coordinators of Negotiating Groups (Part 1 & 2) convened by COP 25 Presidency and the COP 26 incoming Presidency, in collaboration with the Chairs of the Subsidiary Bodies	Virtual/ UN Climate Change Secretariat	Aug 4, 2021
	Ch Srinivasa Rao	ICAR SOC Meeting	Virtual/ICAR	Aug 5, 2021
	Ch Srinivasa Rao	NABARD Regional Advisory Group (RAG) – First meeting of Stakeholders for FY 2021-22 on the theme Telangana: Prototype State for Agri digital revolution – Usage of IT/Digital platform in Agriculture	Virtual/NABARD	Aug 6, 2021
	Ch Srinivasa Rao	Twenty-eighth Annual General Body Meeting of NAAS	Virtual/NAAS	Aug 9, 2021
	Ch Srinivasa Rao	50 th Annual Convocation of Acharya N.G. Ranga Agricultural University as Board Member	Tirupati/ANGRAU	Aug 10, 2021
	Ch Srinivasa Rao	Chief Guest for Book release function (Podarillu – Organic Terrace Gardening) and Awareness Programme on Terrace Gardening, Balcony Gardening, Kitchen Gardening, Growing vegetables in Hydroponics method	Hyderabad/	Aug 14, 2021
	Ch Srinivasa Rao	Meeting to discuss the progress of the different activities especially the National and Sectoral campaigns under Azadi Ka Amrut Mahotsav under the Chairmanship of DG, ICAR.	Virtual/ICAR	Aug 18, 2021
	Ch Srinivasa Rao	ICAR Lecture on Post Covid Reforms by Krishnamurthy V Subramanian, Chief Economic Adviser, Government of India as a part of Azadi ka Amrut Mahotsav Lecture Series	Virtual/ICAR	Aug 19, 2021

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September	P Venkatesan	46 th Foundation Day of ICAR-NAARM	ICAR-NAARM, Hyderabad	Sep 1, 2021
	Bharat S Sontakki	46 th Foundation Day of ICAR-NAARM	ICAR-NAARM, Hyderabad	Sep 1, 2021
	Bharat S Sontakki	ICAR-NAARM Foundation Day Lecture by Dr. Krishna Ella	ICAR-NAARM, Hyderabad (Online)	Sep 1, 2021
	Surya Rathore	NAHEP review meeting under the chairmanship of National Director, NAHEP.	ICAR – NAARM, Hyderabad	Sep 1, 2021
	Surya Rathore	ICAR – NAARM, Hyderabad foundation day celebrations	ICAR – NAARM, Hyderabad	Sep 1, 2021
	Surya Rathore	Board of Studies Meeting	ICAR – NAARM, Hyderabad	Sep 2, 2021
	Surya Rathore	Launch workshop of Empaneled IT firms.	ICAR – NAARM, Hyderabad	Sep 6, 2021
	N Srinivasa Rao	Workshop on Empanelment of IT Firms with NAARM	ICAR - NAARM	Sep 6, 2021
	Alok Kumar	International Webinar on Winning the game of publishing research papers	Research Marketing, APAC, Wiley Publishers	Sep 7, 2021
	Alok Kumar	International Webinar on Reorganizing Extension Services to Meet the Demands of New Agriculture Laws in India : Lessons from Global Experiences	Directorate of Extension Education, Maharana Pratap University of Agriculture &	Sep 10, 2021
	Sivaramane N	Invited as expert in Panel Discussion on procedure for estimation of demand for wood in one day virtual workshop of AICRP#12 - Assessment of Demand and Supply of Timber, Fuel-wood and Fodder in India	ICFRE, Dehradun	Sep 13, 2021
	Ch Srinivasa Rao	Review Meeting with Government of Telangana by Madam Shobha Karandlaje, MOS (A&FW) in Hyderabad.	Telangana State Secretariat /Govt of Telangana	Sep 13, 2021
	Ch Srinivasa Rao	XXVII Meeting of ICAR Regional Committee No VIII under the Chairmanship of DG, ICAR	Virtual/ICAR	Sep 14, 2021
	Ch Srinivasa Rao	Meeting by SMD under the Chairmanship of DDG with Director, JD, Admin Head and In-charges of Finance Wing on quarterly basis to discuss issues that require urgent attention	Virtual	Sep 15, 2021
	D Thammi Raju	National Stakeholder Consultation on - Draft National Human-Wildlife Conflict Mitigation Strategy and Action Plan & Draft Guidelines on key species and issues	ICAR NAARM and GIZ India	Sep 18, 2021
	Ch Srinivasa Rao	Guest of Honor in the Inaugural Programme of Nutri-Cereal Startup Convention 3.0	HICC, Hyderabad/ organized jointly by FAO, India and IIMR	Sep 18, 2021
	Ch Srinivasa Rao	संसदीय राजभाषा समिति की दूसरी उप समिति द्वारा केंद्रीय सरकार के कार्यालयों का निरीक्षण कार्यक्रम/ Inspection of Central Government Offices/Undertakings located at Hyderabad by Second Sub-Committee of Committee of Parliament on Official Language	Hyderabad	Sep 18, 2021
October	S K Soam	India-Sweden Agri-Food Tech Webinar 2021	Embassy of Sweden in India	Sep 29, 2021
	Ch Srinivasa Rao	First Online Roundtable Discussion. (This discussion is in preparation of side event on Sustainable Livestock Farming at COP26	Virtual/ Governments of the Netherlands, New Zealand, Kenya and Colombia and the International Livestock Research Institute (ILRI) and Wageningen University and Research (WUR)	Sep 30, 2021
	S K Soam	Meeting of Consultative Group of Geographical Indications, Government of India for GI application number: 706 Tandur Red Gram	Hyderabad and Government of India	Sep 30, 2021
	Ch Srinivasa Rao	Board of Management Meeting of PJTSAU	Virtual/PJTSAU	Sep 30, 2021
	S Senthil Vinayagam	Second Advisory Committee Meeting on Entrepreneurship Development through Farmer Led Innovations (FLIs)–A Case Study in Plantation Sector under NASF	ICAR-NAARM, Hyderabad	Oct 1, 2021
	I Sekar	Virtual NASF Advisory committee meeting	NASF, ICAR	Oct 1, 2021
	Bharat S Sontakki	Attended and delivered special remarks in the Inaugural function of SANKALP 6.0, the annual Business Festival organized by PGDM-ABM students of ICAR-NAARM (Online mode)	ICAR-NAARM, Hyderabad	Oct 3, 2021
	Alok Kumar	ISEE National Seminar on Transforming Indian Agriculture through pluralistic & innovative extension approaches for self-reliant India	BHU Varanasi	Oct 4, 2021



Month	Name of staff	Name of the Programme	Venue / Organized by	Period
October	Bharat S Sontakki	Chaired Technical Session (online) in the National Seminar of the Indian Society of Extension Education on Transforming Indian Agriculture through Pluralistic & Innovative Extension Approaches for Self Reliant India	BHU, Varanasi	Oct 4-6, 2021
	S K Soam	Meeting of Consultative Group of Geographical Indications, Government of India for GI application number: 715-Admachini Chawal, 716-Banaras Langda Aam, 717-Ramnagar Bhanta, 730- Banaras Pan Leaf	Varanasi and Government of India	Oct 5, 2021
	G R Ramakrishna Murthy	6th National Conference on E-learning and E-learning Technologies	Online	Oct 7, 2021
	Umesh Hudedamani	Review and Sensitization Workshop of ZTMUs/ITMUs/PMES under NAIF Scheme ICAR- ZTMU [Engg] & ZTMU [Education]	IPTM Unit, ICAR	Oct 8, 2021
	Ch Srinivasa Rao	NAAS Sectional Committee meetings for election of Fellows/ selection of Associates and Young Scientists	NAAS	Oct 9, 2021
	S K Soam	Meeting of Institute Management Committee of ICAR-NAARM, Hyderabad	Varanasi and Government of India	Oct 10, 2021
	Bharat S Sontakki	61 st Meeting of Institute Management Committee (Online)	ICAR-NAARM, Hyderabad	Oct 11, 2021
	Ch Srinivasa Rao	Chief Guest and to deliver a talk in the Celebration of World Standards Day based on the theme Shared Vision for a Better World identified by International bodies namely ISO, IEC	Hyderabad	Oct 14, 2021
	Surya Rathore	Webinar on 'Feature Funding Opportunities for NGOs and Academic Institutions	Navjeevan Centre for Development, Mehsana	Oct 15, 2021
	Ch Srinivasa Rao	An Evening with World Food Laureates - Discussion on Global Food and Nutrition Security to meet the SDGs during and after Covid 19 Pandemic on the occasion of World Food Day under the Chairmanship of T. Mohapatra, Secretary (DARE) & DG (ICAR) through virtual mode. Five World Food Laureates had participated in the Programme	Virtual/ICAR	Oct 16, 2021
	D Thammi Raju	Sensitization workshop on ARMS	IASRI, New Delhi	Oct 18, 2021
	MB Dastagiri	Programme of Sensitization Workshop on ARMS and Monthly Reporting in ICAR-Institutions	IASRI, New Delhi	Oct 18, 2021
	Surya Rathore	Sensitization workshop on ARMS & monthly reporting in ICAR institutions	ICAR – IISWC, Dehradun	Oct 18, 2021
	Ch Srinivasa Rao	Invitee for the panel discussion, Enabling Food and Nutrition Security in Drylands, organized as a virtual side event during the 2021 International Borlaug Dialogue (World Food Prize Foundation)	Virtual/ICRISAT	Oct 19, 2021
	Ch Srinivasa Rao	NAAS Brainstorming Session on Secondary Agriculture: Challenges, Opportunities and Way Forward under the Convenorship of S.N Jha, FNAAS	Virtual/NAAS	Oct 21, 2021
	Ch Srinivasa Rao	Meeting to review the progress of Compendium on Impact of Agricultural Research by Chairman of Task Force, NAAS	Virtual/NAAS	Oct 23, 2021
	Ch Srinivasa Rao	120 th Executive Council Meeting of NAAS	Hybrid Mode/NAAS	Oct 23, 2021
	S Ravichandran	International webinar on Fighting the hunger using Smart Technology	ICAR-IIOPR	Oct 26, 2021
	Ch Srinivasa Rao	Chief Guest in the International webinar on Fighting the Hunger using Smart Technology	Virtual/ ICAR- Indian Institute of Oil Palm Research, Pedavegi	Oct 26, 2021
	B Ganesh Kumar	Brainstorming Session on	ICAR-Central Avian Research Institute, Izatnagar	Oct 26, 2021
	Ch Srinivasa Rao	Review meeting on special campaign on Swachhta and pending matters under the Chairmanship of Hon'ble Agriculture Minister Shri Narendra Singh Tomar	Virtual/ICAR	Oct 27, 2021
	Bharat S Sontakki	Virtual Interaction Programme of Secretary, DARE and DG, ICAR with Scientists of ICAR	ICAR, New Delhi	Oct 28, 2021
	P Venkatesan	Virtual Interaction Programme of Secretary, DARE and DG, ICAR with Scientists of ICAR	ICAR, New Delhi	Oct 28, 2021

Month	Name of staff	Name of the Programme	Venue / Organized by	Period
November	Ch Srinivasa Rao	NAAS Meeting (Regional Chapter)	Hybrid/ jointly organized by Indian Institute of Rice Research, Hyderabad and Hyderabad Regional Chapter of NAAS	Nov 1, 2021
	Ch Srinivasa Rao	ICAR SOC Meeting	Virtual/ICAR	Nov 2, 2021
	M Balakrishnan	International Webinar on Fostering Global Collaboration among Agricultural Higher Education Institutions	ICAR-NAARM & ICAR-IASRI (Virtual) under NAHEP Component -2	Nov 8-9, 2021
	V V Sumanth Kumar	NAHEP International Webinar on Fostering Global Collaborations Among Agricultural Higher Educational Institutions.	ICAR-NAARM	Nov 8-9, 2021
	S K Soam	Meeting of Consultative Group of Geographical Indications, Government of India for GI application number: 696 Bihar Makhana	Patna and Government of India	Nov 8, 2021
	S Ravichandran	International Webinar on Fostering Global Collaboration among Agricultural Higher Education Institutions	ICAR-NAARM & ICAR-IASRI (Virtual) under NAHEP Component -2	Nov 08-09, 2021
	Surya Rathore	International webinar on Fostering global collaboration among Agricultural Higher Education	ICAR – NAARM, Hyderabad	Nov 08-09, 2021
	D Thammi Raju	International Webinar on Fostering Global Collaborations Among Agricultural Higher Educational Institutions	ICAR-NAARM	Nov 8-9, 2021
	Ch Srinivasa Rao	Interaction with Students of CAU, Imphal on Climate Change	CAU, Imphal/	Nov 9, 2021
	D Thammi Raju	External Advisory Panel Meeting of NAHEP	ICAR-NAARM	Nov 10, 2021
	Surya Rathore	NAHEP External Advisory panel meeting	ICAR – NAARM, Hyderabad	Nov 10, 2021
	D Thammi Raju	NAHEP External Advisory Meeting	Online	Nov 10, 2021
	B Ganesh Kumar	Training on Quality Improvement Measures Committee (TQIMC) meeting	NIRD&PR, Hyderabad	Nov 10, 2021
	Ch Srinivasa Rao	Information lecture under Azadi Ka Amrit Mahotsav on Talent Search for Manning Agriculture TREE (teaching, research, extension and Education) by CD Mayee, Formerly Chairman, ASRB & Ag Commissioner, Min of Agriculture, Govt of India, New Delhi	Virtual/ICAR	Nov 11, 2021
	Ch Srinivasa Rao	Co-Chaired the Technical Session 1: Concepts and Approaches for Optimizing Food-Energy-Water Security Nexus in Sub Session 2: Climate Energy Nexus in Agriculture under the Theme: Food-Energy-Water-Climate Security Nexus and the Challenges in XV Agricultural Science Congress & ASC Expo - Energy & Agricultural Challenges in 21st Century	BHU, Varanasi/	Nov 14, 2021
	Vijay Avinashilingam	Fifth International Agronomy Congress on Agri-Innovation to Combat Food and Nutrition Challenges	PJTSAU, Hyderabad	Nov 23-27, 2021
	M Ramesh Naik	Exhibition in the Fifth International Agronomy Congress on Agri-Innovation to Combat Food and Nutrition Challenges	PJTSAU, Hyderabad, Telangana	Nov 23-27, 2021
	Ch Srinivasa Rao	Inaugural Programme of Fifth International Agronomy Congress (IAC) Agri-Innovations to Combat Food and Nutrition Challenges	PJTSAU, Hyderabad/ Indian Society of Agronomy, PJTSAU and ICAR-NAARM	Nov 23, 2021
	Ch Srinivasa Rao	Co-Chaired Symposium 1: Climate Resilient Agriculture and Ecosystem Services in Fifth International Agronomy Congress (IAC) Agri-Innovations to Combat Food and Nutrition Challenges	PJTSAU, Hyderabad/ Indian Society of Agronomy, PJTSAU and ICAR-NAARM	Nov 23, 2021
	Ch Srinivasa Rao	Meeting on draft 'IPCC AR6 Working Group - II Report' Hosted by MOEFCC	Virtual/MoEF&CC	Nov 23, 2021
	Ch Srinivasa Rao	Read the Preamble of the Constitution along with the Hon'ble President	Virtual/ICAR	Nov 26, 2021
	Ch Srinivasa Rao	Participated as ICAR SOC Member in Webinar titled Kartavya and Jan Bhagidari under Indian Constitution organized as part of Constitution Day Celebrations	Virtual/ Department of Administrative Reforms & Public Grievances (DARPG)	Nov 26, 2021
	I Sekar	National workshop on Farmers' income and impact assessment of Agricultural Research	NIAP, New Delhi	Nov 29, 2021



Month	Name of staff	Name of the Programme	Venue / Organized by	Period
November	Ch Srinivasa Rao	Second Divisional Meeting of Education Division to review the status/progress of Divisional activities	Virtual/Education division, ICAR	Nov 29, 2021
	S K Soam	Meeting of Consultative Group of Geographical Indications, Government of India for GI application number: 669-Nagri Dubraj, 699-Sharbatu Gegu (wheat)	Gwalior and Government of India	Nov 29, 2021
	Bharat S Sontakki	Rural Business Incubation Centre (RBIC) – Review cum Exposure.	Ag Hub, PJTSAU, Hyderabad	Nov 30, 2021
December	S K Soam	Meeting of Institute Management Committee of ICAR-NIAP, New Delhi	ICAR-NIAP	Dec 4, 2021
	Ch Srinivasa Rao	Meeting of the Fellows of the National Academy of Sciences, India during the 91st Annual Session of NASI and Symposium on 'interface between biological and physical sciences towards Atmanirbhar Bharat'	Virtual/NASI	Dec 5, 2021
	Ch Srinivasa Rao	Annual General Body Meeting of the Academy (NASI)	Virtual/NASI	Dec 5, 2021
	Ch Srinivasa Rao	ICAR SOC Meeting	Virtual/ICAR	Dec 6, 2021
	S K Soam	ITMU meeting of ICAR-Indian Institute Millets Research, Hyderabad	ICAR-IIMR	Dec 8, 2021
	Bharat S Sontakki	Virtual Interaction Programme of Secretary, DARE and DG, ICAR with Young Scientists of ICAR (as Nodal Officer)	ICAR, New Delhi	Dec 8, 2021
	Ch Srinivasa Rao	Brainstorming Session on Road Map to Rehabilitate 26 million ha Degraded Lands in India by 2030? under the Convenership of Drs. Ch. Srinivasa Rao, J.C. Katyal and Anil K. Singh	Virtual/NAAS	Dec 9, 2021
	Ch Srinivasa Rao	XXV Meeting of ICAR Regional Committee No. III comprising the States of Assam, Sikkim, Mizoram, Arunachal Pradesh, Nagaland, Tripura, Meghalaya and Manipur.	Virtual/ ICAR Research Complex for NEH Region, Umiam - Meghalaya	Dec 11, 2021
	B Ganesh Kumar	XXV Meeting of ICAR Regional Committee No. III comprising the States of Assam, Sikkim, Mizoram, Arunachal Pradesh, Nagaland, Tripura, Meghalaya and Manipur through Video Conferencing	ICAR Research Complex for NEH Region, Umiam – Meghalaya	Dec 11, 2021
	S Senthil Vinayagam	TQIMC Meeting	NIRDPR, Hyderabad	Dec 15, 2021
	Ch Srinivasa Rao	Participated in the Conference on Natural Farming (Zero Budget Natural Farming), being organized as a part of Vibrant Gujarat Programme addressed by Hon'ble Prime Minister of India, Shri Narendra Modi (virtual mode)	Virtual/ICAR	Dec 16, 2021
	Bharat S Sontakki	55th Meeting of Institute Management Committee of ICAR-CTRI, Rajahmundry (Online)	ICAR-CTRI, Rajahmundry	Dec 20, 2021
	Bharat S Sontakki	National Conference on Transformation of Agricultural Extension-Strategies for Effective Reformation organized in virtual mode Co-chaired Technical Session 1 (Investigation interventions)	SV College of Agriculture (ANGRAU), Tirupati	Dec 22, 2021
	Ch Srinivasa Rao	Meeting to discuss and finalize Agri Startup Recommendation (NAAS Policy Paper) with President, NAAS	NAAS, New Delhi/NAAS	Dec 22, 2021
	G R Ramakrishna Murthy	107th Meeting of the Academic Council of ANGRAU	ANGRAU, Guntur (AP)	Dec 22, 2021
	Surya Rathore	National Farmers' Day as part of Swacchta Pakhwada	ICAR – NAARM, Hyderabad	Dec 23, 2021

5.3.2 International

Month	Name of Staff	Programme	Date (From – To)	Venue / Organized / Sponsored by
April	Ch Srinivasa Rao	Informal Consultations on Koronivia Joint Work on Agriculture	Apr 13, 2021	UN Climate Change Secretariat/ MoEF&CC, Govt. of India
	Ch Srinivasa Rao	UNFCCC Meetings – SBI Consultations on matters related to the Fourth Review of the Adaptation Fund and First Expert Group Meeting on ACE	Apr 14, 2021	UN Climate Change Secretariat/ MoEF&CC, Govt of India
	Ch Srinivasa Rao	UNFCCC Informal Consultations on Santiago Network on Loss and Damages	Apr 15, 2021	UN Climate Change Secretariat/ MoEF&CC, Govt of India
	MB Dastagiri	First Nobel Prize Summit 2021 on OUR PLANET, OUR FUTURE	Apr 26-28, 2021	Nobel Foundation and organized by the U.S. National Academy of Sciences in partnership with the Potsdam Institute for Climate Impact Research, and the Stockholm Resilience Centre/Beijer Institute.
May	Ranjit Kumar	5-7 th Virtual Expert Workshop of RySS-CIRAD-FAO Foresight Study AgroEco2050	May 24, 2021	RySS- CIRAD-FAO
	Ch Srinivasa Rao	G77 & China Coordination Meetings of UNFCCC	May 28 & 31, 2021	UN Climate Change Secretariat/ MoEF&CC, Govt of India
	Ch Srinivasa Rao	Inaugural Programme of UN Climate Change Conference on Koronivia Joint Work on Agriculture and UNFCCC 2021 Informal SB Sessions	May 31 – Jun 17, 2021	UN Climate Change Secretariat/ MoEF&CC, Govt.of India
June	Ch Srinivasa Rao	G77 & China Coordination Meetings of UNFCCC	Jun 2-14, 2021	UN Climate Change Secretariat/ MoEF&CC, Govt of India
	Ch Srinivasa Rao	Webinar on Adaptation Communications: draft supplementary guidance for voluntary use by Parties in communicating adaptation information through Adaptation Communications,	Jun 30, 2021	UN Climate Change Secretariat/ MoEF&CC, Govt. of India
	Ch Srinivasa Rao	UNFCCC Informal Consultations on Santiago Network on Loss and Damages	Jun 28, 2021	UN Climate Change Secretariat/ MoEF&CC, Govt of India
	Ranjit Kumar	5– 7th Virtual Expert Workshop of RySS-CIRAD-FAO Foresight Study AgroEco2050	Jun 29, 2021	RySS- CIRAD-FAO



Month	Name of Staff	Programme	Date (From – To)	Venue / Organized / Sponsored by
July	Ch Srinivasa Rao	Asia Pacific Climate Week (APCW) Side Event on KJWA, Transformation and Ambition in Food Systems in Asia and the Pacific.	Jul 09, 2021	FAO Regional Office for Asia and Pacific, Bangkok, Thailand
	Ch Srinivasa Rao	Meeting of Quad Climate Working Group on Adaptation, Resilience and Preparedness Working Level Discussions	Jul 21, 2021	United States/ MoEF&CC, Govt.of India
August	Ch Srinivasa Rao	Multilateral Heads of Delegations Consultations on Loss and Damage with Chairs and Coordinators of Negotiating Groups (Part 1 & 2) convened by COP 25 Presidency and the COP 26 incoming Presidency, in collaboration with the Chairs of the Subsidiary Bodies (SBs)	Aug 04, 2021	UN Climate Change Secretariat/ MoEF&CC, Govt.of India
	Ch Srinivasa Rao	Release of first of its three-part 6 th Assessment Report (AR6) by The Intergovernmental Panel on Climate Change (IPCC) - the UN body responsible for climate science, on August 9, 2021.	Aug 09, 2021	IPCC/ MoEF&CC
September	MB Dastagiri	U.N. Foreign Policy	Sep 21, 2021	76th annual United Nations General Assembly (UNGA)
	MB Dastagiri	The United Nation's Food Systems Summit (UNFSS)	Sep 23, 2021	Organized by The Global Panel on Agriculture and Food Systems for Nutrition of 76th United Nations General Assembly
	Ch Srinivasa Rao	First and Second Online Roundtable Discussion. (This discussion was in preparation of side event on Sustainable Livestock Farming at COP26 being organized by the Governments of the Netherlands, New Zealand, Kenya and Colombia and the International Livestock Research Institute (ILRI) and Wageningen University and Research (WUR).	Sep 30, 2021	Dutch Ministry of Agriculture, Nature and Food Quality

Month	Name of Staff	Programme	Date (From – To)	Venue / Organized / Sponsored by
October	Ch Srinivasa Rao	First and Second Online Roundtable Discussion. (This discussion was in preparation of side event on Sustainable Livestock Farming at COP26 being organized by the Governments of the Netherlands, New Zealand, Kenya and Colombia and the International Livestock Research Institute (ILRI) and Wageningen University and Research (WUR).	Oct 28, 2021 & Oct 30, 2021	Dutch Ministry of Agriculture, Nature and Food Quality
	MB Dastagiri	UN-FAO Global Conference on Green Development of Seed Industries - Plenary Sessions	Nov 04-05, 2021	UN-FAO AM Amsterdam, Berlin, Rome, Stockholm, Vienna
November	MB Dastagiri	UNFSS (UN Food System Summit) Transforming food systems and combating climate change: A virtual event post-COP26 and beyond; Special Envoy at COP26; the latest on Summit Dialogues.	Nov 18, 2021	UNFSS (UN Food System Summit)

6.1 Special Events at NAARM

6.1.1 Visionary Innovations in Statistical Theory and Applications (VISTA-2021)

Visionary Innovations in Statistical Theory and Applications – 2021 (VISTA 2021) in conjunction with 23rd Annual Conference of Society of Statistics, Computers and Applications (SSCA) an online event, was organized by the Academy during 24-28, February 2021. The Conference was inaugurated by Dr. (Mrs.) Pankaj Mittal, Secretary General, Association of Indian Universities. The inaugural programme was chaired by Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM and the Guests of Honor included Prof. V.K. Gupta, former National Professor, ICAR and President of the SSCA and Dr. R.C. Agrawal, DDG (Education). Dr. Mittal during her inaugural address, asked for revamping statistical sciences courses to align with the new Educational Policy by incorporating experiential learning and association with industries. She stressed that the teachers should encourage students to find different solutions to the same problems by themselves. Dr. Ramana V Davuluri, Professor of Statistics, Brooks University, New York, Prof. B.L.S. Prakasa Rao, INSA Senior Scientist, Prof. C.R. Rao Advanced Institute of Mathematics, Statistics and Computer Science, Hyderabad delivered keynote address and Prof. M.N. Das Memorial Lecture, respectively. The entire conference was held in virtual mode and this allowed several speakers to join from different parts of the globe. To ease the participations in concurrent sessions, the break-out room feature of online platform was used so that everyone can join in a single meeting and join the concurrent sessions of their interest or move from one session to another without leaving the meeting.

During the conference, there were 50 invited Talks, 7 plenary talks, 9 talks in Prof. Aloke Dey Memorial Session and 5 talks in Financial Statistics and 32 contributed papers by eminent speakers from statistical sciences. The conference covered wide range of topics in inferential statistics, official statistics, financial statistics, Big Data and Computational Statistics, Bioinformatics, applications of statistics in different domains. There were 170 registered delegates and more than 30 special invitees participated in the conference. During the valedictory session on 28 Feb 2021, a special resolution was passed by the society appreciating the efforts made by ICAR-NAARM. While addressing the valedictory session, Dr. Ch. Srinivasa Rao has emphasized the importance of involvement of statisticians in helping biological scientists to solve emerging problems. Dr. Rao also stressed on the importance of training agricultural scientists to handle large volume of data in the fields of climate change research, bioinformatics etc. Dr. Ch. Srinivasa Rao, Director was the Chairman of organizing committee and Dr. Soam was the convener. Dr. A. Dhandapani and Dr. S Ravichandran were the Organizing Secretaries of the conference.



Fig. 6.1: Visionary Innovations in Statistical Theory and Applications

6.1.2 Scoping Meeting on Building Synergies & Exploring Areas of Partnerships between ICAR-NAARM & GIZ

The Academy organized a virtual Scoping-cum-Brainstorming meeting with GIZ-India on May 21, 2021, to deliberate on the potential areas of partnership between ICAR-NAARM and GIZ-India. During the meeting, the initiatives undertaken by GIZ-India specifically in addressing the human wildlife conflict for developing strategy document, policy brief and in mainstreaming through capacity building initiatives was presented by Dr. Neeraj Khara, Team Leader-Human Wildlife Conflict Mitigation Project, GIZ India. Dr. Ch Srinivasa Rao, Director, ICAR-NAARM made a detailed presentation on various achievements of the Academy in research, policy advocacy and capacity building and highlighted the close linkages between the larger mandates of both organizations and the benefits that would flow to the sectors because of the partnership. He suggested that NAARM would partner with GIZ-India for infusing innovations in its capacity building and assessing their impacts. Dr. Neeraj mentioned that GIZ-India would look forward to the Academy in developing methodological framework for assessing crop damages in human-wildlife conflict (HWC) related compensations, policy brief on effective use of crop and livestock-based crop insurances to reduce vulnerabilities of communities to HWC and rapid research surveys in specific areas.



Fig. 6.2: Brainstorming meeting with GIZ-India

6.1.3 Second Meeting of VIII Research Advisory Committee (RAC)

The second meeting of the VIII RAC was held on June 11, 2021 on online mode. The Meeting was chaired by Prof. G. Padmanaban, President NASI, Ahmedabad and Former Director, Indian Institute of Science (IISc), Bengaluru, wherein all the members, including Dr. RC Agrawal, DDG (Edu.) participated. Dr. Ch Srinivasa Rao, Director, ICAR-NAARM introduced all the members of RAC and welcomed them for the meeting. Dr. P Krishnan, Principal Scientist and Member Secretary, VIII RAC of NAARM, presented the action taken report on the recommendations given during the 1st meeting of the Committee held during July 2020. Dr. G Venkateshwarlu, Joint Director, ICAR-NAARM presented the major activities relating to capacity building and academic programs of the Academy. The Director presented a comprehensive overview of the Academy's activities related to research and think tank/policy advocacy including number of projects currently underway, new grants won, number of research papers and policy papers published, copyrights registered, Start-ups supported, MoUs signed, and the networks established during the preceding year. The Committee complimented the Director, and all the Scientists for their significant efforts during the pandemic. The Members expressed satisfaction that the Academy has raised up to the challenge and has justifiably delivered on its new role as think tank of ICAR, as witnessed by the number of policy papers published by the Academy on various topical issues concerning the overall governance of NARES and the farm sector. RAC recommended that NAARM shall continue to build new need-based programs based on the evolving needs and expand its reach to different clientele, using technology and contribute to policy advocacy on agricultural research and higher education and particularly to



such areas which would significantly strengthen the implementation of Gol initiative.



Fig. 6.3: Second Meeting of VIII Research Advisory Committee

6.1.4 25th Institute Research Council Meeting

The 25th Institute Research Council (IRC) meeting was conducted virtually during July 13-15, 2021. During his address, Dr. Ch Srinivasa Rao, Director, ICAR-NAARM appreciated the faculty for their contribution to various capacity building programmes, academic and research activities as per the mandate of the Academy even during the pandemic situation.



Fig. 6.4: 25th Institute Research Council Meeting

During this IRC meeting, a total of 22 in-house, 10 extra-mural and 3 new in-house projects were discussed in seven technical sessions including the plenary, each headed by the Chair and Co-Chair with rapporteur support. Decisions taken or suggestions made during each technical session were presented by the respective Session Co-Chair during the Plenary. The project duration of two extra mural projects were completed and six in-house projects have been extended during the IRC period in order to accomplish the project

objectives. Only one new in-house project was approved during this IRC. Therefore, out of the 31 projects after this IRC, 23 are in-house and 8 are extra-mural funded. Number of research projects per scientist after this IRC has remained at 0.91 which was same as after 24th IRC also.

6.1.5 National Summit on Roadmap for Organized Sheep and Goat Meat Sector in India

A National Summit on “Building organized sheep and goat meat sector in India towards ensuring *Atmanirbhar* Bharat” was jointly organized by ICAR-NAARM, ICAR-NRC on Meat, NAAS-Hyderabad Chapter and Indian Meat Science Association at the Academy in a hybrid mode on September 3, 2021 under the aegis of *Azadi Ka Amrit Mahotsav*. Dr. Bhupendra Nath Tripathi, DDG (Animal Sciences), ICAR graced the occasion as the Chief Guest and Dr. Praveen Malik, Animal Husbandry Commissioner, Govt. of India and Dr. G Venkateshwarlu, Joint Director, ICAR-NAARM were the Guests of Honour. Dr. SB Barbuddhe, Director, ICAR-NRC on Meat set the context for the summit. Dr. Ch Srinivasa Rao, Director, ICAR-NAARM and Convener, NAAS Hyderabad chapter expressed hope that the summit will help to address the core concerns of the sector and complemented the NRC on Meat Team for their initiatives. The Summit was attended by over 125 experts comprising retired and serving ADGs, Directors of ICAR Institutes, University Officers, Fellows of National Science Academies, researchers and academicians from NARES, central and state department officials, representatives from federations, associations, cooperatives and industry. Experts from across India delivered lead talks on various aspects of developing road map for sheep and goat meat sector in India, breeding and reproduction policies to augment production, strategies for linking sheep and goat farmers to market and role of private sector and e-commerce in organizing the small ruminant sector. This was

followed by round-table discussions on selected themes viz., breeding, production & health; animal markets, transport & supply chain; slaughtering & processing and entrepreneurship & policy issues.



Fig. 6.5: National Summit on Roadmap for Organized Sheep and Goat Meat Sector in India

6.1.6 NAHEP Project Review Meetings

Under National Agricultural Higher Education Project Component 2A, 4 review meetings and two External Advisory Panel Meetings were organized.

6.1.6.1 Virtual Review Meeting of Career Development Centres and Faculty Development Centres:

The virtual review meeting of Career Development Centres(CDCs) and Faculty Development Centre(FDC) was held on 30-04-2021 under the chairmanship of Dr. S K Soam, CCPI, NAHEP Component 2. This review meeting was held to discuss about the progress & expenditure of Financial Year 2020-21 and action plan for 2021-21 of five CDCs and one FDC. Dr. Tej Pratap, Vice Chancellor, GBPUAT and Chairman, FDC; the Nodal Officers and Coordinators of five CDCs presented their progress. To strengthen CDCs and FDC, an online monitoring system has been developed by the project team, which was demonstrated as live demo. Dr. Tej Pratap informed that GBPUAT is further expanding the FDC as 'College of Faculty Development'. Dr. N Srinivasa Rao and Dr. D. Thammi Raju, Principal Scientists & Co-PIs, NAHEP Component 2 coordinated this review meeting.

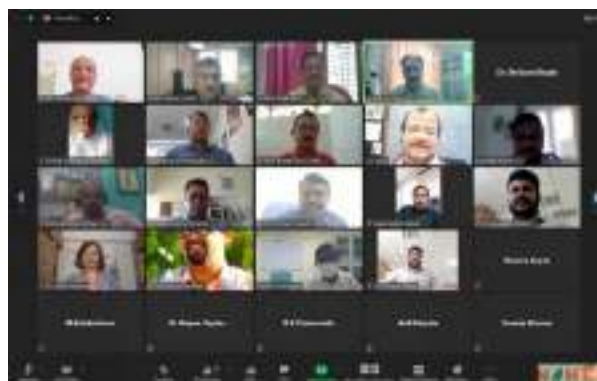


Fig. 6.6 : Dr. Tej Pratap, VC, GBPUAT and other participants during virtual review meeting of Career Development Centres and Faculty Development Centre

6.1.6.2 Project Review Meeting of NAHEP-Component 2

A Review meeting of NAHEP- Component 2 "Investments in Indian Council of Agricultural Research Leadership in Agricultural Higher Education" was held on 19th May 2021 at ICAR-NAARM Hyderabad virtually to discuss the progress and action plan for the year 2021-22. Dr. Sudeep Marwaha, Principal Investigator, NAHEP Component 2, IASRI reviewed the progress of ICAR-NAARM activities. Dr. S K Soam, CCPI, NAHEP Component 2 & Dr. D Thammi Raju, Co-PI, NAHEP Component 2 presented the quarter wise progress during FY 2020-21 and action plan for FY2021-22. Co-PIs from IASRI, New Delhi and ICAR-NAARM, Hyderabad attended this review meeting. In view of Covid-19, the targets and alternate action plan was finalized during the meeting.



Fig. 6.7: Project Review Meeting



6.1.6.3 Project Review Meeting by National Director, NAHEP

1. Project Review Meeting by National Director, NAHEP: The Project Review Meeting of NAHEP, Component 2 with National Director NAHEP, National coordinator, PI & Co-PI of NAHEP Component 2 held on 07 January, 2021 under the chairmanship of Dr. R.C. Agrawal National Director, NAHEP and DDG (Agril. Edu). Dr. D Thammi Raju, Co-PI of NAHEP Component 2 invited the chief Guest of the programme Dr. R.C. Agrawal, National Director, NAHEP and DDG (Agril. Edu). He welcomed all the participants. After that, briefed about the objective-wise progress of activities which was conducted during this financial year under NAHEP Component 2. He explained about quarter wise activities and current quarter/upcoming activities to be conducted up to March 2021. The activities done by CDCs and FDC from all 5 Agricultural Universities and FDC of G. B. Pant University of Agriculture and Technology, Pant Nagar has also explained by him.



Fig. 6.8: Participants of Review Meeting of NAHEP, Component 2

2. The review meeting for the project “Investments in Indian Council of Agricultural Research Leadership in Agricultural Higher Education” under National Agricultural Higher Education Project (NAHEP), Component 2A was

held on 01-09-2021 under the Chairmanship of Dr. R.C. Agrawal, National Director NAHEP and DDG (Edn), ICAR at ICAR-NAARM, Hyderabad. The objective of this meeting was to discuss about progress of the project activities and the action plan for the year 2021-22.

At the outset Dr. D Thammi Raju Principal Scientist and Co PI NAHEP welcomed Dr. R.C. Agrawal, National Director NAHEP and DDG (Edn), Dr. Ch Srinivasa Rao Director, NAARM, Dr. G Venkateshwarlu Joint Director, NAARM, team members and invitees. Later Dr. S. K. Soam, CPI, NAHEP made a presentation on the progress of project activities, upcoming targets and pending issues.

6.1.6.4 External Advisory Panel of NAHEP

Under the NAHEP two meetings of the External Advisory Panel were organized during the reporting period.

1. The NAHEP External Advisory Panel (EAP) Meeting of the World Bank and ICAR sponsored project “Investments in Indian Council of Agricultural Research Leadership in Agricultural Higher Education (Under NAHEP Component 2)” was held in the forenoon of 1st February 2021 at ICAR-NAARM Hyderabad to discuss the progress and upcoming events of the project. Dr. J C Katyal, Member, NAHEP External Advisory Panel & Former DDG (Agril. Education) and Dr. B Venkateswarlu, Member, NAHEP External Advisory Panel & Former VC, VNMKV, Parbhari were present during this meeting. This event was coordinated by Dr. D Thammi Raju, Co-PI of NAHEP, Component 2, ICAR-NAARM. He welcomes the members & special invitees of the panel and briefed them about the major recommendation of the last external advisory panel meeting held on 13 October 2020. Dr. S K Soam, CC-PI, NAHEP Component 2, ICAR-

NAARM briefed about the activities of NAHEP Component 2 completed till date.



Fig. 6.9 : Dr. J C Katyal and Dr. B. Venkateswarlu, members of EAP interacting with project team

2. The Academy organized 3rd Meeting of External Advisory Panel constituted under NAHEP to appraise the progress and formulate the way forward for the achievement of the overall objective of NAHEP. The panel comprised renowned education and agricultural experts – six from India, three from the USA and one from Australia. In his remarks Dr. R.C. Agrawal, National Director, NAHEP and DDG (Agril. Edu.) requested the External Advisory Panel to guide ICAR pertaining to issues viz. how to bring transformative changes in agricultural education in tune with National Education Policy, interaction of global universities for globalization of agricultural education, collaboration with national, international and government departments, private sector, and sustainability of NAHEP efforts. During the meeting the External Advisory Panel members viz. Dr. Brajesh Singh, Dr. Glen C Shinn, Dr. K V Raman, Dr. P L Gautam, Dr. J C Katyal, Dr. P K Joshi, Dr. B Venkateswarlu, Dr. Tanuja Nesari participated and gave valuable suggestions and guidance in the improving the agricultural education in the country. The special invitees viz. Dr. Rajender Parshad, Director, ICAR-IASRI and Dr. G Venkateshwarlu, JD, ICAR-NAARM suggested various measures. The panel

provided high-quality inputs and suggestions to take the NAHEP further for excellence.



Fig. 6.10: 3rd Meeting of External Advisory Panel of NAHEP

6.1.7 Stakeholder Consultation on National Action Plan for Human-Wildlife Conflict

A National Stakeholder Consultation was organized by the Academy jointly with GIZ-India; Wildlife Institute of India (WII), Dehradun; and Ministry of Environment, Forest and Climate Change (MoEFCC), New Delhi in hybrid mode at Bengaluru on September 18, 2021 to elicit the views of various stakeholders / experts on the National Human Wildlife Conflict Mitigation Strategy and Action Plan (HWC-NAP) and Guidelines for Key Species. The draft documents were developed by GIZ and MoEFCC under the Indo-German Technical Cooperation Project on 'Human-Wildlife Conflict Mitigation in India'. In his opening remarks, Dr. P Krishnan contextualized the significance of this theme to the agricultural sector and highlighted that the agricultural researchers can contribute in strengthening the NAP on HWC in terms of developing and popularizing crop varieties for different conflict situations and developing methodological framework for assessing the crop/livestock damage due to wildlife. Dr. Neeraj Khera, Team Leader, Indo-German Project on



Fig. 6.11: Stakeholder Consultation on National Action Plan for Human-Wildlife Conflict

HWC Mitigation, GIZ provided an overview of the reports, and indicated the expected outcomes of the consultation. The meeting was attended by about 75 members comprising subject matter experts, agricultural researchers from different ICAR institutes and agricultural universities, senior officials from the forest department of different states, field officers and members from the NGOs. Shri Vijayakumar Gogi, IFS, Principal Chief Conservator of Forest (Wildlife) & Chief Wildlife Warden (CWW), Karnataka Forest Department was the Guest of Honour and Sh. Soumitra Dasgupta, Additional Director General (Wildlife), MoEFCC, GoI chaired the consultation and moderated the deliberations.

6.1.8 International Webinar on “Fostering Global Collaboration among Agricultural Higher Educational Institutions”

An International Webinar on “Fostering Global Collaboration among Agricultural Higher Educational Institutions” was organized during

November 8-9, 2021 in collaboration with the ICAR-Indian Agricultural Statistics Research Institute, New Delhi under the National Agricultural Higher Education Project. In his address, Dr. R.C. Agrawal, Deputy Director General (Agricultural Education), ICAR & National Director, NAHEP emphasized on the internationalization of agricultural education, aligning to the National Educational Policy, Human Resource Development and multi-global practices, etc. Shri Bekzod Shamsiev, Task Team Leader, World Bank suggested for utilizing the World Bank’s Network across different countries to foster the global collaborations. Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM and Dr. Rajender Parsad, Director, ICAR-IASRI, New Delhi were also present during the occasion. The Webinar registered participation from more than 200 participants from the State Agricultural Universities, National Coordinators, NAHEP Component - 2 team members and other ICAR Institutes.



Fig. 6.12: Webinar on “Fostering Global Collaboration among Agricultural Higher Educational Institutions

6.1.9 Webinar on Aroma Biology and Livelihood Security

The Academy organized a Webinar on “Aroma Biology and Livelihood Security” on November 01, 2021 in association with ICAR-Indian Institute of Rice Research (IIRR), Hyderabad at IIRR-Hyderabad in hybrid mode, which was also as part of Azadi ka Amrut Mahotsav. The speaker, Dr. Ajit K Shasany, Director, ICAR-National Institute of Plant Biotechnology (ICAR-NIPB), New Delhi, presented a panoramic view of aroma biology and the research facets regarding secondary metabolites in plants. He informed about the huge market and export value of aroma. Indian perfume industry has high prospects as India is the biggest producer and consumer of aroma. He gave a brief account of research work carried out at CIMAP. He also emphasized the need for the extension of new aromatic/medicinal varieties, package of practices, demonstrations, value addition to the farmers across the country. He enlightened about livelihood security and the profits made by farmers using such varieties and their byproducts. A holistic approach for increasing farmers’ income through collaborative partners across India with the involvement of farmers and young agri-innovators was presented. Dr. P. Ananda Kumar, Emeritus Scientist, ICAR-IIRR and NAAS Fellow presented his views on the possible prospects

of using promoters for imparting resistance against various pests and diseases. Earlier, Dr. R.M. Sundaram, Director, ICAR-IIRR welcomed the speaker, all the dignitaries, scientists, NAAS fellows and associates. In his Opening remarks, Dr. Ch. Srinivasa Rao, Director, ICAR–NAARM and Convener, NAAS Hyderabad Chapter emphasized that aroma is a critical component of agriculture that plays significant role in food security, nutrition, and agriculture export. Dr. P. Krishnan, ICAR-NAARM presented a formal vote of thanks. Dr. Ch. Srinivasa Rao and Dr. R M Sundaram also thanked Dr. Ajit Kumar Shasany and all the scientists for the fruitful discussions. The webinar was greatly appreciated and attended by scientists, NAAS fellows, and NAAS associates.



Fig. 6.13: Webinar on Aroma Biology and Livelihood Security

6.1.10 Brainstorming Session on Agri-startups in India: Opportunities, Challenges and Way Forward

On behalf of the National Academy of Agricultural Sciences (NAAS), the Academy conducted a one-day brainstorming session on ‘Agri-startups in India: Opportunities, Challenges, and Way Forward’ on November 5, 2021 under the Chairmanship of Dr. T. Mohapatra, President, NAAS. The session was co-chaired by Prof. V. Ramgopal Rao, Director, IIT Delhi. Dr. P.K. Joshi, Secretary, NAAS welcomed the participants and briefed regarding



the objectives of the session. In the beginning, the convenors made base presentation giving glimpse of the existing agri-startup ecosystem in India and scope of deliberations. Dr. Ramgopal Rao shared the initiatives taken at IIT Delhi to encourage and support the entrepreneurial ideas. He also gave example of FIRE (Faculty Innovation and Research-driven Entrepreneurship) scheme under which faculty are given funding support to have their startups. The Chairman emphasized upon the urgent need for NARES to bring necessary changes in the working environment to encourage the agri-students to take up startup and entrepreneurship as career choice. While chairing the Viewpoint session, Dr. RC Aggarwal, DDG (Edn), ICAR conveyed that about 100,000 students graduate (UG to PhD) every year in agricultural sciences from 74 agricultural universities as well as several other private universities and affiliated colleges. Efforts have been made to mainstream the entrepreneurship in the course curriculum through Experiential Learning Program (ELP) under Student READY course in the agricultural universities. Later, three experts Mr. Hemendra Mathur, Dr. Manish Diwan and Sh. Ram Kaundinya presented their views on how to scale up the agri-startups and the existing opportunities in the country.



Fig. 6.14: Brainstorming Session on Agri-startups in India: Opportunities, Challenges, and Way Forward

It was followed by a panel discussion with five agri-startups' co-founders, namely Mr. Tauseef Khan (Gramophone), Mr. Nikhil (DeHaat), Mr. Babbar Singh (Delmos Research), Mr. Lovlesh Nagori

(Sampkal Business) and Dr. Renuka Karandikar (Bioprime Agrisolutions). A total of 42 participants attended the online workshop.

6.1.11 Online Monitoring System for Career Development Centres and Faculty Development Centers

ICAR-NAARM, under National Agricultural Higher Education Project (NAHEP) Component 2, established five Career Development Centers (CDCs) & one Faculty Development Centre (FDC). All the CDCs are situated at different geographical locations. This has helped in simplifying the information collection and analysis process. This system was developed to monitor CDCs and FDC activities. This system can ensure that all the CDCs are working on same standards. This system has ensured public availability of CDC activities; standardization of activities; Effective monitoring; Structure data organization; Collecting timely and untampered feedback; Standardization in reporting. Looking to wider online application and real time data visualization capabilities, the application for copyright registration has been in principle approved by ITMC of the Academy. Another two Career Development Centres(CDCs) are established at Kerala Veterinary and Animal Sciences University (KVASU), Kerala and Anand Agricultural University, Anand, Gujarat. Experiencing the advantages as an innovative initiative in earlier five Agricultural Universities(AUs), few more SAUs are coming up to create CDCs in their campuses. NAARM is providing technical support in establishment of Career Development Centres at these two universities, and operational and monitoring of activities through NAARM copyrighted software. Dr.T.S.Rajeev, Associate Professor, KVASU and Dr. Dinesh H Patel, Director of Student's Welfare, AAU are the Nodal officers of CDCs at KVASU and AAU respectively.

URL:

Faculty Development Centres Monitoring System:
<https://nahep.naarm.org.in/fdc>

Career Development Centres Monitoring System:
<https://nahep.naarm.org.in/cdc>



Fig. 6.15: Screenshot of the online monitoring system for CDCs and FDC

6.1.12 111th Foundation Course for ARS Scientists concluded

The 37 ARS probationers belonging to 15 agricultural disciplines, who were undergoing their Foundation Course at ICAR-NAARM since

05 Oct. 2020, completed their training program and received their completion certificates during the Valedictory Program held on 02 Jan 2021. Dr. R.S. Paroda, Former Secretary, DARE and DG, ICAR was the Chief Guest of the event, which was attended by many eminent researchers and research managers from across NARES through the online platform. Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM gave an overview of the foundation program and highlighted the steps taken to re-orient the program under the new normal. Dr. R.S. Paroda, in his address, particularly recognized that the young scientists have traversed across the country for acquiring higher education in agriculture and have now received postings far away from their home state. This, he hoped that would help them understand the diversity of research needs across the country and contribute to the clientele based on the skills they acquired in research and management. He called upon them to align their research with that of the



Fig. 6.16: 111th Foundation Course for ARS Scientists

vision set by the Indian Council of Agricultural Research. Dr. Paroda congratulated all the winners of various events conducted during the program and complimented the Team NAARM for successfully completing the course during the pandemic. During the event, various publications of the Academy were released by the Chief Guest. Earlier, Dr. S.K. Soam, Joint Director, ICAR-NAARM, introduced the Chief Guest and welcomed all the participants who attended the event physically and through online platform.

6.2 Visits and Interactions of Dignitaries at NAARM

6.2.1 Dr. RC Agrawal, National Director, NAHEP and DDG (Education), visits ICAR - Disaster Recovery Centre

Dr. RC Agrawal, National Director, National Agricultural Higher Education Project and Deputy Director General (Agricultural Education) visited ICAR-Disaster Recovery Centre (DRC) at the Academy on September 1, 2021. With the total outlay of about Rs.

9.00 Crore, the DRC has been set up by IASRI and NAARM under component 2 of World Bank supported National Agricultural Higher Education Project (NAHEP). Dr. SK Soam, Head, Information and Communication Management and Collaborating Centre Principal Investigator (CCPI) explained about the current status of applications of DRC, and maintenance by Hitachi. Dr. N Srinivasa Rao, Co-PI, NAHEP Component 2 & Principal Scientist explained the technical specifications. Dr. Agrawal discussed about the technical capabilities of DRC and suggested the incorporation of all ICAR websites at DRC and also to have its own NKN line. Dr. Ch Srinivasa Rao, Director, ICAR-National Academy of Agricultural Research Management discussed about improving the CCTV network, civil and other aesthetic works. During the visit of DRC, Dr. Agrawal put his comments on 'Visitors Book'. Dr. VV Sumanth Kumar, Co-PI and Senior Scientist; Dr. D. Thammiraju, Co-PI and Principal Scientist; Dr. G. Venkateshwarlu, Joint Director, NAARM were also present.



Fig. 6.17: Dr. RC Agrawal, National Director, NAHEP and DDG (Education) visits ICAR - Disaster Recovery Centre at NAARM

6.2.2 संसदीय राजभाषा समिति की दूसरी उप समिति ने भाकृअनुप-नार्म (ICAR-NAARM) की समीक्षा की

संसदीय राजभाषा समिति की दूसरी उप समिति के माननीय सदस्यगणों ने 29 सितंबर 2021 को हैदराबाद में स्थित केंद्र सरकार के अन्य कार्यालयों के साथ-साथ भाकृअनुप-राष्ट्रीय कृषि अनुसंधान प्रबंध अकादमी (ICAR-NAARM), हैदराबाद की राजभाषा कार्यान्वयन की समीक्षा की गयी और कार्यालय में राजभाषा का अनुपालन एवम उन्नति पर संतोष व्यक्त किया। इस संसदीय राजभाषा समिति की दूसरी उप समिति की अध्यक्षता माननीय सांसद प्रो (श्रीमती) रीता बहुगुणा जोशी ने की जिसमें उप समिति के माननीय सदस्य माननीय सांसद श्री सुशील कुमार गुप्ता, श्री प्रदीप टम्टा और श्रीमती रंजनबेन भट्ट भी शामिल थे।

समीक्षा के दौरान यह सुझाव दिया गया कि विदेशी छात्रों को अकादमी में प्रशिक्षण प्रदान करने के लिए अकादमी को आवश्यक प्रयास करने चाहिए। विभिन्न भारतीय भाषाओं के साथ-साथ हिंदी सीखने के लिए भाषा अध्ययन प्रयोगशाला के प्रयास की सराहना की गई। समिति ने राजभाषा की त्रैमासिक बैठकें आयोजित करने और नगर राजभाषा कार्यान्वयन समिति की बैठक में भाग लेने के निदेशक के प्रयासों पर प्रसन्नता व्यक्त की गयी। प्रो. रीता बहुगुणा जोशी ने अकादमी को प्रशंसा पत्र प्रस्तुत करते हुए अकादमी में राजभाषा के कार्यान्वयन पर संतोष व्यक्त किया। अकादमी से डॉ. सीएच. श्रीनिवास राव, निदेशक, डॉ. जी. वेंकटेश्वरलू, संयुक्त निदेशक, श्री. बी डी फंसल, संयुक्त निदेशक (प्रशासन), श्री. जेड एच खिलजी, मुख्य वित्त एवं लेखा अधिकारी, श्री. प्रदीप सिंह, सहायक निदेशक (राजभाषा) और भाकृअनुप-डेयर से डॉ. सीमा जग्गी, सहायक महानिदेशक (एचआरएम), श्री ए.के. जगदीशन, उप निदेशक (आरबीए) और श्री ओमप्रकाश जोशी, सहायक मुख्य तकनीकी अधिकारी ने अकादमी की समीक्षा बैठक में भाग लिया।



चित्र 6.18: संसदीय राजभाषा समिति की दूसरी उप समिति ने भाकृअनुप-नार्म (ICAR-NAARM) की समीक्षा की

6.3 National Campaigns and Celebrations

6.3.1 Mahila Kisan Diwas

A one-day webinar on Understanding the Journey of Women Agripreneurs was organized on 15 October 2021 on *Mahila Kisan Diwas*. On this occasion, a poster competition was also organized on the themes; role of women in agriculture, women empowerment, nutrition, and income generation followed by an e-exhibition of the submitted posters. Dr. Ch Srinivasa Rao, Director, ICAR-NAARM highlighted the critical role of women in Indian agriculture, agri-business management, agri education and research. Ms. Maya Vivek CEO & Co-founder, Oorvi Sustainable Concepts, Pvt. Ltd. Hyderabad discussed about her journey as a woman agripreneur and shared the efforts she made towards flower-based entrepreneurship development and environmental conservation by means of holy waste (Flower Waste) recycling in her company activities. Her deliberations would be meant to open doors for innovation among students as well as our farmers and farm women to take up agri innovations for better agriculture development in coming times. Around 130 participants including farm women, students and staff and faculty members attended the programme.



Fig. 6.19: Webinar on Understanding the Journey of Women Agripreneurs @ Mahila Kisan Diwas

6.3.2 World Water Day

The Academy celebrated World Water Day on 22nd March 2021 with the theme “Valuing Water”. On this occasion, Academy organized various events including farmers visit to technology demonstration, essay competition for staff and drawing competition for students emphasizing the value of water. During his address to the farmers, staff and students, Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM emphasized the value of water in rain fed agriculture and stressed the need for policy initiatives for crop planning based on available water resource and economic maximization for the farmers. He also explained the various water conservation and harvesting measures implemented at the academy like roof water harvesting, Integrated Farming System, Micro irrigation systems etc. The Chief Guest of the function, Dr. D. Raji Reddy, retired Director of Extension, PJTSAU, Hyderabad elaborated various methods of water conservation and need for crop diversification towards oilseed or pulse crop in place of paddy cultivation. The winners of the various events organized on this occasion were felicitated by the Chief Guest. Dr. P. Ramesh, Dr. Ramesh Naik, Dr. Venkatesan, Dr. Bharat Sontakki coordinated with programme along with other staff of the academy.



Fig. 6.20: World Water Day Celebrations

6.3.3 Farmers Awareness Campaign on “Balanced Use of Fertilizers”

The Academy organized a Farmers Awareness Campaign on “Balanced Use of Fertilizers” in virtual mode on 18th June 2021, as a part of Celebration of 75 Years of Indian Independence. The campaign was attended by about 85 members including 60 farmers from 6 different villages and 4 districts of Telangana state. Also 25 members of PGDMA-ABM students, faculty, officers and staff of the Academy has participated. Shri Gone Shyamsunder Rao, Progressive Farmer and Member, RAC and IMC of NAARM was the Chief Guest on this occasion. Shri Shyamsunder lucidly explained the importance of Balanced Fertilizer Use to the audience citing his own personal experiences as a practicing farmer. Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM, in his address, highlighted the critical importance of soil as a living resource for the sustenance of farming



Fig. 6.21: Farmers Awareness Campaign on “Balanced Use of Fertilizers”

and why balanced use of fertilizers by farmers is essential. He urged the participant farmers to share the messages with other farmers in their village and other villages so as to rapidly spread knowledge on Balanced Use of Fertilizers. Dr. G. Venkateswarlu, Joint Director, ICAR-NAARM, in his remarks, spoke on the remarkable progress made in agriculture since Independence and also the need to preserve soil for posterity with proper and balanced fertilizer use.

6.3.4 Kisan Diwas

The Academy celebrated the Kisan Diwas (Farmers' Day) by organizing an interactive meeting between Innovative Farmers with an emphasis on "Farmers Perspective on Agri-Preneurship" and the young PGDM-Agri Business Management (PGDM-ABM) students of the Academy. Dr. G. Venkateswarlu, Joint Director, ICAR-NAARM hailed the selfless contributions of farmers of the country in achieving record agricultural production amidst the serious challenges imposed by COVID-19 pandemic and other natural calamities. He specifically called upon the students of the Academy to rededicate their services for the welfare of farming communities. Three innovative farmers' namely Shri Gone Shyamsunder Rao, Shri

Vallabhaneni Asha Kiran and Shri. Vasanthapu Venkateswarlu were felicitated on this occasion. These farmers spoke on the need for improving natural resource management, market linkages, better pricing and other issues that need attention of researchers and policy makers. Critical role of Agriculture Business Management models for the welfare of the farming community of the country was discussed by the Innovative Farmers with ABM students.

6.3.5 National Science Day

As per the directives received from Department of Science and Technology (DST), National Science Day- 2021 was organized at the Academy on Feb. 28, 2021. The theme to mark the occasion was "Science, Technology and Innovation (STI): Impacts on Education, Work and Skills". Accordingly, essay writing and poster presentation competitions were conducted to the students of Post Graduate Diploma in Agricultural Management (PGDMA) on Feb. 26, 2021 for the first year students and on Feb. 27, 2021 for the second year students. Many of Students participated in these competitions with much enthusiasm and fervor. Theme is well addressed through their written word and also



Fig. 6.22: Participants of Kisan Diwas celebrations



through pictorial representations in posters. Dr. Kareemullah, Dr. Ganesh Kumar, Dr. Surya Rathore, and Dr. Ramesh Naik acted as judges to these competitions. The closing ceremony of National Science Day was organized on Feb. 28, 2021. Dr. Ch. Srinivasa Rao, Director, NAARM gave away the cash prizes and certificates to the first, second and third prize winners of these competitions.

6.3.6 Agriculture Education Day

The Academy organized Agricultural education day on 3rd December, 2021 to commemorate the birth anniversary of Bharat Ratna Dr. Rajendra Prasad, the first president of Independent India and first Union minister of Agriculture. A total of 105 Students from College of Agriculture, University of Agricultural Sciences, Bengaluru actively participated in the celebrations. Marking the occasion, elocution and quiz competitions were organized by the Academy for the students. Dr. Srinivas Rao, Director, ICAR-NAARM, the chief guest on the occasion, addressed the students and emphasized on the importance of the agriculture, entrepreneurship and overall development of the sector. He encouraged students to contribute actively through their entrepreneurship by creating employment to others instead of job seeking. Dr. G. Venkateswarlu, Joint Director, ICAR-NAARM explained about the importance of agricultural education in the country, and role of

the Academy in higher agricultural research and education. He advised the students to serve the farming community in future.

6.3.7 Foundation Day Celebrations

The Academy celebrated 46th Foundation Day on September 1, 2021 in hybrid mode (Zoom Platform and live streaming through YouTube and Facebook). The event was conducted with limited number of invitees by following all the COVID-19 precautions. Dr. Krishna Ella, Chairman and Managing Director, M/s. Bharat Biotech International who has been instrumental in the development of indigenous corona vaccine-Covaxin delivered a lecture entitled 'Innovations for Transformation' commemorating the 46th Foundation Day of the Academy and Azadi Ka Amrit Mahotsav. Dr. T. Mohapatra, Secretary DARE and DG, ICAR highlighted the innovations of ICAR in the areas of animal vaccines and complemented the speaker for being a role model for the youth in India, especially agriculture graduates and others associated with ICAR by his innovations par excellence in global market. In the forenoon session of Foundation Day celebrations, the Chief Guest Dr. RC Agrawal, DDG (Agriculture Education) released the publications on 'Policy brief on Circular Carbon Economy in India' and 'Key Outcome of Farmer FIRST Program 2016-2020 - Transforming the Lives and Livelihood



Fig. 6.23: Agriculture Education Day

of Indian Farmers' and presented the awards for different categories of staff in scientific, technical, administrative, supporting staff and young professional categories. The innovative farmers across India having close linkage with NAARM, best performing Start-ups and Farmer Producer Organizations (FPOs) that got trained from NAARM and Print and electronic media who have active engagement with NAARM have been felicitated on the occasion. Dr. RC Aggarwal appreciated the activities, achievements of the Academy and congratulated the award winners and called upon the others to get motivated from the award winners. He inaugurated new facilities such as Agro-biodiversity Park; Herbal Garden, Common Reception Centre and Maintenance Office (Civil) and appreciated the

ambience of the academy in the midst of green cover and beautiful landscape. Dr. Ch Srinivasa Rao, Director, ICAR-NAARM welcomed all the guests and highlighted the salient achievements of the Academy in the areas of capacity building, research and think tank and acknowledged the roles of QRT, RAC and IMC members for their guidance and motivation. Guest of Honour, Dr. Seema Jaggi, ADG (HRD) virtually participated and commended the activities of the Institute in her address. Dr. G Venkateswarlu, Joint Director, ICAR-NAARM proposed the vote of thanks. The program was attended by former Directors of NAARM, Directors of various ICAR Institutes, Vice Chancellors from SAUs and other staff of the SAUs and ICAR.



Fig. 6.24: Foundation Day Celebrations

6.3.8 Sankalp 6.0 – Annual National Level B-fest

Sankalp 6.0, a month-long National Level Business Fest was organized by the Post Graduate Diploma in Management-Agribusiness Management (PGDM-ABM) students of the Academy. The finale of the event was organized on October 03-04, 2021. The 6th edition of the event organized in online mode received 6400+ registrations from about 75 premiere institutes across India, including IIMs, several engineering colleges, business schools and agricultural universities. While welcoming all the guests and participants, Dr. Ranjit Kumar, Chairman of Organizing Committee informed the audience about the seven competitive events: Adhishtatha- The Best Manager Competition, Samadhaan- The Case Study Competition, Advitiya- The B-Plan Competition, Darpan- The Business Quiz Competition, Flashgun- The Photography Competition, Fintra- The Trading Event and Strategia- The Business Strategy Competition. Around 200 teams were shortlisted for the finale. The inaugural session was graced by Dr. Binaya Kumar Parida, AVP and Chief Agronomist from Coromandel International Ltd. and Mr. Anil Pannu, Business Manager, India Operation, Verdesian Lifesciences USA. Dr. N. Sivaramane, CEO, a-IDEA highlighted the role of incubator centre in nurturing agri-startups. Dr. Pradeep Kumar, Senior VP- HR from

Coromandel International Ltd in his address expressed that the event has acted as a bridge between the institute and the company in identifying the talent pool present in the field and appreciated the efforts of the NAARM students. Mr. RK Goyal, Business Director (SA, SEA & AUZ) was also the Guest of Honor for the valedictory session. He congratulated the efforts in organizing such mega-event and told how agri-input industry has evolved and technology has transformed the farming methods in India. During the valedictory session, results of winners and runner ups across different events were announced and total cash prizes of ₹1.0 lakh were distributed. Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM in his presidential address appreciated the efforts of the students and emphasized on how the strategy plays an important role in deploying the evolving technology in growing agribusiness. He also urged the stakeholders to consider sustainability issues as well as farmer's income in the business decisions. He congratulated the winners and encouraged the young minds to think innovatively.

The fest was sponsored by Coromandel International Ltd and Verdesian Life Sciences & Cytozyme Labs USA as the Title Sponsors; a-IDEA as the Diamond Sponsor; Rallis India Ltd as the Gold Sponsor; Heranba Industries Ltd as the co-sponsor, and NAARM PG Alumni Association as the Knowledge Partner.



Fig. 6.25: Sankalp 6.0 – Annual National Level B-fest

6.3.9 Vigilance Awareness Week

The Academy celebrated Vigilance Awareness Week during October 26-November 01, 2021 on the theme “Independent India @ 75: Self Reliance with Integrity”. The activities such as Display of Banners, e-posters display in meeting hall and LED display in hostels and other places, Integrity Pledge by all Employees, Essay Competition among faculty/officers/staff members, Session on Vigilance Management by Shri. Sandeep Mukherjee, Former Deputy Director (Vigilance), ISTM, DoPT and Session Independent India @ 75: Self-Reliance with Integrity by Dr. G. Vinod Kumar, Former Head, Dept. of law, Osmania University were carried

out as per the CVC guidelines besides Elocution Competition among PGDMA students. The prizes were given to competition winners of PGDMA students namely, Mr. B Pranay Kumar, Mr. Akhil C & Ms. Krishnashree and academy staff namely, Smt. N. Vijayalakshmi, Smt. V. Shailaja & Shri C. Julius Samuel.



Fig. 6.26: Vigilance Awareness Week

6.3.10 Swachhta Campaign

The Academy celebrated Swachhta Campaign for disposal of various references and other matters during 2-31 October. During the campaign, the Academy undertook several activities within the campus and outside adapted villages (under SC-Sub Plan) namely Agraharam Potlapally, Rajapur Mandal of Mahbubnagar District on 12th October, 2021 by involving the Officers of NAARM, 150

farmers (Men, Women & Youth) and other stakeholders.

The main focus of the campaign was on Waste to Wealth in which Kitchen Waste to Compost, Farm Waste to Compost, Use of EKCEL De-Composeure Capsule and *In-Situ* Decomposition Technology for Rice and Wheat Residue which was displayed at village in the form of display banners.



Fig. 6.27: Swachhta Campaign

6.3.11 Swachta Pakhwada

The Swachhta Pakhwada was observed at the Academy during 16-31 December 2021. Swachhta Pakhwada was kick-started on 16th December 2021 with the displaying of banners at both the main gates and near the bus shelter of the Academy. Swachhta Pledge was also taken by 20 invited farmers from Agraharam potlapally village, and staff of the Academy near the photo stand area. The farmers visited Agri-Bio Diversity Park and they were explained about making of

vermicompost from litter and dry waste. Regular cleaning, sweeping of lawns and main roads, moping of the floor of all office and guesthouse buildings, removal of waste like stones boulders, scrap, cutting of weeds in the campus and cleaning the surroundings of guest house buildings was observed during this pakhwada. Regular fogging for the control of mosquitos and as a part of waste out of wealth, shifting of leaf litter & grass clippings to vermicomposting unit for making vermicompost was done during this period.



Taking swachhta pledge(oath) by the farmers and staff under the guidance of respected Director



Cleaning of a-IDEA block



Moping of floor at Auditorium



Collection of grass clippings at IGH 1 lawn



Fogging operation near the main security room

Fig. 6.28: Swachta Pakhwada

6.3.12 Tree plantation programmes in 2021

A tree plantation programme was organised at the Academy on 16th July 2021 as a part of National tree plantation programme on the occasion of 93rd ICAR Foundation Day. Taking into consideration of rainy weather prevailing in the Academy, the area between Type IV & V qtrs. was assigned for tree planting. In view of the Pandemic situation limited members were invited to participate in



Fig. 6.29: Director, Joint Director, faculty & staff during tree plantation on the occasion of 93rd ICAR Foundation Day

the programme. Chief guest Dr. Ch Srinivasa Rao, Director, NAARM inaugurated the programme by planting a tree followed by Dr. G. Venkateshwarlu, Joint Director. The other faculty and officers who participated in tree planting included Dr. B. S. Sontakki, Head XSM division, Sh. B. D. Phansal, JD(AR), Sh. Z.H. Khilji, CFAO, Dr. M. Ramesh Naik, Scientist & Chairman FAC, Dr. M.A.Basith, OIC, Farm, Dr. P.Vijender Reddy, OIC, Training unit, and Sh D. Murahari, Technical Asst, Training Unit.



6.3.13 ICAR-NAARM Post Card Released by Department of Posts, Govt of India

As part of the endeavour to highlight and to commemorate special events as well as features of the rich heritage and culture of Hyderabad City and its research and academic institutions, Department of Posts, Telangana Postal Circle, this year, released Picture Postcards on the premier Scientific Research Institutions working on diverse fields and contributing to the development of scientific temper among citizens and development of the nation. Department of Posts, Telangana Postal Circle issued a Picture Post Card on ICAR-NAARM on March 10, 2021 at release function held at Dak Sadan, Hyderabad, Telangana. The Picture Post Card was released by the Chief Post Master General, Telangana Circle. Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM and Dr. VK Singh, Director ICAR-CRIDA graced the occasion. Picture Post Cards of the other scientific

research institutes viz ICAR-CRIDA, ICRISAT, NIN, IICT, CCMB, NGRI, NRSC, CDFD, and NIPER were also released on the occasion, which was attended by Heads of respective organizations.



Fig. 6.30: ICAR-NAARM Post Card Released by Department of Posts, Govt of India

6.4 Yoga and Meditation

The Academy celebrated the “International Day of Yoga” on June 21, 2021 with the theme “Yoga for well-being”. The benefits of practicing Yoga in our health, happiness and well-being were



Fig. 6.31: Celebration of International Day of Yoga

explained and demonstrated on online mode, as per the guidelines issued by Ministry of AYUSH, Government of India. All the faculty members, Staffs and retired employees of ICAR located at Hyderabad were guided to perform yoga at home with family. Dr. A Debnath, Consultant Physician and Yoga Practitioner, conducted the Yoga session starting with warming-up exercises, *pranayama* and *asanas*.

6.5 General Meetings

6.5.1 Zonal Technology Management Committee Meeting

The 1st meeting of the Zonal Technology Management Committee (ZTMC) representing the all the ITMUs of the Education Division was held on 28 January, 2021 through online mode. The meeting saw deliberations and discussion on the roles of ITMUs, apprehensions related to IP filing and commercialization of technologies. Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM and Chairman, ZTMC impressed upon the members for aggressively pursuing IP filings with special emphasis on copyrights. Dr. SK Soam and Dr. K

Srinivas stressed on the need for all education division ITMUs to work collectively to harness the strength of the all the institutes. ITMU in charges from IASRI, New Delhi (Dr. Ajit), CIWA, Bhubaneswar (Dr. Lipi Das) and NIAP, New Delhi (Dr. Ankita Kandpal) participated in the deliberations.

6.5.2 Second Advisory Committee Meeting on Entrepreneurship Development Through Farmer Led Innovations

The Second Advisory Committee Meeting on 'Entrepreneurship Development through Farmer Led Innovations (FLIs)–A Case Study in Plantation Sector' under NASF was organized on October 01, 2021 under the Chairmanship of Dr. Anupam Mishra, Vice Chancellor, CAU, Imphal. The Advisory Committee Members, CIs and CCPCIs of partnering institutions namely CPCRI, Kasaragod and IIPM, Bangalore Participated in the meeting. The committee suggested that while upscaling of innovations, preparation of business plan for potential innovations etc. may be worked out; and also to identify the existing competitors for selected farmers innovations. Dr. S Senthil Vinayagam, PI of the project coordinated the event program.



Fig. 6.32: Second Advisory Committee Meeting on Entrepreneurship Development through Farmer Led Innovations

6.5.3 Review-cum-Action Plan Workshop on Entrepreneurship Development Through Farmer Led Innovations

A Review-cum-Action Plan Workshop on 'Entrepreneurship Development through Farmer Led Innovations – A Case Study in Plantation Sector' was organized under National Agricultural Science Fund during October 22-23, 2021. The project team of cooperating institutes namely IIPM, Bangalore and CPCRI, Kasaragod attended the meeting wherein the action plan for October 2020 to March 2021 was prepared. The project team discussed about the framework on analysis of Agribusiness Climate, Economic Potential and Market potential.



Fig. 6.33: Review-cum-Action Plan Workshop on Entrepreneurship Development through Farmer Led Innovations

6.5.4 Admin & Finance Staff Meeting

An Interactive meeting with Administrative Officers and Finance & Accounts Officers of institutes located at Hyderabad was conducted by Sri GP Sharma, Director(Finance), ICAR, New Delhi

at the Academy on December 10, 2021. Twenty Officers of Administrative and Finance Cadre from different institutes participated in the meeting. Dr. Ch Srinivasa Rao Director inaugurated the function and gave a roadmap for common issues of the institutes located at Hyderabad. Director (Finance) interacted on the issues raised and suggested practical tips.



Fig. 6.34: Admin & Finance staff meeting

6.6 Other Major Activities

6.6.1 Participation in International Agronomy Congress

The Academy participated in the Exhibition organised during the Fifth International Agronomy Congress on 'Agri-Innovation to Combat Food and Nutrition Challenges' during November 23-27, 2021 at PJTSAU, Hyderabad. Drs. M Ramesh Naik, NA Vijay Avinashilingam, PC Meena and Mr. D Murahari participated in the Exhibition.



Fig. 6.35: Participation in International Agronomy Congress



6.6.2 Operational Success of Empanelled IT Firms

The Academy had empanelled four IT firms with the aim of co-developing IT solutions in the farm sector by leveraging the partnership with reputed IT firms. To mark the successful partnership, a virtual meeting was convened on April 30, 2021 by the Empanelment Operationalization Committee of NAARM with the empanelled firms. Dr. I Sekar, Chairman of the Committee welcomed the representatives from the firms and the Committee members. He recognized that the partnership could make a good beginning with few IT projects initiated and some in advanced stage of development, while expressing hope that there is much greater scope to expand. The representatives acknowledged that the fullest potential of the collaboration is yet to be achieved and reasoned that impact of pandemic in the past one year. Dr. P Krishnan, Member Secretary of the Committee briefly outlined the next phase of empanelment initiative by NAARM and invited all to participate in the process and contribute to take forward the association.



Fig. 6.36: Operational Success of Empanelled IT Firms

6.6.3 ICAR-NAARM Empanels Eight IT Firms

The Academy has taken a pioneering initiative to empanel private firms to co-develop IP-enabled IT solutions to address issues related to agricultural research, education, development, and governance based on mutual expertise. A launch meeting was

organized on Sep 06, 2021 wherein the Academy signed agreements for Joint Product Development with eight empanelled firms viz., M/s. ADEPTO Geo Informatics Private Limited, M/s. Centre for Good Governance, M/s. Neemus Software Solutions Private Limited, M/s. Tecdatum Info Services Pvt Ltd, and M/s. Threshold Software Solutions Private Limited, Hyderabad; M/s. NCDEX e-Markets Limited, Mumbai; M/s. Nippon Data Systems Limited, Delhi and M/s. Skillmine Technology Consulting Private Limited, Bangalore. Dr. G. Venkateshwarlu, Joint Director, NAARM highlighted the key roles of the Academy and expressed hope that the partnership will provide an opportunity for advancing the interests of the Academy, through mutually beneficial partnership. Earlier, Dr. Ch Srinivasa Rao, Director, NAARM hoped that this arrangement would serve as a platform for the scientists from different ICAR institutes to network with NAARM and the empanelled IT firms for jointly developing technological solutions in their respective domains. The program was conducted in hybrid mode and representatives from all ICAR institutes based at Hyderabad and PME Cell In-charges from different ICAR institutes joined virtually.



Fig. 6.37: ICAR-NAARM Empanels Eight IT firms

6.6.4 Successful Completion of 11th Batch of PGDM-ABM (2019-21)

The 11th batch of PGDM-ABM successfully completed their 2-year residential programme at the Academy. All the 48 students of PGDM-ABM (2019-21) have been placed successfully in 20 reputed agribusiness companies. The package offered ranged between Rs. 6.5 lakhs and 12

lakhs per annum (lpa) with an average CTC of Rs. 8.5 lpa. A send-off ceremony was organized on this occasion on Mar 31, 2021. Speaking on the occasion, Dr. Ch. Srinivasa Rao, Director, NAARM congratulated and wished the students for successful career ahead. He also reminded the students that they carry the image of the Academy on their shoulders. Therefore, they should always strive to perform with honesty, humility and keeping farmers' welfare in the centre.



Fig. 6.38: Faculty with 2019-2021 batch of PGDM(ABM) students

6.6.5 Diploma in Educational Technology and Management (DETM)

Online contact classes for DETM candidates were organized during May 22-29, 2021. Thirty-three candidates from 2019, 2020 & 2021 batches attended the classes virtually. All the seven courses of the DETM programme were covered by various resource persons. Besides contact classes, Internal Assessment was also organized during 01-11 June 2021 and Term-end examination during 23-30 June 2021 for all the three batches.



Fig. 6.39: DETM Online Sessions

6.6.6 Joint Advisory Committee Meeting

The 10th JAC for Diploma in Education Technology Management (DETM) was held on 22 September 2021 virtually. The meeting was organized under the Chairmanship of Dr. G Venkateshwarlu, Joint Director, ICAR-NAARM to finalize the dates for contact classes, Internal Assessment and Term-end examination for 2019, 2020 and 2021 batches and to finalize the 4th batch admission process.



Fig. 6.40: JAC Meeting

6.6.7 Exposure Visit of PGDM-ABM Students

First year students of PGDM-ABM attended fifth International Agronomy Congress on "Agri-Innovation to Combat Food and Nutrition Challenges", which was held during November 23-27, 2021 at PJTSAU, Hyderabad, Telangana. The students visited various stalls which were installed during the agronomy congress. During their visit they also interacted with different stakeholders as well as several private companies.



Fig. 6.41: PGDM (ABM) students visiting Agronomy Congress



6.6.8 Online Classes for PGDETM

Online contact classes for PGDETM candidates was organized from November 25-28, 2021. Twenty-eight candidates attended from 2021 batch. First semester and second semester final examination were organized during Dec 20-23 and Dec 27-30, 2021 respectively.



Fig. 6.42: PGDETM Classes

6.6.9 Annual Meeting of the Institutional Complaints Committee (ICC)

The annual meeting of the Institutional Complaints Committee (ICC) was held at the Academy on August 18, 2021, under the chairmanship of Dr. Surya Rathore with other members from the Academy as well as outside member. Mrs. P. Padmavathi, Secretary, Kasturba Gandhi National Memorial Trust, Hyderabad. It was decided in the meeting that Display boards to be depicted with the members' names and contact numbers in all the buildings. Awareness programmes regarding the rules and provisions of sexual harassment of women at workplace [prevention, prohibition, and redressal] Act 2013 for various categories of the employees. A session on Prohibition of Sexual Harassment at workplace to be included in the Academy's training programmes such as pre-RMP, MDPs, EDPs etc. to create awareness.

6.7 Visits of Students/Trainees/Farmers from various Agricultural Universities and other institutes

Sl. No.	Name of the College and University	Degree/ designation	Date of Visit	No. of Students
Virtual visits				
1.	College of Forestry, Ponnampet, University of Agricultural and Horticultural Sciences, Shivamogga	B. Sc. (Forestry)	Jun 15, 2021	72
2.	University of Agricultural Sciences, GKV, Bengaluru	B.Sc (Agricultural Marketing, Co-operation & Business Management)	Jul 12, 2021	65
3.	College of Forestry, Dr. Y.S.P. University of Horticulture and Forestry, Nauni, Solan	B.Sc (Forestry)	Jul 16, 2021	139
4.	College of Community Science, Acharya NG Ranga Agricultural University, Lam, Guntur, Andhra Pradesh	B.Sc (Community Science)	Jul 22, 2021	68
5.	College of Agriculture, UAS, Dharwad	B.Sc (Community Science)	Aug 9, 2021	68
6.	College of Agriculture, Acharya NG Ranga Agricultural University, Bapatla, Andhra Pradesh	M.Sc (Agricultural Extension)	Aug 24, 2021	13

Sl. No.	Name of the College and University	Degree/ designation	Date of Visit	No. of Students
7.	College of Agriculture, Padannakkad, Kerala Agricultural University	B.Sc (Hons)	Sep 2, 2021	92
8.	College of Agriculture, Vellayani and College of Agriculture, Ambalavayal, Waynad, Kerala Agricultural University	B.Sc (Agri. Hons)	Sep 13, 2021	242
In-person visits				
9.	College of Agriculture, UAS, GKVK, Bengaluru	B.Sc Hons (Agri)	Dec 1, 2021	33
10.	College of Agriculture, UAS, GKVK, Bengaluru - Batch 1	B.Sc Hons (Agri)	Dec 2, 2021	37
11.	College of Agriculture, Hassan, Karnataka - Batch 1	B.Sc Hons (Agri)	Dec 3, 2021	33
12.	College of Agriculture, Hassan, Karnataka - Batch 2	B.Sc Hons (Agri)	Dec 3, 2021	103
13.	Lovely Professional University, Punjab	B.Sc Hons (Agri)	Dec 4, 2021	6

The Academy is established in a green serene and sprawling campus of about 50 hectares. It has various facilities like Faculty Center, Academic Block, Administrative Block, Library, Medical and Health Centre, Sports Complex, Children Park, Guest houses and Trainees Hostel, Housing and Residential Quarters.

7.1 Library

The NAARM Library has more than 31,456 books, 11 international journals, 3 International online journals, and 6 online databases / software, i.e. EBSCO Business Source Premier, Indiatat, Commodities, Economic Outlook, iThenticate License, Orbit.com. The library has many of its publications in digital formats like CD, VCD and its own digital repository (EPrints@NAARM). The Academy is also a member of CeRA and has access to journals and databases through CSIRO, Springer and Open J-Gate etc. Library management has been completely digitized through the open source software "Koha". The Academy also has a rich repository of online patent search providers, GIS software and other statistical software.



Fig. 7.1: Library at ICAR-NAARM

7.2 Health Center

NAARM Health Center provides preventive and curative treatment for the employees and their dependents, trainees, students and pensioners

and their dependents. On average, 40 patients are seen each day, and medications are given to them. Other ICAR institutes' staff members and their relatives receive emergency treatment. More than 750 ICAR retirees have signed up for regular OPD treatment and referral services at the NAARM health center. Health camps are held at the health facility on a regular basis, with investigations and specialized doctor consultations. Throughout the year, numerous camps were held for NAARM employees, their families, trainees, and retirees. Health talks by the reputed specialist doctors of recognized corporate hospitals are conducted at the academy from time to time. Periodical blood donation camp are organized by health center with the help of Red Cross blood bank. The details activities undertaken during 2021 are as follows-

- Total 76 covid patients were treated under home isolation.
- Because of increasing number of covid cases, the pensioners and other patients were requested for video consultations.
- Diabetic camps were conducted periodically at health center for the benefit of pensioners and other diabetic patients of the Academy.
- Entire staff of the Academy and their dependents, PGDM students were vaccinated for covid 19 (both 1st and 2nd dose) at corporate hospitals and Rajendranagar PHC.
- Vaccination camp was also organized at health center particularly for 2nd dose. As many as 96 persons were vaccinated in the camp.
- Dr. Chandrakala of Rajendranagar PHC, nurse Padma and pharmacist Rajyalakshmi were felicitated by Director, NAARM on the occasion of Independence Day.



Fig. 7.2: COVID testing of Students

7.3 Sports Complex

The sports complex at NAARM has stadium for outdoor sports such as cricket, football, lawn tennis, and a variety of sporting events. There is a badminton court, table tennis, billiards/snooker, and a gymnasium in the indoor complex. This facility is used by NAARM personnel and trainees.



Fig. 7.4: Sports Complex at NAARM

7.4 Agro-technology Park

The Agro-technology Park is a demonstration farm consisting of orchard block, floriculture block, avenue plantations, landscape gardens, nursery, poly house, herbal garden, Agro-biodiversity park etc. The total farm area is about 31 hectares. The Agro-technology Park is developed to demonstrate different agro-technologies in different aspects of the farm. This will act as a show window of agro-technologies developed by ICAR and SAUs for the benefit of visitors, students, trainees and farmers.



Fig. 7.3: COVID vaccination of NAARM employees

7.5 Orchard Block

The orchard crops viz. mango (2.0 ha), sweet lime (1.5 ha), sapota (0.25 ha), aonla (0.25 ha), custard apple (1.0 ha), phalsa (0.5 ha) and other miscellaneous fruits were grown in an area of 6.5 hectares. The orchard plot (H-7) was cleaned and established with nutritionally important fruit crops like drumstick, papaya, aonla, rose apple, dragon fruit, carambola etc. During this year, gap filling with improved varieties of mango and custard apple was taken up in the existing orchards and the plants are well established. Intercropping of groundnut (var: Kadiri-7) crops was taken up in the orchard demonstration plots. The fresh pods were sold to the staff members.

7.6 Floriculture Block

The Floriculture block consists of rose garden, Jasmine plantation & seasonal flower beds viz. gladiolus, gaillardia, cosmos, marigold cultivated in an area of 1.0 ha. Latest named varieties of HT roses, Indian Varieties, floribundas, miniatures were planted in the rose garden of the Academy. The varietal names are labelled. During this year, new planting was taken and 50 plants of 12 new rose varieties were planted and well established in the rose garden extension block. From this floriculture block, farm has provided its services of floral arrangement and floral bouquets during various office functions.



Fig. 7.5: Roses blooming in the Rose Garden of the Academy

7.7 Campus Beautification

Lush green lawns along with foliage and flowering plants were maintained around the office buildings and guest houses. The foot path plantations were established all along the main road of the campus. Several species of foliage and flowering plants were propagated at the nursery and poly house. This helped in the development of potted plants which were used in decorating the office buildings.

7.8 Vermicompost Units

The farm wastes were used for the preparation of vermicompost at different locations viz. Nursery, Rose garden, Sweet lime orchard and Agro-biodiversity park.



Fig. 7.6: Farm wastes are segregated and converted into vermicompost

The vermicompost produced was applied as manure to orchards and nursery plants. The surplus vermicompost was sold to the staff members.

7.9 Agro-biodiversity Park

An area of about 4 acres has been developed as agro-biodiversity park. The land was prepared, native flora was retained, pits digging for planting completed, a water harvesting pond was dug. The planting of suitable plants including trees, flowering and ornamental plants, fruit and fragrant plants, climbers and grasses was taken up.



Fig. 7.7: Agro-biodiversity Park established in the Academy



Fig. 7.8: Plants Flowering in Agro-Biodiversity Park



Fig. 7.9: DDG (Education), Dr. R.C. Agarwal visiting the Agro-biodiversity Park on 1st September 2021 after Inauguration

7.10 Herbal Garden

A herbal garden has been established in the campus with 50 species of medicinal and aromatic plants. Display boards depicting the common and scientific names and information regarding usable

plant parts and utility are installed for all the sub plots. This garden has demonstration of medicinal and aromatic plants at one place.



Fig. 7.10: Medicinal and Aromatic Plants in Herbal Garden

7.11 Orchard at NAARM

The orchard crops viz. mango, sweet lime, sapota, amla, custard apple, guava, fig, pomegranate, phalsa, and other miscellaneous fruits are grown in an area of 6.5 hectares with an objective of demonstrating new agro-

technologies to the FOCARS, trainees, PG students, visiting farmers and students. Latest agro-technologies like INM, IPM, mulching practices, micro-irrigation system etc. are practiced and demonstrated.



Fig. 7.11: Sweetlime and Amla in Fruiting Stage

8.1 Recognitions

8.1.1 Participation in International Events Ch Srinivasa Rao, Director, ICAR-NAARM participated in the following virtual international meetings as Indian delegate:

- 30th BASIC (Brazil, South Africa, India and China) negotiators meetings – host India, held virtually on April 07, 2021, organized by MoEFCC, Govt. of India.
- Informal consultations on Koronivia joint work on agriculture held virtually on April 13, 2021, organized by UN climate change secretariat.
- UNFCCC meetings - SBI consultations on matters related to the fourth review of the adaptation fund and First expert group meeting on ACE on April 14, 2021, organized by UN climate change secretariat.
- UNFCCC informal consultations on Santiago network on loss and damages held on April 15, 2021 and on June 28, 2021.
- Inaugural programme of UN climate change conference on Koronivia joint work on agriculture discussions in the UNFCCC 2021 SB session scheduled during May 31-June 17, 2021 as Indian delegate, organized by UN climate change secretariat.
- Webinar on adaptation communications: draft supplementary guidance for voluntary use by parties in communicating adaptation information through adaptation communications, organized by UN climate change secretariat on June 30, 2021.
- Asia pacific climate week (APCW) side event on KJWA, transformation and ambition in food systems in Asia and the Pacific, organized virtually by FAO regional office for Asia and Pacific, Bangkok, Thailand on July 09, 2021.
- 1st Meeting of quad climate working group (USA, Japan, Australia and India) on adaptation, resilience, preparedness pillar, held on July 13, 2021, invited by MoEFCC, Govt. of India.
- Quad working group on climate change meeting of the adaptation, resilience and preparedness working level discussions, invited by MoEFCC, Govt. of India, held on July 21, 2021.
- Opening ceremony of 54th session of intergovernmental panel on climate change organized by Global Strategic Communications Council (GSCC), held on July 26, 2021.
- Invited speaker for the COP26 dialogue on prioritizing adaptation and resilience in India's climate action organised virtually by United Nations Development Programme (UNDP) with British High Commission, India, held on July 28, 2021.
- Multilateral heads of delegation's consultations on loss and damage with chairs and coordinators of negotiating groups (Part 1 & 2) convened by COP 25 presidency and the COP 26 incoming presidency, in collaboration with the chairs of the subsidiary bodies, organized by UN climate change secretariat, held on August 04, 2021.
- Release of first of its three-part 6th assessment report (AR6) by the Intergovernmental Panel on Climate Change (IPCC) - the UN body responsible for climate science, on August 09, 2021.

- First online roundtable discussion organized virtually by Dutch Ministry of Agriculture, nature and food quality, held on September 30, 2021. (this discussion is in preparation of side event on sustainable livestock farming at COP26 being organized by the governments of the Netherlands, New Zealand, Kenya and Colombia and the International Livestock Research Institute (ILRI) and Wageningen University and Research (WUR).
- Participated in the second online roundtable discussion organized virtually by Dutch Ministry of Agriculture, nature and food quality, held on October 30, 2021. (this discussion is in preparation of side event on sustainable livestock farming at COP26 being organized by the governments of the Netherlands, New Zealand, Kenya and Colombia and the International Livestock Research Institute (ILRI) and Wageningen University and Research (WUR).

8.1.2 NAARM Foundation Day Awards

Awards for the Decade	
Best Faculty	K H Rao, Principal Scientist, HRM Division
Technical Staff category	M Basith, Chief Technical Officer D Rajagopal Rao, Technical Officer
Administrative and Finance Staff category	G Raj Reddy, Assistant
Skilled Supporting Staff category	L Sathyanarayana
CLTS Category	P Amrutha
Awards for the Year	
Scientist	S K Soam, Head, ICM Division
Faculty	Bharat S Sontakki, Head, XSM Division
Trainer	Ranjit Kumar, Head, ABM Division
Non-Officer Admin and Finance category	P Srinivasu, Assistant; M Sridhar, Assistant
Technical Officer (T1-T5 category)	N Prabhakar Rao, STA (Civil)
Technical Officer (T6 and above category)	M Ravi, ACTO (Photographer)
SSS (Male category)	M Satyanarayana (Farm Section)
CLTS (Male category)	T Srinivas, JDAR Office
CLTS (Female category)	L Chandramma, Farm Section
Institution Building	P Krishnan, Principal Scientist; A Dhandapani, Principal Scientist
COVID 19 Special Recognition	S Narasimha, SSS- Health Centre

8.1.3 Other Recognitions Received by the Faculty of NAARM

- Alok Kumar acted as chairperson, Co-chairperson in sessions & convener during ISEE national seminar on transforming Indian agriculture through pluralistic & innovative extension approaches for self-reliant India organized by ISEE, New Delhi, BHU, Varanasi, BUAT, Banda, India and GCRA, Sydney, Australia during October 04-06, 2021.
- Bharat S Sontakki appointed as editor of Indian Journal of Oilseeds Research, published by Indian Society of Oilseeds Research, ICAR-IIOR, Hyderabad.
- Bharat S Sontakki appointed by BIRAC as expert for review and evaluation of BIG 19 call proposals.
- Bharat S Sontakki appointed by MANAGE, Hyderabad as jury member for evaluation of M.Sc. thesis in the subject of agricultural



extension and member of committee for selection of best agri startups from North East Region for MANAGE-Samunnati Best Agri-Startup Award 2021.

- Bharat S Sontakki nominated by ICAR as member of the institute management committee of ICAR-Indian Institute of Millets Research, Hyderabad and ICAR-Central Tobacco Research Institute, Rajahmundry.
- Bharat S Sontakki nominated to the extension education council of University of Agricultural Sciences, Raichur.
- Bharat S Sontakki served as expert for evaluation of Ph.D. and M.Sc. thesis of PJTSAU, Hyderabad; MPKV, Rahuri and UAS, Raichur.
- Ch Srinivasa Rao served as member, academic council, Professor Jayashankar Telangana State Agricultural University, (PJTSAU), Hyderabad, Telangana.
- Ch Srinivasa Rao served as member, advisory board, The Journal of Research – PJTSAU.
- Ch Srinivasa Rao served as member, advisory board, The Journal of Research – ANGRAU, Andhra Pradesh from April, 2018 onwards.
- Ch Srinivasa Rao served as member, Asian long term Experimental Network for Agriculture (ALTENA), Tsukuba, Japan (2017 – till date)
- Ch Srinivasa Rao served as Member, Board of management, Acharya N G Ranga Agricultural University, Guntur, Andhra Pradesh (as ICAR Nominee).
- Ch Srinivasa Rao served as member, Board of Management, Professor Jayashankar Telangana State Agricultural University (PJTSAU), Hyderabad, Telangana (as ICAR Nominee).
- Ch Srinivasa Rao served as member, Committee to revise and explore the possibility of filing APAR online.
- Ch Srinivasa Rao served as member, executive board of the International Dryland Development Commission (IDDC), Cairo, Egypt. (2016 – till date)
- Ch Srinivasa Rao served as member, executive council, National Council for Cooperative Training (NCCT), Department of Agriculture, Cooperation & Farmers Welfare, Ministry of Agriculture & Farmers' Welfare, Govt. of India.
- Ch Srinivasa Rao served as member, Food and Agriculture Division Council (FADC), Bureau of Indian Standards, Ministry of Consumer Affairs, Food and Public Distribution, Govt. of India.
- Ch Srinivasa Rao served as member, ICAR committee on development of tech dome for central vista project.
- Ch Srinivasa Rao served as member, ICAR committee to review Career Advancement Scheme (CAS) and score card for scientists of ICAR.
- Ch Srinivasa Rao served as member, ICAR steering committee for implementation of ICAR research data management guidelines.
- Ch Srinivasa Rao served as member, ICT steering committee of ICAR (2020 onwards)
- Ch Srinivasa Rao served as member, management committee of Extension Education Institute (southern region), Department of Agriculture, Cooperation & Farmers Welfare, Ministry of Agriculture & Farmers' Welfare, Govt. of India.
- Ch Srinivasa Rao served as member, Member, Regional Advisory Group (RAG), NABARD.
- Ch Srinivasa Rao served as member, NAAS organizational committee to frame guidelines relating to organizational, personnel, finances and investments as well as delegation of powers.

- Ch Srinivasa Rao served as member, NAAS sectional committee in natural resource management.
- Ch Srinivasa Rao served as member, of 'performance review committee' on recommendations of working group report on finalization of parameters of activity plan of the faculty members of VAMNICOM/RICMs/ICMs under NCC.
- Ch Srinivasa Rao served as member, of chief advisory committee for preparation of integrated horticulture development perspective plan, Dept. of Horticulture and Sericulture, Govt. of Telangana.
- Ch Srinivasa Rao served as member, PGDM committee for post graduate diploma in management - agribusiness & management of Vakunth Mehta National Institute of Cooperative Management (VAMNICOM), (Ministry of Agriculture & Farmers Welfare, Govt. of India), Pune, Maharashtra.
- Ch Srinivasa Rao served as member, to the reconstituted ICAR committee on agricultural research service for a period of two years (2021-23).
- Ch Srinivasa Rao served as member,, advisory committee, World Bank funded ICAR-NAHEP component-2 project on Investment in ICAR Leadership for Agriculture Higher Education.
- Ch Srinivasa Rao served as member,, selection-cum-standing committee for emeritus scientist scheme of ICAR (2019-22).
- Ch Srinivasa Rao served as member, Academic Council, Acharya N G Ranga Agricultural University, Guntur, Andhra Pradesh.
- Ch Srinivasa Rao served as member,r, ICAR steering committee to review and monitor the performance of National Agricultural Innovation Fund (NAIF) Scheme.
- Ch Srinivasa Rao, accredited by PJTSAU for teaching and guiding its M.Sc. and Ph.D. students in the field of Soil Science and Agricultural Chemistry for a period of five years.
- Ch Srinivasa Rao, Chairman, ICAR committee to develop targeted roadmap for the creation of technology incubation centres in all institutes including KVKs (2021).
- Ch Srinivasa Rao, Chairman, ICAR national convention for industry-institute interface for food processing technologies, held during April, 2021.
- Ch Srinivasa Rao, joined as Editor, Journal of Climate Change & Environmental Sustainability.
- Ch Srinivasa Rao, joined as Editor, Springer Journal.
- Ch Srinivasa Rao selected as member of expert committee on agro-biodiversity, national biodiversity authority, (MoEFCC, Govt. of India), Chennai, Tamil Nadu for a period of one year from October, 2021.
- D Thammi Raju acted as a member of the selection committee for assistant professor cadre in the discipline of fisheries extension, statistics and economics in PVNR Telangana Veterinary University, Hyderabad.
- D Thammi Raju acted as an expert for the evaluation of the concepts of national level hackathon on precision and economical farming organized by NAHEP for south zone.
- D Thammi Raju evaluated the proposals under the incubation programme of krishiboot of a-Idea, Hyderabad on October 7, 2021.
- Ganesh Kumar as member, Training Quality Improvement Measures Committee (TQIMC) of NIRD & PR, Hyderabad.



- M B Dastagiri, ISDS Fellow by International Society for Development and Sustainability (ISDS), Japan.
- N Srinivasa Rao nominated by Agricultural Scientists Recruitment Board (ASRB) as the chairman of the assessment committee for considering the assessment promotion of technical personnel under the functional group field / farm technicians of CMFRI, Kochi.
- N Srinivasa Rao nominated by Agricultural Scientists Recruitment Board (ASRB) as an expert member of the selection committee for assessing the promotion cases of technical category (III) of IIRR, Hyderabad.
- N Srinivasa Rao nominated by DDG (Education), ICAR as a member of the national level IT committee "KRISHI MEGH Operations & Management Committee" for management, compliance and operational management of data center, ASHOKA and KRISHMEGH.
- P C Meena as member in direct promotion committee at Central Research Institute for Dryland Agriculture for promotion of two technical officers from T7/8 to T9.
- P Krishnan as a key speaker in the regional capacity building program on biotechnological tools in aquatic genetic resource management and ex-situ conservation organised by ICAR-NBFGR, Lucknow in collaboration with APARI on December 17, 2021 and delivered a talk on "spatial information systems for aquatic resources management & governance regional fisheries resource management: case studies for bridging gaps in policy with science."
- P Krishnan as an expert in the National Agriculture Disaster Management Plan (NADMP): release workshop organized by National Institute for Disaster Management (NIDM), Ministry of Home Ministry, New Delhi on September 24, 2021.
- P Krishnan as member of the Planning Board of Tamil Nadu J Jayalalitha Fisheries University (TNJFU), Nagapattinam, Tamil Nadu for a period of three years from January, 2021.
- P Venkatesan received Best Oral Presentation Award – for his oral presentation of paper titled "Farmer first programme: an approach to improve the rural economy" in Indian Society of Extension Education (ISEE) national seminar on transforming in Indian agriculture through floristic and innovative extension approaches for self-reliant India, organized by Banaras Hindu University, Varanasi during October 04-06, 2021.
- P. Krishnan invited as a key speaker in the expert consultation meeting on "cross-learning for thematic regional/ sub-regional challenges in aquaculture and fisheries" organized by SAARC Agriculture Centre (SAC), Dhaka, Bangladesh on August 4, 2021 and delivered a talk on, regional fisheries resource management: case studies for bridging gaps in policy with science".
- Ranjit Kumar as external evaluator and examiner for Ph.D. thesis and viva voce of student of Institute of Agri-Business Management, SKRAU, Bikaner.
- Ranjit Kumar as external examiner for M.Sc. thesis viva voce of students of MBA (Agribusiness) and MBA (Rural Management) of Dr RP CAU, Samastipur.
- Ranjit Kumar as member of expert panel of an explorative research study "AgroEco2050: foresight on natural farming in Andhra Pradesh" led by RySS, CIRAD and FAO.
- Ranjit Kumar as member of the study management committee of the project on "Co-Benefits of Large-scale Organic Farming on Human Health (BLOOM): an evaluation

of Andhra Pradesh community-managed natural farming” by the Global Academy of Agriculture and Food Security, the University of Edinburgh in partnership with the Public Health Foundation of India (PHFI).

- S K Soam as a convener for 23rd annual conference of Society of Statistics, Computer and Applications (SSCA) titled “Visionary Innovations in Statistical Theory and Applications (VISTA-2021)” organized by ICAR – National Academy of Agricultural Research and Management, Hyderabad during February 24-28, 2021
- S K Soam as a member, National Advisory Committee National Conference on ICT based e-resources for smart agriculture: a journey towards atmanirbhar bharat post covid-19 pandemic situation during March 16-17, 2021.
- S K Soam as a nominated member attended ITMU meeting of ICAR-Indian Institute of Oil Palm Research, Pedavegi on August 16, 2021, discussed patenting and commercialization of ‘abration tool’ and suggested copyrights for software
- S K Soam as an expert reviewer evaluated the business proposals submitted to 18th call of Biotechnology Ignition Grant (BIG) under Biotechnology Industry Research Assistance Council (BIRAC) scheme of Department of Biotechnology, Govt. of India.
- S K Soam as Guest of Honour for hands-on-training cum workshop on ‘development of presentation skills, organised by career development center, SKN agricultural university, Jobner on February 7, 2021.
- S K Soam as nodal officer extended the MoU with Biotech Consortia India Limited (BCIL) for the project ‘orientation-cum-awareness and implementation of action of Access and Benefit Sharing (ABS) guidelines of ICAR. The

project started under the MoU signed on December 12, 2019.

- S K Soam chaired a session on agriculture transformation through institutional strengthening, skilling and entrepreneurship development during national agricultural conference on public-private-industrial centric approach to transform agriculture in virtual mode organized by school of agricultural sciences, G D Goenka University with the support of NABARD on March 13-14, 2021
- S K Soam chaired a session on technology driven ideas and their application for promoting agribusiness opportunity during 6th national youth convention on “innovation and agricultural reforms for farmer’s prosperity” jointly organized by AIASA, ICAR and PJTSAU, Hyderabad on February 20-21, 2021
- S K Soam nominated by Agricultural Scientists Recruitment Board (ASRB) as a chairman on the assessment committee of technical staff of ICAR-IIMR, chaired a meeting on September 24, 2021 at IIMR, Hyderabad
- S K Soam was Guest of Honour at the ‘virtual training for university students to improve their performance in competitive examination’, organized by SVVU, Tirupati, under CDC of NAHEP, about 500 students attended, V Padmanabha Reddy, Vice Chancellor, SVVU inaugurated the event.
- Sivaramane N as examiner to act as co-opted/ external member for the conduct of the qualifying viva-voce examination of a Ph.D. student of IARI, New Delhi.
- Sivaramane N participated as expert in the discipline of agricultural economics for Career Advancement Scheme (CAS) assessment for principal scientist at ICAR-Sugarcane Breeding Institute, Coimbatore on August 3, 2021.

- Surya Rathore as special guest for online training programme on 'development of soft skills for entrepreneurship among agri-graduates' of Birsa Agricultural University, Ranchi on October 22, 2021.
- Surya Rathore awarded with Reviewer Excellence Award as a reviewer of NAAS rated journal, "Legume Research" (6.53) from the editors of ARCC journals in recognition of significant and outstanding contribution to the journal and reviewing the articles on September 13, 2021.
- Surya Rathore served as invited expert in UN-FAO & MANAGE consultative workshop on 'development of training module package on nutrition sensitive agriculture in Extension Advisory Services (EAS) on August 24, 2021, at MANAGE, Hyderabad.
- Surya Rathore served as member, Departmental Promotion Committee (DPC) meeting for regularization of Casual Labourers with Temporary Status (CLTS) to the post of Skilled Support Staff (SSS) on August 25, 2021, at ICAR – NAARM, Hyderabad.
- Surya Rathore won the team award for "most impactful training delivery" during the teach-back of GIZ – Dale Carnegie training of trainers programme hosted by Indira Gandhi National Forest Academy, Dehradun, on May 14, 2021.
- V V Sumanth Kumar successfully completed one-year Post Graduate Diploma in Education Technology and Management (PGD-ETM) course in distinction conducted by ICAR-NAARM in collaboration with University of Hyderabad with enrollment number 35ETM00044-19.

8.1.4 ICAR-NAARM in Media

AGRI UDAAN® Food and Agribusiness Accelerator 3.0 Launched

Dr. Trilokesh Mahapatra, Secretary (OAR) & D.O (ICAR) launched the Agri UDAAN, Food and Agribusiness Accelerator 3.0 at the ICAR-National Academy of Agricultural Research Management, Dr. Mahapatra wished AGRI UDAAN 3.0 a grand success in which scaling up of innovative enterprises is envisaged. He emphasised on the need for keeping the farmers' interest during the whole process.

Shri Hitesh Kumar Mittal, Advisor, Member Secretary, National Science & Technology Entrepreneurship Development Board & Member Secretary, Technology Development Board, Department of Science & Technology, Government of India stressed upon taking the timely action on this knowledge for the good entrepreneurs.

While highlighting the Entrepreneurial ecosystem in Telangana, Shri Jayesh Ranjan (IAS), Principal Secretary, Departments of Industries & Commerce (S&C) & Information Technology, Electronics and Communications (ITEC), Government of Telangana urged for taking the maximum benefit from the State Government's policies.



Shri Hemant Sengodkar, Chief General Manager, OT Farm Development Department, National Bank for Agriculture and Rural Development, Mumbai briefed about the NABARD's efforts to promote the entrepreneurship for the rural development. He also emphasised on the need for convergence and networking for a vibrant entrepreneurial ecosystem.

Dr. P. Appalarao, Vice-Chancellor University of Hyderabad mentioned that the incubators of UoH are ready for co-working with all the prominent stakeholders like NAARMs, TBI or IDEA.

Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM briefed about the ICAR-NAARM's activities and a-IDEA. He also outlined the detailed timeline and plan for AGRI UDAAN 3.0. More than 250 Delegates from Academia, Corporates, Research Organizations, Investors, Government Functionaries, Trainers and start-up Entrepreneurs were present during the occasion.

148

ఎగుమతికి
గిరాక్ ఉన్న

- మూల అవస్థలో అభివృద్ధి చెల్లినట్లు
- అభివృద్ధి చెందే అవకాశం
- మృత్యువు వచ్చే వరకు అభివృద్ధి చెందే అవకాశం
- అభివృద్ధి చెందే అవకాశం

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పంటలే పండించాలి

இவாசியைக் காலத்தில் தவறாமல் மூலம் மூலம் மூலம்

And the other side of the coin is that the more you know about the world, the more you know about yourself. It's a two-way street. The more you know about the world, the more you know about yourself. It's a two-way street.

- **How much time** will you have to spend on the project?
- **How much money** will you have to spend on the project?
- **How much risk** will you have to take on the project?
- **How much effort** will you have to put into the project?
- **How much time** will you have to spend on the project?
- **How much money** will you have to spend on the project?
- **How much risk** will you have to take on the project?
- **How much effort** will you have to put into the project?



2000

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HYDERABAD

निम्नलिखित विभाग

వ్యవసాయ శాస్త్రవేత్తలకు కేరాఫ్ నార్మ్

● నేడు 46వ వ్యవస్థాపక బహిరంగం

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नार्म ने मनाया 46वाँ स्थापना दिवस



संक्षेप रूपेण विवेचनार्थं प्रस्तुत के गणना विवरण (अंश) के अंतर्गत विवरण प्रस्तुत है।
संक्षेप रूपेण विवेचनार्थं प्रस्तुत के गणना विवरण (अंश) के अंतर्गत विवरण प्रस्तुत है।

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The first step in the process is to identify the problem. This is often done by the patient or the doctor. The next step is to collect information about the problem. This is often done by the doctor. The third step is to analyze the information. This is often done by the doctor. The fourth step is to make a diagnosis. This is often done by the doctor. The fifth step is to develop a treatment plan. This is often done by the doctor. The sixth step is to implement the treatment plan. This is often done by the doctor. The seventh step is to evaluate the results. This is often done by the doctor. The eighth step is to make adjustments. This is often done by the doctor. The ninth step is to follow up. This is often done by the doctor. The tenth step is to provide ongoing care. This is often done by the doctor.



United News of India
India's Multi-Media News Agency

Published: May 2, 2011 | View: 2011 FreEST

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వ్యవసాయ ఆధారిత రంగాల్లో వార్డ్ కృషి అమోఘం

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15420 • J. Neurosci., July 26, 2006 • 26(30):15415–15420

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• **Wages:** The amount of money paid to workers for their labor.

सत्र में ऑनलाइन पाठ्यक्रम 1 से

January 4, 1992 (Friday)

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per volume of water and dissolved oxygen; a lower rate means the water quality is not as good as the water in the river; and a high rate means the water is good. The water quality is good if the rate is 1.0 or higher and the water is bad if the rate is 0.5 or lower.

► అవీల్డెన్ నెమరూ బోదవస్తు 'నూర్' శిక్షణ

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సాంకేతిక విద్యా, భోధనపై

అన్ లెన్ శిక్షణ తరగతులు

రంగు రెడ్డి, సీటీ టిమ్ : తననా కాలంలో

అనే టైన్ కెళ్లగా ఎంతో అవసరమని నానీ పంపిన
నలు కోజుల కెళ్లగా కార్మికులపార్టీ ఏర్పడ్డా ముప్ప
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చెల్లించాలని మోసించారు

नाम द्वारा विधोशिकृत काउन्सिल ऑफ
एड्युकेशन टेक्नालॉजी आनलाइन कोर्स । जन से

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पूरा समाज में नये अधिकांशों ने बताया कि उन्होंने कोई सक्रिय भूमिका नहीं निभाई, उन्होंने अलग-अलग, फैलावती चिन्ताएँ बाँधी थीं और समाज में सक्रिय नहीं हैं, उनके पास उपयोगिता है एवं वे इस सोचों को पूरा का पूरा छोड़ देते हैं। विचार, विचारण का समाज में। साथ ही संवेदित समाज का भी परिवर्तन देखा जा सकता है।

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సాంకేతికతపై శిక్షణ

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చేసి వారు <https://clearing.narm.org.in> వారు తమ పేర్లు
నిమిషం లేకుండా వారు

అంకురాలకు దన్ను. అగ్రి ఉడాన్

- పదకొండో శతాబ్దం ఎంతో మేరం
- ఒక రకం ప్రాచీనతరం



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8.1.5 Invited Lectures Delivered by the Faculty and Staff

Lecture	Programme/ Organized by
Alok Kumar	
Building team spirit	Training workshop on “building team spirit among students for career development”, SKNAU, Jobner, Rajasthan.
Human resource interventions for institutional innovation	ISSE national seminar on transforming Indian agriculture through pluralistic & innovative extension approaches for self-reliant India.
Training & Development	e-Training Course on ICAR JRF/SRF, NET/ ARS Exam Preparation organized by Mahatma Phule Krishi Vidyapeeth (MPKV) Rahuri Dist. Ahmednagar, Maharashtra
A Dhandapani	
Statistical analysis of plant health experimentation	PGDPHM programme, NIPHM, Hyderabad.
Analysis of multi-location trials	Statistical designs and analytical methods for multifactor experiments ICAR-CMFRI, Kochi.
Analysis of repeated measures data	Statistical designs and analytical methods for multifactor experiments ICAR-CMFRI, Kochi.
Introduction to R software: data structures and acquisition of data hands-on session on handling of time series data using R software	Time series techniques for forecasting in agriculture, ICAR-IARI, New Delhi.
B Ganesh Kumar	
Business Opportunities and Source of Information for New Entrepreneurs in Livestock and Veterinary Sector	Online Faculty Development Program on “Entrepreneurship in Livestock and Veterinary Sciences”
Integrating Livestock Farmers with Market	Online Training Programme on “Promotion of FPO in livestock sector: Opportunities and Challenges”
Backward and Forward Linking Livestock Farmers to Markets	Online Training Programme on “Market Led Extension in Animal Husbandry Sector”
PPP in Animal Husbandry Sector -Challenges/Issues; Working Models and Experiences	Online Training Programme on “Promotion of Public Private Partnership in Agri and Allied Sectors”
Precision Dairy Farming	Online training programme on ‘Precision Dairy Farming’
Bharat S Sontakki	
Motivation, positive thinking and communication skills in research work	Proficiency development on quality evaluation, processing and product development of natural fibres for technical staff” (January 20-22, 2021) ICAR-NINFET, Kolkata.
Motivation, positive thinking and communication skills in research work	Proficiency development on quality evaluation, processing and product development of natural fibres for technical staff” (January 28-30 2021) ICAR-NINFET, Kolkata.
Research in behavioural sciences: status and prospects	Online collaborative training on research methods in behavioral sciences (February 2-5, 2021) KAU, Thrissur.

Lecture	Programme/ Organized by
How to prepare winning research proposal?	Training programme on “enhancing teaching and managerial skills among the faculty of TANUVAS” (March 8-12, 2021) MANAGE, Hyderabad.
How to prepare winning research proposal?	Training programme on “enhancing teaching and managerial skills among the faculty of TANUVAS” (March 15-19, 2021) MANAGE, Hyderabad.
Improving communication skills	Online orientation training for newly recruited subject matter specialists of KVKs of ATARI zone VIII (May 3-5, 2021) ICAR-ATARI Zone VIII, Pune.
Monitoring and evaluation of virtual farmer field schools	Capacity development programme on virtual farmer field schools UAHS, Shivamogga and ICAR-ATARI (Zone XI), Bengaluru.
How to prepare winning research proposal?	Online training programme on “enhancing teaching and managerial skills among the faculty of SAU and SVU” (28-30 June 2021) MANAGE, Hyderabad.
Participatory technology development in agriculture and allied sectors	National refresher course on “reorienting extension education and advisory services for sustainable development of farming community” KVAFSU, Bidar & NADCL, Baramulla.
Student-centred teaching and learning	HRD programme on “booming teaching and scientific professionalism” for the faculty of College of Sericulture (UAS, Bengaluru), Chintamani.
Soft skills for effective teaching and briefing on the exercises	HRD programme on “booming teaching and scientific professionalism” for the faculty of College of Sericulture (UAS, Bengaluru), Chintamani.
Micro-teaching exercise	HRD programme on “booming teaching and scientific professionalism” for the faculty of College of Sericulture (UAS, Bengaluru), Chintamani.
Concept note preparation: orientation and hands-on exercise	HRD programme on “booming teaching and scientific professionalism” for the faculty of College of Sericulture (UAS, Bengaluru), Chintamani.
Concept note presentation exercise	HRD programme on “booming teaching and scientific professionalism” for the faculty of College of Sericulture (UAS, Bengaluru), Chintamani.
Ch Srinivasa Rao	
Innovative technologies adopted in stress management of dryland agriculture crops	National webinar on “stress management in crops for food and nutritional security – role of fertiliser policy - way forward” organized by FAI.
Sustainable natural resource management for enhancing pulse production and nutritional security in India	National web conference on “sustaining pulse production for self sufficiency and nutritional security” (Pulse WebCon 2021), ICAR-IIPR, Kanpur.
Integrated Nutrient Management (INM) for climate smart agriculture	DST sponsored online national training programme on “soil & water conservation technologies for climate smart agriculture in the context of extreme weather events” organized by ICAR-Indian Institute of Soil and Water Conservation, Research Centre, Udhagamandalam.
Impacts of climate change and weather forecasting on agriculture and food systems in coastal ecosystems of India	ICSA webinar - international symposium on “transforming coastal zones for sustainable food & income security”, organized by Indian Society of Coastal Agricultural Research (ISCAR).
Agritech incubators - scope and opportunities for students and farmers	Webinar on ‘agritech incubators - scope and opportunities for students and farmers’ Vignana University, Guntur, Andhra Pradesh.



Lecture	Programme/ Organized by
Prioritising adaptation and resilience in India's climate action	COP26 dialogue on prioritising adaptation and resilience in India's climate action, United Nations Development Programme (UNDP) with British High Commission, India.
Agri entrepreneurship development in India	15 Days international online training on "advances in agripreneurship and skill development for reshaping future of Indian agriculture" organized jointly by just Agriculture-the magazine and AEEFWS, Punjab.
Food security and sustainable agriculture in India	Online training programme on "climate change: challenges and response" for women scientists & technologists, LBSNAA, Mussoorie.
Soil carbon sequestration in India: research, technologies and policy needs	International virtual workshop on "soil carbon for sustainable crop production and soil health management", Sri Karan Narendra Agricultural University, Jobner.
Climate Change Mitigation: Role of Technology, Science, Economic and Policy to the participants of	Online training programme on "role of technology in community level disaster mitigation for scientists & technologists" working in government sector organized by LBSNAA, Mussoorie
Soil carbon dynamics and cropping practices	Technical session 1 : Energy smart tillage and nutrient management for soil health and higher productivity under Theme 2 : Energy smart resource use in the XV agricultural science congress & ASC expo - energy & agricultural challenges in 21st century NAAS and Institute of Agricultural Sciences, Banaras Hindu University.
Use of AI and ML in engineering and agriculture education	Online training programme AICTE/ISTE induction/orientation programme held during 2-7- Dec, 2021 by Guru Tegh Bahadur Institute of Technology, New Delhi.
Climate change mitigation: role of technology, science, economics and policy	DST sponsored five days online training programme on "climate change: challenges and response (for scientists & technologists)" LBSNAA, Mussoorie.
D Thammi Raju	
Digital teaching initiatives and methodologies	Online digital teaching techniques organised by UAS, Bangalore on February 18, 2021.
Re-imagining higher agricultural education in india on the face of challenge from COVID-19 pandemic - strategies for adapting to the new normal	Higher education going online, as a panel speaker in Central University of Tamil Nadu on April 04, 2021.
Online education system	Online programme on innovative extension strategies for sustainable development conducted by TANUVAS, Chennai on April 29, 2021.
Online education system as innovative extension strategy	Online programme on reorienting extension education and advisory services for sustainable development of farming community conducted by KVAFSU Bidar, Karnataka and National Agriculture Development Co-operative Ltd. Baramulla on July 13, 2021.
Concept and use of Massive Open Online Courses (MOOCs) for agriculture extension	Online programme on ICTs for agricultural extension: new concepts conducted by college of agriculture, Bawal of CCSHAU and MANAGE, Hyderabad on August 25, 2021.

Lecture	Programme/ Organized by
G R Ramakrishna Murthy	
Potential of technology in teaching-learning process	Virtual faculty induction programme for assistant professors /associate professors from degree colleges and universities by Maulana Azad National Urdu University, Hyderabad on August 14, 2021.
M Balakrishnan	
Advanced bioinformatics tools and its applications in agriculture	Workshop on advances in bioinformatics, department of biotechnology, NIT Warangal.
ICT tools- e-learning, web technologies and information kiosks	Online tutorial classes on ICAR NET/ARS examination for students, MPKV, Maharashtra.
Types of network information system and challenges in the use of ICT	Online tutorial classes on ICAR NET/ARS examination for students, MPKV, Maharashtra.
Scope of artificial intelligence research in agriculture sector	Training on applications of analytics and prediction in farming world G B Pant University of Agriculture, Uttarakhand.
Big data and the future of agriculture	Training on applications of analytics and prediction in farming world G B Pant University of Agriculture, Uttarakhand.
Biological databases and bioinformatics tools and software's	Online Faculty Development Programme (FDP) advances in bioinformatics BIT, Anna university Campus, Trichy.
Bioinformatics in agriculture –challenges and opportunities	Online Faculty Development Programme (FDP) advances in bioinformatics BIT, Anna university Campus, Trichy.
Sequence alignment and phylogenetic study with hands on session	Online Faculty Development Programme (FDP) advances in bioinformatics BIT, Anna university Campus, Trichy.
N Srinivasa Rao	
Content development through blogs	Online training programme on “Content Development and Management” organised by TANUVAS, Chennai on April 21, 2021.
Open source tools for building personal websites	UAS, Raichur organised online training programme on “e – resources, e-learning and IoT on August 26, 2021.
Content development through Blogs	Training programme on “Digital Content Development and Management” conducted by TANUVAS, Chennai on September 18, 2021.
Open source tools for building personal websites	Online training programme on “e – resources, e-learning and IoT conducted by UAS, Raichur on December 16, 2021.
P Krishnan	
De-covidization of (fisheries) higher education: leveraging enhanced readiness to integrate digital technology	Online seminar on towards knowledge economy and innovation society – role of professional educational institutions in Kerala on June 17, 2021.
Science of adaptation & methods and tools: impacts, vulnerability & adaptation	Fast-track learning in climate change and marine life organized by in Sathyabama University in association with Earth Science and Technology Cell, Ministry of Earth Sciences (ESTC – MoES) and IUCN Commission on Ecosystem Management (IUCN – CEM) on June 19, 2021.
Seaweed culture & processing <i>Policies, Laws & Opportunities</i>	CSIR integrated skill initiative: skill development program in seaweed cultivation and processing technology (SEA-CPT) organised by CSIR-Central Salt and Marine Chemicals Research Institute (CSMCRI), Bhavnagar (Online) on February 24-26, 2021.



Lecture	Programme/ Organized by
Impact of commercial cultivation of <i>Kappaphycus alvarezii</i> in palk bay on coral reefs of Gulf of Mannar, South India.	ISCA webinar: international symposium on coastal agriculture: transforming coastal zone for sustainable food and income security conducted by Indian Society of Coastal Agricultural Research (ISCAR), ICAR-CSSRI, Canning Town, West Bengal (online) on March 16-19, 2021.
Biological Diversity Act, 2002 – implementation for agricultural research	Module on biodiversity governance in India for IFS officers from IGNFA, Dehradun on July 15, 2021.
Seaweed culture & processing: Policies, Laws & Opportunities	CSIR-Central Salt and Marine Chemicals Research Institute (CSMCRI), Bhavnagar organised integrated skill initiative on 'Seaweed Cultivation and Processing Technology (SEA-CPT)' on July 23, 2021.
Regional fisheries resource management: case studies for bridging gaps in policy with science	Expert consultation meeting on "cross-learning for thematic regional/ sub-regional challenges in aquaculture and fisheries" conducted by SAARC Agriculture Centre (SAC), Dhaka, Bangladesh on August 04, 2021.
Tips for research project formulation & science communication	Kerala Agricultural University organised DOE – online foundation course on research, education and extension management on August 06, 2021.
Frameworks for ranking research and innovation in agricultural universities: adapting to the new normal	Sri Konda Laxman Telangana State Horticultural University, Mulugu, organized webinar on higher education on research, innovation and ranking on August 09, 2021.
Covid vis-à-vis fisheries education: emerging opportunities & adapting to new normal	KUFOS alumni meet on October 02, 2021.
Seaweed culture & processing: Policies, Laws & Opportunities	CSIR-CSMCRI, Bhavnagar, Gujarat organized integrated skill initiative entitled 'Seaweed Cultivation and Processing Technology (SEA-CPT)' on November 12, 2021.
Writing research papers and breaking minds for science communication	PMFGR Centre, ICAR-NBFGR, Kochi conducted workshop on "Cutting Edge Technologies in Fisheries and Aquaculture for Food and Nutritional Security (CETFAQ-2021)" on December 22, 2021.
Spatial information systems for aquatic resources management & governance	ICAR-NBFGR, Lucknow, jointly with APARI organized regional capacity building program on biotechnological tools in aquatic genetic resource management and ex-situ conservation on December 17, 2021.
P Ramesh	
Psychological principles of teaching and learning	Faculty induction program
Emotional intelligence	Faculty development programme
P D Sreekanth	
Use of GIS in Integrated Watershed Management	IFS probationers 2019-21 batch organised by Indira Gandhi National Forest Academy (IGNFA), Dehradun on May 26, 2021.
Remote sensing, GIS and GPS with special focus on marketing and insurance in agriculture	Digital marketing strategies for effective marketing of agri and allied products, Extension Education Institute, Hyderabad on October 07, 2021.
Climate smart extension – applications of remote sensing and GIS in agriculture	Extension strategies for climate smart extension in agri and allied sectors, Extension Education Institute, Hyderabad on October 21- November 26, 2021.

Lecture	Programme/ Organized by
RS, GIS & GPS – digital way forward in marketing	Digital applications for promotion of marketing in agriculture and allied sectors,
P Supriya	
Micro RNA analysis	Workshop on “molecular data analysis through bioinformatics tools” S V Agricultural College, Tirupati.
P Venkatesan	
KVK as a centralized hub for rural mission at district level: bridging the rural urban divide	Rurban mission ideation workshop, NIRD& PR.
Qualitative analysis : thematic analysis/item analysis- NVIVO 12	FDP- qualitative data analysis in social sciences, school of agricultural innovations and advanced learning (VAIAL), VIT University Vellore, Tamil Nadu.
Training objective and designing of training program	Workshop on training of trainers for in-service SFS officers of state forest department, central academy for state forest service, Coimbatore.
Training need analysis	Workshop on training of trainers for in-service SFS officers of State Forest Department, Central Academy for State Forest Service, Coimbatore
Effective training skills	For in-service SFS officers of State Forest Department , Central Academy for State Forest Service, Coimbatore.
Scope towards implementation of farmer FIRST programme by Mahatma Phule Krishi Vidyapeeth, Rahuri	Programme on kharif Season Planning, Mahatma Phule Krishi Vidyapeeth, Rahuri.
Ranjit Kumar	
eNAM – Helping in better price discovery for small farmers	Online International Training Program on Innovative Marketing Models for Smallholder’s Agriculture
Business Plan and Goal Setting to Launch a Successful Start-up	Sensitization e-Workshop on ‘Startup Opportunities for Agricultural Graduates’
Entrepreneurship among students	Training on Intellectual Property Rights and Technology Commercialization in Agriculture
Ecosystem and policies promoting FPOs in India	Webinar
Developing synergies between FPO and Agristartups (Chief Guest’s Address)	Webinar
Essential ingredients for agri-startups and agripreneurs	Webinar
R V S Rao	
Enhancement of communication skills	ICAR-IISR, Lucknow.
Motivation	ICAR-IISR, Lucknow.
Leadership development	Faculty development programme
Sanjiv Kumar	
Time Management for Achieving Goals	Webinar as part of NAHEP IDP



Lecture	Programme/ Organized by
How to assess prefeasibility of project through market survey	DST sponsored online Faculty Development Program on “Entrepreneurship in Livestock and Veterinary Sciences
Presentation of data using MS Excel & Tableau Software	Online training program on “Agriculture for Economic Development: Policy Planning, Policy Analysis and Techniques of Economic Analysis
Design Thinking in Livestock Sector	Training program on Innovative Ideas for Entrepreneurship Development in Livestock Sector
ICT Enabled Marketing in AH Sector	Market led extension for the extension officers of Animal Husbandry and Dairy Development
Design Thinking in Business	Guest lecture and webinar series
Design Thinking for Agribusiness and Agri Startups	Online Refresher Training Programme on Transforming Agribusiness into Agri Startups
Digitalization and ICT Application in FPOs in Animal Husbandry	Online training programme on promotion of FPO for officers of Animal husbandry and dairy development departments
S Senthil Vinayagam	
Perspectives of extension management for academic leadership	Off-campus programme on teaching competency enhancement of middle and senior faculty of AAU, Assam on January 19, 2021.
Lessons from ATMA	Online training programme on promotion of innovative extension strategies for effective extension delivery organised by EEI, Hyderabad on January 30, 2021.
Oral communication	University of Agricultural Sciences, Raichur organised online training programme on e-resources, e-learning and IoT on February 15, 2021.
Competency enhancement for digital teaching	University of Agricultural Sciences, Bangalore organised training programme on digital teaching techniques on February 17, 2021.
Entrepreneurship and start-ups	Online training programme on e-resources, e-learning and IoT organised by University of Agricultural Sciences, Raichur on February 18, 2021.
Role of innovators & start-ups in rural transformation	Azadi ka Amrit Mahotsav 2021 organised by NIRD&PR Hyderabad on March 26, 2021.
ICT applications in agriculture	Online refresher course on research management organised by CCSHAU, Hisar, Haryana on April 14, 2021.
Overview of ET in agricultural education	Online training programme on “content development and management” organised by TANUVAS Chennai on April 16, 2021.
Overview of innovations in educational technology	Online training programme on “facilitative teaching technologies organised by TANUVAS, Chennai on May 17, 2021.
Skillset for entrepreneurship	Online sensitization programme for entrepreneurship development in agriculture conducted by Rani Lakshmi Bai Central Agriculture University, Jhansi on June 01, 2021.
Repositioning agricultural extension to the new challenges of DFI	National online training program on agricultural extension strategies for doubling farmers’ income by MANAGE, Hyderabad on June 07, 2021.
Agri-clinics and agri-business centres	Online training programme on entrepreneurship development for agricultural graduates of Sri Konda Laxman Telangana State Horticulture University, Hyderabad on June 18, 2021.
Competency for project appraisal	Online training programme on project management for agricultural projects conducted by MANAGE, Hyderabad on July 16, 2021.

Lecture	Programme/ Organized by
Extension strategies for enhancing market access	Online reforms in agricultural marketing organised by Rani Lakshmi Bai Central Agricultural University, Jhansi on July 28, 2021.
Core competencies in research, teaching, and technology transfer	Online Foundation Course on Research, Education and Extension Management organised by Kerala Agricultural University on 10 August 2021
Entrepreneurial potential of grass-root innovations	TANUVAS Chennai organized an online training programme on intellectual property rights, copyright, industrial design and plagiarism on August 27, 2021.
Overview of educational technology in agricultural education	TANUVAS Chennai organised online training programme on digital content development and management on September 14, 2021.
Youth in agripreneurship for sustainable agricultural development	Annamalai University, Coimbatore organised virtual international conference on “extension management strategies for sustainable agriculture” on September 21, 2021.
Innovations in educational technology	CCS HAU, Hisar conducted refresher course on educational technology on September 23, 2021.
Current trends in extension management	AAU, Assam organised FDP on competency enhancement in agricultural research and education on September 29, 2021.
Current trends in extension management	Off-campus FDP on competency enhancement in agricultural research and education conducted by CAU, Imphal on November 11, 2021.
Strategies for promotion of agri-clinics and agribusiness centre	Dr Y S R Horticulture University, Andhra Pradesh conducted virtual training programme on entrepreneurship development on November 23, 2021.
Flagship programmes of the Ministry of Agriculture & Farmers Welfare	Unnat Bharat Abhiyan (UBA) programme of the Ministry of Education for 30 Regional Coordinating Institutions (RCIs) conducted by NIRD&PR, Hyderabad on December 03, 2021.
S Ravichandran	
Chairman of a technical session on data analytics	National conference of the “Society of Statistics and Computer Applications”.
S K Soam	
AI and IoT in agriculture: capacity building of NARS	International virtual seminar on artificial intelligence in agriculture.
Applications of analytic hierarchy process (AHP) in extension research	KAU- DOE - CTI- MANAGE funded collaborative training on research methods in behavioral sciences for social scientists during from February 02-05, 2021
Intellectual property rights	Webinar on “ethics in agricultural entrepreneurship.
Copyright and geographical indications	Awareness workshop on intellectual property rights in agricultural research.
Intellectual property rights with special reference to copyright and trademark	Webinar on intellectual property rights with special reference to copyright and trademark.
IPR issues	Capacity building programme for all the staff of ICAR-IISR, Lucknow from March 04-06, 2021.



Lecture	Programme/ Organized by
Preparation of winning project proposals	Enhancing the efficiency of Krishi Vigyan Kendras. Organized by Maharana Pratap University of Agriculture & Technology, Udaipur & MANAGE, Hyderabad.
Geographical indicators and its implications for Indian farmers	Training program on WTO and its implications on Indian agriculture organized by MANAGE, Hyderabad.
Agriculture and trade related intellectual property rights	Training program on WTO and its implications on Indian agriculture organized by MANAGE, Hyderabad.
Agricultural GIs in India: role of research organizations in protection and product development	National webinar on IPR & SMEs (special focus on agriculture & food sector): strengthening bonds in prime times"
IP and outcome management	International online training on advances in agripreneurship and skill development for reshaping future of Indian agriculture.
Research methodology and paper writing	National webinar on research methodology and paper writing
IPR and patent filing	Webinar on IPRs and patent filing
Effective online communication techniques during	FDP on effective communication and advanced presentation skills
Surya Rathore	
Vocational scope of home science extension & communication management	National webinar on paradigm shift in home science education during February 18-20, 2021, Home Science Association of India (HSAI), Kerala chapter.
Successful journey of home science extension & communication management	One-day national webinar on inspiring young minds in community sciences: journey of successful personalities, Maharana Pratap University of Agriculture & Technology, Udaipur under IDP, NAHEP.
Enhancing the managerial skills of senior scientists & heads of KVKs for their efficient functioning	Maharana Pratap University of Agriculture & Technology, Udaipur & MANAGE, Hyderabad during April 22-24, 2021.
Role of media in facilitating the unheard farm women voice their concerns	National webinar on facilitating the unheard farm women voice their concerns. MANAGE, Hyderabad.
Social safeguards in institutions	National level capacity building workshop for agricultural LIS professionals under NAHEP (IG) during November 22-27, 2021, PJTSAU, Hyderabad.
Communication skills for personality development	FDP on imparting soft skills for success at workplace from December 27-29, 2021, MPUA&T, Udaipur.
Blue ocean strategy for professional excellence	FDP on imparting soft skills for success at workplace from December 27-29, 2021, MPUA&T, Udaipur.
Gender in agriculture value chain	ATMA directorate of SAMETI, Gandhinagar, SAMETI, Gandhinagar & MANAGE, Hyderabad.
जैविक पोषण वाटिका की उपयोगिता	e-Kisan choupal on climate resilient organic farming, KVK, Turki, Muzaffarpur, RAU, Pusa, Samastipur (Bihar).
Communication skills for career development	Birsa Agricultural University, Ranchi (Bihar).

Lecture	Programme/ Organized by
Applications of ICT in agricultural extension	One-day training on “role of ict in up-scaling extension mechanisms” under NAHEP-IDP Junagarh Agricultural University, Junagarh (Gujarat).
Digital extension in the new normal Post – Covid - 19	Webinar on digital extension the new normal post Covid - 19 Punjab Agricultural University, Ludhiana (Punjab).
Sivaramane N	
Incubation Process for Agripreneurs	Collaborative Online Training Programme on “Promoting Women Agripreneurship through Crop-Livestock-Fisheries Technologies”
V V Sumanth Kumar	
Cloud and AI	Training programme on e-resources, e-learning and IoT for students of UAS, Raichur held during February 15-19, 2021.
AI & cloud in education	Online training program on “e-resources, e-learning and IoT” for faculty of UAS, Raichur from March 23-27, 2021.
Cloud based LMS for education	Online training program on “content development and management” for faculty of TANUVAS, Chennai from April 16-21, 2021.
Cloud based digital teaching	Online training program on “Facilitative Teaching Technologies” by UAS, Raichur on May 21, 2021.
Business plan preparation	Entrepreneurship development for agricultural graduates (for students of Sri Konda Laxman Telangana State Horticulture University) SKLTSHU, Hyderabad, from June 14 –18, 2021.
AI & cloud in education	Online training programme on “e -resources, e-learning and IoT” for faculty of UAS, Raichur during August 23-27, 2021.
Cloud based LMS”	Online evaluation through LMS in the digital content development and management” for TANUVAS faculty, Chennai during September 14-18, 2021.
Vijay Avinashilingam	
Innovative schemes, insurance, credit and government support for farmers	Sustainable farmers friendly transferable technologies to enhance income of farmers in arid zone, ICAR-CAZRI & MANAGE, Hyderabad.

8.2 Publications

8.2.1 Research Papers in Refereed Journals

- Akhila, K., Naga Sindhuja, P.V., Senthil Vinayagam, S., Krishnan, M., Yashavanth, B.S. and Badri Narayanan, G. (2021) Growth, Instability and Trends in Area, Production and Productivity of Coconuts in India. ASERC Journal of Socio-Economic Studies, 4(2), 151-163
- Anu, S.S., Azhar, A., Subhas, P.S., Sathyan, A.R., Vijayan, D., Kaechele, H., Ranjit, K. and Muller, K. (2021). Flood vulnerability and food security in eastern India: A threat to the achievement of the Sustainable Development Goals. International Journal of Disaster Risk Reduction, Vol 66. Published online: 20 Sept 2021. P9. <https://doi.org/10.1016/j.ijdr.2021.102589>
- Bal, S.N., Sandeep, V.M., Vijaya, K.P., SubbaRao, A.V.M., Pramod, V.P., Manikandan, N., Srinivasa Rao, Ch., Singh, N.P. and Bhaskar, S. (2021). Assessing impact of dry spells on the principal rainfed crops in major dryland regions of India. Agricultural and Forest Meteorology, 313, 108768.



- Balakrishnan M., Soam S.K., Sumalatha K. and Ch. Srinivasa Rao (2021). Molecular Modelling and Docking of Tomato Beta Galactosidase enzyme-7 (A2JGX1). *Research Journal of Biotechnology*. 16(1), 36-41.
- Dastagiri, M.B. (2021) "Paradoxical theory of agriculture suffer plenty of production in India: Frame of theory, proof, policy advocacy, and prophecy". *International Journal of Agricultural Economics and Extensions*, 9 (2), PP 1-7.
- Dastagiri, M.B. (2021) Global Farmers Welfare and Neoliberal Agricultural Trade during WTO Regime: Global Trade Analysis, *Agrociencia Journal*, 55(6), 84-115.
- Dastagiri, M.B. and Naga Sindhuja, P.V. (2021) Global agricultural prices and policies during WTO regime: Explorative research to price policy advocacy. *World Food Policy*, <https://doi.org/10.1002/wfp2.12023>.
- Dastagiri, M.B. and Naga Sindhuja, P.V. (2021) Space economy: boon for socio-economic-nature development for future mankind, *International Journal of Current Research*, 13(01), 15438-15448.
- Dastagiri, M.B. and Naga Sindhuja, P.V. (2021). Agriculture trade policy measures and geopolitics in major regional trading blocs: Policy advocacy. *World Food Policy*, <https://doi.org/10.1002/wfp2.12024>.
- Dastagiri, M.B., Rao, S., Mishra, S. and Bhavigna, L. (2021) International Trade Economic Analysis for Indian Agriculture during WTO Regime: Effects on Production, Exports, Imports and Poverty. *European Journal of Applied Sciences*, 9(3), 551-567.
- Dubey, R.K., Dubey, P.K, Chaurasia, R., Srinivasa Rao, Ch. and Abhilash, P.C. (2021). Impact of Integrated Agronomic Practices on Soil Fertility and Respiration on the Indo-Gangetic Plain of North India. *Agronomy*, 11(2), 402. <https://doi.org/10.3390/agronomy11020402>.
- Ganesh Kumar, B., Sivaramane, N. and Srinivasa Rao, Ch. (2020). Finger Millet, the Smallholders' Livelihood and Urban Consumers' Health-food in India, *Productivity*, 61 (2), 206-218.
- Gopi, M., Jeevamani, J.J.J., Goutham, S., Nina-Tabitha, S., Deepak-Samuel, V., Abhilash, K.R., Robin, R.S., Hariharan, G., Muruganandam, R., Krishnan, P., Purvaja, R. and Ramesh, R. (2021) Status of health and conservation classification of tropical coral reefs in Lakshadweep archipelago, *Wetlands. Ecol. Manage.*, <https://doi.org/10.1007/s11273-021-09801-z>.
- Jayaraman, S., Sinha, N.K., Mohanty, M., Hati, K.M., Chaudhary, R.S., Shukla, A.K., Shirale, A.O., Neenu, S., Naorem, A.K., Rashmi, I., Biswas, A.K., Patra, A.K., Srinivasa Rao, Ch. and Ram C. Dalal (2021). Conservation Tillage, Residue Management, and Crop Rotation Effects on Soil Major and Micro-nutrients in Semi-arid Vertisols of India. *Journal of Soil Science and Plant Nutrition*, 21(523-535).
- Kareemulla, K., Krishnan, P., Ravichandran, S., Ganesh Kumar, B., Sharma, S. and Bhatta, R. (2021) Spatiotemporal Analysis of Size and Equity in Ownership Dynamics of Agricultural Landholdings in India Vis-à-Vis the World. *Sustainability*, 13, 10225. <https://doi.org/10.3390/su131810225>
- Karunakaran D. and M. Balakrishnan (2021). Prediction of Fish Production in Tamil Nadu Reservoirs Using Artificial Neural Network (ANN). *International Journal of Advanced Research in Computer and Communication Engineering*. 10(2), 44-52. DOI 10.17148/IJARCCCE.2021.10205.

- Kiruba-Sankar, R., Krishnan, P., George, G., Kumar, K.L., Angel, J., Saravanan, K. and Roy, S.D. (2021). Fisheries governance in the tropical archipelago of Andaman and Nicobar–opinions and strategies for sustainable management. *Journal of Coastal Conservation*, 25(1), 1-13.
- Krishnan, P., Abhilash, K.R., Sreeraj, C.R., Deepak-Samuel, V., Purvaja, R., Anand, A., Mahapatra, M., Sankar, R., Raghuraman, R. and Ramesh, R. (2021). Balancing livelihood enhancement and ecosystem conservation in seaweed farmed areas: A case study from Gulf of Mannar Biosphere Reserve, India, *Ocean and Coastal Management*, 207: 105590.
- Krishnan, P., Praharaj, C.S., Kantharajan, G., Bhoomaiah, D., Sekar, I., Soam, S.K. and Srinivasarao, Ch. (2021). A Scientometric Analysis of Research on Pulses in India During 2000–2017. *Agricultural Research*. <https://doi.org/10.1007/s40003-021-00595-3>.
- Kujur, S., Vinayagam, S.S., Awasthi, H.K. and Murthy, G.R.K. (2020) Teacher-Learner Training Needs Assessment of Agricultural Universities - A Comparative Analysis. *Indian Journal of Extension Education*. 56 (1), 84-87.
- Kumar, M., Mitra, S., Amit, B. and Ramesh Naik, M, (2021) Energy use pattern and carbon footprint of different cropping systems under nutrient and crop residue management practices in eastern Indo-Gangetic plain, *Journal of Environmental Biology*, 42, 1053-1061.
- Kumari, S., Sharma F.L. and Rathore, S. (2021). Construction of knowledge test to measure the knowledge of Rashtriya Krishi Vikas Yojana (RKVY) beneficiary farmers about recommended interventions of maize crop. *Indian Journal of Extension Education*, 57(1), 101-107.
- Kumuri, A., Vinayagam, S. and Akhila, K. (2021) Assessment of learning environment through instructional tools during COVID-19, *International Journal of Education & Management*, 11(1), 45-49.
- Munawar, W. and Vinayagam, S.S., (2020), Technology Enabled Education in Agriculture – An Initiative of MANAGE, *The Andhra Agricultural Journal* 67 (Spl. II), 19-21, 2020.
- Murthy, G.R.K., Seema Kujur, S., Senthil, Vi., Yashavanth, B.S., Srinivasa Rao, Ch., Pandey, P.S., Vanita, J. and Indradevi, T. (2021) Study of Learners' Behavior in Massive Open Online Course using Cluster Analysis, *Indian Research Journal of Extension Education*, 21(4), 30-38
- Murthy, G.R.K., Seema, K., S. Senthil Vinayagam, Yashavanth, B.S., Srinivasa Rao, Ch., Pandey, P.S., Vanita, J. and Indradevi T. (2020). Analysis of learning behavior and pattern of online learners on a MOOC platform, *Pantnagar Journal of Research*, 18(3), 338-346.
- Nataraj, K.C., Balaguravaiah, D., Srinivasa Rao, Ch., Giridhara Krishna, T., Ramu, Y., Kumari, P. (2021). Biochar production through drum method and characterization for soil amendment qualities. *The Pharma Innovation Journal*, 10(6), 544-551.
- Nataraja, K.C., Balaguruvaiah, D., Srinivasa Rao, Ch., Giridhara, K.T., Reddi, R.Y. and Lavanya, K.P. (2021). Comparative analysis of chemical composition and spectral properties of Biochar produced from Pigeonpea and Cotton residues. *J. Res. ANGRAU*, 49(2), 23-34.
- Nikhila, V.A. and Ranjit, K. (2021) Floriculture sector in India: Current status and export potential. *The Journal of Horticultural Science and Biotechnology*. Published online: 27 Mar 2021. P9.
- Payal, D., Senthil Vinayagam, S. and Shrivastava, K.K. (2020). Descriptive Assessment of



- Entrepreneurial Attributes of Women's Groups in Raipur District of Chhattisgarh. *Indian Journal of Extension Education*, Vol. 56(4), 109-114.
- Pratibha, G., Rao, K.V., Srinivas I, Raju, B.M.K., Arun, K., Shanker, Madhavi, M., Indoria, A.K., Srinivasa Rao, M, Keshavamurthy, Sammi Reddy, K., Srinivasa Rao, Ch., Biswas, A.K. and Chaudhari, S.K. (2021) Weed shift and community diversity in conservation and conventional agriculture systems in pigeonpea- castor systems under rainfed semi-arid tropics. *Soil & Tillage Research*, 1-9. <https://doi.org/10.1016/j.still.2021.105075>.
- Ramesh, P. (2020). Personality, Emotional Intelligence and Happiness: A study of Scientists and Non-Scientists. *The IUP Journal of Organizational Behavior*, 19 (4), 22 – 39.
- Rao, R.V.S., Alok, K., Rao, K.H. Yashavanth, B.S., Indu Priya, M., Anwer M.M. and Srinivasa Rao, Ch. (2021) Leadership Styles of the Professionals from the National Agricultural Research and Education System, *Indian Journal of Extension Education*, 57 (1), 105-109.
- Rathore, S., Barghe, H.B. and Kaur, M. (2021) Towards entrepreneurial orientation: Lessons from Nainital farms, *Indian Journal of Extension Education*, 57(1): 115-119.
- Rathore, S., Chandola, M., Raghuvanshi, R., Kaur, M. and Singh, K.V. (2021) Indigenous Pest Management Practices of Indian Hill Farmers: Introspecting their Rationale and Communication Pattern for Secure Ecosystems, *Sustainability*, 13(21), 11608 doi.org/10.3390/su132111608
- Rathore, S., Samala, A., Poonam and Singh, K.V. (2021) 'Digital tools impacting the administrative procedures in Agricultural colleges of northern India, *International Journal of Current Microbiology and applied sciences*. 10(2), 1890-1897.
- Ravi Shankar, K., Nagasree, K., Nirmala, G., Rama Rao, C.A., Raju, B.M.K., Anshida Beevi, C.N., Jagriti Rohit, Pankaj, P.K., Ramana, B.V., Srinivas, I., Vijaya Kumar, S., Sindhu, K. and Srinivasa Rao, Ch. (2020). Farmers' Perceptions, Attitudes and Adaptions towards Climate Change in Selected Semi-Arid Districts of India: Implications from an Adaptation View. *Current Journal of Applied Science and Technology (British Journal of Applied Science & Technology)*. 39(48), 379-395.
- Ravindra, K.P., Senthil Vinayagam, S. and Akhila, K. (2020), Enhancement of Economic Status through Entrepreneurship - A Case Analysis in Honey Processing, *Agricultural Update*, Vol. 15(4), pp: 438-441.
- Satish, S., Ranjit, K. and Koshta, A.K. (2021). Challenges to overcome by the members of the NARMUL dairy cooperative. *Asian Journal of Agricultural Extension, Economics & Sociology*, 39(10), 75-82.
- Seema, K., Murthy, G.R.K., Indradevi, T. and Senthil Vinayagam, S. (2020). Learners' Participation Analysis in MOOCs during COVID-19. *Journal of Global Communication*, 13(2), 55-63.
- Seema, K., Senthil Vinayagam, S., Awasthi, H.K. and Murthy, G.R.K. (2020). Enhancing the Quality of Classroom Teaching by Using Educational Technology Tools, *The Andhra Agricultural Journal*, 67, 1-7.
- Shukla, A.K., Behera, S.K., Prakash, C., Patra, A.K., Srinivasa Rao, Ch., Chaudhari, S.K., Das, S., Singh, A.K. and Green, A. (2021). Assessing Multi-Micronutrients Deficiency in Agricultural Soils of India. *Sustainability*, 13, 9136. <https://doi.org/10.3390/su13169136>.

- Shukla, A.K., Behera, S.K., Prakash, C., Tripathi, A., Patra, A.K., Dwivedi, B.S., Trivedi, V., Srinivasa Rao, Ch., Chaudhary, S.K., Das, S. and Singh, A.K. (2021). Deficiency of phyto-available sulphur, zinc, boron, iron, copper and manganese in soils of India. *Scientific Reports*, 11:19760. <https://doi.org/10.1038/s41598-021-99040-2>
- Shukla, A.K., Behera, S.K., Tripathi, R., Prakash, C., Nayak, A.K., Suresh, K.P., Chitdeshwari, T., Kumar, D., Nayak, R.K., Surendra, B.P., Katkar, R.N., Subbarayappa, C.T., Mahorana, K.C., Patra, A.K., Srinivasa Rao, Ch., Chaudhari, S.K., Subba Rao, A., Singh, A.K. and Das, S. (2021) Evaluation of spatial spreading of phyto-available sulphur and micronutrients in cultivated coastal soils. *PLOS ONE* 16(10): <https://doi.org/10.1371/journal.pone.0258166>
- Singh, N.P., Anand, B., Singh, S., Srivastava, S.K., Srinivasa Rao, Ch., Rao, K.V. and Bal, S.K. (2021). Correction to: Synergies and trade-offs for climate-resilient agriculture in India: an agro-climatic zone assessment. *Climatic Change*, 165(48) <https://doi.org/10.1007/s10584-021-03067-3>.
- Sreekanth, P.D., Krishnan, P., Rao, N.H., Soam, S.K. and Srinivasa Rao, Ch. (2021) Mapping surface-water area using time series lands at imagery on Google Earth Engine: a case study of Telangana, India. *Current Science*, 120(9), 1491-1499.
- Srinivasa Rao, Ch., Ramesh Naik, M., Umesh, H., Ranjit Kumar, G., Surendra Babu, P. and Manasa. (2021). Soil-plant-animal and human health continuum. *Indian Journal of Agronomy*. 66, S118-S129.
- Srinivasa Rao, N., Soam, S.K. and Srinivasa Rao, Ch. (2021). Applications of Artificial Intelligence in Precision Agriculture. *Gradiva review journal*. 7(7), 174-177
- Srinivasa Rao, Ch. (2021) Programmes and Policies for Improving Fertilizer Use Efficiency in Agriculture. *Indian Journal of Fertilizers*, 17(3), 226-254.
- Srinivasa Rao, Ch., Kundu, S., Yashavanth, B.S., S. Rakesh, S., Akbari, K.N., Sutaria, G.S., Vora, V.D., Hirpara, D.S., Gopinath, K., Chary, G.R., Prasad, J.V.N.S., Bolan, N.S., and Venkateswarlu, B. (2021). Influence of 16 years of fertilization and manuring on carbon sequestration and agronomic productivity of groundnut in vertisol of semi-arid tropics of Western India. *Carbon Management*, 12(1), 13-24.
- Srinivasa Rao, Ch., Rakesh, S., Ranjith Kumar, G., Manasa, R., Somashekar, G., Subha Lakshmi, C. and Sumanta K. (2021). Soil degradation challenges for sustainable agriculture in tropical India. *Current Science*, 120(3), 492-500.
- Srinivasa Rao, Ch., Ramesh Naik, M.C., Subha Lakshmi, Ranjit Kumar, G., Manasa, R., Rakesh, S., Sumanta, K. and Prasad, J.V.N.S. (2021). Economic and Environmental Benefits of Integrated Nutrient Management in Indian Agriculture. *Indian Journal of Fertilisers*, 16(11), 1124-1137.
- Srinivasa Rao, Ch., Singh, S.P., Sumanta, K., Vikas, A., Rattan, L., Abhilash, P.C., Chary, G.R., Pravin, B., Thakur, Prasad, J.V.N.S., Venkateswarlu, B. (2021). Integrated nutrient management improves soil organic matter and agronomic sustainability of semiarid rainfed Inceptisols of the Indo-Gangetic Plains. *Journal of Plant Nutrition and Soil Science*. DOI:1-11. 10.1002/jpln.202000312.
- Srinivasa Rao, Ch., Subha Lakshmi, C., Kundu, S., Ranjit Kumar, G., Somashekar, G., Manasa, R., Prasad, J.V.N.S., Narayanaswamy, G., Krishnan, P., Sivaramane, N., Mrunalini, K., Pratibha, G. (2021). Indigenous Technical Knowledge for



- Enhanced Agronomic Productivity and Soil Health of Small Holder Farmers in Tropical India. *Climate Change and Environmental Sustainability*, 9(1), 1-13.
- Srinivas Rao, Ch., Prasad, J.V.N.S., Rao, K.V., Kiran, B.V.S., Ranjit, M., Girija Veni, V., Priya, P., Abhilash, P.C. and Choudhari, S.K. (2021). Land and Water Conservation Technologies for Building Carbon Positive Villages in India. *Land Degradation & Development*. <https://doi.org/10.1002/ldr.4160>
- Tanwar, S., Kaur, M. and Rathore, S. (2021) "Attitude of wheat growers towards Pradhan Mantri Fasal Bima Yojana" *Indian Journal of Agricultural Sciences*, 91 (6), 38-42
- Thammiraju, D., Murthy, G.R.K., Khade, S.B., Padmaja, B., Yashavanth, B.S., Ajay Kumar, S., Soam, S.K. and Srinivasarao, Ch. (2021). Understanding Learner Behaviour in Online Courses through Learning Analytics, *Asian Journal of Agricultural Extension, Economics & Sociology*, 39(10), 381-390.
- Umamaheswari, T., Sugumar, G., Krishnan, P., Ananthan, P.S., Anand, A., Jeevamani, J.J.J., Mahendra, R.S., Amali-Infantina, J. and Srinivasa-Rao, Ch. (2021) Vulnerability assessment of coastal fishing communities for building resilience and adaptation: Evidences from Tamil Nadu, India, *Environmental Science and Policy*, 123, 114-130.
- Venkatesan, P., Sundaramari, M. and Ahire Laxman, M. (2021) Adoption of Rationale Indigenous Crop Protection Measures by "Malayali" Tribes of Kolli Hills, Tamil Nadu. *Journal of Community Mobilization and Sustainable Development*, 16(1), 316-322.
- Venkatesan, P., Sundaramari, M. and Sivakumar, P.S. (2021) Scientific Rationality and Adoption of Indigenous Italian Millet (*Panicum Italicum* L) Cultivation, *Multilogic in Science*, 10(36), 1685-1687.
- Vinayagam, S.S. and Akhila, K. (2021) Educational Management through Digital Teaching in Agricultural Education" published during International Conference of Agricultural Librarians & Users Community (ICALUC-2021) on Management of Knowledge Resource Centres in the Networked Digital Environment: Trends, Challenges and Opportunities held on February 25-26, 2021, pp: 186-188.
- Vinayagam, S.S. and Akhila, K. (2021), Assessment of student's knowledge level on e-learning, e-resources and IoT, *Pantnagar Journal of Research*, 19(1), 77-80.
- Vinayagam, S.S., Akhila, K., Bhat, P.S., Jacob, N.M., Mahesh, A., Manojkumar, T.S. and Venkateswaran, K. (2020), Interdisciplinary approaches for translating rural innovations into agripreneurship, *Agriculture Update*, 15(4), 301-307.
- Yashavanth, B.S., Sanjiv, K. and Sharath Kumar, M.P. (2021) Forecasting Buffalo Milk Production in India: Time Series Approach, *Buffalo Bulletin*, 40(2), 335-343.

8.2.2 Books and Monographs

- Ashok, K.S., Venkatesan, P., Sivaramane, N., Bharat, S.S., Chahal, P.P. and Srinivasa Rao, Ch. (2021) Key Outcome of Farmer FIRST Programme 2016-2020- Transforming the Lives and Livelihood of Indian Farmers, Indian Council of Agricultural Research, New Delhi, p136, ISBN:978-81-7164-225-0.
- Ashok, K.S., Venkatesan, P., Sivaramane, N., Bharat, S.S., Chahal, V.P. and Srinivasa Rao, Ch. (2021) फार्मर फर्स्ट कार्यक्रम 2016-2020 के प्रमुख परिणाम - भारतीय किसानों के जीवन एवं आजीविका का

- रूपांतरण करना, Indian Council of Agricultural Research, New Delhi, p136.
- Meena, V.S., Meena, S.K., Rakshit, A., Stanley, J. and Srinivasarao, Ch. (eds). (2021). Advances in Organic Farming: Agronomic Soil Management Practices. Wood head Publishing Series in Food Science, Technology and Nutrition, Elsevier, p272.
- Rathore, S., Vijayalakshmi, B. and Kumar, V.V. (2021) Digital technologies in Agriculture, Biotech Books, New Delhi. 254p. ISBN 978-81-7622-470-3.
- Singh, A.K., Venkatesan, P., Sivaramane, N., Sontakki, B.S., Chahal, V.P and Srinivasarao, Ch. (2021). Key Outcome of Farmer FIRST Program 2016-2020 – Transforming the Lives and Livelihood of Indian Farmers, Indian Council of Agricultural Research (ICAR), New Delhi, p136
- Srinivasa Rao, Ch., Balakrishnan, M., Krishnan, P., Sumanth Kumar, V.V. (2021) Agricultural Research, Technology and Policy: Innovations and Advances, ICAR – National Academy of Agricultural Research Management (NAARM), Hyderabad, Telangana, India, p442.
- सीएच. श्रीनिवास राव, तच्चा श्रीनिवास, आलोक कु मार, एन. श्रीनिवास राव, आर.वी.एस. राव, पी. कृष्णन एवं एस.के. सोम (2021), जलवायु परिवर्तन और भारतीय कृषि : चुनौतियां अनुकूलन और शमन रणनीतियाँ, भाकृ अनुप-राष्ट्रीय कृषि अनुसंधान प्रबंध अकादमी, हैदराबाद-500 030, तेलंगाना, भारत, ISBN 978-93-5493-809-2, पृष्ठ: 1-257
- National Academy of Agricultural Research Management, Hyderabad, Telangana, India, Pp.191-227.
- Alok, K. and Rao, N.S. (2021). Artificial Intelligence Technologies Driven Smart Agriculture. In Book: Agricultural Science: Research and Reviews (Volume III) (ISBN: 978-93-91768-13-3), Published by Bhumi Publishing, Nigave Khalasa, Kolhapur 416207, Maharashtra, India, pp 23-34.
- Aman Kumar, S. and Dastagiri, M.B. (2021). “International Trading Prices of India’s Cotton: Growth Rates, Elasticities and Foreign Trade Policy”, Texas A&M International University, 25th Annual Western Hemispheric Trade Conference Proceedings, held on April 14-16, 2021 in Laredo, Texas, USA.
- Bharati, P., Ritwika, D., Balakrishnan, M., Supriya, P. and Sumanth Kumar, V.V. (2021) Recent Advances in Proteomics for Crop Improvement. In: Ch. Srinivasa Rao *et al.*, (Eds). Agricultural Research, Technology and Policy: Innovations and Advances, ICAR-National Academy of Agricultural Research Management (NAARM), Hyderabad, Telangana, India, pp.369-386.
- Dastagiri, M.B. (2021) Global Farmers Welfare and Neoliberal Agricultural Trade during WTO Regime: Global Trade Analysis, Texas A&M International University, 25th Annual Western Hemispheric Trade Conference Proceedings, held on April 14-16, 2021 in Laredo, Texas, USA. PP 127-145.
- Ghosh, S., Prasad, J.V.N.S, Ravisankar, N., Ramesh Naik, M. and Srinivasarao, Ch. (2021). Farming System Research for Resource Conservation and Improved Livelihoods in India. In: Srinivasa Rao, Ch. *Et al.*, (Eds.). Agricultural Research, Technology and Policy: Innovations and Advances, ICAR-National Academy

8.2.3 Book Chapters

Abinash, D., Mishra, R., Rani, K., Kundu, S., Jayaraman, S. and Srinivasa Rao, Ch. (2021). Improving Nutrient Use Efficiency: Research, Technology and Policy. In: Srinivasa Rao, Ch. *et al.*, (Eds.). Agricultural Research, Technology and Policy: Innovations and Advances, ICAR-



- of Agricultural Research Management, Hyderabad, Telangana, India, Pp.163-190.
- Krishnan, P. (2021) Bridging Gaps in Policy with Science: Contributions to Fisheries Resource Management, In: Abstracts of Presentations by Fellows Elected (2021), National Academy of Agricultural Sciences. New Delhi, p9.
- Krishnan, P. (2021) De-covidization of (Fisheries) Higher Education: Leveraging Enhanced Readiness to Integrate Digital Technology, Paper presented in Online Seminar on Towards Knowledge Economy and Innovation Society – Role of Professional Educational Institutions in Kerala as part of 55th State Conference organized by Kerala Gazetted Officers Association on 17 June 2021.
- Madhu Babu, K., Ahire, L.M. and Anand Singh, K. (2021) Innovations in Agricultural Extension, Michigan State University Extension, East Lansing, MI 48824. 17-1 to 17.10.
- Meena, R.S., Kumar, S., Sheoran, S., Jhariya, M.K., Bhatt, R., Yadav, G.S. and Srinivas Rao, Ch., and Rattan Lal (2021). Soil Organic Carbon Restoration in India Programs, Policies and Thrust Areas. DOI:10.1201/9781003102762.
- Mrunalini, K., Jayaraman, S., Srinivas Rao, Ch., Praharaj, C.S., Singh, N.P. and Patra, A.K. (2021). Impact of Conservation Agriculture and Residue Management on Soil Properties, Crop Productivity Under Pulse-Based Cropping Systems in Central India. In: S. Jayaraman et al. (eds.) Conservation Agriculture: A Sustainable Approach for Soil Health and Food Security, Springer Nature Singapore Pte. Ltd. DOI: 10.1007/978-981-16-0827-8_6, Pp117-137.
- Patel, A., Ajaykumar, K., Kumar, M., Rao, K.V. and Srinivasa Rao, Ch. (2021). Sustainable Land and Water Management for Reducing Soil Erosion in Tropical India. In: Srinivasa Rao, Ch. et al., (Eds.). Agricultural Research, Technology and Policy: Innovations and Advances, ICAR-National Academy of Agricultural Research Management, Hyderabad, Telangana, India, Pp.333-347.
- Rathore, S. (2021) Digital tools and their evaluation for impactful learning. In Digital technologies in Agriculture (Eds.) Surya Rathore, Vijayalakshmi, B. and Sumanth Kumar, V.V., Biotech Books, New Delhi. Pp. 235-249.
- Rathore, S. and Kumar, V.V.S. (2021) Cyber extension tools and models In Digital technologies in Agriculture (Eds.) Surya Rathore, Vijayalakshmi, B. and Sumanth Kumar, V.V., Biotech Books, New Delhi. Pp. 9-17.
- Ravikishore, M., Supriya, P. and Johnson, M. (2021). Gender Mainstreaming: Unlocking the Potential of Women in Agriculture and Allied Sectors In Anitha Kumari, Letha Devi G, Ashaletha S, Adhiguru P (Eds.), Gender and Agriculture: An Indian Perspective. pp. 31-47.
- Shravya, K. and Dastagiri, M.B. (2021). "International Trading Prices of India's Ginger: Growth Rates, Elasticities and Foreign Trade Policy", Texas A&M International University, 25th Annual Western Hemispheric Trade Conference Proceedings, held on April 14-16, 2021 in Laredo, Texas, USA. PP 19-28.
- Shweta, B., Jamaludheen, A., Biswajit, S., Suresh, A., Krishnan, P. (2021). Agricultural research policy for food security & sustainability in India. In: Ch. Srinivasa Rao et al., (Eds.). Agricultural Research, Technology and Policy: Innovations and Advances, ICAR-National Academy of Agricultural Research Management (NAARM), Hyderabad, Telangana, India, pp.271-290.
- Singh, K.K., Srinivasa Rao, Ch., Singh, V.P. and Pathak, A.D. (2021) (Hindi). Jalvayu samrudhyansheel dalhani fasalei. In:

- Srinivasarao, Ch. *et al.*, (Eds.). *Jalvayu Parivarthan Aur Bharatiya Krishi: Chunoutiya, Anukulan aur Shaman Rananeetiya*, ICAR-National Academy of Agricultural Research Management, Hyderabad-500030, Telangana, India. Pp29-48.
- Sneha, M., Soumya, S., Balakrishnan, M., Sumanth Kumar, V.V., Supriya, P. and Srinivasa Rao, Ch. (2021). Recent Advances in Proteomics for Crop Improvement. In: Ch. Srinivasa Rao *et al.*, (Eds.). *Agricultural Research, Technology and Policy: Innovations and Advances*, ICAR-National Academy of Agricultural Research Management (NAARM), Hyderabad, Telangana, India, pp.387-410.
- Srinivasarao, Ch., Krishnan, P. and Ragupathy, V. (2020). Science-led Agricultural Development. In: *Gandhian Thoughts. Mahatma Gandhi's Vision of Agriculture: Achievements of ICAR*. Indian Council of Agricultural Research, 2020. New Delhi. p228.
- Srinivasarao, Ch., Kundu, S., Rakesh, S., Lakshmi, C., Kumar, R., Manasa, R., Somashekar, G., Swamy, G.N., Muranilini, K., Jayaraman, S., Mohanty, M., Venkatesh, G., Pratibha, G. and Prasad, J.V.N.V. (2021). Managing Soil Organic Matter under Dryland Farming Systems for Climate Change Adaptation and Sustaining Agriculture Productivity, In: *Advances in Soil Science - Soil Organic Carbon and Feeding the Future* (Ed. Rattan Lal), CRC Press, Taylor and Francis Group, DOI: 10.1201/9781003243090-10. Pp.219-251.
- Srinivasa Rao, Ch., Somasekhar, G., Prasad, J.V.N.S. and Chaudhary, S.K. (2021). (Hindi). *Bharat mei jalvayu gaav: Jalvayu parivarthan ke anukulan evam prashamn ke liye bhavishya*. In: Srinivasarao, Ch. *Et al.*, (Eds.). *Jalvayu Parivarthan Aur Bharatiya Krishi: Chunoutiya, Anukulan aur Shaman Rananeetiya*, ICAR-National Academy of Agricultural Research Management, Hyderabad-500030, Telangana, India. Pp11-28.
- Sudhakara, N.R., Yadav, B., Mohanty, M., Pratibha, G. and Srinivasa Rao, Ch. (2021). Soil Management for Mitigating Greenhouse Gas Emissions in Agricultural Sectors. In: Srinivasa Rao, Ch. *et al.*, (Eds.). *Agricultural Research, Technology and Policy: Innovations and Advances*, ICAR-National Academy of Agricultural Research Management, Hyderabad, Telangana, India, Pp.229-269.
- Supriya, P., Balakrishnan, M., Soam, S.K. and Srinivasa Rao, Ch. (2021) Trends and Application of Data Science in Bioinformatics. In: Rautaray SS, Pemmaraju P, Mohanty H (eds) *Trends of Data Science and Applications. Studies in Computational Intelligence*, Vol 954. Springer, Singapore. https://doi.org/10.1007/978-981-33-6815-6_12.
- Vijay, A., Meena, P.C., Sumant Kumar, V. and Srinivasa Rao, Ch. (2021). Critical Needs of Social Science Based Capacity Building for Sustainable Agriculture Development in India, In *Current Trends in Agricultural Sciences*, a Pre-Congress Publication of Indian Social Science Congress at Vidisha, Madhya Pradesh, March 16-19, 2021.
- आलोक कुमार, तच्चा श्रीनिवास एवं सीएच. श्रीनिवास राव (2021) जलवायु परिवर्तन में कृषि प्रसार की भूमिका व प्रतिबद्धता इन: सीएच. श्रीनिवास राव एवं साथी (संपादित) *जलवायु परिवर्तन और भारतीय कृषि: चुनौतियाँ, अनुकूलन और शमन रणनीतियाँ*, भाकृअनुप-राष्ट्रीय कृषि अनुसंधान प्रबंध अकादमी, हैदराबाद -500 030, तेलंगाना, भारत, पृष्ठ: 209-225.
- एन.एस. राव, चंदन कुमार देब एवं सुबीश ए (2021), जलवायु परिवर्तन से होने वाली पादप घटनाओं (फंक्शनलॉजी) में परिवर्तन हेतु कृत्रिम बुद्धिमत्ता (आर्टिफिशियल इंटेलिजेंस) आधारित मॉडलिंग इन: सीएच. श्रीनिवास राव एवं साथी (संपादित) *जलवायु*



परिवर्तन और भारतीय कृषि: चुनौतियाँ, अनुकूलन और शमन रणनीतियाँ, भाकृ अनुप-राष्ट्रीय कृषि अनुसंधान प्रबंध अकादमी, हैदराबाद -500 030, तेलंगाना, भारत, पृष्ठ: 189-198.

एस.के. सोम, बी.एस. यशवंत एवं सीएच. श्रीनिवास राव (2021). भारत में जलवायु परिवर्तन एवं इससे बचाव की रणनीतियों पर किसानों की अनुभूति इन: सीएच. श्रीनिवास राव एवं साथी (संपादित) जलवायु परिवर्तन और भारतीय कृषि: चुनौतियाँ, अनुकूलन और शमन रणनीतियाँ, भाकृ अनुप-राष्ट्रीय कृषि अनुसंधान प्रबंध अकादमी, हैदराबाद -500 030, तेलंगाना, भारत, पृष्ठ: 227-236.

तत्त्वा श्रीनिवास, आलोक कुमार एवं सीएच. श्रीनिवास राव (2021) जलवायु परिवर्तन के सामाजिक-आर्थिक प्रभाव इन: सीएच. श्रीनिवास राव एवं साथी (संपादित) जलवायु परिवर्तन और भारतीय कृषि: चुनौतियाँ, अनुकूलन और शमन रणनीतियाँ, भाकृ अनुप-राष्ट्रीय कृषि अनुसंधान प्रबंध अकादमी, हैदराबाद -500 030, तेलंगाना, भारत : 199-208.

पी. सुप्रिया, एम. बालाकृष्णन, एस.के. सोम, सतेन्द्र के. मंगरोथिया एवं सीएच. श्रीनिवास राव (2021) कृषि में जलवायु परिवर्तन के लिए जैव सूचनाप्रणाली अनुप्रयोग: संभावनाएँ एवं चुनौतियाँ इन: सीएच. श्रीनिवास राव एवं साथी (संपादित) जलवायु परिवर्तन और भारतीय कृषि: चुनौतियाँ, अनुकूलन और शमन रणनीतियाँ, भाकृ अनुप-राष्ट्रीय कृषि अनुसंधान प्रबंध अकादमी, हैदराबाद -500 030, तेलंगाना, भारत, पृष्ठ: 181-188.

राज कुमार योगी, ए. मोहनासुन्दरम, मो. मोनोबरुल्लाह, राजन चौधरी, आलोक कुमार, निर्मल कुमार एवं केवल कृष्ण शर्मा (2021) भारत में प्राकृतिक राल एवं गोदों का उत्पादन एवं जलवायु परिवर्तन: वर्तमान स्थिति, चुनौतियाँ एवं संभावनाएँ इन: सीएच. श्रीनिवास राव एवं साथी (संपादित) जलवायु परिवर्तन और भारतीय कृषि: चुनौतियाँ, अनुकूलन और शमन रणनीतियाँ, भाकृ अनुप-राष्ट्रीय कृषि अनुसंधान प्रबंध अकादमी, हैदराबाद -500 030, तेलंगाना, भारत, पृष्ठ: 53-68.

सीएच. श्रीनिवास राव, तत्त्वा श्रीनिवास, आलोक कुमार, एन. श्रीनिवास राव, आर.वी.एस. राव, पी. कृष्णन एवं एस.के. सोम (2021) जलवायु परिवर्तन के अनुकूलन एवम शमन के लिए अग्रसारित दिशा, इन: सीएच. श्रीनिवास राव एवं साथी (संपादित) जलवायु परिवर्तन

और भारतीय कृषि: चुनौतियाँ, अनुकूलन और शमन रणनीतियाँ, भाकृ अनुप-राष्ट्रीय कृषि अनुसंधान प्रबंध अकादमी, हैदराबाद -500 030, तेलंगाना, भारत, पृष्ठ: 249-257.

8.2.4 Popular and Newspaper Articles

Deepranjan, S., Soam, S.K., Rakesh, S. and Amitava, R. (2021). Land Restoration during the UN Decade on Ecosystem. Agriculture World, Vol 7, Issue 07, July 2021, pp 8-11. ISSN: 2455-8184

Mooventhana, P., Singh, S.R.K., Venkatesan, P. and Singh, U. (2021), Waste decomposer to improve soil and plant health: a success story from the tribal belt of Chhattisgarh under the "Farmer FIRST" initiative, ICAR-IISS E-M@GAZINE, Harit Dhara, Vol: 4 Issue 1, 7-9

Ramesh Naik, M. (2021). Hydroponics: The Future of Farming. Agriculture Letters. 2 (7): 15-17.

Ramesh, N.M. and Hemalatha. (2021) Precision Nitrogen Management Using Leaf Colour Chart. Chronicle of Bioresource Management. 5(2), 059-062.

Singh, K.V. and Rathore, S. (2021) IOT Use in Agriculture: Solution to Farmers' problems. Agrobios Newsletter, 19 (8), 103-104.

Soam, S.K. (2021). 3D Printing: Scope and Potential in Agribusiness. Agriculture Today, Vol XXIV, Issue 11, November 2021, 56-57. Published by M/S Concept Agrotech Consultants Ltd, New Delhi- 110001.

Soam, S.K., Raghuvanshi, R. and Raghupathi, B. (2021). Agricultural Higher Education in India: Status and Scope. Just Agriculture e-magazine. Vol 2(1): pp 25-29, e-ISSN: 2582-9149.

Srinivas Rao, Ch. (2021) Climate resilient agriculture systems: The way ahead. Down to Earth. Fortnightly Magazine. <https://www.downtoearth.org.in/blog/agriculture/climate-resilient-agriculture-systems-the-way-ahead-75385>.

- Sumanth, V.V., Rakesh, S., Soam, S.K., Sudeep, M. and Prabhat, K. (2021). Agricultural Students' Alumni Network Implemented by NAHEP. In Magazine- AGRICULTURE WORLD: India's Food Systems Approach to be Reimagined. November issue. 7(11): 16-19. ISSN 2455-8184. Published by Krishi Jagran Media Group, New Delhi- 110 016.
- Sweety, S., Srikanth, P.D. and Soam, S.K. (2021). Horticultural Education in Entrepreneurship Development. Just Agriculture, 2(3), Nov 2021: 1-7 e-ISSN: 2582-8223
- Vijay Avinashilingam, N.A., Ola, C.M. and Pratibha, T. (2021) Oyster mushroom se kisan ki aay vridhhi, Phal Phool, ICAR Publications, 42(4), 27 – 28
- कुमुद शुक्ला, मनमीत कौर, सूर्या राठौर एवं दीक्षा शर्मा “बदलते परिवेश में जैविक प्रमाणीकरण की अनिवार्यता” अभिनव कृषि, वर्ष ३, अंक २, जून २०२१, कृषि विश्वविद्यालय, कोटा, ३६-३८
- डी थम्मिराजु, श्रीकांत खाडे, स्वीटी शर्मा एवं एस के सोम (2021). कोविड-19 महामारी का पशुधन क्षेत्र पर प्रभाव एवं न्यू नार्मल के लिये पहल. In Magazine- पशुधन प्रकाश के बारहवाँ अंक वर्ष 2021. प्रष्ठ संख्या (Page no) 40-42. ISSN: 0976-4569. भाकृअनुप-राष्ट्रीय पशु आनुवंशिक संसाधन ब्यूरो, करनाल द्वारा प्रकाशित.
- सूर्या राठौर और मनमीत कौर (२०२१) “स्वयं सहायता समूह: महिला सशक्तिकरण का माध्यम” राजस्थान खेती, अंक-११, मई, पृष्ठ २५-२६.
- सूर्या राठौर और ममता मीणा ‘महिलाएँ आत्मनिर्भर कैसे हो? सितम्बर २०२१, अंक - ३, प्रसार दूत, भारतीय कृषि अनुसन्धान संस्थान, कृषि तकनीकी सूचना केंद्र, नई दिल्ली पृष्ठ ३५-३६।
- 8.2.5 Training Manuals/ Technical Bulletins**
- Balakrishnan, M. (2021). Training Manual Application of Bioinformatics in Agricultural Research and Education. pp.1-277.
- Balakrishnan, M., Sumanth Kumar, V.V. and Krishnan, P. (2020). Training Manual for the Training Programme on Foundation courses for Agricultural Research Services pp.1-30.
- Bharat, S. and Sontakki. (2021). Soft Skills for Extension Professionals: Resource Material for Online Training Programme organized during 21-30 October 2021, ICAR-NAARM, Hyderabad. p135.
- Jat, N.K., Vijay Avinashilingam N. A., P.R. Meghwal, A K Patel, V P Chahal and C.M.Ola (2021). Sustainable livelihood intervention for enhancing small land holder's income in western Rajasthan. Indian Farming, 71(10):54-57.
- Krishnan, P., Kareemulla, K., Sekar, I., Soam, SK., Srinivasarao, Ch., Rajagopalan, V. (2020) CapacityBuildingofSeniorOfficialsfromCentral and State Governments on Mainstreaming Biodiversity in Agriculture Sector – Trainers' Guide, Ministry of Environment, Forest and Climate Change (MoEFCC), United Nations Development Program (UNDP) and ICAR-National Academy of Agricultural Research Management (NAARM), p64.
- Krishnan, P., Kareemulla, K., Sekar, I., Soam, SK., Umesh, H., Ramesh-Nayak, M., Srinivasarao, Ch., Rajagopalan, V. (2021) SNRM: A Framework for Mainstreaming Biodiversity Conservation in the Development Schemes of the Agriculture Sector in India, Ministry of Environment, Forest and Climate Change (MoEFCC), United Nations Development Program (UNDP) and ICAR-National Academy of Agricultural Research Management (NAARM), p166.
- Murthy, G.R.K., (2021). Training Manual for the Digital Content Development and Management. 1-149.



Murthy, G.R.K., Senthil Vinayagam, S. and Balakrishnan, M. (2021). Training Manual for the E-Resources E-Learning and IoT. 1-155.

Rathore, S., Soam, S., Raju, D.T.R., Kumar, A., Rao, N. S. and Rao, Ch.S.R. (2021). Preparing Agri graduates beyond curriculum: new dimensions in Human Resource Development, ICAR – National Academy of Agricultural Research Management, Hyderabad, India.

Ravichandran, S. and Sivaramane, N. (2021). "Analysis of Experimental Data using Statistical Analysis System (SAS)", pp.271

Sumanth Kumar, V.V. (2021). Manual on Facilitative Teaching Technologies. ICAR-National Academy of Agricultural Research Management, Hyderabad. Pp. 1-117.

Thammi Raju, D., Yashavanth, B.S., Sweetey, S. and Soam, S.K. (2020). Agricultural Education: Career Opportunities. ICAR–National Academy of Agricultural Research Management, Rajendranagar, Hyderabad, Telangana, India, pp 38.

Vinayagam, S.S. (2021). Training Manual for Facilitative Teaching Technologies for TANUVAS, Chennai. 1-155.

Vinayagam, S.S. (2021). Training Manual for the Innovations in Educational Technology. 1-118.

Yogi RK, Kumar Alok and Singh AK 2020. Lac., Plant resins and gums statistics: At a Glance. ICAR-Indian Institute of Natural Resins and Gums, Ranchi (Jharkhand), India. Bulletin (Technical). No. 5/2020. 1-74 pp.

नार्म, 2021. एनुअल रिपोर्ट (वार्षिक प्रतिवेदन) (2020), भाकृअनुप-राष्ट्रीय कृषि अनुसंधान प्रबंध अकादमी, राजेन्द्रनगर, हैदराबाद - 500 030

8.2.6 Papers Presented in Conferences/ Seminars

Krishnan, P. (2021) De-covidization of (Fisheries) Higher Education: Leveraging Enhanced Readiness to Integrate Digital Technology, Paper presented in Online Seminar on Towards Knowledge Economy and Innovation Society – Role of Professional Educational Institutions in Kerala as part of 55th State Conference organized by Kerala Gazetted Officers Association on 17 June, 2021.

Krishnan, P. (2021). Bridging Gaps in Policy with Science: Contributions to Fisheries Resource Management, In: Abstracts of Presentations by Fellows Elected (2021), National Academy of Agricultural Sciences. New Delhi, p9.

Seema, K., Indradevi, T., Murthy, G.R.K., Vinayagam, S.S. and Akhila, K. (2021). "Knowledge Management Opportunities through MOOCs" published during International Conference of Agricultural Librarians & Users Community (ICALUC-2021) on Management of Knowledge Resource Centres in the Networked Digital Environment: Trends, Challenges and Opportunities held on February 25-26, 2021, 32- 37.

Sontakki, Bharat, S. (2021, December). Transformation of Agricultural Extension: Research Needs and Options. Lead paper presented in the National Conference on Transformation of Agricultural Extension-Strategies for Effective Reformation organized by SV Agriculture College (ANGRAU), Tirupati in virtual mode during 22-23 December.

Sontakki, Bharat, S. and Akhila, S. (2021, October). Extension for Risk Management in Agriculture-Need of the hour. Lead paper presented at the National Seminar on Transforming Indian

Agriculture through Pluralistic & Innovative Extension Approaches for Self Reliant India. Organized by the Indian Society of Extension Education, New Delhi; Banaras Hindu University, Varanasi; Banda University of Agriculture and Technology, Banda, and Global Communication Research Association, Paraguay during 4-6 October 2021 at BHU, Varanasi.

Venkatesan, P. (2021). "Farmer FIRST Programme: An Approach to improve the rural economy" in Indian Society of Extension Education (ISEE) National seminar on Transforming in Indian Agriculture through floristic and innovative extension approaches for self-reliant India, organized by Banaras Hindu University, Varanasi during 4-6 October.

Venkatesan, P. (2021). Presented Lead Paper on "Farmer FIRST Approach: Towards Sustainability of Agricultural Extension" in International Virtual Conference on "Transforming Agricultural Advisory Services to mitigate the effects of pandemic for farmers' welfare (ICTAAS-2021)" organized by Vellore Institute of Technology, TANUVAS, Chennai, Bulehora University, Ethiopia and Al Neelein University Sudan on 12.11.2021.

Vinayagam, S.S. and Akhila, K. (2021) "Educational Management through Digital Teaching in Agricultural Education" published during International Conference of Agricultural Librarians & Users Community (ICALUC-2021) on Management of Knowledge Resource Centres in the Networked Digital Environment: Trends, Challenges and Opportunities held on February 25-26, 2021, 186-188.

Yashavanth, B.S. (2021). An application of topic modelling for discovering themes in the

queries raised at farmers' call center" during 23rd Annual Conference of SSCA (online) on "Visionary Innovations in Statistical Theory and Applications (VISTA-2021)" during 24 – 28 February, 2021 at ICAR - National Academy of Agricultural Research Management (ICAR - NAARM), Hyderabad.

8.2.7 Research Project Report and Working Papers

Kareemulla, K. and Ravichandran, S. (2021). DST revised project report on "Agricultural Research and Development Infrastructure in Andhra Pradesh and Telangana".

Krishnan, P., Kareemulla, K., Sekar, I., Soam, S.K., Umesh, H., Ramesh-Nayak, M., Srinivasarao, Ch., Rajagopalan, V. (2021). SNRM: A Framework for Mainstreaming Biodiversity Conservation in the Development Schemes of the Agriculture Sector in India. Ministry of Environment, Forest and Climate Change (MoEFCC), United Nations Development Program (UNDP) and ICAR-National Academy of Agricultural Research Management (NAARM), p166.

Krishnan, P., Kareemulla, K., Sekar, I., Soam, S.K., Srinivasarao, Ch., Rajagopalan, V. (2020). Capacity Building of Senior Officials from Central and State Governments on Mainstreaming Biodiversity in Agriculture Sector – Trainers' Guide. Ministry of Environment, Forest and Climate Change (MoEFCC), United Nations Development Program (UNDP) and ICAR-National Academy of Agricultural Research Management (NAARM), 64.

Ravichandran, S. and Sivaramane, N. (2021). "Farmerless Farming – Myth or a reality?"

8.2.8 Policy Briefs

Krishnan, P., Ananthan, P.S., Soam, S.K., Parsad, R., Srinivasarao, Ch. (2021). Ease of Doing



Research (EoDR): A Methodological Framework for Agricultural Research Organizations, Indian Council of Agricultural Research (ICAR), New Delhi. 30.

Krishnan, P., Kareemulla, K., Sekar, I., Soam, S.K., Srinivasarao, Ch. and Rajagopalan, V. (2021). Mainstreaming Biodiversity in Public Finance - A Case for Agriculture Sector in India, Ministry of Environment, Forest and Climate Change (MoEFCC), United Nations Development Program (UNDP) and ICAR-National Academy of Agricultural Research Management (NAARM), 20.

Samra, J.S. and Srinivasarao, Ch. (2021). Circular Carbon Economy in India: Efficient Crop Residue Management for harnessing Carbon, Energy and Manure with Co-benefits of Greenhouse Gases (GHGs) Emissions Mitigation. Policy

Paper, ICAR-National Academy of Agricultural Research Management, Hyderabad. 20.

Srinivasa Rao, Ch. and Singh, A.K. (2021). Strategies for Enhancing Soil Organic Carbon for Food Security and Climate Action, NAAS Policy Paper 100.

Thammi Raju, D., Soam, S.K., Srinivasa Rao, N., Kumar, A., Sumanth Kumar, V.V., Kumar, S., Rathore, S., Vinayagam, S.S., Balakrishnan, M., Yashavanth, B.S., Krishnan, P., Sudeep Marwaha., Prabhat, K., Venkateshwarlu, G., Srinivasa Rao, Ch. and Agrawal, R.C. (2021). Curriculum Development Framework for Agricultural Education, National Agricultural Higher Education Project – Component 2, Indian Council of Agricultural Research (ICAR). National Academy of Agricultural Research Management (NAARM), Hyderabad-12.

8.2.9 Copyrights

Title	Registration Number & Date	Authors
Multifunctional technology enhanced learning centre	2664/2020-CO/SW	G R K Murthy, D Thammi Raju, S Senthil Vinayagam, Ch Srinivasa Rao, S K Soam, P S Pandey and Vanita Jain
Comprehensive management of massive open online course (MOOC) through moodle	L-97699/2020	G R K Murthy, D Thammi Raju, S Senthil Vinayagam, Ch Srinivasa Rao, S K Soam, P S Pandey and Vanita Jain
Impact of massive open online courses (MOOCs) on capacity building	L-102452/2021	D Thammi Raju, G R K Murthy, S Senthil Vinayagam, M Krishnan and Ch Srinivasa Rao
Vinutna: a holistic digital ecosystem for foundation courses	24845/2021-CO/SW) Filing Date: 13 th October, 2021	V.V. Sumanth Kumar, M Balakrishnan, P Krishnan, S K Soam and Ch Srinivasa Rao
Methodological framework of spatial decision support system for managing agricultural experimental farms	SW-14613/2021 07/06/2021	P D Sreekanth, S K Soam and Ch Srinivasa Rao
Research survey management system	SW -14683/2021	P Krishnan, Bharath Kumar, B Ganesh Kumar, M Balakrishnan, V.V. Sumanth Kumar, S K Soam and Ch Srinivasa Rao
Digital technologies in agriculture[book]	ISBN 978-81-7622-470-3 dated Jan 2021	Surya Rathore, B Vijayalakshmi and V.V. Sumanth Kumar

Title	Registration Number & Date	Authors
AGRI-PRIORITIZE [mobile app]	SW-14949/2021 and 29/09/2021	N Sandhya Shenoy, P Venkatesan, B Vijayalakshmi., V K Jayaraghavendra Rao, Bharat S Sontakki, Surya Rathore, S K Soam and Ch Srinivasa Rao
SQI CAL	SW-14175/2021	M Mohanty, N S Sinha, R N Sahoo, H Yadav, M Prabhakar, K Sammi Reddy, G Ravindra Chary, A K Patra, S K Chaudhari, and Ch Srinivasa Rao
Low cost portable kiln for biochar production	L-100062/2021.	G Venkatesh, B Venkaeswarlu, K A Gopinath, Ch Srinivasa Rao, G R Korwar and B Sanjeeva Reddy
Prototype for Socio-economic vulnerability index (SVI): national and regional assessment framework.	14649/2021.	P Krishnan, Bharat Kumar, I Sekar, S K Soam and Ch Srinivasa Rao

9.1 Budget Allocation and Expenditure during Jan-Dec 2021 (Rs. in lakhs)

Sl. No.	Head of Account	Budget	Expenditure
A. Grant in Aid - Capital			
1	Works	161.39	50.54
2	Equipments	71.73	60.94
3	Information Technology	22.04	10.59
4	Library Books & Journals	22.50	0.00
5	Vehicles & Vessels	0.00	0.00
6	Furniture & Fixtures	31.59	21.60
Total Capital		309.25	143.67
Grants in Aid - Salaries (REVENUE)			
1	Establishment Expenses		
2	i. Establishment Charges	2163.71	2115.86
3	ii. Wages	146.68	86.66
4	iii. OTA	0.25	0.13
Total Salaries		2310.64	2202.65
B. Grant in Aid - General (REVENUE)			
1	B. Pension & Other Retirement Benefits	3429.11	3429.11
2	A. Domestic TA / Transfer TA	38.62	18.57
3	Research & Operational Expenses	187.35	187.35
4	Administrative Expenses	2399.39	2399.39
5	A.HRD	38.62	1.35
6	Miscellaneous Expenses	203.13	124.96
Total Grants in Aid - General		6296.22	6160.73
Grand Total (Capital + Salaries + Revenue)		8916.11	8507.05

9.2 Resource Generation

9.2.1 Sponsored Research Projects Budget (Rs. in lakhs)

Sl. No.	Name of the Project	Fund received	Expenditure
1	Agril.Res.&Dev.Infra.in Select Indian States under NSTMIS Scheme of DST	1.16	1.16
2	IFPRI Sponsored Project - ASTI in Indian NARES- ASTI Project	7.78	7.78
3	ICAR-NAARM-NIPHM Project "Plant Biosecurity through MOOC Platform"	1.06	0
4	ICAR-NAARM-MANAGE Project "Agri-Warehousing Management"	5.80	1.40
5	ICAR-NAARM-PJTSAU Digital Content for MOOCS	3.75	1.40
6	Profiling the Resources Devoted in Agricultural R&D Dev. of National framework for Evidence-based S&T Governance in India Scheme of DST	0.43	0.43

Sl. No.	Name of the Project	Fund received	Expenditure
7	Skill Development in Bioinformatics Infrastructure Facility on Agril. Bio Technology under DBT	12.16	0.75
8	Adoption of Natural Farming and its effect on Crop Yield and Farmers Livelihood in India NITI Ayog Project	12.42	12.42
9	ICAR-NAARM-NDRI, Karnal for developing MOOC Courses on Commercial Dairy farming & Milk processing & Value Edition	4.58	3.67
10	ICAR-NAARM-NIPHM Project developing online course on "Rodents & Household Pest Management"	1.13	0.46
11	IFPRI Sponsored Project-Developing a National Dashboard for Evidence-Based Governance of Agril. Res. India	14.96	14.96
12	ICAR-NAARM-BCIL, New Delhi " Main Streaming Biodiversity Conservation in the Development Schemes of Agricultural Sector in India" (SNRM)	9.11	9.11
13	ICAR-NAARM-BCIL, New Delhi "Management Development Programme on Orientation cum awareness and Implementation of ABS guidelines (SNRM)	7.96	7.96
14	DST-NSTMIS Project "Profiling Resources Devoted to R&D in Agriculture Sector"	14.00	0.00
15	International Consultancy Project -ICAR-NAARM-GIZ "Developing and Pilot testing Specialized Knowledge Products on Human Wildlife Conflict (HWC) Mitigation for Agriculture and Veterinary Sector in India	31.93	0.00

9.2.2 Off-Campus and Sponsored Programmes from Jan-2021 to Dec 2021 (Rs. in lakhs)

Sl. No.	Name of the Programme	Total Budget received
1	M/s. Coromandal International Ltd., CIL-NAARM Collaborative Training Programme for NEAT-II from 11-26 th September, 2020 & NEAT-I from 15-19 th February, 2021	7.49
2	M/s. Coromandal International Ltd., CIL-NAARM Collaborative Training Programme on MORE 16-20 th November, 2020	5.65
3	E-Resources, E-Learning & IoT for students & Faculty of UAS, Raichur 15-19 th February 21 & 23-27 th March, 2021	3.43
4	Online Training Programme on "Intellectual Property Rights, Copy Right, Industrial design and Plagiarism" for staff of TANUVAS, Chennai from 22-26 th March, 2021	1.77
5	Online Training Programme on "Content Development and Management" for staff of TANUVAS, Chennai from 16-21 st April, 2021	1.77
6	Online Training Programme on "Facilitative Teaching Technologies" for staff of TANUVAS, Chennai from 17-21 st May, 2021	1.77
7	Online Training Programme on Advanced Bio-Informatics tools and its applications in Agril. For the Faculty of SVPUAT, Meerut from 22-24 th February, 2021	0.99
8	Kerala University of Fisheries & Ocean Studies, Kerala	2.64
9	Design Thinking in Research & Education for faculty members of RAJUVAS, Bikaner (2 Programmes)	2.95



Sl. No.	Name of the Programme	Total Budget received
10	M/s. Coromandal International Ltd., CIL-NAARM Collaborative Training Programme for NEAT-II (8-12 th Feb. 21) & NEAT-II (18-22 May, 21)	6.40
11	Induction training for newly appointed scientists of ICFRE-FRI, Dehradun	2.25
12	Virtual Training Programme for Students on “Entrepreneurship Development for Agriculture students of SKLTSHU, College of Horticulture, R. Nagar from 14-18 th June, 2021	1.73
13	For conducting of 5 Training programmes for Teaching & Non-Teaching staff of Dr. YSR Horticulture University, Venkataramanagudem, A.P.	17.50
14	Online Training Programme On “E-Resources, E-Learning & IoT” for faculty of UAS, Raichur from 23-27 th August, 2021	1.83
15	Online Training Programme on “Intellectual Property Rights, Copy Right, Industrial design and Plagiarism” for staff of TANUVAS, Chennai from 24-28 th August, 2021	1.79
16	Online Training Programme on “Content Development and Management” for staff of TANUVAS, Chennai from 14-18 th September, 2021	1.79
17	Online Training Programme on “Facilitative Teaching Technologies” for staff of TANUVAS, Chennai from 26-30 th October, 2021	1.80
18	Short Certificate Course on E-Resources, E-learning and IoT for the faculty of UAS, Raichur from 13-17 th December, 2021	3.60

9.2.3 Resource Generation from Educational Programmes (Rs. in lakhs)

Sl. No.	Educational Programmes	Receipts
1	PGDMA 2019-21	44.78
2	PGDM (ABM) 2020-22	225.44
3	PGDM (ABM) 2021-23	193.18
4	PGDTMA 2020	0.13
5	DTMA 2021	5.53
6	PGTETM 2020	0.16
7	DETM 2021	5.93

9.2.4 Resource Generation from other Activities (Rs. in lakhs)

Sl. No	Net Resource Generation from other activities	Receipts
1	Receipts from Farm Produce	3.37
2	Receipts from Guest House and Quarters	20.66
3	Receipts from Institute Training Programmes	21.80
4	Miscellaneous Receipts	219.85

9.2.5 Net Receipts from Sponsored Research Projects (Rs. in lakhs)

Sl. No	Sponsored Research Projects	Receipts
1	Agril. Res. & Dev. Infra. in Select Indian States under NSTMIS Scheme of DST	1.15
2	Profiling the Resources Devoted in Agricultural R&D Development of National framework for Evidence-based S&T Governance in India Scheme of DST	0.43
3	NASF, New Delhi - Entrepreneurship Dev. Through Farmer Led Innovations	1.43
4	Adoption of Natural Farming and its effect on Crop Yield and Farmers Livelihood in India NITI Ayog Project	7.75
5	Net receipts from Educational Programme 2019-21	200.00
6	Net receipts from IRGS Programmes	3.49
7	ICAR-NAARM-BCIL, New Delhi " Main Streaming Biodiversity Conservation in the Development Schemes of Agricultural Sector in India" (SNRM)	0.85
8	ICAR-NAARM-BCIL, New Delhi "Management Development Programme on Orientation cum awareness and Implementation of ABS guidelines (SNRM)	3.58

9.3 Plan Schemes (Rs. in lakhs)

Sl. No.	Name of the Scheme	Fund / Budget	Expenditure
1	National Agril. Innovation Fund (Component-I)-ZTMC	11.63	9.47
2	National Agril. Innovation Fund (Component-II)-ABI	1.60	0.23
3	XII Plan EFC Main Scheme KRISHI Project	11.27	7.84
4	Management & Impact Ass. of Farmer First Project	10.10	9.46
5	NASF, New Delhi- Entrepreneurship Dev. Through Farmer Led Innovations	9.88	8.21
6	Short Courses	7.87	0.00
7	NAHEP	122.06	62.59

Human Resources at the Academy (as on 31.12.2021)

Category	Sanctioned Strength	In position	Vacant
Scientific - RMP	02	02	0
Scientific - Faculty	60	32	28
Technical	29	24	05
Administrative	45	31	14
Skilled Support Staff	33	32	1
Skilled Support Staff (Additionally Sanctioned)	16	15	1
Total	185	136	49

Research Management Positions

1. Dr. Ch. Srinivasa Rao, Director
2. Dr. G. Venkateshwarlu, Joint Director

Scientific Staff

Agribusiness Management Division

1. Dr. Ranjit Kumar, Head
2. Dr. Tavva Srinivas, Principal Scientist
3. Dr. B. Ganesh Kumar, Principal Scientist
4. Dr. P.C. Meena, Principal Scientist
5. Dr. N. Sivaramane, Principal Scientist
6. Dr. Sanjiv Kumar, Scientist

Education Systems Management Division

1. Dr. S. Senthil Vinayagam, Head (I/c)
2. Dr. G.R. Ramakrishna Murthy, Principal Scientist
3. Dr. D. Thammi Raju, Principal Scientist

4. Dr. V.V. Sumanth Kumar, Senior Scientist

Extension Systems Management Division

1. Dr. Bharat S. Sontakki, Head (I/c)
2. Dr. Surya Rathore, Principal Scientist
3. Dr. P. Venkatesan, Principal Scientist
4. Dr. N.A. Vijay Avinashilingam, Principal Scientist

Human Resources Management Division

1. Dr. R.V. S Rao, Head (I/c)
2. Dr. P. Ramesh, Principal Scientist
3. Dr. K.H. Rao, Principal Scientist
4. Dr. Alok Kumar, Principal Scientist

Information and Communication Management Division

1. Dr. S. K. Soam, Head (I/c)
2. Dr. A. Dhandapani, Principal Scientist
3. Dr. S. Ravichandran, Principal Scientist
4. Dr. M. Balakrishnan, Principal Scientist
5. Dr. N. Srinivasa Rao, Principal Scientist
6. Dr. P.D. Sreekanth, Principal Scientist
7. Dr. B.S. Yashavanth, Scientist
8. Dr. Purru Supriya, Scientist

Research Systems Management Division

1. Dr. I. Sekar, Head (I/c)

2. Dr. K. Kareemulla,
Principal Scientist
3. Dr. M.B. Dastagiri,
Principal Scientist
4. Dr. P. Krishnan,
Principal Scientist,
(On deputation w.e.f.
29.12.2021)
5. Dr. Umesh Hudedamani,
Scientist
6. Dr. Ramesh Naik, Scientist

Administrative staff

Joint Director (Admn.) & Registrar

Sh. B. D. Phansal

Chief Finance & Accounts Officer

Sh. Z. H. Khilji

Senior Administrative Officer

Sh Mukul Raj Singh

Assistant Administrative Officer

1. Sh. M. Dinesh
2. Smt. K. K. Rukmani Ammal
3. Dr. K.R. Ghanshyam

Assistant Finance and Accounts Officer

Smt. N. Vijayalakshmi

Private Secretary

1. Sh. N. Raghunath
2. Smt. A. Mercy
3. Smt. T. Vanisri

Assistants

1. Sh. T. Srinivas
2. Sh. G. Raj Reddy

3. Sh. C. Phani Raj
4. Smt. B. Padma Saroja
5. Sh. P. Srinivasu
6. Sh. R. Chandra Babu
7. Sh. C. Julius Samuel
8. Sh. M. Sridhar
9. Smt. Y. Gayathri

Personal Assistants

1. Sh. T. V. Ramadas
2. Smt. Y. Anuradha
3. Smt. S. Sessa Sai
4. Smt. V. Shailaja
5. Smt. S. Shanthi

Upper Division Clerks

1. Sh. M. K. Samson
2. Smt. Rajashri Bokde
3. Sh. K.Suryanarayana
4. Sh. P. Swamy
5. Sh. M. Narsing Rao

Lower Division Clerk

1. Sh. M. Ashok
2. Sh. C. Bikshapathi

Technical Staff

Grade T-9 (Category-III)

1. Dr. M. A. Basith,
Chief Technical Officer
(Jr. Farm Superintendent)
2. Sh P .Namdev, Chief Technical
Officer (Artist)
3. Sh. P. Vijender Reddy,
Chief Technical Officer
(Research Assistant)
4. Sh. Sohail Ahmad Khan,
Chief Technical Officer
(Junior Engineer-Civil)

Grade T (7-8) (Category-III)

1. Ms. G. Aneeraj,
Assistant Chief Technical
Officer (Assistant Editor)

2. Sh. M. Shekhar Reddy,
Assistant Chief Tech. Officer
(Dark Room Assistant)
3. Dr. Ahire Laxman Maharur,
Assistant Chief Technical
Officer (Horticultural Technical
Assistant)
4. Sh. M. Ravi,
Assistant Chief Technical Officer
(Photographer-cum-Artist)
5. Sh. Sham Bahadur,
Asst. Chief Technical Officer
(Catering in-charge)
6. Sh. P. Mohan Singh,
Asst. Chief Technical Officer
(Computer Assistant)
7. Ms. Savithri Murali,
Asst. Chief Technical Officer
(Catering in-charge)

Grade T-6 (Category-III)

1. Sh. Ravi Viswanathan,
Sr. Technical Officer (Editor-
cum-Information Officer)

Grade T-5 (Category-II)

1. Sh. K. Shivaiah,
Technical Officer (Technician)
2. Sh. D. Rajagopal Rao,
Technical Officer
(Media Operator)
3. Sh. S. Rajukumar,
Technical Officer (Electronic
Computer Operator)
4. Sh. T. Laxman,
Technical Officer (Driver)
5. Sh. A. C. P. Rama Nageswara
Rao, Technical Officer
(Lineman)

Grade T-4 (Category-II)

1. Sh. Pitla Srinivas, Sr. Technical
Assistant (Proof Reader)
2. Sh. N. Ashok, Sr. Technical
Assistant (Driver)



3. Sh. N. Prabhakar Rao, Sr. Technical Assistant (Plumber)
4. Sh. M. Srinivasa Rao, Sr. Technical Assistant (Pump Driver)
5. Sh. B.K. Venkatram, Senior Technical Assistant (Pump Driver)

Grade T-3 (Category-II)

1. Sh. R. Siva Prasad, Technical Assistant (Driver)
2. Sh. Danam Murahari, Technical Assistant (A. V. Operator)

Skilled Support Staff

1. Sh. Sirigiri Venkatesham
2. Ms. S. Shakuntala
3. Sh. Kumba Satyanarayana
4. Sh. K. Pentaiah
5. Sh. P. Yadaiah
6. Sh. G. Pentaiah
7. Sh. J. Chandraiah
8. Sh. R. Sattaiah
9. Sh. L. Satyanarayana
10. Ms. C. Kausalya
11. Sh. M. Ganesh Kumar
12. Sh. Chillumula Venkatesham
13. Sh. Sirigiri Narsimha
14. Sh. Kyasam Satyanarayana
15. Sh. B. Premdas
16. Smt. V. Saroja
17. Sh. S. Nayab Rasool
18. Sh. B. Ashok
19. Sh. K. Daniel
20. Sh. M. Satyanarayana
21. Sh. D. Srisailam
22. Sh. B. Ratnaiah
23. Sh. D. Babaiah

24. Sh. L. Narasimha
25. Sh. K. Samson
26. Sh. K. Shamulu
27. Sh. K. Satyanarayana
28. Sh. K. Narasimha
29. Smt. K. Andalu
30. Sh. M. S. Ravi
31. Sh. N. Krishna
32. Smt. P. Amrutha
33. Smt. K. Lakshmi
34. Smt. L. Chandramma
35. Sh. G. Stayanarayana
36. Sh. B. Raju
37. Sh. T. Srinivas
38. Sh. Md. Sadat
39. Smt. E. Yadamma
40. Smt. S. Lakshamma
41. Sh. P. Venkatesh
42. Sh. G. Ramulu s/o Laxmaiah
43. Sh. P. Yadaiah S/o Kistaiah
44. Smt. J. Pramila
45. Smt. M. Iaxmamma
46. Sh. G. Ramulu S/o Danaiah
47. Smt. M. Sharada

Appointments / New Joinings

1. Sh. Shamulu, CLTS as Skilled Support Staff (SSS) w.e.f. 29.01.2021 on regularization of his services
2. Sh. K. Satnarayana/ CLTS as Skilled Support Staff (SSS) w.e.f. 29.01.2021 on regularization of his services
3. Sh. K. Narasimha, CLTS as Skilled Support Staff (SSS) w.e.f. 29.01.2021 on regularization of his services
4. Smt. V. Lakshmi, CLTS as Skilled

Support Staff (SSS) w.e.f. 30.01.2021 on regularization of her services

5. Smt. K. Andalu, CLTS as Skilled Support Staff (SSS) w.e.f. 30.01.2021 on regularization of her services
6. Sh. M. S. Ravi, CLTS as Skilled Support Staff (SSS) w.e.f. 30.01.2021 on regularization of his services
7. Dr. Vijay Avinashilingam N. A., joined the post of Principal Scientist (Agricultural Extension) w.e.f. 18.02.2021 on his transfer from ICAR-CAZRI, Jodhpur
8. Dr. G. Venkateswarlu, joined the Academy as Joint Director w.e.f. 29.04.2021
9. Sh. N. Krishna, CLTS as Skilled Support Staff (SSS) w.e.f. 09.08.2021 on regularization of his services
10. Smt. P. Amrutha, CLTS as Skilled Support Staff (SSS) w.e.f. 09.08.2021 on regularization of her services
11. Smt. K. Lakshmi, CLTS as Skilled Support Staff (SSS) w.e.f. 30.08.2021 on regularization of her services
12. Smt. L. Chandramma, CLTS as Skilled Support Staff (SSS) w.e.f. 30.08.2021 on regularization of her services
13. Sh. G. Satyanarayana, CLTS as Skilled Support Staff (SSS) w.e.f. 30.08.2021 on regularization of his services
14. Sh. B. Raju, CLTS as Skilled Support Staff (SSS) w.e.f. 30.08.2021 on regularization of his services

15. Sh. Md. Sadath,
CLTS as Skilled Support Staff
(SSS) w.e.f. 30.08.2021 on
regularization of his services

16. Sh. T. Srinivas,
CLTS as Skilled Support Staff
(SSS) w.e.f. 30.08.2021 on
regularization of his services

17. Smt. E. Yadamma,
CLTS as Skilled Support Staff
(SSS) w.e.f. 30.08.2021 on
regularization of her services

18. Smt. S. Laxmamma,
CLTS as Skilled Support Staff
(SSS) w.e.f. 30.08.2021 on
regularization of her services

19. Sh. P. Venkatesh,
CLTS as Skilled Support Staff
(SSS) w.e.f. 30.08.2021 on
regularization of his services

20. Sh. G. Ramulu (S/o. Laxmaiah),
CLTS as Skilled Support Staff
(SSS) w.e.f. 30.08.2021 on
regularization of his services

21. Sh. P. Yadaiah,
CLTS as Skilled Support Staff
(SSS) w.e.f. 30.08.2021 on
regularization of his services

22. Smt. J. Pramila,
CLTS as Skilled Support Staff
(SSS) w.e.f. 30.08.2021 on
regularization of her services

23. Smt. M. Laxmamma,
CLTS as Skilled Support Staff
(SSS) w.e.f. 30.08.2021 on
regularization of her services

24. Sh. G. Ramulu (S/o. Danaiah),
CLTS as Skilled Support Staff
(SSS) w.e.f. 30.08.2021 on
regularization of his services

25. Smt. M. Sharada,
CLTS as Skilled Support Staff
(SSS) w.e.f. 30.08.2021 on
regularization of her services

Promotions / Financial upgradations under MACPS Promotions

1. Sh. V. V. Sumanth Kumar,
Sr. Scientist has been
promoted under CAS to the
grade of Principal Scientist
(RGP 10000) w.e.f. 16.04.2020.

2. Sh. Mukul Raj Singh,
Administrative Officer has
been promoted to the post of
Senior Administrative Officer
w.e.f. 29.06.2021

3. Smt. N. Vijaya Lakshmi,
Junior Accounts Officer has
been promoted to the post of
Assistant Finance & Accounts
Officer w.e.f. 28.06.2021

Retirement list from 1st January to 31st December, 2021

1. Sh. K. V. Kumar,
T-9 Chief Technical Officer
(T-3 Statistical Assistant)
superannuated on 28.02.2021

2. Sh. G. Muthyalu,
T-4 Senior Technical Assistant
(T-1 Driver) superannuated on
31.03.2021

3. Sh. Naresh Kumar N,
T-5 – Technical Officer (T-2
VCR-cum-Tele Cine Operator)
retired on superannuated
w.e.f.31.05.2021 (AN)

4. Sh. Chandramouli C,
Skilled Support Staff (SSS)
retired on superannuated
w.e.f.31.06.2021

5. Smt. Laxmi V,
Skilled Support Staff (SSS)
retired (on superannuation)
w.e.f. 31.07.2021 (AN)

6. Smt. Bharatamma I,
Skilled Support Staff (SSS)
retired (on superannuation)
w.e.f. 31.08.2021 (N)



List of Committees

1. Higher Education Committee

1. Dr. Ch. Srinivasa Rao, Director - Chairman
2. Dr. G. Venkateshwarlu, Joint Director - Member
3. Dr. Bharat Sontakki, Head, XSM Division - Member
4. Dr. S. Senthil Vinayagam, I/C Head, ESM Division - Member
5. Dr. I. Sekar, Head, RSM Division - Member
6. Dr. Ranjit Kumar, Head, ABM Division - Member
7. Dr. R V S Rao, Head, HRM Division - Member
8. Dr. P. Krishnan, Pr. Scientist - Member
9. Dr. P. Ramesh, Principal Scientist & Controller of Examinations (CoE) - Member
10. Coordinators of PGDM-ABM, DTMA, DETM Co-opted - Members
11. Dr. P. C. Meena, Pr. Scientist & OIC, P.G.S Unit - Member Secretary

2. Board of Studies

1. Dr. Ranjit Kumar, Head, ABM Division-Chairman
2. Dr. S.K. Soam Head, ICM Division - Member
3. Dr. Bharat S Sontakki, Head, XSM Division - Member
4. Dr. K. H. Rao, Principal Scientist & Advisor, Placement Cell - Member
5. Dr. A. Dhandapani, Pr. Scientist, ICM Division - Member
6. Dr. P Ramesh Pr. Scientist & Controller of Examination - Member
7. Dr. D. Thammi Raju, Pr. Scientist, ESM Division - Member
8. Dr. B. Ganesh Kumar, Pr. Scientist, ABM Division - Member
9. Shri. B. D. Phansal, Joint Director (Admn.) & Registrar - Member
10. Shri. Z. H. Khilji, CFAO - Member

11. Coordinators of PGDM-ABM, TMA, ETM - Co-opted Members

12. Dr. P.C. Meena, Pr. Scientist & OIC, P.G.S Unit - Member Secretary

3. Works Committee

1. Dr. R. V. S. Rao, Pr. Scientist and I/C Head, HRM Div. - Chairman
2. Dr. S. Ravichandran, Pr. Scientist - Member
3. Dr. G.R. Ramakrishna Murthy, Pr. Scientist - Member
4. Dr. M. Balakrishnan, Pr. Scientist - Member
5. Dr. P. D. Sreekanth, Pr. Scientist - Member
6. Joint Director (Admn.) & Registrar - Member
7. Chief Finance & Accounts Officer - Member
8. Sh. Sohail Ahmad Khan - Member Secretary

4. Anti-Ragging Committee

1. Joint Director / Dean - Chairman
2. Dr. I. Sekar, Head, RSM Division - Member
3. Joint Director (Admn.) & Registrar - Member
4. Academic Coordinators, PGDM (A) Programme - Member
5. Dr. P. C. Meena, Pr. Scientist & OIC, P.G.S Unit - Member Secretary

5. Anti-Ragging Squad

1. Dr.(Mrs.) Surya Rathore - Chairman
2. Dr. B. Ganesh Kumar, Pr. Scientist, ABM Division - Member
3. Dr. Umesh Hudedamani, Scientist - Member
4. Chief Finance & Accounts Officer - Member
5. Officer Incharge, Hostel - Member Secretary

6. Grievance Committee

1. Dr. G. Venkateshwarlu, Joint Director - Chairman
2. Dr. Bharat Sontakki, Head, XSM Division - Member
3. Sh. Z. H. Khilji, CFAO - Member
4. SAO - Member
5. Dr. V. V. Sumanth Kumar, Pr. Scientist - Member
6. Sh. T. Laxman, Technical Officer - Member
7. Sh. P. Swamy, LDC - Member
8. Smt. Saroja, SSS - Member

9. Sh. M. Dinesh, AAO - Member Secretary

7. Farm Advisory Committee

1. Dr. Mude Ramesh Naik - Chairman
2. Dr. B. S. Yashavanth, Scientist - Member
3. Dr. P. Vijender Reddy, CTO - Member
4. Dr. L. M. Ahire, ACTO - Member
5. Dr. P. Mohan Singh, ACTO - Member
6. Sh. M. Shekhar Reddy, ACTO - Member
7. Sh. T. Laxman, TO - Member
8. Sh. Danam Murahari, TA - Member
9. Dr. M. A. Basith, ACTO - Member Secretary

8. Library Advisory Committee

1. Dr. R. V. S. Rao,
Pr. Scientist and I/C Head, HRM Div - Chairman
2. All HoDs - Member
3. Dr. P. Krishnan,
Pr. Scientist, RSM Division - Member
4. Dr. V. V. Sumanth Kumar, Pr. Scientist - Member
5. Joint Director (Admn.) & Registrar - Member
6. Chief Finance & Accounts Officer - Member
7. Dr. N. Srinivasa Rao,
Pr. Scientist - Member Secretary

9. Newsletter Committee

1. Dr. Ranjit Kumar, Head, ABM Division - Chairman
2. Dr. Tavva Srinivas, ABM Division - Member
3. Dr. Sanjiv Kumar,
Scientist, ABM Division - Member
4. Dr. B. S. Yashavanth, Scientist, ICM Division
Member Secretary

10. Annual Report Committee

1. Joint Director - Chairman
2. Head, ICM Divn - Member
3. Head, RSM Divn - Member
4. Head, HRM Divn - Member
5. Head, ESM Divn - Member
6. Head, ABM Divn - Member
7. Head, XSM Divn - Member
8. Dr. Tavva Srinivas, Head, PME Cell - Member

9. Dr. Sanjiv Kumar, Scientist, ABM Division-Member

10. Dr. BS Yashavanth, Scientist, ICM Division-Member

11. Dr. Alok Kumar,
Pr. Scientist, HRM Division - Member Secretary

11. Campus Residents Welfare Committee

1. Dr. S.K. Soam, Head, ICM Division - Chairman
2. Dr. Umesh Hudedamani, Scientist - Member
3. Dr. Mude Ramesh Naik, Scientist - Member
4. Sh. Z. H. Khilji, CFAO - Member
5. Dr. B. S. Yashavanth, Scientist - Member Secretary
6. Sh. Shitantsu Kumar - Out Side Member
7. Sh. Ch. Suresh Kumar,
Asst, ICAR-IIMR - Out Side Member

12. Examination Unit

1. Dr. P. Ramesh,
Pr. Scientist & Controller of Examination
2. SAO - Member
3. Sh. Pitla Srinivas, STA - Member
4. Mrs B. Padam Saroja, Assistant - Member
5. Sh. B Ashok, SSS, Training Unit - Member

13. Hostel Management Committee

1. Dr. M. Balakrishnan, Pr. Scientist - Chairman
2. Dr. Umesh Hudedamani, Scientist - Member
3. Joint Director (A) & Registrar - Member
4. Chief Finance & Accounts Officer - Member
5. Smt. Savithri Murali, Sr. Tech Officer - Member
6. Sh. A. C. P. R. Nageshwara Rao,
Tech Assistant - Member
7. Sh. Sham Bahadur, ACTO - Member Secretary

14. Institute Technology Management Committee

1. Dr. Ch. Srinivasa Rao, Director - Chairman
2. Joint Director - Member
3. Dr. S. Senthil Vinayagam,
I/C Head, ESM Division - Member
4. Dr. G. Rama Krishna Murthy, Pr. Scientist - Member
5. Dr. V. V. Sumanth Kumar, Pr. Scientist - Member



6. Joint Director (Admin) Registrar - Member
7. CFAO - Member
8. Dr. R.M. Sundaram,
Pr. Scientist, IIRR, Hyderabad - Member
9. Dr. Umesh Hudedamani, Scientist - Member
Secretary

15. Sports Complex Maintenance Committee

1. Dr. P. Vijender Reddy, CTO - Chairman
2. Dr. M. A. Basithm, CTO - Member
3. Sh. Sohail Ahmad Khan, ACTO - Member
4. Dr. Laxman M Ahire, ACTO - Member
5. Sh. M. Shekar Reddy, ACTO - Member
6. Sh. P. Mohan Singh, ACTO - Member
7. Sh. Sham Bahadur, ACTO - Member
8. Sh. M. K. Samson, UDC - Member
9. Dr. A. Debnath, Medical Consultant - Member
10. Smt. K. K. Rukmani Ammal, AAO - Member
Secretary

16. Campus Canteen Committee

1. Dr. Laxman M Ahire, ACTO - Chairman
2. Smt. Y. Gayathri, UDC - Member
3. Representative from FOCARS - Member
4. Two Representatives each from PGDMA I & II -
Member
5. Secretary, IJSC - Member
6. Member, CJSC - Member
7. Sh. N. Prabhakar, STA - Member Secretary

17. Internal Complaints Committee

1. Dr. (Mrs.) Surya Rathore, Pr. Scientist - Chairman
2. Smt. G. Aneja, ACTO - Member
3. Smt. T. Vanisri, Pvt. Secretary - Member
4. Smt. M. Vijayalaxmi, AFAO - Member
5. Sh. Mukul Raj Singh, SAO - Member
6. Smt. K. K. Rukmani Ammal, AAO - Member Secretary

18. Biometric and Performance Monitoring Committee

1. Dr. Ch. Srinivasa Rao, Director - Chairman
2. Joint Director (Admin) & Registrar - Member

3. CFAO - Member
4. Sh. O.S. Rajukumar,
Nodal Officer, AEBAS - Member Secretary

19. Academic Committee Meeting Committee

1. Dr. Ch. Srinivasa Rao, Director - Chairman
2. All Faculty - Member
3. Dr. A. Dhandapani,
Pr. Scientist, Member Secretary-1
4. Dr. Alok Kumar,
Pr. Scientist, Member Secretary-2

20. Purchase Advisory Committee (Stores)

1. Dr. A. Dhandapani,
Pr. Scientist, ICM Division - Chairman
2. Dr. P. Ramesh, Pr. Scientist - Member
3. Dr. Umesh Hudedamani, Scientist - Member
4. Joint Director (Admn.) & Registrar - Member
5. CFAO - Member
6. Dr. P. Vijender Reddy, CTO - Member
7. Sh. M. Sridhar, Assistant - Member
8. Sh. Mukul Raj Singh, SAO - Member Secretary

21. Technical Advisory Committee (Stores)

1. Dr. GRK Murthy, Pr. Scientist - Chairman
2. Dr. V. V. Sumanth Kumar, Pr. Scientist - Member
3. Sh. S. Rajukumar, Technical Officer- Member Secretary

22. Placement Committee for PGDM-ABM

1. Dr. K. H. Rao, Pr. Scientist - Placement Advisor
2. Dr. B. Ganesh Kumar,
Pr. Scientist - Placement Coordinator - 1
3. Dr. N. Sivaramane, Pr. Scientist Placement
Coordinator - 2

23. Priority Setting, Monitoring and Evaluation Committee

1. Dr. Tavva Srinivas, Pr. Scientist - Head
2. Dr. P. Krishnan, Pr. Scientist - Member
3. Dr. Umesh Hudedamani, Scientist - Member

24. Official Language Committee

1. Dr. Ch. Srinivasa Rao, Director - Chairman

2. Dr. N. Srinivasa Rao, Pr. Scientist - Member
3. Dr. S. K. Soam, Head, ICM - Member
4. Dr. R.V.S Rao, Pr. Scientist - Member
5. Joint Director (Admn.) & Registrar - Member
6. Sr. Administrative Officer - Member
7. Sh. Pradeep Singh, AD(OL) - Member Secretary

25. Admission Committee PGDM-ABM / DTMA / DETM (2021-23)

1. Dr. Ranjit Kumar, Head, ABM Division - Chairman
2. Dr. Bharat S. Sontakki,
Head, XSM Division - Member
3. Dr I. Sekar, Head, RSM Division - Member
4. Dr. R. V. S. Rao, Head, HRM Division - Member
5. Dr. S. Senthil Vinayagam, Head, ESM - Member
6. Dr. A. Dhandapani,
Pr. Scientist, ICM Division - Member
7. Dr. N. Sivaramane,
Pr. Scientist, ABM Division - Member

8. Shri. B. D. Phansal,
Joint Director (Admn.) & Registrar - Member
9. Dr. P.C. Meena,
Pr. Scientist & OIC, P.G.S Unit Member - Secretary

26. Academic Coordinators-PGDM-ABM

1. Dr. B. Ganesh Kumar, Pr. Scientist,
ABM Division-Principal Course Coordinator
2. Dr. P.C. Meena, Pr. Scientist,
ABM Division - Academic Coordinator
3. Dr Sanjiv Kumar, Scientist,
ABM Division- Academic Coordinator

27. Academic Coordinators-DTMA

1. Dr. K. Kareemulla, Pr. Scientist, RSM Division
2. Dr. Umesh Hudedamani, Scientist, RSM Division

28. Academic Coordinators-DETM

1. Dr. S. Senthil Vinayagam, Head, ESM Division
2. Dr. N. S. Rao, Pr. Scientist, ICM Division

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भाकृअनुप-राष्ट्रीय कृषि अनुसंधान प्रबंध अकादमी

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